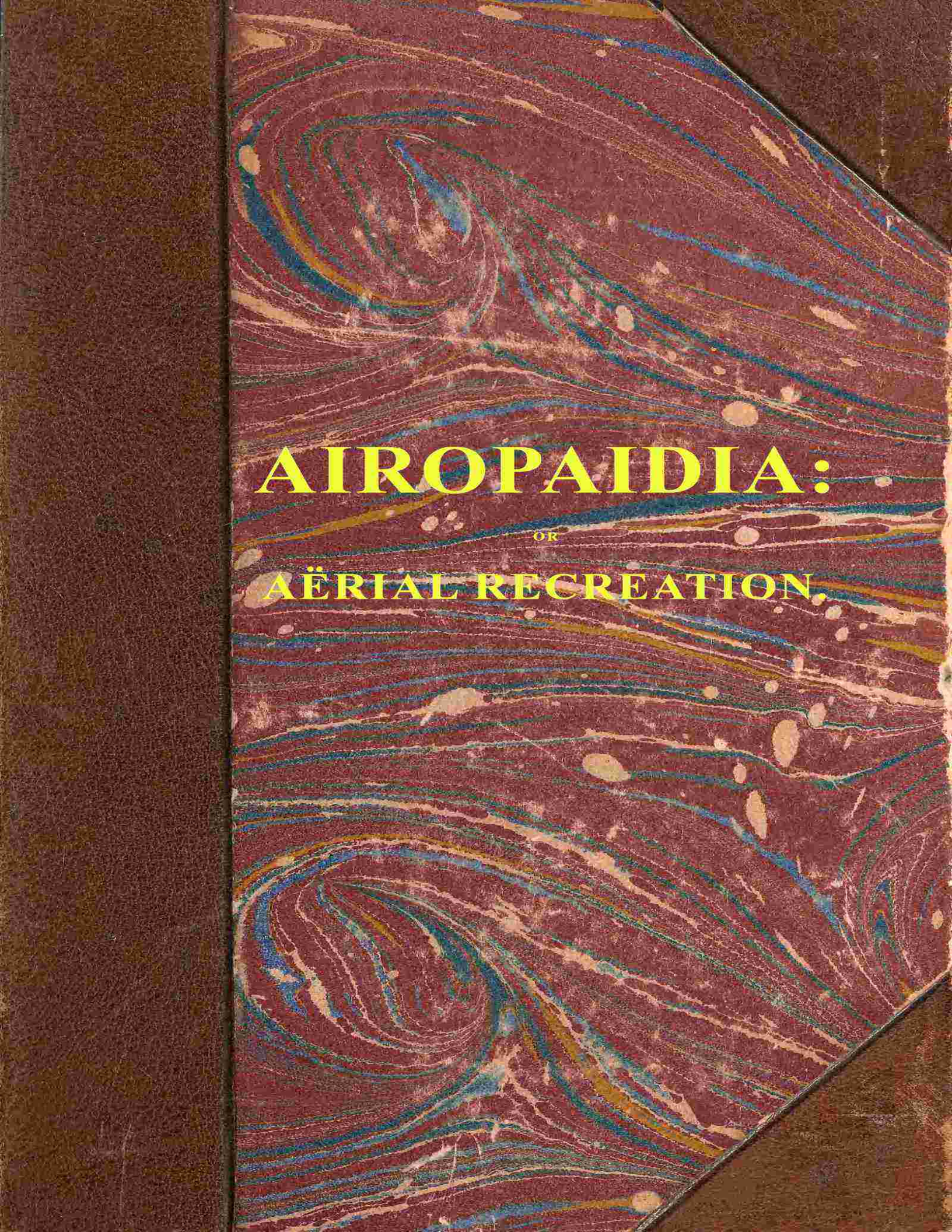
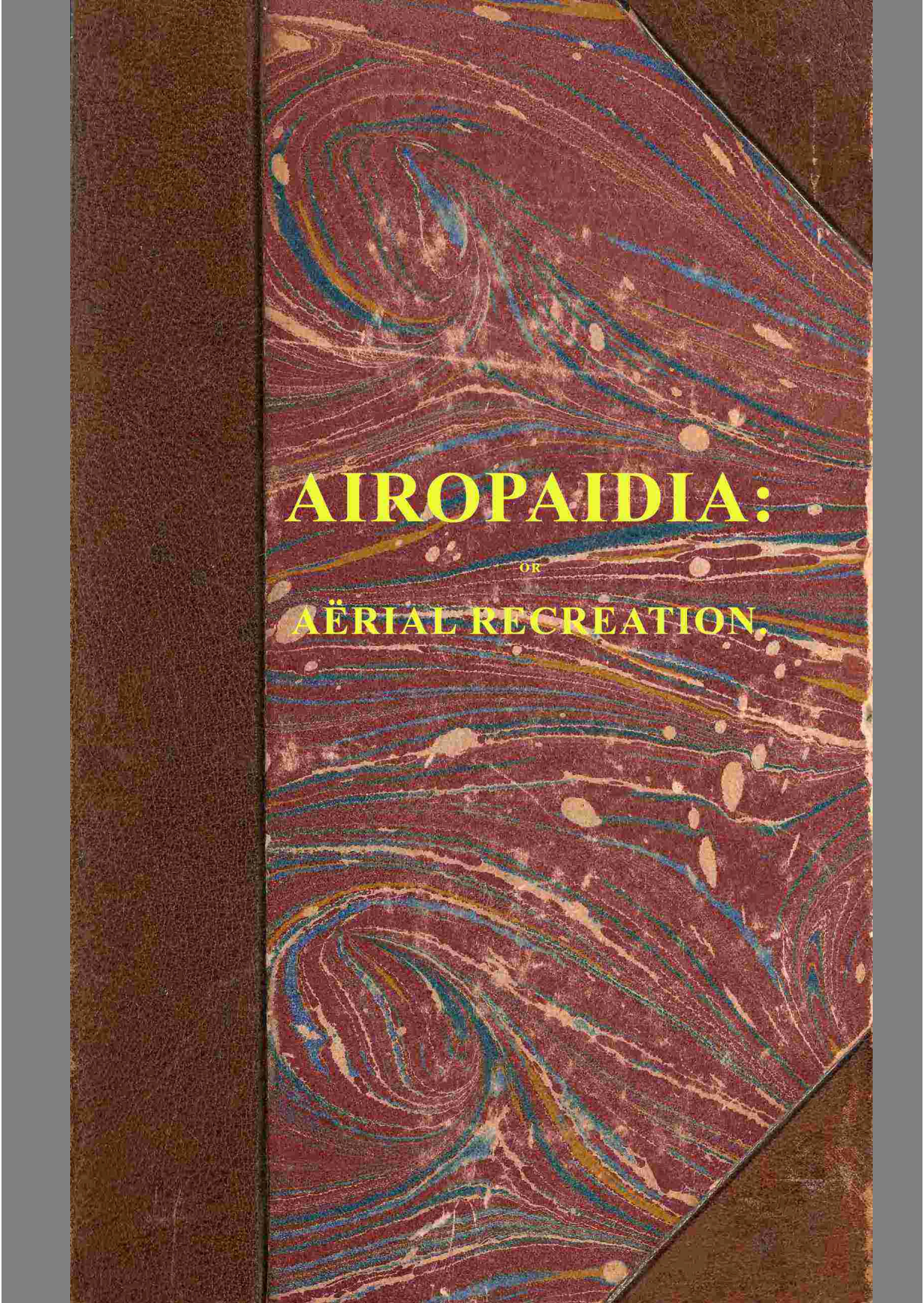


The image shows the front cover of a book. The spine and the four corners are bound in a dark brown, textured material, likely leather or cloth. The central part of the cover is decorated with a marbled paper pattern. This pattern consists of swirling, wavy lines in shades of deep red, blue, green, and yellow, set against a lighter, cream-colored background. The marbling has a classic, organic feel. Centered on the marbled area is the title of the book in a bold, yellow, serif typeface.

AIROPAIDIA:

OR

AËRIAL RECREATION.

The image shows the front cover of a book. The spine and the four corners are bound in a dark brown, textured material, likely leather or cloth. The central part of the cover is decorated with marbled paper. The marbling pattern consists of swirling, wavy lines in shades of deep red, blue, green, and yellow, set against a lighter, cream-colored background. The text is printed in a yellow, serif font. The main title 'AIROPAIDIA:' is in a large, bold font. Below it, the word 'OR' is in a smaller font. The subtitle 'AËRIAL RECREATION.' is in a medium-sized font, matching the main title's style. The overall appearance is that of an antique or vintage book.

AIROPAIDIA:

OR

AËRIAL RECREATION.

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Author: Thomas Baldwin

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*** START OF THE PROJECT GUTENBERG EBOOK AIROPAIDIA

AIROPAIDIA:

O R

AËRIAL RECREATION.

DESCRIPTIONS of the aërial Scenes are illustrated with
ENGRAVINGS, by the best Masters: two of which are COLOURED.

The one, a circular View from the Balloon at its greatest Elevation; the City of Chester appearing in the Center.

The other, a Specimen of Balloon Geography: being a Prospect from above the Clouds, of the Country between Chester, and Warrington in Lancashire, with the Track of the Balloon in the Air.

A third represents the Balloon over Helsbye-Hill in Cheshire, with a beautiful View of the adjacent Country.

AIROPAIDIA:
CONTAINING
THE NARRATIVE OF A
BALLOON EXCURSION
from CHESTER, the eighth of September, 1785,
taken from MINUTES made DURING the Voyage:
H I N T S
ON THE IMPROVEMENT OF BALLOONS,
AND MODE OF INFLATION BY *STEAM*:
MEANS TO PREVENT THEIR DESCENT OVER WATER:
OCCASIONAL ENQUIRIES
INTO THE STATE OF THE ATMOSPHERE,
FAVOURING THEIR DIRECTION:
WITH VARIOUS PHILOSOPHICAL
OBSERVATIONS AND CONJECTURES.
TO WHICH IS SUBJOINED,
MENSURATION OF HEIGHTS BY THE BAROMETER,
MADE PLAIN:
WITH EXTENSIVE TABLES.
The WHOLE serving as an INTRODUCTION to
AËRIAL NAVIGATION:
WITH A COPIOUS INDEX.

BY THOMAS BALDWIN, Esq. A. M.

- - - - Addita NAVIGIIS sunt
Multa. *Lucretius De Rerum Nat. L. 5, V. 335.*

Nihil PERFECTUM simul ac INCEPTUM.

USUS UNI REI deditus, et NATURAM et ARTEM sæpe vincit. *Cicero.*

C H E S T E R:

Printed for the Author, by J. Fletcher; and sold by W. Lowndes, No. 77,
Fleet-street, London; J. Poole, Chester; and other Booksellers.

1786.

Price, in Boards, 7s. 6d.

TO THE

PRINCIPAL INHABITANTS

OF

C H E S T E R:

For their POLITE ATTENTION on the Day of Ascent, and Preservation of ORDER during the INFLATION: on which, the Success of aërial Experiments so much depends, and throu' the Want of which, so many have already failed; for the kind Anxiety manifested during his Absence; and for their friendly CONGRATULATIONS, on his safe Return; the following Account of the Balloon-Excursion, written at their Request, is, by their Permission, with all Gratitude, Esteem and Respect,

DEDICATED,

by their most obliged,
and most obedient Servant

THE **AIRONAUT.**

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AN ACCOUNT OF THE PLATES; WITH DIRECTIONS FOR PLACING THEM.

1st. *An Account of the Plates.*

1. (a) A *Circular View from the Balloon at its greatest Elevation, ([Page 58.](#))* The Spectator is supposed to be in the Car of the Balloon, suspended above the *Center* of the View: looking *down* on the Amphitheatre or *white* Floor of Clouds, and seeing the *City of Chester*, as it appeared throu' the *Opening*: which discovers the Landscape *below*, limited, by surrounding Vapour, to something less than *two* Miles in Diameter.

The Breadth of the *blue* Margin defines the *apparent* Height of the Spectator in the Balloon (viz. 4 Miles) *above* the *white* Floor of Clouds, as he hangs in the Center, and looks *horizontally* round, into the *azure* Sky.

2. (b) *The Balloon over Helsbye-Hill in Cheshire, at half past II. on Thursday the 8th of September, 1785. ([Page 78.](#))*

It is seen in the *South-west-Quarter*.

The View was taken in a *high* Field, at the End of Sutton-Causeway.

Helsbye-Hill, tho' upwards of 600 *Feet high*, appeared from the Car of the Balloon, to be on the *same Level* with the *Grounds below*.

3. (c) A *Balloon-Prospect from above the Clouds*, ([Page 154](#)), or *Chromatic View* of the Country between Chester, Warrington and Rixton-Moss in Lancashire: shewing the whole Extent of the aërial Voyage; with the *meandering* Track of the Balloon throu' the Air.

4. *The Explanatory Print* (d), ([Page 155](#);) which elucidates the former by giving the Names of the principal Places mentioned in the Excursion.

N. B. *The Circular View* is seen to the best Advantage, when placed *flat* on a Table or Chair, and *rather* in the Shade: the Eye looking *directly* down upon the Picture.

Whoever will be at the Trouble of viewing *distinct* Parts of the *Balloon-Prospect*, throu' a very small Opening, made by rolling a Sheet of Paper into the Form of a hollow Tube, and applying it close to either Eye, at the same Time shutting the other; or by looking throu' the Hand, held a little open, and close to the Eye; may form a very accurate Idea of the Manner, in which the *Prospect below* was represented *gradually in Succession*, to the Aironaut; whose Sight was bounded by a Circularity of Vapour, as in Section [79](#), [221](#).

2d. Directions for placing them.

Place the *Top* of the CIRCULAR VIEW, *even* with the *Top* of the Page.

The Plate will then lye over at the Bottom, and at the *right* Side of the Page.

Fold the Bottom up into the Book, even with the Margin: and the *right* Side in like Manner.

Observe to place the *Bottom* of each of the other Plates, even with the *Bottom* of the Page.

The Plate will then lye *over* at the Top, and at the *right* Side of the Page.

Fold the Top *down*, into the Book, *even* with the Margin: and the right Side, in like Manner.

The circular View, to *face* Page 58.

The Balloon over Helsbye-Hill, to *face* Page 78.

The Balloon Prospect, to *face* Page 154.

The explanatory Print, to be placed *on* Page 155: and, when unfolded, to be seen *along with* the Balloon-Prospect.

Literal and other Errors proper to be examined, and corrected with the Pen, before the Book is read.

Page.

6. [Note \[1\]](#)—εἰαδεν *write* εὐαδεν.
18. [Note \[6\]](#)—*Cube* of the Velocity, &c. *write* (as in some Copies) *Square* of the Velocity, &c. the Resistance will be as $3 \times 3 = 9$.—See Chambers's Dictionary, under RESISTENCE.
23. Section 21. *Blot out* [Signs] of Currents.
26. *Before* All Things being thus prepared, *insert* [Section] viz. 25.
35. Line 13.—I o'Clock, *write* I. o'Clock.
54. Section 52.—an Extent above them of 77 Miles, *write* an Extent of 102 Miles. See the Occasion of this Mistake in [Note \[18\]](#) Calculation SECOND, which makes the Answer 102 Miles, 1 Quarter, 320 Yards; and the Ans. to the PROBLEM being
- | | | |
|--|---|----------|
| 102 | 1 | 307 |
| <hr/> | | |
| gives the Prospect | | 13 Yards |
| less than <i>that</i> over the Clouds. | | |
84. Line 4.—great Turnpike-Road, *write* great public Road.
84. [Note \[26\]](#) *After* See Moore's Practical Navigator, *insert* See Page 98 [34].
98. *After* [Note \[34\]](#) *add* See Section 84, Note [26].
118. Line 5.—from a vertical Situation only, to be seen, *write* to be seen from a vertical Situation only:.
174. Line 1.—excessine Diminution *write* excessive Diminution.
177. Line 9.—contain *write* contains.
202. The Sections 259, 260, 261, are repeated.
234. Line 6.—a Yard, *write* two Yards.
236. Line 3. *After the Words* in Danger of breaking; *add* the Bottom of the Balloon must be opened, or the upper Valve drawn. *And erase the Remainder of the Sentence.*
237. Line 4.—which is a Sign that the Balloon descends, *write* (which is a Sign that the Balloon descends).
242. Line 21.—supercede, *write* supersede.

Page.

263. Line 5.—commonly: ascend, *write* commonly ascend:.
266. Line 21.—their Passage *write* its Passage.
271. Line 14.—each 4 Feet *write* each 4 Inches.
278. Lines 15 and 18.—third Tables *and* third Table, *write* fourth Tables *and* fourth Table.
283. [Note \[119\]](#).—more than the three first Decimals *write* more than the four first Decimals.
288. Line 5. *After* .0000076, *insert* which, being divided by .1, gives a Cypher less.
288. Line 11.—with 4° on .25, *write* with 4° on 25.
290. [Note \[120\]](#)—at low Water.) *write* at low Water.
292. Line 23.—there will remain the greater Height, *write* there will remain, *SECONDLY*; (see Section 367) the greater Height.
303. Line 6.—(viz. the 8,) *write* (viz. the .8,).
309. Line 10. Marginal Note.—*7th Step in Section 366. write 7th Step in Section 368.*
310. *After* Line 23, *insert* Air-Thermom. 56°.
311. Line 1.—By the Practice of the first Example, *write* Practice of the second Example.
312. Line 29. *After* The Answer, &c. *insert*, made by rejecting a Cypher,.
317. Before the last Line but two, *insert* END OF THE FIRST STAGE.
318. Line 23.—the 2d Tenth, *write* the 1st Tenth.
319. Line 13.—, gives 7. *write*, gives 97.
322. last Line but two.—and the remaining Feet *write* the remaining Feet.

AIROPAIDIA:

CHAPTER I.

Section 1. **T**HE Public have, for a considerable Time, been entertained with Accounts of aërial Voyages.

Such Accounts are, in many Respects, vague and unsatisfactory: by no Means adequate to the Expectations and Wishes, which have been formed by those, who have not yet penetrated the profound Heights of the Atmosphere.

2. The Voyagers have, now and then, been pretty accurate in Regard to Time Place Distance and Velocity: Circumstances highly worthy of Remark, in order to estimate the Improvement already made in this wonderful Discovery, and point out its Use: but neither ought the *several Occasions of Failure* in the Experiments to be omitted; as they will be found to arise more from a Want of Prudence and Foresight in the Managers, than from any Defect in the Machine, or the Principle on which it acts. Such Failure ought therefore to throw an additional Light and Credit on the Art: and give a Spur to Ingenuity, which, it is not to be doubted, will continue to drive forwards with the same rapid Success; nor rest, till the Art itself is brought to the highest Degree of Perfection; till airostatic Ships make the Circuit of the Globe: a NAVIGATION which, from its Novelty and Importance, deserves to be considered in a separate Treatise.

3. Balloon-Voyagers have likewise been particularly defective in their Descriptions of aërial Scenes and Prospects: those Scenes of majestic Grandeur which the unnumbered Volumes of encircling Clouds, in most fantastic Forms and various Hues, beyond Conception glowing and transparent, portray to a Spectator placed as in a Center of the Blue Serene above them: contemplating at the same Instant, and *apparently* at some Miles Distance immediately below, a most exquisite and ever-varying Miniature of the *little Works of Man*, heightened by the supreme Pencil of Nature, inimitably elegant, and in her highest Colouring.

Such are the Scenes which, Ballooners all allow, constitute the true Sublime and Beautiful: inspire Ideas of rational Humiliation to a thinking Mind, and raise the most careless Mortal to an unknown Degree of enthusiastic Rapture and Pleasure.

Every Beholder is a Judge of the Scenery around him: and no one, it is presumed, ever ascended into the Atmosphere on a *mild Day*, with a sound and well ballasted Balloon, that did not wish to taste the Luxury of a second Voyage.

4. Yet notwithstanding, as Ignorance is known to be the Parent of Fear, the Bulk of Mankind, which are by far the greater Number, will long continue to entertain absurd Apprehensions concerning it; to oppose and ridicule the Invention; as they will oppose every other Discovery, which they have neither Talents *Inclination* or Leisure to understand.

This Reflexion should, on the contrary, rather excite than check the Ardor of the Skilful and Scientific, to cherish and promote the Art.

In the History of Airostation, each Event is yet new and *uncompared*. Every Circumstance ought therefore to be carefully recorded: since it would be unfair to fix Bounds to Science; or argue, that such Inferences, as shall demonstrate the great Utility of the Invention, may not be drawn from Circumstances which Inattention might pronounce to be most trifling and minute.

5. The Reader is requested to observe that, this Account being addressed to the Generality, and not to the Curious and Philosophic only; many Circumstances are added, which would otherwise have been considered as superfluous: and some it was thought proper to repeat, in order to connect the Thread of the Narration, without the Necessity of frequent Reference to the Sections.

6. An Agreement having been made with Mr. Lunardi, that he should resign his Balloon to Mr. Baldwin on Wednesday the 7th of September; an Advertisement to that Purpose appeared in the Chester Paper: and on Wednesday Morning, a great Number of Spectators assembled in the Castle-yard of the City of Chester: where many waited till half past IV in the Afternoon; Mr. Lunardi declaring that, on Account of the Violence and Unsteadiness of the Wind which blew from the South and South-West, it was dangerous to attempt the Inflation of his Balloon; and Mr. Baldwin continuing to assert that, if it could be filled, he was willing to go up.

The Weather was *then* moderate: but Mr. Baldwin, thinking the Hour too late to begin the Inflation, which, judging from the two former Inflations, could not probably have been completed till after *Sunset*; made a Proposal to Mr. Lunardi, that he should postpone the Exhibition till the next^[1] Day. The latter, after some Reluctance, arising from a Fear lest the Public should disapprove his Conduct, politely complied with his Request, on Mr. Baldwin's saying that he would take the Blame on himself.

CHAPTER II.

PREPARATIONS FOR THE VOYAGE.

Section 7. **O**N Thursday the 8th of September 1785, at IX in the Morning, one of the Cannons (a Six-pounder) was first fired in the Castle-yard, to inform the City and Neighbourhood, that the necessary Preparations were making to inflate the Balloon.

Till VIII that Morning, the Air had been hazy: but was then clear, bright and calm *below*, with an upper Tier of light Clouds in the Zenith moving from South-West by West, and dense ones rising in the Horizon.

8. At X o’Clock, the Process began with the Inflation of an airostatic Globe eighteen Feet in Circumference, of Silk Tiffany, made the latter End of the Year 1783, and decorated with Painting, Mottoes and Devices: in the Performance of which little Work, Mr. Baldwin was (in the modern Phrase) the sole Projector, Architect Workman and Chymist.

9. The Airostat was presently liberated by the Hands of Mr. Lunardi; and continuing to turn gently the same Way round its own Axis, afforded a beautiful Spectacle to the Beholders: remaining in Sight about half an Hour. It was intended to serve as a Sort of Pioneer, to delineate the Track of the great Balloon.

10. It fell at some Miles Distance, ’tis said unfortunately on a Hedge, and was presently torn to Pieces by the Eagerness and Avarice of the Pursuers, who expected and undeservedly obtained the Reward promised in the Letter appended to it.

11. At XII the Cannon fired a second Time, to announce that the Process was in a proper Degree of Forwardness.

At this Time Mr. Baldwin went, with some Friends, to take an early Dinner: he also recapitulated the Articles, to be certain that Nothing was omitted.

12. The following Inventory, with which he ascended, may be of Use to future Aironauts; to whom *only* it is addressed.

The Cable and Grapple are considered as Part of the Balloon. (See [Section 13.](#))

12. Article 1. A portable Barometer,^[2] with a common Syphon or Bulb, (purchased at Lausanne.)

12. 2. Martin’s Thermometer,^[3] with Farenheit’s Scale^[4] for the Degrees of Temperature.

12. 3. Mariner’s Compass in a double Box, to be used when the Sun is intercepted from the View by Clouds, in order to discover whether the Balloon turns round.

12. 4. Down, or small Feathers, to be loose in the Pocket, and thrown out, when enshrined in Clouds; or at any other Time, to shew the Rise or Fall of the Balloon.

12. 5. An Asses’ Skin Patent Pocket-book; as Wet spoils Paper.

12. 6. Two *red* Lead Pencils: each Pencil ready pointed at both Ends, to save Time and Trouble: preferable to Ink, which may be spilt or frozen.

The Strokes with *red* Lead are not so easily obliterated, as when made with a *black* Lead Pencil.

12. 7. A small sharp Knife pointed, and ready open, or which will open easily. A Pair of Scissars.

12. 8. A *wicker* Bottle of Brandy and Water, only three Parts full, half and half: such Bottles are more secure: and such Mixture will not soon freeze. The cochuc or elastic Bottle is still better. A Cork-screw.

12. 9. *Compact* Provisions, which do not soil the Fingers or Pocket-book, as Confectionaries, Fruit, Biscuit, Bread.

12. 10. A *boarded* Map of the Country over which the Aironaut may be supposed to pass: the Back serving as a Table.

12. 11. Two Needles with large Eyes: the *raw* Silk put through, and tyed on a Knot at the Ends to prevent the Needles from being lost: to be ready at the Instant wanted, to sew up any Holes within Reach, in the Balloon; the Holes being first tyed up with Twine.

The Needles to be stuck into Parchment, containing a small Hank of *raw* Silk: the Needle Silk run round the Parchment, to keep the Hank dry.

The whole Hank to be tyed by one End to the Side of the Car; when above all Clouds, to shew, by the Divergency of the Threads, the Electricity of the Air.

12. 12. A few Yards of Dutch Twine, loose in the Pocket, to tie the Neck of the Balloon in descending.

12. 13. For easy Experiments; 1st, Dutch Twine, half a Mile long, on a Reel, or Pulley, or two Lengths on different Reels: also to each Reel a Flag, made of white Linen, a Yard square; and stretched by a slender Lath; one Side of the Flag being bound and stitched round it: also a Piece of Twine, two Yards long, is to be fastened by its Ends to the Ends of the Lath: a Loop is to be made in the Middle of the Twine: and to the Loop is to be applied round the Middle of the Lath another Piece of Twine, which will prevent the Lath from being bent; and will keep the Flag always stretched.

By this Apparatus, Observers from below may be enabled to estimate the Height of the Balloon, as will be shewn in its proper Place.

12. 14. 2dly, To try the Density of the Air, at different Heights, *above* the freezing Point with Water; *below* it, with Brandy.

In a Basket take two Pint-bottles, one full of Water, the other of Brandy; and six or eight empty ones: also a small Metal Tunning-dish.

Let one End of a String be tyed round the Neck of each Bottle: and the other End sealed to the Top of a large Cork much tapered, to enter the Mouth easily. Round each Neck, tie a Parchment Label, large enough to contain in abbreviated Characters the Number of the Bottle; Time of Observation, Heights of the Barometer and Thermometer, while on the Ground.

When an Experiment is made in the Air; pour off a full Bottle into an empty one: put the Cork into the emptied Bottle, and mark again the Time, Barometer and Thermometer: which are to be compared with an Eudiometer below, to discover the Rarity and Purity of the Atmosphere.

12. 15. A third white Linen Flag, made as above, and tyed to the upper Hoop of the Balloon, so as to hang in Sight, will give Notice of a Change in the Wind.

12. 16. A Yard of thin Ribbon, two Inches broad, tyed to the lower Hoop, will mark the Rise and Fall Of the Balloon.

(12. 17. A Magnet and Iron Filings in a thin Pewter Dish with a Cover; Also

The Prism and large Telescope were left, as too heavy.) And the Sextant or Quadrant could not be procured in Time. They woud, have been of little Use, as no Horizon of the round Earth was seen during the Excursion: and it is presumed, that the circular Horizon is seldom visible, when the Balloon is at any considerable Height; the Accumulation of Vapour between the Eye and Horizon preventing it: tho' such Vapour remains invisible to Spectators from below.

12. 18. Eight Bladders, each above half blown, and differently coloured for Ornament, tyed round the *upper* Part of the Car, Breast high when the Aironaut stands upright: in Case the Balloon fall into Water.

12. 19. Speaking Trumpet: also a live Pigeon, in a small Basket of Matting.

12. 20. Pepper, Salt, Ginger; to try the Effects of Tastes, which have been said to become insipid on the Peak of Teneriffe.

CHAPTER III.

ADDRESSED TO AIRONAUTS.

Section 13. **T**HE following Anchor and Cable, for greater Safety and some particular Uses, are recommended as an Improvement.

A *strong* Iron double Grapple, moving on a Swivel, fastened to a *Rope*,^[5] half a Mile, or better *a Mile* long; and, if not all; a Part of which at least, at the Distance and for the Length of ten Yards from the Grapple, should be of Silk, as a non Conductor: also other ten Yards, at its upper End, counting from the Reel or Pulley to which the Silk should be tyed.

The Reel or Pulley being at least eighteen Inches in Diameter, and fixed vertically in the Center of the upper Hoop, seven Feet above the Bottom of the Car; by Means of three or four Iron Rods fastened in the Bottom of the Car, and meeting together above the Reel: the Rods so strong as to prevent the Shock which otherwise the Aironaut would receive in alighting on the Ground.

The Reel should have one, or two Iron Winches or Handles, one at each End of the Reel; with moveable Handles of Wood round them. The Reel may be furnished with sudden Checks; or gradual Clamps, as in a Mil, to retard the Velocity.

SIGNS TO BE OBSERVED, WHEN IN THE AIR.

14. The two Extremes to be avoided are, too lofty an Ascent: and too precipitate a Fall.

The former is to be apprehended when Balloon has swelled considerably, and strains as if ready to burst; from the Shape of an inverted Cone, or Children's Top, changed to that of an oblate Spheroid, or Turnep.

It is therefore necessary to look up at the Balloon from Time to Time: and either open the Mouth, or as it is sometimes called the Neck, *for an Instant*; or draw the Valve; which is done by pulling a Cord fixed at the Top of the Machine and running thro' it to the Hand, till the Balloon only appears full without straining.

These Operations are to be occasionally repeated during the *Ascent*.

If it is required to rise still higher; gradually throw out Ballast, and repeat the Operations.

The proposed Quantity of Ballast being thrown out, the Balloon will have acquired its utmost Height, and become stationary, i. e. neither rise nor fall.

The self Descent of the Balloon is only in Proportion, as the inflammable Air or Gass escapes thro' imperceptible Holes in the Silk or Seams.

2dly. TO PREVENT TOO PRECIPITATE A FALL.

15. 1st. Tye, or compress the Mouth of the Balloon, for a Moment; which must always be opened, on observing that the Balloon is again risen to so great a Height as to *strain*, or be distended as above mentioned.

2d. In descending, throw out Ballast, when the Balloon is within a Quarter of a Mile of the Ground, but not before, i. e. at 26 Inches by the Barometer: and, if the Fall is precipitate, not less than 25 Pounds Averdupoise, Pound by Pound, or at once, if there should be Occasion.

3d. In Case of Accident, as the Escape of Gass; or if the Balloon be not furnished with AN EQUATORIAL HOOP; prepare to throw out all the Ballast at the above Height, but not before; as the more forcible the Fall,^[6] the greater the *Resistance* from the Air: cut away Ends of Cords; tear off Ornaments: part with Shoes, Cloaths. All which must be made *loose* and *ready* to throw out, at the Moment the Balloon begins to descend. Before the Landing, particular Care must be taken, that the Weight of the Aironaut be sustained, by grasping the Hands round the OPPOSITE Sides of the upper Hoop; so that the Feet may not touch the Bottom of the Car. The Knees should likewise be bent. Repeating the above, at each Rebound of the Balloon, if any; the Aironaut will alight in the gentlest Manner: and probably the Balloon may act as a Parashute or Umbrella, which *alone* will, at all Times, ensure an easy Descent.

SIGNS WHEREBY TO JUDGE WHETHER THE BALLOON IS RISING OR FALLING.

SIGNS OF RISING.

16. 1. When the Aironaut perceives a Pressure upwards against the Soles of his Feet.
2. When some Objects, on the Surface of the Earth immediately below, diminish, and others disappear.
3. When an upper Cloud approaches or involves the Balloon.
4. When a lower Cloud leaves the Balloon.
5. When Rain Snow or Hail beat VIOLENTLY against the Top of the Balloon.
6. When Feathers, Balloon-Flag, or Ribbon seem to be drawn forcibly downwards.
7. When Objects on Earth, or among Clouds below the Balloon, rise and present themselves *beyond those*, which, the moment before, were thought most distant.
8. When the Balloon appears broader and shorter; also fuller at the Bottom; being more distended than at the first Ascent.

SIGNS OF DESCENT.

17. 1. When the Aironaut perceives the Bottom of the Car withdrawing itself from the Pressure against the Soles of his Feet.

2. When Objects on Earth, and surrounding Prospects encrease in Magnitude and Number.

3. When a lower Cloud approaches or involves the Balloon.

4. When an upper Cloud leaves the Balloon.

5. When Weather beats against the Bottom of the Car or Balloon.

6. When Feathers, Balloon-Flag, or Ribbon appear to be drawn upwards.

7. When the most distant Objects *set*, and disappear.

8. When the Balloon seems taller; and its lower Hemisphere less distended, tho' continuing *tight*.

SIGNS OF PROGRESSIVE HORIZONTAL MOTION.

18. These are equivocal and deceitful.

When the Aironaut has lost Sight of the Earth by intervening Clouds; the Balloon seems at Rest, and only the lower Clouds appear to move: whereas the contrary may be true, the Clouds may rest, and only the Balloon move.

In this Case, Attention must be paid to the half Mile white Flag, whose Situation and Motion must be observed, with respect to the Balloon, and to the Earth before the Cloud intervened. If the Flag retains its Situation with Respect to the Balloon, it may be inferred that no Change in the Direction has happened: if its Situation alters, the Sun or Compass is to be observed: and an Estimate made of the new Current of Air by which the Balloon is affected: its Velocity, Sound, Temperature, &c.

19. But to acquire a Certainty of course, it will be proper to descend below the Cloud: or move by Compass, Map, and a Knowledge of the Country: or try the long Cable ([Section 13.](#))

20. It is likewise necessary to know the *Signs of Wind*, or Currents of Air.

SIGNS OF NEW AND SUDDEN HORIZONTAL CURRENTS.

When the Feathers, Balloon-Flag, or Ribbon, compared with Sun or Compass, take a new and sudden horizontal Direction.

21. OF CURRENTS FROM ABOVE: properly named *Waves Torrents and Tide of Air*.

They are very frequent, and require to be guarded against: are sometimes of long Continuance, at other Times momentary: against the first throw out Ballast at the Height of a Quarter of a Mile, but not before, or *as hereafter directed*: when momentary, and above that Height, Nothing is to be apprehended: the Balloon will appear broader and recover its Form.

CHAPTER IV.

PREPARATIONS FOR ASCENT.

Section 22. **B**EFORE half past I, Mr. Lunardi had inflated his Balloon in the finest Manner; and having, with the most obliging and spirited Attention, made such Preparations, and taken such Precautions, as he thought were necessary to ensure the Success of the Expedition; sent to inform Mr. Baldwin (who continued purposely absent, that he might not disturb or precipitate the Process; but that every Circumstance should be conducted with Deliberation and without Hurry) that all Things were ready for his Departure.

23. And Mr. Baldwin takes this Opportunity of returning his best Thanks to his Friends and the Public, on the Day of Ascent, for keeping THE SMALL CIRCLE CLEAR, by strictly adhering to the Words of the Advertisement, which declared, “that in order to prevent an interruption of the Process in the Inflation of the Balloon, no Persons *were* to be admitted WITHIN THE CIRCLE, except those Gentlemen who politely undertook IN TURN to hold the Lines which detained the Balloon.”

24. It may be proper to mention that Mr. Baldwin being resolved to prevent the disagreeable Circumstances of being *weighed* in the Presence of Thousand spectators, at a Time when it is uncertain whether the Balloon has *acquired* a sufficient Degree of Levity to raise his own Weight, together with the Instruments, Provisions, Ballast, and other Articles, all which are known or easily calculated; finding some Days before, his own Weight, and having calculated the rest as under^[2]; he ordered his Servant, on the Day of the Excursion, to bring Lead Weights equal to the *Sum total*, with an overplus Weight of 10lb. for Levity of Ascent, and place them *gradually* in the Car, attached for that Purpose to the Balloon, soon after the Inflation began. By which Means the Gentlemen who held the Cords were quite at Ease: nor was there Occasion to tye the Lines during the Inflation, to Posts fixed in the Circumference of the Circle; nor consequently to *cut them* afterwards.

But it will be seen that Mr. Lunardi inflated the Balloon in a superior Manner.

25. All Things being thus prepared, Mr. Baldwin stepped into the Car: and finding, that, besides his own Weight, the Provisions, Articles, Ballast, &c. the Balloon woud support an additional Weight, and *still* rise with superior Levity; Mr. Lunardi put in 12lb. of additional Ballast, and *guessed* the encreased Levity at 10lb. more.

Additional	{ Ballast	12
	{ Levity	10
		—
		22
Added to the		234
		—
Make the Sum		256lb.

All which added to the Weight of the Balloon, *by Information only*, as follows:

Balloon varnished	113
Netting and Cords	18
Car and Hoops	24
Mended and added Parts	5
Grapple and Cable	4
	164
With the	256

Make the total Levy of the Gass to produce an Equilibrium, equal to 420lb.

The Weight of a Quantity of Air equal in Bulk to the Balloon, being secluded; and the Gass substituted in its Room.

26. The Calculation of the Weight of Articles was, as follows:

Articles.	Pounds Averd.	Ounces.
1. Eight coloured Bladders ^[8] (Section 13 , Art. 18)	1	— 0
2. Preparations against extreme Cold.		
A WINTER DRESS.		
Flannel or woollen Socks	}	
Cap		
Gloves		
Drawers		
Under Stockings		
——— Waistcoat		0 — 14
3. Brandy, Water, Flask, and Refreshments	1	— 8
4. Barometer (portable)	0	— 12 1/2
5. Thermometer	0	— 3
6. Dial-Compass (a Mariner's Compass in a double Box, will traverse better)	0	— 3 1/2
7. Two white Flags, with Dutch Twine on two Reels furnished with Swivels	0	— 4
8. Asses Skin Pocket Book, Blank Cards, Pencils, Knife and Scissars	0	— 4 1/2
9. Map of Cheshire boarded, the superfluous Parts cut away	0	— 3
10. Speaking Trumpet	0	— 8 1/2
11. Mr. Lunardi's Flag	3	— 8
12. Basket and eight Pint Bottles labelled, one full of Brandy, another of Water	8	— 3
	20	— 0

27. The Ballast consisted of three Bags of dry Sand, and two red grit Stones, taken while in the Car, *additional*.

1st Bag tyed up weighed 12lb.

2d Ditto	12
3d untyed Ditto	20
1st red Grit	7
2d red Grit	5
In all	56lb.

CHAPTER V.

ASCENT WITH 20lb. OF LEVITY.

Section 28. **A**T 40 Minutes past I, the Balloon having a Levity which not less than 20 Pounds Weight woud counterpoise, Mr. Baldwin was liberated by the Hands of Mr. Lunardi, who suffered no one to approach the Car: and he ascended, amidst Acclamations mixed with Tears of Delight and Apprehension, the Misgivings of Humanity, and other usual Sensations of Surprise, which, in a brilliant and numerous Assembly, will long continue to accompany a Spectacle so novel interesting and awful, as that of seeing a Fellow Mortal separated in a Moment from the Earth, and rushing to the Skies.

29. The Balloon well inflated, tower'd aloft in an upright and perpendicular Direction, with a quick Motion, and an accelerated Velocity.

The Aironaut having stood up, for a Minute or two, waving his Hat in the left, and saluting the Spectators with Mr. Lunardi's coloured Flag in the right Hand; put on his Hat, and having fastened the Flag-Staff horizontally among the Lines of the Balloon, immediately betook himself to different Employments, before he woud indulge in looking over the Brink of the Car; lest the Novelty of the Prospect shoud call off his Attention from *Things of Moment*.

30. The Force of Ascent was, from the first, plainly *palpable*: the Sensation being that of a strong Pressure from the Bottom of the Car, upwards against the Soles of the Feet.

31. His first Point being to guard against a Deluge of acidulous Liquor, which, he was told, had fallen, to the Quantity of three Quarts, on the Head and Shoulders of a former Aironaut, from the Trunk or Bottom of the Balloon, which ended in a wide circular Opening of eighteen Inches Diameter; he found that when the Weight either of himself, or of the Ballast, was not exactly in the Center of the Car; the Opening of the Balloon woud, without any Trouble, hang so as to lie on the Outside of the Car: but he did not perceive more than a few Drops issue from the Mouth: which happened a few Minutes after he arose.

32. This Difficulty vanishing; he changed his erect into an inclined Posture between sitting and kneeling; sometimes with the right Knee near the Bottom and Center of the Car: and having both Hands quite free, the Balloon being subject to no *sensible Motion*; he reconnoitred all the Lines and Cords: coiled the Rope or Cable to which the Anchor or grappling Iron was fixed: tyed fast its proper End to the upper Hoop: observed and felt the superior Thickness of the Cord leading to the Valve: coiled it, in order that it might be free to act: placed the untied Bag of Ballast near the Outside of the Car: also the tied Bags at proper Distances to preserve the Equilibrium: unwrapped one of the white Flags, tyed it to the String on one of the Reels, and just threw it an Inch or two over the Side of the Car:

then placed his Watch, *open* Knife, Scissars, Thermometer and Compass on his right Hand: the Barometer being swung above in Sight towards the left.

33. He then stood on his Feet, with a Design to look down: but his Attention was drawn to the Opening of the Balloon, which began to breathe out by Intervals a visible *reddish Vapour*; in Form like that which is seen at the Top of a Brewery, only that the under Surface was not jagged but smooth, altho' wavy and uneven. The Particles which composed it were so large as to be distinctly visible: and appeared, as if endued with a very strong repelling Power, from the great and seemingly equal Distances, of about half a Quarter of an Inch, from each other.

It was observed by a scientific Spectator from below, that the Parts of the Balloon, which reflected the Sun's Rays, appeared of a bright Copper-Colour: but the *reddish* Vapour issuing from its Mouth put on the Form of a lambent Flame. A similar Appearance had been observed by him, in a former Ascent of the same Balloon, the Neck or Mouth being then likewise open; and also by others, who declared they saw the Balloon on Fire.

The Change of the RED into Flame-Colour, when seen at a great Distance, may it not be owing to this, that the direct Rays, being mingled with those which are intercepted between the Eye and the Object, became in Part absorbed, and in Part refracted; and therefore could not reach the Sight?

34. This gentle Evaporation of inflammable Air, or Gass, continued: disappearing at the Distance of four and five Inches below the Opening: nor did it offend the Smell; not descending within its Influence.

35. He then looked upwards at the Balloon, and perceived that it was considerably swelled in its Dimensions: and that the Distention had raised the Bottom-Opening of the Balloon half way between the two Hoops: i. e. from his Hip to his Shoulder, as he stood upright. The Height from the Bottom of the Car (which was a thin circular Board four Feet and a half, Diameter, placed on a strong Netting, and covered with green Bays) to its Top or the lower Hoop, was three Feet; with the Netting continued round between the lower and upper Hoop.

36. He was aware that the Swelling of the Balloon, and copious Vapour then issuing from it, denoted the Moment when it began to lose its ascensional or elevating Power; and that its accelerated Motion was diminishing.

He therefore looked at his Barometer and Watch, which was 53 Minutes past I.^[9]; took up his Pencil, and on a Card (marked before he left the Earth, as follows:

Chester-Castle-Yard. Thursday, the 8th of Sept. 1785, I. o'Clock, Barometer 298/10, Therm: 65 in the Shade towards the North;) he wrote "Rose at 40 Minutes past I." He then looked again at the Barometer, which continued falling for some Minutes, and fluctuating up and down within the Space of an Inch or more. It first began to rest at 231/4, and a little after at 231/2. Having looked again at his Watch, he put down "57 Minutes past I. became stationary: Barometer 231/4: Therm: still 65, sometimes lying in the Shade, and sometimes exposed to the Sun: the Balloon turning round frequently thro' East to South."

37. The Fluctuation of the Barometer, he imagined to arise from continued Exertions of the Gass within the Balloon, opposed by the atmospheric Air, which varying in Density and Temperature woud give an unequal Resistance to the Balloon: and both Gass and Air being elastic, the Power of Ascent would act by Intervals, and communicate its Pulsations to the Quicksilver in the Tube. His own irregular Motions in the Car would increase the Fluctuation.

38. The Compass likewise traversed backwards and forwards, pointing due North, and unaffected by the Turns of the Balloon: but was useless, as the Sun shone bright the whole Time of the Excursion.^[10]

39. Thing's taking a favourable Turn, he flood up, but with Knees a little bent, more easily to conform to accidental Motions, as Sailors when they walk the Deck: and took a full Gaze before, and below him.

But what Scenes of Grandeur and Beauty!

A Tear of pure Delight flashed in his Eye! of pure and exquisite Delight and Rapture; to look down on the unexpected Change already wrought in the Works of Art and Nature, contrasted to a Span by the NEW PERSPECTIVE, diminished almost beyond the Bounds of Credibility.

Yet so far were the Objects from losing their Beauty, that EACH WAS BROUGHT UP in a new Manner to the Eye, and distinguished by a Strength of Colouring, a Neatness and Elegance of Boundary, above Description charming!

The endless Variety of Objects, minute, distinct and separate, tho' apparently on the same Plain or Level, at once linking the Eye without a Change of its Position, astonished and enchanted. Their Beauty was unparalleled. The Imagination itself was more than gratified; *it was overwhelmed.*

The gay Scene was Fairy-Land, and Chester Lilliput.

He tried his Voice, and shouted for Joy. His Voice was unknown to himself, shrill and feeble.

There was no Echo.

40. He then returned to an Employment which, tho' irksome, he imagined would contribute to the Amusement and Information of Spectators below, if it could be completed while he continued in Sight; as it woud furnish them with Ideas of Height and Distance, altogether new and interesting, *as will be seen in their proper Place*: and unwound half the Reel; the white Flag hanging out to the Length of 440 Yards or a Quarter of a Mile.

41. The circular Motion of the Balloon was communicated to the Loop in the Middle of one Side of the Lath or Reel, round which from End to End the Twine was wrapped, and by which it hung on his Finger, and pressed it to a Degree of Pain.^[11]

The Work was again suspended.

He coud not long withstand the Temptation of indulging his Eye with a View of the glorious and enchanting Prospect.

42. But the Beautiful among the Objects below was still more attractive than the Sublime among those around.

On looking down South by West, the Balloon often turning gently to the right and left, and giving the Aironaut an Opportunity of enjoying the circular View without a Change of Attitude; innumerable Rays of Light darted on the Eye as it glanced along the Ground: which, tho' of a gay green Colour, appeared like an inverted Firmament glittering with Stars of the first Magnitude.

43. This splendid Appearance was owing to the Rays of the Sun reflected from certain Pits or Ponds of Water, of which there is one at least in most Fields or Inclosures throughout the County: but particularly in the low Grounds of Leach-Eye and Dodleston.

The Object that next drew his Attention, *while ascending*, was the Overley Turnpike-Road, which is remarkably wide, (resembling the Emilian Way across the Atrian Fens, between Bononia and Ferràra in Italy) raised over Saltney Marsh, leading to North-Wales and Holyhead: composed of Sea-Sand cast up above high Water Mark. This appeared like a narrow Foot-Path well trodden, of a *white* Colour, and strait as if drawn by a Line.

44. Nothing however raised his Curiosity more than the Change in Colour of the River Dee, Avon ddû, (i. e. *Thee*) which in the British Language signifies the *black River*, from the Appearance of its Waters, when seen from an Eminence running in their deep Channel between the Mountains of Wales; but which glides by Chester with a Silver Stream. This River,—Thanks to the cool Climate; not like the *green* Mincius of Virgil!—had now acquired the unvaried Colour of *red Lead*. Nor could he discover even the Appearance of Water; but merely that of a *broad red* Line, twining in Meanders infinitely more serpentine than are expressed in Maps.

Whether the Change arose from the Transparency of its Waters, when seen at the Height which was *apparently* 7 Miles, *as will be noticed hereafter*, though the Barometer made it scarcely a Mile and Half, is uncertain. He was at first inclined to think, that the Rays, having suffered a double Refraction, were reflected to the Eye, from the reddish Sand which forms their Bottom, tho', at the Depth of 7 Yards at an Average, above the Cause-Way, or *artificial* Cascade near Chester Bridge: or possibly the Water of Rivers when seen at a certain Distance, may act as Water composing Clouds when view'd from below, at a certain Height and Angle; reflecting only the *red* Rays: the rest being refracted, or absorbed.

The Colours of Objects shone more brilliant and lively at that amazing Height, than if seen on a Level with themselves.

Nor did the Eye seem to want the Aid of Glasses: as every Thing, that could be seen at all, was seen distinct.

45. The *Redness* of the River Dee was curiously contrasted by a Change equally novel but more pleasing, in the Colour of the City of Chester, when seen directly from above, on a Scale not larger than the Plan of it, in Burdett's Map.

The Town was entirely *blue*.

The highest Buildings had no apparent Height: their Summits were reduced to the common Level of the Ground. Nor was the Cathedral distinguished; nor any Tower or Spire discerned.

The Whole had a beautiful and rich Look; not like a Model, but a coloured Map.

The Roofs of all the Houses appeared, as if covered with *Lead*, in the most elegant Taste.

Strangers may wish to be informed, that in most of the Northern Counties, the Buildings are covered with *blue Stones* called SLATES^[12] found in the Mountains; instead of artificial *red Tiles*, as in London, and the South of England.

CHAPTER VI.

BALLOON VERGING TO THE SEA.

Section 46. **B**EFORE a farther Description of aërial Scenes is attempted, it would be improper not to mention a Circumstance which happened on the first Ascent of the Balloon: and too strongly called forth the tender sympathetic Feelings, by raising, in the Minds of the Spectators, *alarming Apprehensions* for the Safety of the Aironaut, on seeing the Balloon move gently towards the Sea.

They were however, in a great Measure, soon relieved from their Anxiety: for, by rising into another Current, he escaped the Danger: skirting the Coasts of the River Mersey; which could not be seen from the Balloon at the Distance of little more than a League, tho' the Sun was supposed to shine the whole Time on the Water.

The upper Current was, in Fact, rendered visible to the aërial Traveller, for more than two Hours before, and at the Time of his Ascent; by lofty Clouds of the second Stratum, flying in a safe Direction.

CHAPTER VII.

Section 47. **A** Few Seconds of Time before the Balloon had attained its greatest Height; the Velocity of Ascent being every Instant retarded by the Escape of Gass thro' the Opening;—the *remarkable Stillness* which prevailed in so elevated a State of the Atmosphere, *apparently* many Miles above all visible Vapour, far beyond the Sight of every living Creature, and where the human Voice was no longer heard from below; the larger Objects, with which the Surface of the distant Earth was covered, as Rivers Woods Inclosures, diminishing to the View, yet encreasing in their Beauty;—could not but make a lively Impression on the Mind of the Aironaut.

The striking Contrast and Novelty of his Situation filled him with unusual and pleasing Sensations.

He had just left, for the first Time, his native Earth, where he had continued for a while the central Object to some *thousand* Spectators; whose Eyes, he knew, were still turned towards him; that he was still the Subject of their Conversation: yet no human Figure met his Sight; no human Sound vibrated on his Ear.

An universal Silence reigned! an empyrèan Calm! unknown to Mortals *upon Earth*.

The Sky was painted with a purer, and more transparent Azure. The Sun shone hot, and with a brighter Lustre. His Beams were *white* and sparkling: not surrounded with Haze or Vapour: but too fierce for the human Eye to look upon a second Time with Pleasure. ^[13]

48. A Cheerful Serenity filled the Breast of the Aironaut.

In an erect Posture, and with the utmost Composure he gazed around: reflecting with Wonder and Delight on a Situation, where the BEAUTIFUL and SUBLIME were seen united, in a Manner perfectly novel and engaging.

If it be allowed, for the Sake of Illustration, to compare *great* Things with *small*; he found himself suspended in the central Concave of an unmeasurable Crater Bowl or Bason; and considerably above the Rim or Margin, so as to peep fairly over it: for by looking *straight before him*, while the Balloon continued gently turning on its vertical Axis, he could see quite round into the BLUE.

The Earth was the *Miniature-Picture* ^[14] painted on the Bottom of the Bowl, on the Inside. The Sides of the Bowl next the Bottom were rather obscure: as the Objects, on the Surface of the Earth not immediately under the Eye, being foreshortened, were indistinct, either on Account of their immense Distance, or by mere Accumulation of Vapours, and mixed with Haze and Cloudiness.

From thence to the Top of the Bowl, was fantastically grouped, spotted, and dash'd with Clouds dense and luminous, in the strangest and most grotesque Forms; still smaller and more numerous, as the Eye was more extended: The Rim or Margin ending, not in a fringed Border; but in a plain smooth Line; to represent the amazing Distance, at which, the upper Surfaces of *Clouds in Perspective* lost all their rugged mountainous and fringed Shapes;

and terminated even and smooth: making a perfect horizontal Ring in the Heavens, somewhat below the Eye of the Observer. The whole formed a glorious *Concave*: and the Imagination was lost in the surrounding distant Azure.^[15]

49. Considering more attentively the Dimensions of this vast Amphitheatre; as he long continued *apparently* in the *same* Spot, and seemed to himself a mere Atom floating *invariably* in the Center of the empty Space; yet as a sole thinking Being there, whose Mind was bent on estimating the Extent of his View, so accurately defined by the circular Horizon of dense accumulated Vapour; and judging, as of other Distances, by *the natural Eye* alone; pointing downwards on Objects which were only distinguishable when immediately below it, frequently no more than the Circuit of a Mile on the Earth's Surface, the *vertical Boundary* of the profound Abyss; all else being obscured by Haze, or removed from Sight by Volumes of intervening Cloud; he could not divest himself of the Idea, but that the *apparent* Depth below him was at least *seven* Miles: *three* from the Earth to the upper Surface of the superior Clouds,^[16] and *four* above them.^[17]

OBJECTION REMOVED.

50. Some may find a Difficulty in conceiving, how the whole Prospect of Clouds and Earth together could put on a *concave* Appearance: both of which were in Reality *convex*, with Respect to the Situation of the Observer in the Car.

A familiar Illustration may help to remove the Objection.

Imagine a Person placed in the Center of a Plain, or Carpet; extended every Way beyond the Reach of the Eye. If in that Situation he was gradually elevated; the *distant* Parts of the Carpet would *seem* to rise with him: and those Figures of the Pattern would alone be distinguished, that lay immediately below the Eye: the more remote becoming dim and faint. The whole would put on the Form of a concave Bowl; as soon as he had risen to so great a Height, as plainly to perceive the Figures of the surrounding Pattern more and more foreshortened, in Proportion to their Distance from the Center of the Carpet.

CHAPTER VIII.

Section 51. **T**HE Perspective of the Clouds was entirely new; and remarkable both for Beauty and Grandeur.

The lowest Bed of Vapour that *first* put on the Appearance of Cloud was of *a pure white*; in detached Fleeces; encreasing as they rose. They presently coalesced, and were aggrandized into A SEA of Cotton, but more *white*; and *dazling*: tufted here and there by the light Play of Air, and gentle Breezes in every Direction: but where undisturbed, the Whole became an extended Firmament or *white* Floor of thin Cloud, thro' whose Intervals the Sun must shine with fiercer Gleam. The upper Surface was quite even: not blended with the Air above, but defined and separated with the utmost Exactness; being condensed by the Coolness, and checked in their Ascent, by the Levity of the superior Regions.

Thro' this *white* Floor uprose in splendid Majesty and awful Grandeur, at great and unequal Distances, a vast Assemblage of *Thunder-Clouds*: each Congeries consisting of whole Acres in the densest Form.

52. Their conglomerate and fringed Tops rising, at different Distances, in circular Order, one above the other, to the Number of *thirty*: till they became imperceptible from their remote Situation: the Eye commanding an Extent of 102 Miles. ^[18]

Their Form was, as if Pieces of Ordnance were discharged perpendicularly upwards into the Air: and that the Smoke had consolidated, at the Instant of Explosion, into Masses of Snow or Hail: had penetrated thro' the upper Surface or *white* Floor of common Clouds, and there remained visible, and at Rest.

Some indeed had not wholly lost their Motion: continuing still to be lifted up. Others ponderous and sleepy, nodded, by mere Weight, their monstrous Heads. It seemed as if they had persisted in mounting upwards, till they could rise no higher: their lower Parts pressing perpendicularly against the upper, which gradually swelled them out on *all Sides*. By partial and temporary Movements of the Air, some broad *unwieldy* Caps lost the *vertical* Direction of their Columns. The Columns likewise underwent a similar and gradual Change: rolling from their Pedestals or spiral Bases; and, at Times, assuming EVERY ORGANIZED SHAPE that Fancy could suggest.

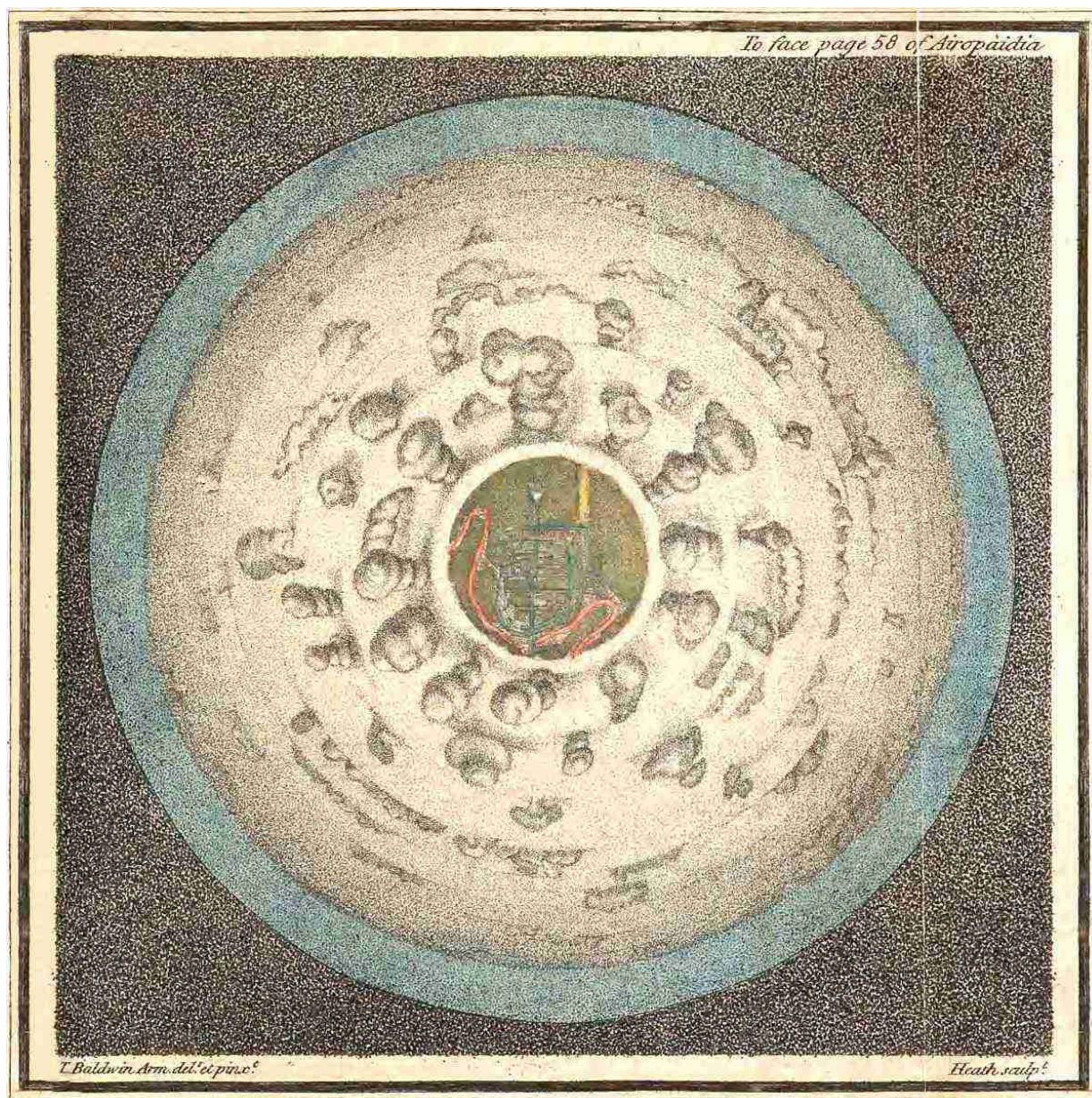
53. The imperceptibly slow yet perpetual Changes they underwent, strongly called to Remembrance, the Opinion of the great Berkeley, ^[19] as well as of the ancient Philosophers, that AIR GIVES FORM TO THINGS: scarcely a Breath of which seemed, however, to disturb their general Order.

The Constitution of these enormous Masses was such as to reflect *some* of the Sun's Rays, and to transmit *others* in a Variety of Colouring.

54. The Parts next the Sun were of a *snowy* Whiteness. Then of a *bright luminous Yellow* melting into a *dusky Sulphur*: afterwards of a *Purple*. The Rays being now shorn; a Degree

of Opacity and Transmission took Place throu' half the Substance of the Cloud, which seemed of a *transparent Blue* like the *Onyx*.

55. These *delightful Tints* must be ever eclipsed to a Spectator on the Surface of the Earth, looking upwards throu' the gross Atmosphere that surrounds it; but highly *interesting* to one who is suspended in a ratified and unencumbered Medium of the etherial Regions, where the Eye darts without Resistance above Clouds, and all visible Vapour.



T. Baldwin Arm. del. et pinx.

Heath sculp.

**A VIEW from the BALLOON at its
GREATEST elevation see [Page III. a.](#)**

Publish'd May 1st. 1786, by T. Baldwin Chester.

Note: the Print, representing a circular View from the Balloon at its greatest Elevation, is taken from a Scene described in the above Chapter.

CHAPTER IX.

OTHER AËRIAL SCENES DESCRIBED.

Section 56. **D**URING the Time that the Balloon from being stationary at 231/4 (corresponding to the Height of about a Mile and a half) began to *decline*, which it must have done with a brisk Motion, imperceptible to the Aironaut at the Time, tho' since recognized, on Account of the great Opening at the Bottom; he traced its *Shadow* over the Tops of Volumes of Clouds below. It was at first small: in Size and Shape like an Egg: but soon encreased to the Magnitude of the Sun's Disk; and woud have made a solar Eclipse to a Spectator looking from the Cloud: still growing larger, as the Balloon descended, or Clouds arose. But his Attention was presently called to another equally novel, but more captivating Appearance; that of an *Iris* encircling the whole *Shadow*, at some Distance round it. The Colours were remarkably brilliant.

This *celestial Phantom* attended the Aironaut for a few Minutes: conforming, as a Vessel at Sea, to the Change of *Surface*; now plainly visible, now indistinct and disappearing; as it passed *throu'* the *luminous* or *shadowy Wave* of Clouds *apparently* at Rest.

57. The Clouds, in which this Phenomenon continued, were of the superior or second Stratum in Height, as in fair Weather; rare; of a transparent *Blue* and purest *White*, alternate. At the End of four Minutes they dispersed, so as to admit an unexpected Sight of the pictured Land thro' THEM, and thro' the Place of *the Balloon-Shadow*; whose Form first vanishing, *the Iris* remained, for a few Seconds, complete, and in resplendent Beauty.

58. *Írides*, of the same Kind, tho' of less vivid Colours, are seen round the Moon, in a mild Evening; as thin light Clouds move slowly under it.^[20]

59. The Sun shone brighter and fiercer, when the Balloon was at its greatest Height: the *Heat* piercing thro' his Cloths, (which were of a *dark* Colour;) while the Aironaut *stood* with his Face from the Light.

The Mouth remaining open, it continued to descend, as appeared by the Barometer which had risen nearly to 24 Inches: at which Instant Mr. Lunardi's *coloured* Flag was thrown out, for the Information of a Friend; and that Spectators below might judge what was nearly the perpendicular Height of a Mile in the Air, according to Halley's Table.

60. The Flag was seen by the Aironaut descend for three Minutes: at which Time it became invisible. It fell, *not* perpendicularly; but in large Spirals, and by Jerks; darting first on one Side, then on the other. The Resistance of the Air made it act as a Parashute. The Flag was instantly pursued, and taken up in a Field one Mile distant from Chester. The Descent of the Balloon must have been retarded, being four Pounds and a half lighter.

61. The Pigeon was then taken out of the Basket of Matting: Thermometer 54; Barometer 253/10. It trembled much. Being turned loose, it looked frequently up at the Car; but flew downwards in cylindrical Gyration eight or ten Yards in Diameter, according to the Turn

of its Head to the right, which seemed to rest in an oblique Attitude: the Wings and Tail continuing extended as much as possible, but without Motion, during its Descent. The Bird was out of Sight in a few Minutes: but continued, as *the Owner* observed, full half an Hour, in the Air.

CHAPTER X.

Section 62. **A**T 10 Minutes and a half past II. o’Clock, the fourth or last Cannon, a Six-pounder (to announce, by preconcerted Agreement, that the Balloon began to be invisible to Spectators in Castle-Yard, Chester) was distinctly heard by the Aironaut; but had no Effect on the Balloon: did not agitate it in the least: the contrary of which was expected.

For the same Cannon, discharged the third Time at the Distance of 30 Yards from the Balloon, when it had risen a few Feet from the Ground; affected it so strongly, that the Aironaut was THEN ONLY obliged to keep himself upright, by holding the Cords with his Hands.

63. At 17 Minutes past II. was heard the Sound of a Number of Voices, which it was then imagined came from Chester, as the farewell Salute after the last Cannon: but it was afterwards known that the Balloon did not become wholly invisible, till that Shout.

64. From an Observation made by a Spectator in the Castle-Yard, just half an Hour intervened between the Discharge of the third, and of the last Cannon; as therefore the Report was half a Minute, or 30 Seconds^[21] longer in reaching the Balloon; the Distance of the Balloon at the Time of the Report was *nearly* six Miles and a half.

64. The single thin white Cloud of the first or lowest Order in Height that rendered Chester invisible to the Aironaut, was observed several Minutes before, *apparently* to pass under the Balloon, retire from it, to approach, and expected to envelope, the *blue* City of Chester: which for a long Time had been kept in View, and seen *obliquely*, under the *common* Perspective, with a small Degree of Elevation above the Level of the Ground: suggesting to his Mind the curious and complete Model of Paris, exhibited some Years ago on a *small Table*, in many Towns of Europe.

The Cloud appeared, four Miles Distance at least from the Aironaut; below; and as if touching the City. The contrary Supposition, it seems, took Place, among the Inhabitants there: who thought, a Cloud, a Mile above them, had surrounded and enveloped the Balloon.

65. The Pepper Salt and Ginger were tasted, and found to retain their usual Pungency: contrary to what Travellers have reported to happen on the Peak of Teneriffe.

The small Hank of yellow raw Silk tyed to the upper Hoop, and hanging down from it, appeared *rough*, as if electric: and, tho’ drawn thro’ the Hand, continued *furred* as before.

66. It was now thought a proper Time to finish the original Work of unwinding the remaining Part of the half Mile of Twine: which proved equally tedious, as at the first; and took up a considerable Time. When completed, the *white* Flag was extended exactly half a Mile from the Car.

67. Perceiving that the Balloon was descending very *briskly*, by the Appearance of Cattle in the Corner of a Field; first, *one* of the two solid Weights was cast down: then the *other*.

A Return of Sound to the Balloon, from the lighter which weighed five Pounds, was heard in 130 Countings of a Watch, which made 120 of the same full Beats in a Minute.

Before the Weight became invisible; it *appeared* to move a good Deal out of the Perpendicular: owing either to an under Current; or to a Deception of Sight, respecting the horizontal Motion of the Balloon in a different Direction, during the Descent of the Stone.

The other must have fallen in soft Grass, or otherwise: as it was not heard.

CHAPTER XI.

Section 68. **A**T 28 Minutes past II. the solid Weights before mentioned were thrown out.

At 29 Minutes the Barometer had fallen to 25 Inches.

A Handful of Feathers were sent adrift, which fell quick: demonstrating *likewise* the Ascent of the Balloon, a second Time: but, tho' 12 Pounds lighter, it did not seem to regain its original Height: judging MERELY from this Circumstance, that no more Gass escaped *visibly* from the Mouth.

69. It is somewhat remarkable, that, on repeated Enquiries from unprejudiced Persons, the *white* Flag, when suspended from the Car above 440 Yards, appeared 4 Yards long: and when at the end of the *half Mile* Twine, seemed about 8 Yards long, to Spectators from below, in different Places: that sometimes it appeared before, and sometimes behind the Balloon: while to the Observer in the Car, it seemed regularly to follow the Balloon: unless when a *new* Motion was impressed upon the latter: at which Time the *white* Flag was situated almost under the Car: or when the Balloon changed its Direction; the Flag being *then* not always *discoverable*.

When seen *edgewise* or *foreshortened*; it woud *appear* to be *nearer* the Car than it really was.

70. As there was a Peculiarity attending the Situation of the *half Mile* Flag, which may prove of singular Use in Airostation; it ought not to be passed over in Silence.

The *half Mile* Flag hanging loosely from the Car; not perpendicularly under, but following it, frequently at an Angle of about 45 Degrees; shews that the Flag met a Resistance from the Air, unfelt by the Balloon: which out strip'd it, in Proportion to the *greater* Surface which the Balloon exposed to the Wind.

Taking also into the Account, that the Balloon remained in Equilibrio; while the Flag was subject to the Force of Gravity: which Force was restrained from Exertion, otherwise than as a *Vis Inertiæ*, to keep it always in a perpendicular Situation.

The Resistance of the Air, acting in an horizontal Direction against the *Vis Inertiæ* of the Flag, must have a Tendency to drive it back: which being ineffectual; the Flag must consequently *rise*; and in rising *will retard the Balloon*.

A Power may therefore be communicated to a Balloon, in the Direction of the Wind, which shall *retard* its Progress throu' the Air: a Subject which seems capable of farther Prosecution.

CHAPTER XII.

Section 71. FROM 28 Minutes after II, till the Balloon had passed over the *Forest* of DELAMERE, and the steep Crag of HELSBYE-HILL; thin light semi-transparent Vapours, which seemed to be collecting at a *vast Depth below*; moving slowly in all Directions; rising to great Heights, falling, melting away, and again condensing;—(the Land, one while covered with a *WHITE Veil*; then *caught thro' Openings* for a few Seconds; the Objects appearing more distinct and *coloured*, from being seen in detached Groupes and single *Pictures framed and enshrined* in fleecy VAPOUR; *now again* discovered by a Glance of the Eye, and *then* repeatedly escaping from the Sight;—WONDERFULLY heightened the Grandeur, Gaiety and *inimitable* Beauty of the *ever varying* Prospects.

An Illustration taken from *Scenes abroad*.

72. Appearances of a similar Kind are frequent in the *noble and venerable* Structures appropriated for divine Worship *abroad*: whose Walls are decorated with the *finest Paintings*; the Subjects solemn and *engaging*; suited to inspire a *cheerful Devotion*.

While the inferior Clerics perfume the Garments of the *Priests* officiating and offering Incense before the *high Altar*; which is ornamented with *full-Length Portraits* in the richest Drapery, of Persons whether male or female, reputed of sound Morals and exemplary Piety; accompanied by Guardian Saints and happy Angels;—Columns of *white* Smoke, wafted from *Silver Censers*, rise to a certain Height in slow *majestic* Movement, before the Eyes of the *kneeling* Suppliants, who are *instantly shut out* from the *enchanting* View; till the Clouds dispersing, shew by *Intervals*, a Glympse of the *celestial* Prospect, and of the *higher* Orders of Beings, who *look down* with *Complacency* upon them; and seem *actually* DESCENDING throu' Openings of the Clouds which *appear* at Rest.

CHAPTER XIII.

Section 73. **A**T 33 Minutes after II, the Balloon-Shadow was *again* the Center of a brilliant *Iris*, painted at some Distance round it on Clouds below.

74. One of the Pint-Bottles for light Air was prepared (as in [Article 14](#), of Section 12;) and dropped from the Car.

The *Water* it contained was poured down, to observe the Effects of *Air* and *Light* on the Drops.

The Air did *not* at that Height oppose a Resistance sufficient to break the Stream into *small* Drops. Nor did they seem to coalesce: remaining, while they continued in Sight, of the same Size; some very large, others less so; and at the same *relative* Distance, as when they first left the Bottle.

The Colours seemed *stronger* than usual.

It may be *here* observed that none of the Bottles were returned; tho' found, and a Reward promised.

The Country People, as soon as they saw a Bottle; imagining it must contain some Liquor, immediately contrived to open it: by which Procedure, the Intention of the Experiment was frustrated.

The Bottles, which are *dangerous* Companions even without Liquor, should, notwithstanding, be left in the Car: at least till the Time of *landing* the Balloon.

75. While the Balloon was *first* rising; a gentle Motion of the lower Current of Air carried it immediately towards the Sea. ([Section 46.](#))

At which Time, the Aironaut by a *Glance* discovered the Mouth of the River Dee, four and five Miles wide, *yawning before him*: the Prospect extending to the Sea, as far as the *Smoke* from the Lead-Works near a Place called Flint on the Welch Coast; and to Burton-Head on the Wirral Side; distant ten Miles from Chester.

He has since been informed; that the Balloon seemed to *REST*, for a few Minutes, in the Air: and then *return slowly* over Chester.

It is therefore more than probable, that as the Balloon continued to ascend; it was *becalmed* in a *quiescent Stratum* or Bed of the Atmosphere, which existed for a certain Depth or Thickness, between the lower and upper Current: and that the Direction of the Balloon was changed; the Instant it arrived within the Influence of the *upper* Current.

Consequently, with a proper Apparatus to ascend and descend *at Will*, without Loss of Gass or Ballast; the Balloon woud have remained suspended *invariably* at the same Height, and *vertically* over the same Spot of Earth: or, with propulsive Machinery; might, on the same Level, have been *rowed* to any Point of the Compass.

76. In passing *only* ACROSS Trafford Meadows, three Miles from Chester; the Balloon lost its usual progressive Motion over the Country: for more than a Quarter of an Hour, following the Course of the River Goway to the West North-West, and towards the Sea, as at Chester: turning gently backwards and forwards round its own Axis, near the Villages of Great and Little Barrow: and making Curves over the Meadows, whose Breadth at those Places was about a Mile.

The Balloon then returned into its former Direction: inclining, *again*, towards a Brook and Meadow near Alvanley: passed Eastward a little to the left of Manley (*white*) Mill: crossed the Forest of Delamere, and Crag of Helsbye, (about twice the Height of Shooter's Hill, near London;) whose lofty Summit was *apparently* reduced to a common Level with the Valley made by the River Wever, and with the adjacent Sea Marsh. Nor could it have been distinguished by a Stranger, as an *Eminence*.

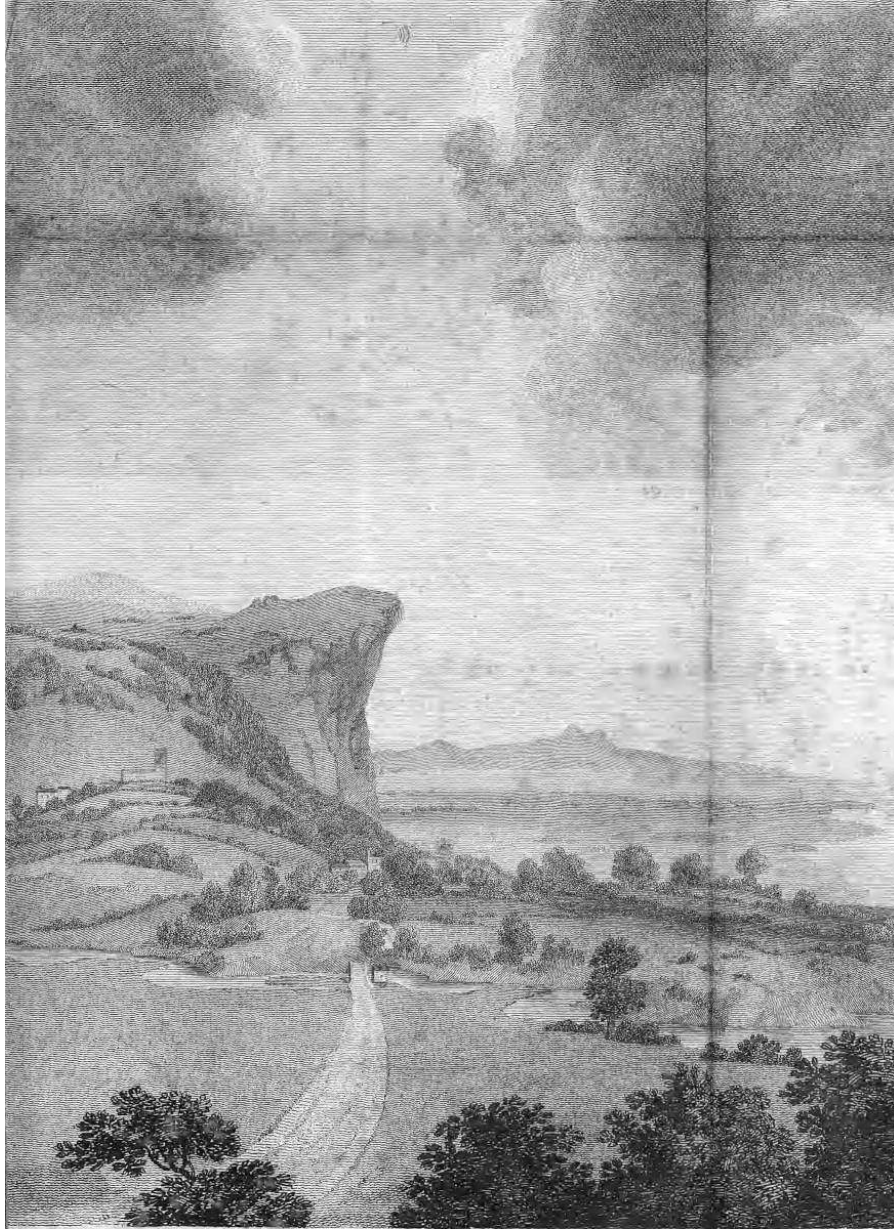
Indeed, the Wood near Kingsley, which grows on a sloping Ground, skirting the Hill, and *from* the Sun, put on a *dusky* Hue; and the Tops of the Trees a *darker Green*: this Difference of *Colour*, conveyed the *faint Resemblance* of a rising SLOPE. A *real* Knowledge of the Country probably contributed to aid the Imagination in this Distinction.

Note: the Print representing a View of the Balloon over *Helsbye Crag*, refers to a Scene in the above Chapter.

CHAPTER XIV.

Section 77. **A**T 39 Minutes after II, Thermometer 60, Barometer $23\frac{3}{4}$, corresponding to the Height of a little more than a Mile.^[22] *the Vapours dispersing*, discovered the Town of Frodsham, and Bridge over the Wever distant from the Town one Mile: the Balloon still continuing at a vast Height; having risen imperceptibly from the Time that the Ballast was thrown down.

From a Conversation held the next Morning at Frodsham, with some intelligent Persons who had descried it gliding gently throu' the Air; the Balloon appeared so extremely *minute*, that it was thought impossible to be the ONE expected the same Day to rise at Chester with an Aironaut.



R. Newton
T. Baldwin Arm. } design et pinx.

Stothard delin.
Sharp sculps.

The BALLOON over HELSBYE HILL in CHESHIRE see [page IIII b.](#)

Published May 1st 1786, by T. Baldwin Chester.

To use their own Expression, “*it could not have been larger than a Bladder, if they had seen it on the Ground.*” The same Persons observed the *white* Flag, like a *Feather* about 8 Yards Distance from the Balloon.

A second Air Bottle was thrown down.

78. The Town of Kingsley being to the East; Frodsham-Bridge half a Mile to the West; the Conflux of the Rivers Wever, and the *wide* Mersey falling into the Sea one Mile farther Westward; the Balloon proceeding in its usual Course over the Country in the *upper*

Current; began to be *impeded*, on its vertical Approach ACROSS the Meadows to the Wever; was actually stopped; and being *entangled* by the *River*, evidently changed its former Direction: imitating, if possible, *its* Meanders; or at least making Gyration in Circles of different Diameters, at the same Time turning different Ways round its Axis: describing Curves, something similar to that of the Moon round the Earth in her Orbit; or of Saturn, Jupiter, and Mars, as those *Curves* are delineated in the *Prints* of Long's Astronomy:^[23] the Course of the River being its *changeable* Center.

79. It is to be observed, that if the Balloon had continued to pursue its *former* Course; no Danger was to be apprehended of its falling on the Sea, or on the broad Branch of the River Mersey towards Warrington.

On the contrary, it must have gone into the Heart of the adjoining County, and passed near Manchester.

It is likewise worthy of remark; that unless a Fragment of light Vapour intervened for a few Seconds; the Country immediately below the Observer was *continually* illuminated by the Sun's Rays: tho' none but the larger Objects were distinguishable at the Bottom of the profound Abyss, *more* than two Miles in Diameter at one View: that being the utmost Boundary of the circular Prospect below.

80. The *Sea* tho' known to be *near* by the Dashing of its Waves upon the Shore, which were plainly heard, was totally eclipsed: as if by Haze or Vapour, which began to be accumulated only at a certain Height *below* the Balloon; yet in such a Manner as *not* to prevent the solar Rays from penetrating throu', and shining bright upon the Water.

81. There was now sufficient Leisure to trace the incredible Variety of most beautiful *Curves*, into which the Stream had worked the Bed of the River Wever in a Course of *Time*, and in the Compass of a few Miles: an Appearance which *demonstrates* the Incorrectness of MAPS.

Some *actual* Clouds presented themselves in detached Groupes over the Land: and the *Land* itself *shone* plainer throu' the Intervals, than in Places near which *no* Clouds appeared.

82. On reconnoitring the scattered Town of Frodsham, which like Chester was of a *light* BLUE; the Balloon moving *by Intervals* round its Axis, the Prospect seemed to *open* on a *sudden*; and the Aironaut could discover the Town of Warrington: the Plan of which was small, neat, but of a *darker Blue*, inclining to *Grey*: the Slates^[24] there used being almost peculiar to the County of Lancaster.

83. From this *Enlargement* of the Prospect over Land, he imagined that the Balloon was either *gently* descending; or that it appeared throu' the *clear* Intervals of *actual* Clouds *below* him.

84. He had Time however to make the following Remarks. Cattle, if grazing in the Meadows, were not distinguishable; or at least were not distinguished. It was in vain to look for Sheaves of Corn, or Hattocks on the Ground: possibly from a Sameness of *Colour* like the growing *Stalks*, and *Field*: or *protruding* but a small Degree of ELEVATION; tho' the *Shadow* even at twelve o'Clock^[25] was something *longer* than the *perpendicular* Height of

EACH Object.^[26] Noises of Carriages along the great public Road; especially Waggon and Carts HEAVILY laden; (the Gratings of whose *Wheels* against the *Stones* seemed uncommonly *harsh*;) were discriminately heard, tho' not *discoverable* by the Eye. Numbers of human Voices were almost CONTINUALLY huzzaeing: except while STATIONARY at the first Rise; when ALL AROUND was wrapt in the Sublimity of SILENCE; which afforded a pleasurable Contrast; —diffusing a DELICIOUS CALM.

A third Bottle of Air was thrown out.

CHAPTER XV.

Section 85. **A**T 4 Minutes past III, the Balloon remained *vertically* over the River, and over the elegant Mansion called Aston.

86. A Wind was heard BELOW for a few Seconds: and the Air felt a *little cool*. Thermometer 55, or Temperate: Barometer $25\frac{1}{2}$, corresponding to the Height of near a Mile.^[27]

87. The Balloon continuing its eccentric Movements from *Side to Side* across the Meadows; yet still gliding *down* the River, in a North-West by North Direction, almost at right Angles to that which it *before* had held; consequently *towards* the Sea, and in a Line which continued must pass throu' the Center of the Channel: some Step it was necessary to take, and SOON. By throwing out Ballast, the Balloon woud instantly rise: but it woud probably, as *before*, rise into A CALM, and therefore *descend* nearly in the same Line: which woud merely *protract* the Time till the Balloon had reached the Center of the Channel: where, having no Resource, the Ballast being then expended; there might be some Risque in waiting for a Vessel, tho' the Balloon woud not for SEVERAL HOURS, have lost its *levitating* Power, so as to have sunk with the Aironaut. To him however it immediately occurred, that there might be an UNDER Current of Air, as usual in the Middle of the Day, blowing from Sea to Land: and, that if the Balloon was made to descend QUICKLY into the Sea-Breeze; it might, in a few Minutes, be carried so *far* within the Country, as to be *soon beyond* the Influence of the *Sea and River*: and THEN, by throwing out some Pounds of Ballast, woud return into the *upper* Current, and pursue a *safe* Course towards Manchester; or even towards Prescot and Liverpool, if an easterly Wind prevailed *above*.

88. In Consequence of these Expectations; he looked downwards *towards* the Sea, *then* wholly *invisible*; tho' the Murmuring of its Waves was *more* plainly heard.

THICK SMOKES were distinguished issuing from *different* Places along the Marsh near the Coast: and *apparently* skirting the Ground, as if impelled by a *brisk* Wind from the Sea.

89. No Time was to be lost.

The Balloon having reached the Cascade; and continuing to move more regularly along the Course of the River, past the *Bridge*, and proceeded to *Rock-Savage*.

90. The Neck or *Mouth* which remained shut, by its own Pressure against the Outside of the upper Hoop, as *it* lay over it; was instantly brought within the Hoop, and set WIDE OPEN in a perpendicular Situation.

Not more than a Couple of Minutes had elapsed before Sounds were more audible and louder.

Cattle and Corn in the Fields became visible.

91. The Observer very deliberately stooping to put down his Card and Pencil; with his *left* Hand grasping the Hoop of the Car, and with his Right holding a Sand-Bag, to throw

over as he approached the Earth; found that the Balloon was *influenced* by an UNDER Current blowing from the Sea: and marked his Progress by the half Mile WHITE Flag; whose Stretcher having acquired a Position parallel to the Plane of the Horizon, placed the Flag in an excellent Point of View: the Balloon *towing* it *apparently* with a *slow* Motion, over the distant Tops of the *dark-green* Trees.

CHAPTER XVI.

BALLOON DESCENDING.

Section 92. **N**O sooner had the Balloon *descended* within the Influence of the *Sea Breeze*, than it became INSTANTLY *condensed* by a certain CHILLINESS which THEN began to prevail.

93. This Height has *since* been considered as the LEVEL of FLEECY VAPOUR, SCUD, or lowest Stratum of Clouds, in BRIGHT and WARM Weather.^[28]

No *visible* Clouds were presented *near* the Spectator. On the contrary, they seemed to *shrink back* to the Distance of a Mile round the Eye; and then *immediately* appear above it, the Balloon continuing to descend. Nor did any *circular* Horizon of the Earth shew itself; till the Balloon had reached below this Level: viz. Barom. 26½, Thermom. 55. i. e. Temperate.

Prospects were most EXTENSIVE and beautiful at this Altitude: which the Barometer estimates at full half a Mile.^[29]

Looking again at the Barometer, scarce a Minute afterwards; it had risen to 27.

94. The Condensation by Chill and Moisture, and quick Contraction of *its* Dimensions acted like a *Charm* on the *Balloon*.

In a Moment; as if dropped from the Clouds, the SEA suddenly presented itself.^[30] It seemed NEAR, and of a RED Colour. *Circular* Landscapes of the *distant* Countries filled the Eye.

Almost the whole Extent of the Channel was a perfect Calm: and rather *dazled* the Sight. But from the Peninsula of Hale to that of Runcorn, and upwards, a *partial* BREEZE from the North-West *ruffled* the Surface (which was there of a *dark* and menacing Complexion;) and seemed in its Course to have reached and *influenced* the Balloon: whose Descent proving *more* rapid than was expected; the Sand-Bag tyed up, weighing 12 Pounds, was opened, and the Sand *dispersed*.

95. The Aironaut continuing as before to *stand* upright in the Car, and having resumed his Card and Pencil; Thermometer *again* at 55°, on finding the Descent not *sufficiently* retarded, wrote swiftly, “NO MORE REMARKS, MIND THE SHIP:” meaning the Balloon: and briskly stooping for the second Bag of Sand, weighing likewise 12 Pounds, *dispersed* it by *Handfulls* in the same Manner.

96. The circular Mouth of the Balloon continuing wide open, at about 18 Inches Diameter; so much *cool* and *moist* Air rushed in during the Descent; that, tho’ its Momentum or acquired Motion was retarded by *Dispersion* of the Ballast, it had not yet recovered an ACTUAL LEVITY: being too near the Ground before the second Bag was discharged.

Presuming however that 24 Pounds Weight of Ballast thrown out, was sufficient to break the Fall, tho' in a cool moist *condensing* Atmosphere of *pure defloguisticated* Air; the Event of *landing* was *waited for*.

It has been *since* imagined that a *heavy* DEPRESSING torrent of *cool* Air took Place from the North-West at a certain Height over the Water, and *assisted* the Descent of the Balloon.

97. In order to judge with what Rapidity the Balloon descended, when so *LOW* as to be within the Influence of the *under* Current, while the *cool moist* Air rushed in at the Bottom, and most probably pressed out the Gass; the following Intelligence has been communicated by a Person of Veracity.

As two credible Farmers were working, with their Servants, in the *HARVEST*; on hearing a hollow, rushing Sound in the Air, which they took to be a *Whirl-wind*, or *distant Thunder*, and which seemed every Moment to encrease and approach them; they all retreated under a large Oak. While there, they first perceived the swift Descent of the Balloon. Two, who were afraid of Thunder, then began to take Courage, boldly exclaiming they should never fear Thunder again, since the *Falling* of a Balloon could be attended with so *terrible* a Noise.

CHAPTER XVII.

BALLOON STILL DESCENDING.

Section 98. **T**HE Car, gliding over Trees in the *farther* Hedge-Row of a Grass Field, glanced on the Ground.

The Aironaut, being *prepared* for the Event, supported a *Part* of the Weight of his Body by his Hands, grasping the *upper* Hoop.

The Balloon *stooping*, and declining from the North-West Breeze, drew the *upper* Hoop out of the Perpendicular: by which Means, the Bottom of that Division of the Barometer-Frame which contained the *Tube*, pressed against the Bottom of the Car on the Ground, was *separated* from the remaining Half of the Frame, and fell on the *Grass*.

The Balloon, then rising with an elastic *Bound*, elevated the Car a few Yards, and descended to the Ground, but *more* gently than before: rose again; and the Aironaut perceiving that the progressive Motion of the Breeze was bringing the Balloon near a *third* Hedge; took up his Knife, (which lay by him *ready open* for Use) and *cut away* the remaining Half of the *Barometer-Frame*; threw out the *Basket with the Bottles*, and *Tunning Dish*; the *Speaking Trumpet*; the *Woollen Gloves*, the remaining half Mile of *Twine on the Reel*.^[31]

The Car *cleared* the Hedge, and *slightly* for an *Instant* touched Ground, the *third* Time.

99. During these Operations; the Aironaut had observed different Persons in Motion towards him, who proved to be several *Farmers* and *Labourers* who had run themselves out of Breath to *overtake* the Balloon.

One asked the Aironaut, whether he intended to alight; and was answered, "*Not for any Time.*"

100. The Car alighted each Time so *smoothly*, that neither the *Watch* nor *Thermometer* that lay near each other on the *green* Bays at the Bottom, were displaced. Nor was the *Glass Tube* containing the Quicksilver, separated from the Division of the Frame in which it was originally fixed: but the whole was brought back, a few Days after, in a perfect State: except a small Hole, made in Consequence of the inverted Situation of the *Mercury* in *Vacuo*, *which* fell against the Top of the exhausted Tube.

The Car *first* landed at 28 Minutes past III, in a Field belonging to a Farm called Bellair, in the Township of Kingsley, near two Miles East by South from the Town of Frodsham, and twelve from Chester.

END OF THE FIRST PART.

AIROPAIDIA.

THE
SECOND PART
OF AN
AËRIAL EXCURSION

FROM CHESTER THE EIGHTH OF SEPT. 1785.

CHAPTER XVIII.

RE-ASCENT OF THE BALLOON.

Section 101. **B**ELLAIR-Meadow: half past III o'Clock:^[32] Thermom. at 55: *bright Sun: few Clouds in Sight.*

The Balloon being now 31 Pound lighter; taking a Direction FROM the Sea-Breeze into the Country, and *again* towards Aston-Hall^[33]; mounted up like a Sky-Rocket, with accelerating Velocity: its upper Parts *nodding* from Side to Side, as if to *shake off* the *resisting* Column of Air immediately above it.

102. There being no proper Opportunity of closing the Mouth of the Balloon on its *near* Approach to the Sea, or during the *Swiftness* of its Descent; tho' there had been *frequent Inclination* to attempt it; this *little* but ESSENTIAL Work was instantly resolved upon. And the more so, as the Mouth had continued open *from* the first: and as Mr. Lunardi did not *happen* to mention this Circumstance: the Utility of which, tho' *too late* to be put in Practice, had, but a few Minutes before, very plainly suggested itself. His Directions were, to *open* the Valve in order to descend: which woud *possibly* have *encreased* the Rapidity of Descent: and, by *introducing* a thorou' Air *upwards*, while the Motion of the Balloon was in a *contrary* Direcion, might have occasioned a *dangerous Rupture* of the lower Parts of the Balloon, *which* actually took Place in a preceding Excursion.

103. The Balloon, tho' rising *quick*, seemed *not* to be wholly disengaged from the Ground, but to have received a Check; and to *lean* a little out of the Perpendicular: particularly the Car, which was evidently drawn a *different Way from* the Balloon.

On perceiving that the half Mile *white* Flag, fastened to the *upper* Hoop of the Car, sensibly impeded the Elevation of the Machine, by *trailing* along the Ground, (the Balloon being yet within the Influence of the Sea-Breeze, or *lower* Current of Air;) the Question was, whether it woud not be imprudent to suffer the Balloon to rise near half a Mile, before the *white* Flag; was *disentangled* and free to follow it.

For as neither the *Twine*, nor the *lower Cords* of the Balloon were of Silk; the Twine having lain on the Trees or MOIST Ground, might become a CONDUCTOR from the Earth to any Stratum of Air that had LESS or MORE than what is called its natural Quantity^[35] of the ELECTRIC Fluid.

Adding to the above, a Wish to rise higher the *second* Time than the first; stooping for the Scissars, the String was *cut*: reserving a Remainder to tye the Neck of the Balloon; which was immediately done by gathering the Parts of the Balloon into the Hand, wrapping a Couple of Yards loosely round, and tying them on a SLIP or BOW Knot: one End of which was PURPOSELY left hanging three Feet downwards, to *untie* instantly on Occasion.

This additional Levity of *nearly* one Pound, gave the whole Quantity of Ballast thrown over in a *few* Minutes, *nearly* 32 Pounds.

104. The intelligent Farmer who stood near the Balloon, when it alighted at Bellair, had observed it for some Time *before* near the Sea, and marked its Return, as coming *apparently* from *Overton*.

At first, which was *more* than five Minutes before it came to the Ground, it seemed to him as if it could not have been *larger* than a Bladder.

He saw it reascend, first *sideways*, then upright; moving from the Sea.

Afterward it rose *rapidly*, and rather *towards* the Sea and Warrington, distant twelve Miles.

He watched it for a Quarter of an Hour: and caught it by Intervals, near and above a Cloud in the *blue* Sky, at so great a Height that it looked like a *Lark*: and at last: so *small* that the People who stood near him could none of them regain a *Sight*, when they had once lost it.

105. The remaining *white* Flag was unfolded, and tyed to one of the Balloon-Cords attached to the *upper* Hoop, at a proper Distance to *play* freely in the Wind: and, notwithstanding all that has been said to the contrary, shewed *instantaneously* and *plainly* the corresponding Changes made by the Wind in different Directions.

And, as the Breeze was accompanied with a Sensation of *Coolness* against the Face of the Aironaut, looking towards that Quarter from whence the Wind came, as indicated by the Flag; (which Quarter was not in a Line with the Path of the Balloon;) the Flag must have shewn that the Change was made by the *Air* in *its peculiar* progressive Direction, and not by *its* Resistance or Progress in the Track of the Balloon.

106. It is probable that the *Momentum* of the Balloon, acquired by its centrifugal or accelerating Force upwards, might have kept it in *one* Direction, while it continued to rise throu' *different* Currents.

CHAPTER XIX.

BALLOON STILL RE-ASCENDING.

Section 107. **T**HE Balloon being now *wholly* unconfined, continued to *rise* with *great* Rapidity: crossed the Meadow in the Sea-Breeze, and remained as *before*, for 6 or 8 Minutes,—by Intervals *gently* turning on its Axis—almost wholly *vertical* over Aston-Hall, but rather more to the *Eastward* of it.

The Country still exhibited *bright gay* and EXTENSIVE Prospects.

108. *Three* Sail of Vessels appeared in the CHANNEL: and *four more* were sailing down the River Wever, *apparently* just under the Balloon, diminishing to *mere Cockle-Shells*, or like Boats which have *no* Rigging.

Shouts continued.

Corn and Cattle were visible in the Fields and Meadows.

Aston, tho' a *large* and *elegant Mansion*, appeared like a *House* which Children *build* with *Cards*.

109. A Chilliness in the Air was again perceived in rising, as he imagined, to the *same* Height at which he felt it in *descending*, indicated by the Thermometer at 55°. ([Sect. 92.](#))

He then found himself inclined to taste the *Brandy and Water*, ready mixed by his Order, and to eat a Biscuit: but on putting the Liquor to his Lips; thought it *too* strong, so drank none, nor eat any Thing.

110. The *three* Sail, and the CHANNEL DISAPPEARED.

The River put on a deep *red* Colour, like the Dee. Its Meanders seemed to *encrease*; as its Width *diminished* to a *broad* Line. Its Water was *lost* to the Sight.

Corn and Cattle were no longer distinguishable.

The House at Aston was yet a beautiful tho' minute Object: the Balloon moving several Times round it; as if loth to quit *that* and the River.

The Cascade was become a *white* Line: and the *fine* Bridge below, a *yellow Straw* crossing the broad *red* Streak.

Of the *four* Vessels in the Wever, *not* an *Atom* visible.

The Shouting entirely ceased.

111. The *blue* scattered Houses, wide public Road called Sutton-Causeway over Frodsham Marsh, the Meadows Fields and Woods, the *lofty* Hills, Helsbye Crag and Halton-Tower, were *reduced* to *one common* Level; and diminished to the Size and Semblance of a *coloured* Map, but it was the *superb* and *finished Colouring* of NATURE.

112. Ceasing to look *down* on the smooth LAWNS *below*, which were *now* of the richest and *fullest* Patterns, seen as throu' the *small* or inverted End of a common Perspective-Glass, and *spun*, as it were, to a *fine Thread*; Pleasure and Delight, tho' of another Kind, fill'd the Imagination of the Beholder: who, raising his Eyes on a Level with himself, so as to look *straight before* him; found that the Balloon had *already*, and almost beyond his own Belief, soared to so amazing a Height in the Atmosphere, as to raise him *far* above the RIM of the immense Bowl or Crater; and that it was still *stealing* with *Rapidity* upwards.

113. During the Contemplation of this magnificent Prospect, a *perfect* Calm took Place, and *soothing Silence* reigned.

And thus; for *a while* detached, *far* detached from Earth, and *all* terrestrial Thoughts; wrapt in the *mild Azure* of the *etherial* Regions; suspended in the Center of a vast and almost endless Concave; come, as a *mere* VISITOR, from *another* Planet; surrounded with the stupendous Works of *Nature*, yet *above* them;—the GLORIOUS SUN except, which enlivened ALL, and shone with pure celestial Lustre;—a peaceful SERENITY of *Mind* succeeded; an ENVIABLE EUROIA.^[36] An Idea of which it is not in the Power of Language to convey, or to describe.

CHAPTER XX.

Section 114. **R**Espiration at so great an Altitude was perfectly free and *easy*: *forced* Trials being made for Information on that Point: a Sensation of Levity seemed *rather* to be communicated by the Air to the Lungs: but this might be the Effect of the Imagination. It was however a *curious* Circumstance to find the Breath *not* visible; the Thermometer rising AGAIN to 60. Nor did the Pulse seem to be quicker than *usual*, in this elevated tho' *inactive* Situation.

115. The Perspective of a vast Series of Thunder-Clouds of a *sulphúreous* and *metallic* Tinge, placing themselves in Ranks, each beyond the other, in *bright* and tremendous Order, and a Sort of *Battle-Array*, beyond Conception *grand yet beautiful*; could not pass *under* him without Notice. The immense circular and visible Distance of the NEBULOUS Horizon, extended NOW 102 Miles *at the least* round the Eye, as already mentioned ([Sect. 52](#)); was a grand Source of the Sublime. Nor did the contracted View of the Landscape below fail, in Turn, to *regain* an Attention to its *indiscriminate yet pleasing* Scenery.

116. ON A SUDDEN he was called back to himself.

Several of the BLADDERS, which were tyed round the Car, in Case the Balloon should *alight on* the Sea, and were DRY on the Outside, began at the same Instant to CRACKLE; being greatly distended by the Air within. When pressed with the Hands and Fingers, they felt extremely hard, and *ready to burst*.

On looking upwards at the Balloon, it appeared GREATLY inflated: the external Pressure of the surrounding Air being *much lessened*, in so elevated and *rarified* a State of the Atmosphere.

117. The Balloon *pressed* in an unusual Manner *throu'* the Meshes of the Net, quite round.

118. The Shape was much altered by this Distention of the Sides: and its *perpendicular* Diameter *shorter* than before.

119. The Neck or *Mouth*, which was *tyed*, had actually risen *upwards*, and was THEN near *eight Feet above* the Bottom of the Car.

120. It was not known till afterwards, that Mr. Lunardi on his second aërial Voyage from Liverpool, had been obliged to cut off the *lower* Part of the Neck, weighing upwards of *two Pounds* and a *half*, in order to lighten his Descent near Tarporley in Cheshire; and that he had not *Silk* sufficient to repair the Loss.

CHAPTER XXI.

Section 121. **I**N vain did the Aironaut strive to reach the *Neck* of the Balloon, from the *Car*. Attempting to put his Feet on the *opposite* Sides of the *lower* Hoop, by grasping the *upper* with his Hands; he could not in *that* Situation raise himself so high *as before*; nor let go his Hold with *either* Hand.

He then stepped *down* into the Car. The Agitation of *which*, brought within the Reach of his Hand, the loose *End* of the Twine (*purposely tyed on a Bow or Slip Knot*) that had stuck to one of the Side-Cords, and held the *Center* of the *Neck* rather *out* of the Perpendicular.

CHAPTER XXII.

BALLOON AT ITS GREATEST HEIGHT.

Section 122. **B**Eing cautious how he suffered the LIGHTEST Gass to escape throu' the Top of the Balloon, which must have happened in drawing down the String of the Valve; yet apprehending the POSSIBILITY of an immediate Rupture at its PRESENT GREATEST Elevation;—glancing his Eyes *around* to take a FAREWEL View;—he *pulled* the Twine, that tyed the NECK.

123. *Instant* Relief was given to the Balloon: which shrunk into the Shape which it had assumed in the former Ascent, when the Gass began to issue in visible Vapour, the *Neck* likewise *lowering* itself to the Height of his Shoulders, as in Section 35.

124. On stooping he found the Time 41 Minutes after III, and the Thermometer 57.

Nor was he surprised that no *visible* Vapour escaped; as he had imagined that much *common* Air had been pressed into the *Mouth* of the Balloon: and which, being *heavier* than Gass, woud *go out first*.

On that Ground he was confirmed in his Resolution *not* to open the Valve at the top, which always emits the *lightest* Gass.

125. As soon however as the Neck of the Balloon reached his Shoulder, he *gathered* the Silk in his Hand, and held it *Air-tight* tho' untyed, to prevent Evaporation of much *real* Gass: presuming that if any *Levity* remained; the Balloon woud presently *rise* again, and *swell*.

And he was pleased to find the Event answer his Expectations.

CHAPTER XXIII.

AIR WARMER ABOVE THAN BELOW.

Section 126. **I**T was a Matter both of Surprize and Pleasure to observe that the Thermometer had risen AGAIN to 60, when the Balloon had soared *above* the Sea-Breeze; as the Aironaut had expected to feel the extreme Rigour of Winter; and had made Preparations against *intense* Cold.

Nor did he find any Difficulty in Respiration *during* the Excursion; which may possibly be accounted for from the *Warmth* of the Air.

That the Breath^[37] was *not* visible at *any* one Time, and particularly while the Balloon was elevated above the *under* Current, might it not be owing to the uncommon DRYNESS of the Air, which woud *dissipate* the Vapour at the Instant of *Exposure*?

127. It was remarked, some little Time *before*, and *during* the *last* Glance of the Prospect taken at the *highest* Elevation, that the House at Aston was still visible, and the *dark coloured* LINE forming its DIMINUTIVE Shadow seemed *thicker* in Proportion to the *Plan*, than when the Mansion was *first* seen before the Re-ascent. And it had a *sensible* Effect in *apparently raising* it above the *common* Level.

128. The Circuit of the *Land-Abyss below* was also greatly *contracted*: and a Haziness inclining to a *dark Green* seemed to cover the *outward* Verge round the *Lawn*.

The *red* River Wever only appeared.

The Channel and *broad* Branch of the River Mersey towards Warrington, had long since vanished.

The Lawn itself, which composed the Ground-View, was full of *innumerable* Enclosures *almost* CLOSE to each other; with *much* Wood:—dwindling to the Pattern of an elegant Turkey-Carpet: which, according to Principles of Mahommedan Faith, tho' wrought in *gay* and *vivid* Colours, is *made* to exhibit NO EXACT^[38] *Resemblance* to the Works either of Art or Nature.

129. The Colours, of which the Ground Work was *principally* formed (except WHITE; also the *roughened* Sea, which *alone* was BLACK; and Shadows, which *constantly* gave a *transparent* VIOLET) were four simple and *primary* ones, viz. RED, YELLOW, GREEN, and BLUE: all which seemed to GLOW, tho' in a *less* Degree, like the Colours of the Prism.

This unmixed *Coloration* of Objects, to be seen from a vertical Situation *only*, to be seen without *Refraction*, is a new singular and *pleasing* Phenomenon.

130. A View, taken *above* the Level of the Clouds, may, from this Circumstance, without Impropiety, be called a CHROMATIC VIEW of the Earth: of which, the *Print* is an Example: delineating the Extent of the aërial Excursion; and placed at the End of the *second* Part, including the Re-ascent.

CHAPTER XXIV.

BALLOON ABOVE THE INFLUENCE OF WATER.

Section 131. **T**HE Balloon pursued its former *gentle* Course in the *upper* Current of Air moving from the South West, and Aston House: and had risen *above* the Influence of the *Waters* and *Sea-Breeze*.

132. In Consequence of having *held tight* the Neck of the Balloon, the Gass *within* began *again* to expand, and the Machine became *more bloated* than when *stationary* at the first Ascent: the Bottom of the Balloon being drawn up to the Height of his Hand, when the Arm was stretched, and himself on Tip-toe.

133. Tho' the late Descent, at the last OPENING of the Balloon, had been rapid; which was known *chiefly* by the Want of Reaction from the Bottom of the Car against the Soles of the Feet; yet being still *far above all* Clouds; fearless of the *Currents*, *Rocks*, and *Shoals*, to which ALL MARITIME *Navigation* is subject; he took the Opportunity of trying the upper Valve; *purposely* to know the Effect. So retaining the Bottom of the Balloon in his *right* Hand, he drew the Valve Cord with his *left*.

Immediately he heard it *click*: which proved that it was quite open, and in good Order.

134. He tried the Valve three Times *smartly*, and deliberately.

The Escape of the inflammable Air or Gass was like the *growling* Sound made in a Mill by the Grinding of the Mill-stones, but by no Means so loud.

CHAPTER XXV.

THIRD BALLOON-IRIS

Section 135. **T**HE successive Operations of untying the Neck, and repeated Trials of the Valve, brought the Observer so low, that he could trace the *Image* of the Balloon on the *upper* Surface of *light silvery* Clouds beneath him.

136. *Iris*, a bright *celestial* Nymph, his *former* Attendant, deck'd in gay Attire as usual for the Bow, made her *third* Appearance: instantly *encircling* the Balloon. Nor was her Stay so short as before; as if to *recompense* the Aironaut for the lost Sight of Earth and all *terrestrial* Objects, which then began to *disappear*.

137. In less than *a Minute* after the *Deflation*; the Neck of the Balloon continuing to be held tight in the Hand; the Balloon *quickly* encreased in Bulk, and soared *aloft*, as before.

138. It continued *rising* as long as the Hand could *reach* to hold the Neck *tight*: and, on loosing it *an Instant*, made a rapid Descent: on Account of the *Gass* which escaped, and of the atmospheric *Air* which rushed in by the *same* Opening at the Bottom.

139. The alternate Play of FAST AND LOOSE, was frequently and *successfully* repeated: the Balloon always rising till it swelled out of the Reach of the Hand: at which Time it was let go: and the Neck (as well as the Balloon) descending; was *presently* caught in the Hand, and made *Air-tight* as before.

140. These Manouvres were performed, at a Height *far* above the Level of *all* Clouds, and in Sight of Numbers of People: some of whom were at least 15 Miles distance: yet could plainly, from an Eminence called *Hoole-Mill* Field, a Couple of Miles from Chester, discover the Balloon at an amazing Height, darting up and down several Times; or as they expressed themselves, "*quivering and warping in the Air.*"

CHAPTER XXVI.

SENSATIONS ACCOMPANYING THE BALLOON.

Section 141. **T**HE alternate Elevation and Descent of the Balloon gave sufficient Leisure to reflect on the SECURITY and PLEASURE of his Situation, thus *wafted* on the *Pinions*, and *merging* in the *Ocean* of Air.

Indeed the whole Excursion was a Continued Scene of Pleasure.

The Eye and the Imagination were beyond Measure delighted.

142. If there had been any Thing to wish for, it was the *living* Pencil of ANGELICA,^[39] or some other celebrated Painter: in order to gratify the World with the *bright Miniatures* and *Colouring* of so much *variegated* Beauty.

143. As it would be difficult, if not impossible, by *mere* Description, to convey an adequate Idea of the different SENSATIONS experienced while in the Car; (for Pleasure is itself unspeakable;) yet the Fancy may possibly, without Censure, be a Moment *indulged*, in its Allusions to such familiar Subjects as approach nearest to THEM: so as not to leave the *public* Mind *wholly* in the Dark, with Respect to the above Points of natural and general Curiosity.

144. Most young People, whenever they have Opportunity, amuse themselves on the SLACK ROPE, or Swing: the Pleasure *encreases* in Proportion to the *Loftiness of Ascent* they are *able* to acquire.

145. In the East, where the Heat of the Climate forbids robust Exercises; the *Swing* is considered as a princely Diversion: and of which the MOGUL himself *condescends* to partake. He is swung by Slaves: and thus enjoys the *pure* Air *without* Fatigue.

146. The Ascent of the Balloon is not unlike what is felt, in the *ascending* half of the Swing: and the Descent is attended with that agreeable Sensation known to those who *sink* throu' the *descending* half.

147. A Diversion similar to the above is peculiar to the *North of Europe*, practised by the Russians, particularly the Inhabitants of Zarsko Zelo; and accompanied with a Sensation so *delightful*, that they seek it in the *open* Air, amidst the *utmost* Severity of the *Frost*. It is a Sort of Boat or *Car*, in which they *glide*, for a considerable Distance, *down* an *artificial* Declivity of *waved* and *polished* Ice: being drawn up by Servants; they launch precipitately forwards, and *down* again as before.

148. *Sledges* drawn *swiftly* over the undulated Surface of a *snowy* Country, a favourite Diversion in many Parts of Germany, in Lapland, and Siberia: Skaiting on *level* Ice; the Motion of a Vessel on *smooth* Water; of a *fleet* Horse; also of Wheel-Carriages rolling over EVEN Gravel, or a *grassy* Plain, are each a *Luxury* of the same Kind; and *grateful* to the Nerves.

149. There is yet another Amusement, which is said to be of *German* Extraction, still frequent in the North of England, called the *vertical* FLYING-COACH.^[40]

Two Persons are required to turn the Machine (when full): which moves like the four Sails of a Windmill: a Seat being placed at the End of each Sail.

150. The *Pleasure* communicated to the *Nerves* during the Descent, is to some Constitutions so *exquisite*, as to be full as much as the human Frame can support: others are affected by it in a *gentler* Manner.

These different Diversions, flowing from the same Principle in common with the Balloon, viz. that of *being carried with a gentle Motion*, are *one or other* suited to all Ranks and Ages.

151. The Pleasure of the double Slack Ropes, when seated in the Car appended between them, is perhaps in itself *superior* to that of most others.

152. The *vertical Flying-Coach*^[41] compleats the *Circle*, of which the Slack Rope describes but the lower *Half*.

153. The Sensations communicated by the Motion of the Balloon, come nearest those of the vertical flying Coach, tho' *more* gentle, and if possible, *more* pleasing.

At Sea, the most experienced Mariner is sometimes *sick* or *giddy*.

154. Nothing of the Kind happens in the Balloon: where an infinite Variety charms the Imagination.

155. The Spirits are raised by the *Purity* of the Air^[43], and *rest* in a *cheerful* Composure.

156. Even when *stationary* above the Clouds, the *Height* conveys with it no *Danger* of *falling*: any more than *when* in a Vessel at Sea, (as off the West-India Islands, for Example) the *Fish* are seen gliding over the clear *white rocky* Bottom, at the Depth of twenty Fathom: as the Aironaut seems perfectly unconnected with the Earth, and unconcerned about it.

157. Nor does the Depth *below the Clouds* give an Idea of *Distance*. On the contrary, the *smooth chequered Lawns* which form the Surface of the Earth, are presented to the Eye, as on a *Level* with the *Clouds* themselves: *at least* COME UP to their UNDERSIDES, and appear so much a Part of *them*; that the *Clouds* occupy the Place of *Earth*: and the Aironaut seems able to descend from the *Car* upon the *Clouds*, and to walk from Side to Side over the *empty* Space, as over a Sheet of *transparent* Ice, across a *River*, whose Depth is equal to the *small* but indefinite Thickness of the Clouds.

158. It is from *frequent* EXPERIENCE only that the *Diminution of Objects* presuppose their *Distance*.

CHAPTER XXVII

USEFUL CONCLUSIONS.

Section 159. **I**T was remarkable that, the lower Parts of the Balloon regularly adopted a *similar* Form at each Descent: not unlike a *Ship's Bottom*; looking up, at the Head or Prow, while on the Stocks: the *Neck* of the Balloon forming a beautiful *central* Pillar; in Shape like that of a *Speaking Trumpet* inverted.

And hence may be derived a Piece of *useful* Information: as the *precise* Time of descending is discovered by bare *Inspection* of the Machine.

160. Another Conclusion seems likewise deducible from the above, that if the Balloon is so burdened, as to *descend* while it retains the Form of an ELLIPTIC SOLID;^[44] it will descend more rapidly, than if it contained less Gass: the Force of Descent in both Cases being supposed the same.

For if the Diminution of Gass be so great as *not* to fill the upper Hemisphere of the *Balloon*; the Resistance of the atmospheric Air *below* woud probably give *it* the Appearance of a *Concave* or Umbrella, which woud greatly *check* the Descent: viz. in Proportion to the Square of the Number of Feet of which the Surface was composed.

161. Hence also the evident Utility of an EQUATORIAL HOOP for Balloons: in Preference to a Parashute, which woud be only an Incumbrance.

CHAPTER XXVIII.

Section 162. **A**T 40 Minutes past III, when the Balloon was *apparently* some Miles above the Level and Summit of the Clouds; a SUDDEN and uncommon *Sound* was heard for three or four Seconds only.

A Sort of *hollow* Wind seemed issuing from a Plain of Clouds in the North-East Quarter, greatly below the Balloon: which as SUDDENLY ceased.

The Instant the Sound was heard; a gentle Motion was *impressed* on the *Balloon*, as if by a Hand touching *it* near the *Top*.

163. Clouds to the North-East appeared, for the first Time, in *rapid* Motion towards the Balloon.

They *sailed* directly *under* it: filled up the Chasm, and drew a *white* Veil over *all terrestrial* Objects.

164. It has been *since* imagined, that a fresh Wind *descended* from the South-West Quarter in the upper Current, and was heard in the North-East, being ecchoed from the upper Tier of Clouds *below*: and that the Balloon, finding less Resistance than the RANGE of Clouds, soon overtook and passed them: particularly as the lower Part of the *white* Flag vibrated only in the usual Direction.

165. The *encreased* progressive Motion of the Balloon was *not* perceived ([Section 18](#)): being considered as at *Rest*, and the *apparent* Motion *referred* to the *Clouds*.

CHAPTER XXIX.

Section 166. **I**N a few Minutes, a Side-Break throu' the Clouds discovered a long ill-formed narrow Line or Ditch, something less than a Foot in Breadth, extending several Ways: and which from its Proximity to Places that were known, and coming into View; viz. the Country about Norton and Halton-Castle; proved to be the Duke of Bridgewater's Canal.

SUDDENLY came in Sight the *spacious* OPEN of the Mersey above Runcorn Gap: which appeared of a RUDDY Colour, and very near: as if the Balloon had again *felt* the Influence of the *River*.

167. A new System, that of *Balloon-Geography* here suggested itself: in which the Essentials of *Proportion* and *Bearings* would be far more accurate, than by the present Method, both for *Maps* and *Charts*, viz. To make Drawings by SIGHT, from the Car of a Balloon with a *Camera Obscura*, aided by a Micrometer applied to the under Side of *the transparent Glass*.

The Season proper for such an aeronautic Expedition, would be *any calm bright Day*: the Wind having blown from the South West Quarter, for some Days before, which is *frequently* the Case: the Air, at *such* Conjecture probably remaining WARM, to the Height of a Mile or more, unless in the very Midst of Winter.

168. And particularly for Charts, which in a *maritime* Country are *most* useful: as Balloons have an extraordinary Predilection to become *stationary* over Channels and Rivers; altho' a very *strong* Gale of Wind, should continue the *whole* Time to blow in an horizontal Course directly UNDER the Balloon.

Of which Event the Writer of this Account was an *Eye-Witness*, in the Case of Mr. Lunardi: who was *detained* above 20 Minutes over the *broad Bend* of the River Mersey, near Ince, in Cheshire, the Day he *landed* between Tarporley and Beeston-Castle, ascending from the New Fort at Liverpool. He quitted his Station by the Escape of Gass, and descended into the *Stream* of Wind, which continued as *violent* as before.

CHAPTER XXX.

Section 169. **T**HE Summer Scenes of Fairy-Land below, being soon eclipsed by the *quick* Intervention of a *Range* of Clouds; the SUDDEN Contrast of which was highly pleasing to the Imagination; a Prospect of MID WINTER instantaneously succeeded.

170. The Earth's *Surface* throu' an immeasurable Crater of Vapour accumulated round the Aironaut, who was suspended, and seemed fixed in the Center above it, *no longer existed*. And, if it will not be allowed, that a *new* Earth, and a *new* SKY appeared; at least, let the Imagery and Resemblance of what was really seen, be taken from that EARTH, which in Fact did *not* appear.

A WORLD of *Clouds*, GREATER than the ONE below, became, for the first Time the *sole* Object that engrossed the Sight. (See [Section 144.](#))

171. The Balloon was *apparently* raised some Miles above the *Surface* of a *concave shallow* Plate, or Shell, or rather an immense Plain, *which* was in general smooth and well defined: but the *dense* tonitruous Masses, rising here and there *above* the Rest, greatly resembled steep and RUGGED MOUNTAINS seen in Perspective, at different Distances from 5 and 10 to at least a hundred Miles. ^[45]

An unvaried deep cerulean and pellucid Azure, without a Cloud above, enclosed the NOVEL EARTH: whose Surface, whether Valley, Plain, or MOUNTAIN *in Appearance*; seemed as if covered to a prodigious Depth, by successive Falls of Snow, driven and polished by the Winds and Frost, and dazzling to the Sight: the Sun still shining above all, with WHITE, unremitting and invigorating Rays. ^[47]

CHAPTER XXXI.

Section 172. A Thunder Cloud in most grotesque Form;—of superior Magnitude, Density, and BRIGHTNESS—a *celestial Colouring*; and whose *Shade* was itself a *Colour* of semi-transparent and transcendent *Blue* and *Violet-Purple*;—remaining for several Minutes, *exactly under* the Balloon, *tempted* the Aironaut to descend into it; and, if possible, investigate its Structure and Composition.

Blanchard, he knew, had passed throu' MANY without Danger: any Fears that might otherwise have been entertained on that Head were therefore groundless: particularly as Gass, i. e. *inflammable Air* and the *electric Fluid* (supposing an electric Atmosphere had surrounded the Thunder Cloud) mutually *repel* each other. He however declined the Trial: among other Reasons which then offered; that the temporary and apparent Rest of both Balloon and Clouds portended *his* Situation to be over the Center of *some Water*: so that if Gass had been let out in order to *descend*; *enough* might *not* have remained to make Choice of a proper Place to *land*.

173. Some Minutes after; on the *Retreat* of the Clouds, or *progressive* Motion of the Balloon; he found himself suspended over the most *enchanted* Meanders of a Rivulet.

Where he could not tell.

CHAPTER XXXII.

Section 174. **H**E *thought* himself again over the Wever.

At 47 Minutes after III, the Prospect *beneath* opened, just wide enough to shew, that he was suspended in the open Space over the Center of some Rivulet.

The Map of the Country which had been so carefully studied, was *now* consulted for the first Time, but could not bring to his Recollection any Traces of the extraordinary Curves which then met his Eye.

They bore not the least Resemblance to any Part of the River Mersey.

No River like that below him had ever presented itself.

Its *Doublings* were so various and *fantastic* as to exceed the Limits of Credibility.

175. He was still stationary, at an immense Height, without the *least* Inclination to descend: having *some Time before* taken the Precaution to tie *again* the Neck of the Balloon, as soon as he had perceived it did *not* inflate, as at first, to any *dangerous* Degree.

No Towns, no Houses appeared. No *public Roads* were discoverable. No Voices were heard.^[48]

The Country beyond the Rivulet began to disclose itself: but was quite *new* to him at that *Altitude*, and seemed as if almost covered with Wood.

176. His *Watch* shewed the Time of the Day, and the Sun alone *sufficiently* indicated the Point of the Compass.

The *white* Flag manifested *no* Change in the Wind.

But whether he was near Liverpool, Wigan, or Manchester, he could not discover.

177. He was entirely *lost* in the *blue Fields* of Air; far above the Summits of the Clouds; tho' the Balloon was in Sight of the Earth, and of Numbers who were gazing at it.

178. The *Colour* of the new Rivulet was full as *red*, as any he had seen before.

He thought it might be an insignificant Brook, which tho' curiously curved, was too small to be inserted in the Map.

Still he continued over it: turning and returning *gently* in small Curves.

179. He presently passed *Northward* of the Rivulet *over* a woody Country, in which he could discover *no* Variety of *Colouring* either in the Ground Work or Enclosures; the whole having a *dark green* Cast.

An Appearance of a very distant and remote *Plain* then presented itself; the Size of a moderate Carpet: of a *ruddy* Colour; and surrounded by a *green* Border. Being an unusual Object it continued to engage his Attention.

180. Not far from the first, another of the same Kind, of a more dusky Cast, but *less* and somewhat nearer, that is *more under* him, then attracted his Notice.

He wished to *decipher them*, but in vain.

181. The Sun shone BRIGHT on both: and in a very few Minutes, the *circular* Prospects *encreased*: which was now become a *regular* and undeniable Signal that the Balloon had begun to *descend*. ([Section 17.](#))

The *latter* Plain appeared, at the first, about the Size of a common *Handkerchief*.

The Balloon continued to descend.

182. In a Couple of Minutes, the Plain appeared intersected CLOSELY every Way, like the Coat of a *ripening Melon*. Descending a little *lower*; it seemed covered with a *Net*, the *Meshes* of which were distinct. And *lower* still; it extended itself *greatly* on *all* Sides: (at which Time a certain Degree of *Chilliness* prevailed:) and was then *again mistaken*, and looked upon as a DRY *Heath*, deeply overrun with Shrubs of the same Name.

183. The Descent of the Balloon being *rather* quicker than was expected, or desired; it was deemed expedient to have Recourse to the *last* Bag of Sand, which lay open, and weighed 20 Pounds.

It was accordingly thrown out, a Handful at a Time.

But that Method not seeming *sufficient* to *check* the Descent, when at the Height of 150 or 200 Yards; *all* the Sand was poured out, and the Bag thrown down.

This had the desired Effect: and the Balloon continuing to descend with a Motion *uniformly retarded*, alighted, as the DOWN of a *Thistle*, in the gentlest Manner, without ANY *Rebound*.

184. There being scarcely a *Breath* of Air abroad, the Aironaut made no Use of his Anchor and Cable: but continued as from the first, *STANDING upright* in the Car; which, having moved a Yard or two *only* along the Ground, rested in a perpendicular Situation.

The Balloon, suspended over him like a *vast Umbrella*, LEVITATED *vertically* in the grandest Manner.

185. He was *alone* when he alighted: but, in a *few* Minutes, found himself surrounded by the Country-People, who had waded *above Ankle-deep*, and came running from all Parts, to see the WONDER, and contribute their *Assistance*.

186. He *landed* exactly, at 7 Minutes before IV: Thermometer 59: but WHERE he could *not* tell.

The first Question was “Pray where am I?” And the Answer;—in *Lancashire*.

On asking the nearest Distance to a Turnpike-Road; the People said he was within *two Fields* of one, and offered to conduct him thither.

He accepted their Offer, and shared *his Liquor* among them.

CHAPTER XXXIII.

Section 187. **T**HE Balloon alighted *near* the *Middle* of a MOSS; called RIXTON-MOSS, a Place he had never before heard of.

It was a large Tract of unenclosed WET Land, above four Miles long and above two broad, intersected by Ditches or Water Courses, which divide the Moss into Fields of a *moderate* Size. The whole is surrounded by *tall* Forest Trees.

This was the *lesser* of the two dusky Plains, which appeared about the Size of a Handkerchief, and which he wished to decipher, but in vain.

188. Rixton-Moss is situated five Miles North North East of Warrington, and a little to the left of the Turnpike Road leading from thence to Manchester, and 25 from Chester.

189. He has since been informed that the other Plain, about the Size of a moderate *Carpet*, was no *less* a Place than CHAT MOSS, a vast *Tract* of barren wet Land, *many* Miles in Extent.

190. Curiosity tempted him to make particular Enquiry concerning the Rivulet over which he hung, *admiring* the Beauty of its serpentine Meanders; and, from a Description given of his Manouvres over *Lymm*, situated to the East of Warrington, and from a peculiar Curve, appearing in the Form of a *true Lover's Knot*, when over the Gunpowder Water-Mills, he was convinced the *Rivulet* could have been no *other* than the broad Branch of the River MERSEY.

191. The AËRIAL EXCURSION was performed in two Hours, and a Quarter, within two Minutes.

The Distance of the Balloon-Course, if traced along the Ground, 30 Miles. Section 130.

192. In comparing the Dates at Bellair and Rixton-Moss; it is certain that the Balloon, excluding the Force of *Ascent*, must have moved *forwards*, during some Part of the Re-ascent, at least at the Rate of 30 Miles an Hour: tho' the Aironaut, for the most Part, imagined he was gliding throu' a serene Atmosphere.

Probably the progressive Motion was encreased, from the Time the unusual Sound was heard, in Section 162.

Note: The Print, representing a CHROMATIC View above the *Level* of the Clouds, of the Country from *Chester* to *Rixton-Moss*, is to front the *left* Page, at the End of this Chapter.

END OF THE RE-ASCENT.

CHAPTER XXXIV.

THE SEQUEL.

Section 193. **T**HE Sequel contains an Account of *several Flights* made, in Presence of the Aironaut, by different Persons, during *three* Hours, in the Car of the Balloon, viz. from the Time he alighted, till *after* SUNSET.

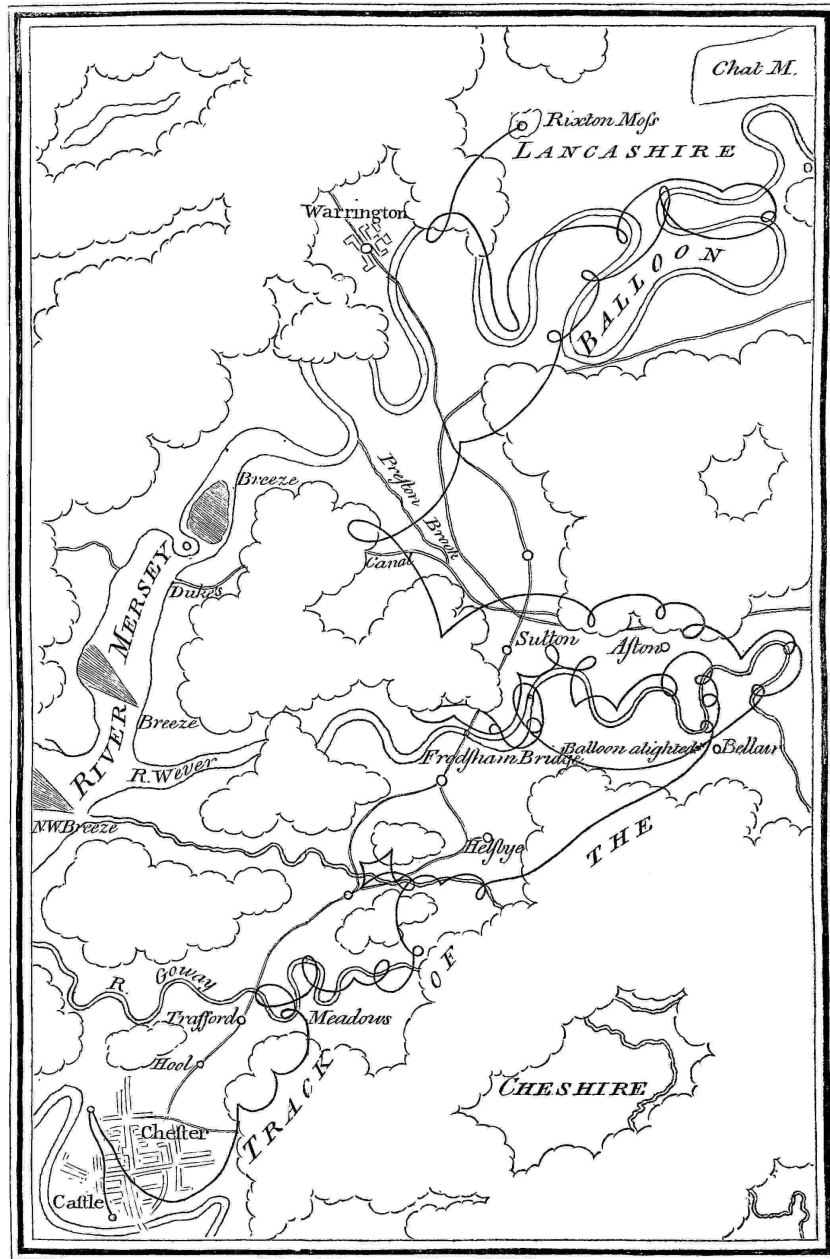


T. Baldwin Arm. del. et pinx.

Angus sculp.

A BALLOON PROSPECT *from* ABOVE *the* CLOUDS see [page IIII c.](#)

Publish'd May 1st. 1766, by T.Baldwin Chester.



The EXPLANATORY Print.

see [Page IIII](#). d.

Rixton-Moss, LANCASHIRE, IV. o'Clock P.M.

The Afternoon being *fine*, the Sun *bright*, and the Air *calm*; finding the Country People remarkably civilized and kind; and having dispatched a Messenger on Foot to return in a Post Chaise from Warrington; the Aironaut was resolved to gratify the Curiosity of his numerous Followers, and give the young People a Taste for Balloons, by treating them successively with an Airing.

194. Indeed it was no inconvenient Method of removing and conducting the Machine: and possibly different Positions of the Balloon might furnish a *useful* Hint.

Having asked aloud *who chose to ride*, several answered in the Affirmative. So having pitched upon a young Fellow of less Weight than himself; bid him get up, between the Cords, over the Hoop, into the Car; stand near the Middle, and hold an opposite Cord in each Hand.

He obeyed with the greatest Alacrity: and seemed to be *a noisy bold* Adventurer.

195. The *Aironaut* then got out; and having suffered the Balloon to rise; fastened the End of the Cable to *central* Meshes of the Net, at the Bottom of the Car: ordering the strongest and tallest Man to hold the Cable, and let it go by Degrees till the Anchor or grappling Iron *alone* remained in his Hand.

The Balloon now rising *above* the Height of the Trees, and giving the Adventurer a new and extensive Prospect of the Country; he became *silent; pale*; his *Countenance* the *Picture* of *Distress*; looking *down* as if for *Help*.

The Conductor repeatedly bid him take Courage. But, in vain.

By lowering the Car *within* the Height of the Trees, he seemed to *recover* from his *Dismay*.

CHAPTER XXXV.

Section 196. **T**HE Route of the Balloon being now throu' a *flat woody* Country, with *tall* Trees growing in the Hedge Rows; a Difficulty occurred, how to conduct the Cable, when the Balloon was *above* or *between* the Trees, without entangling; which gave the Conductor much Trouble, as he was frequently obliged to walk round a Field, the Balloon being held in the Center, before he could espy a proper Opening.

The Procession marched slowly forward: and the young Man was carried among *his Peers* in Triumph through the Air, across the Turnpike-Road, into the Middle of an open Grass Field, where he descended; took a Companion *less heavy*, and left the Car.

This Stripling was a good Deal surprised the Instant he rose above the Trees; but ventured to look around: and appeared on the *whole* much delighted.

197. A great Concourse of People were now collected.

Accidental Carriages halted: joined the Cavalcade, and partook of the Diversion: the *greater* Part following the Balloon throu' the *open* Fields adjoining the Road.

The Conductor *generally* preferring the beaten Track; yet *suspecting* the Balloon with its Adventurer in the Car, might *designedly* be suffered to *escape*, took the Precaution to have the Grapple held by *nearest* Relations to the Person in the Car.

198. The Gass evaporating; a smart young Fellow, who seemed ready for the Jaunt, stepped in: on which the former resigned his Place. But he was no sooner raised a few Yards above his Companions, than the *florid* Colour forsook his Cheeks; he *trembled*; bent himself *double* with Fright; and the Balloon was obliged to be hauled down.

199. A fond Mother then requested that her Child, a fine blooming Girl, might ascend: boasting of her Courage, and comparing it with that of the Person who had none.

The *Venus* smiled, and mounted her Car with great Spirit.

200. Some Ladies and Gentlemen of the Neighbourhood who had watched the Balloon, while it hung at an immense Height over Lymm, and the Gunpowder Works on the River Mersey, came, in their Evening Walk, to meet it: joined the Procession; gave the Aironaut *polite* Invitations to their Houses, and shewed him every possible Civility.

201. The Resistance made by the *Surface* of the Balloon, against the *least Breath* of Air moving *horizontally*, was *frequently* tried by occasionally holding the Grapple: and it was a decided Point, that the *least Motion of the Air* was sufficient, together with the Action of *Levitation*, to prevent the Person, who held the Grapple when the Cable was extended, from transporting the Balloon against the Current: nay it was with Difficulty he could remain in the *same* Place: the Balloon sometimes pulling him forwards, and almost off his Feet.

202. When the Air was perfectly *calm*, which frequently happened while the Balloon migrated with different Passengers, as the Evening was the finest in the World, and the

Country flat and woody in the Hedge-Rows; it was with *Difficulty* that the Conductor could draw the Balloon after him, faster than the Rate of a moderate *Walk*: viz. three Miles an Hour.

CHAPTER XXXVI.

Section 203. **T**HE Sun set at 34 Minutes past VI. and, tho' it was *then* near that Time, the Post-Chaise was *not* arrived.

204. On Enquiry for a dry smooth Meadow, he was recommended to proceed a little farther, to a Place on the Road within three Miles of Warrington.

205. Having by this Time gratified the Curiosity of the Country in admitting Boys and Girls to the Age of six or seven Years, into the Car; and being arrived after Sun-set at the Place appointed, viz. *Milton's Croft-Green*; he ordered the Balloon to be laid on its Side along the Ground: having removed the Car, and opened the Mouth; the inflammable Air or Gass, was soon pressed out by Means of a *long Pole* rolled *across* it by two Men, standing one at each End of the Pole: beginning at the Top or upper Valve, which was held down close to the Ground; and ending at the Mouth or Neck.

It was then rolled up, put into the Car; and the whole Apparatus placed on the Top of the Chaise which arrived the Moment wanted.

206. The Operation was completed at 53 Minutes past VI: the Conductor having accompanied the Balloon on Foot exactly THREE Hours.

207. The Balloon had therefore continued *floating* in the Air, with different Persons, in the whole, for the Space of five Hours and a Quarter.

The Conductor, promising to accept the very polite Invitation offered him by Mr. *Stanton*, a Gentleman who is principally concerned in the Gunpowder-Works upon the Mersey; called at his House, and partook of some Refreshments.

He then drove to Warrington, where he was met by a Person whom Curiosity had inspired to follow the Balloon *on Foot* from Chester, as long as he could keep it *in View*.

208. Mr. Lunardi likewise with great Civility dispatched his Servant to assist the Aironaut in *the Care* of the Balloon; but he did not arrive in Time; not reaching Warrington till VIII. at Night: having lost Sight of the Balloon about *Daresbury*, four Miles from Warrington.

209. Nor was it visible to any, at least very few, of the Inhabitants of that Town, which was equally hidden from the Aironaut: who, *then* ignorant of his Situation, must have remained a considerable Time suspended above the Clouds; which concealed both the Town and River.

He saw Warrington but twice when ABOVE: for a short Time, at a great Distance, and a *mediate* Altitude.

210. The following Day he returned to Chester: was met by the Militia-Music, and ushered with loud Huzzaes into his native City.

On his safe Arrival; besides the private and sincere Congratulations of his Relations and Friends; the Bells rang: his Flags were carried in Procession, and every public Demonstration of Joy was shewn on the Occasion.

**TO THE INHABITANTS OF CHESTER
T H A N K S.**

**END OF THE EXCURSION
THROU' THE AIR.**

AIROPAIDIA.

CHAPTER XXXVII.

OBSERVATIONS, HINTS, AND CONJECTURES,

ON THE

SUBJECT

OF THE

BALLOON AND EXCURSION

FROM CHESTER THE EIGHTH OF SEPT. 1785.

OF THE WEATHER, IN THE VICINITY OF CHESTER, ABOUT THE TIME OF THE EXCURSION.

Section 211. **F**OR more than ten Days *before* the Balloon-Voyage, the Wind had blown (*interruptedly* on Account of the Sea-Breeze) from South and South by West.

Monday the 5th of September:

A Conjunction of the Planet Mercury and the Moon, at ONE in the Afternoon.

Tuesday the 6th:

A violent Hurricane in the South of England, at London, Portsmouth, &c.

The same Day at Chester North-North-West, and distant from London 182 Miles; South-Breeze; Rain most of the Day. Thermometer at Noon in the Shade, 62: and 14 Divisions colder each Night, than the *following* Day, at an Average of five Years. Barometer, below *Much Rain*, viz. at 28 Inches 9/10ths.

Wednesday the 7th:

Violent Squalls from South and South-West, with hazy Air, till half past IV in the Afternoon. Thermom. 58; Barom. Changeable, viz. 29 1/2.

Thursday the 8th, which was the Day of the Excursion:

Much bright Sun. (On Enquiry) calm *below* till half past III in the Afternoon, then West Sea-Breeze: South-West Breeze *above* till half past IV. Calm bright Evening.

Also the upper Stratum of Clouds thin and *white*, in *quick* Motion, when seen from *below* till Noon: at which Time the Sky was almost cloudless: and, from *above* the upper Stratum, were seen, interspersed, Multitudes of detached Thunder-Clouds in large Masses, rising at Intervals, in the *Middle* of the upper Surfaces of white Clouds, and stretching *above* them.

Friday and Saturday moderate: South and South-West Breeze.

Sunday the 11th. The Planet Mercury stationary.

Cloudy Morn. South-West Breeze. Thermom. at 60 at Noon. Barom. *above*, Changeable, viz. at 29½. MUCH THUNDER and Rain in the Afternoon.

212. Quere, Had the *Thunder*-Clouds on Thursday, tho' not remarked by any from BELOW, yet visible to a great Extent from the Balloon *above* them,—any Connexion with the *Thunder* that happened THREE Days after?

Answer: It appears to the Observer, that the *Thunder* was *gradually* collecting in the Air from *Thursday* till Sunday: and if so; will not Balloons, when more *frequent*, prognosticate the Weather, *by Sight*, better than any other known Methods?

CHAPTER XXXVIII.

ON CERTAIN APPEARANCES AT DIFFERENT ALTITUDES OF THE BALLOON.

Section 213. **T**HE highest visible *white* Clouds, often seen in detached Streaks, during the finest and also in the worst Weather, (if not intercepted by lower Clouds) and which, when melting away, are known in some Counties by the common Appellation of Horse-Tails; and, suspended over Great-Britain, are frequently *marbled* or dappled by the Wind; putting on the Appearance of white Waves, like Sea-Sands ruffled and left by a rapid Tide;—had been disturbed, separated, and almost *melted* down by the *Storm* the Day preceding the Excursion.

Two of them *only* were still visible in Streaks, near the Sun's Place, at the first Ascent. They seemed without Motion, and became afterwards *invisible*.

Saussure, the celebrated Professor of Philosophy at Geneva, is very exact in his Definition, Description, and Height of these Appearances: and thinks it *probable*, their Situation may be "*at least fifteen English Miles above the Surface of the Earth.*"

"Car quand je considere ces fines Pommelures, &c." "For when I consider these delicate Dapplings, which, in a Series of fair Weather, begin to cover the azure Vault of Heaven with a white and transparent Gauze, and which portend Rain a long Time before it happens; I am led to believe they occupy a very elevated Situation in the Atmosphere" (Essais sur l'Hygrometrie, P. 271.)

It seems however that *Crosbie*, in his Excursion from Dublin on the 25th of January 1785, pierced throu' and soared above these *fine Webs*, at the Height of 16 Inches by the Barometer in a *frosty* Air.

214. It has been already noted, that at a certain Height, a Kind of CHILLINESS was perceived, not ascertainable by the Thermometer.

The Sensation was *suddenly* impressed four Times, in ascending and descending to and from the same Height, viz. about 26 and 27 Inches, equivalent to between 500 and 1000 Yards above the Surface of the Earth at the first Ascent.

From the Uniformity of Effect at the same Height; the Sensation may be ascribed to the same Cause, viz. the Level of the first or lower Tier of Clouds: altho' the Aironaut did not pass throu' any visible Cloud or Vapour, during the Excursion. See [Section 93](#).

215. At the same Height likewise, tho' the Observations have not been set down at large; the Appearances of the Earth and Clouds were very remarkable.

During the Ascent of the Balloon, between the Altitudes of 26 and 27 Inches; the *circular* Prospects of the subjacent Earth *instantly* contracted, and, during the Descent, about the same Height, *instantly* enlarged themselves to the Eye of the Aironaut.

216. At the same Height mentioned before, the *circular* Prospects of the Clouds appeared on the same horizontal Plane with the Eye: tho' at the Distance of a Mile. See [Section 49](#).

In *this* Situation, the Observer endeavoured to discover the Thickness of the *Stratum* of Clouds: but was always baffled by a Deception of Sight worth recording.

The *Strata* were plainly composed of three or more Heights of Clouds, *sailing* at great Intervals, one above the other: all which regularly *vanished*, as he approached their respective Levels: as if *instantly* thrown into the Circumference of a Circle, whose Radius was a Mile.

During the Ascent, in passing their supposed Level, the Clouds *instantly* appeared *far below* him: and during the Descent, as far *above*.

217. Quere: Is it not from the same Cause, that all Vapour is *generally* invisible to a certain Height and Distance from the Eye?

It being incontrovertible that more Vapours rise about NOON, than at any other Hour, particularly at Sea, while the Sun continues to *shine*; which, notwithstanding, are wholly *invisible*, till arrived at a *certain* Height?

And hence the Visibility of Vapours by *mere* Distance, which contains a sufficient Number of Particles to intercept and refract the Light, without Cold, Condensation, or *actual* Accumulation: viz. by Refrangibility of those primary Rays of Light, which Air and Vapour united are most *apt* to reflect or transmit.

Mons. Saussure has proved by his Horse-Hair comparàble Hygrometer, that “the Air shews Signs of *greatest* Humidity an Hour after Sunrise, and of *least* Humidity, between three and four in the *Afternoon*.” But the Air being *then* also the hottest, will *dissolve* or evaporate the greatest Quantity of Vapours, and raise them *above* the Hygrometer (which by its *Heat* will not retain, but on the contrary repel and *dissipate* them) to great Heights in the Atmosphere.

See “Essais sur l’Hygrometrie, C. 6, P. 315.”

218. In general then:

Is not the *Cause* of the above Deceptions, *not* an *Absence*, but a *Transparency of Vapour* to a certain Distance: (just as the Zenith *appears* CLOUDLESS, when the Air is *overcast* around;) beyond which Distance, the *Number* and relative Proximity of Particles with Respect to the Eye, is such, as to intercept the Rays of Light: *when only*, they put on the *Colour* of Air, and Form of Vapour and Cloud?

And hence the probable Reason, why NO *circular* Horizon of the Earth’s Surface was presented during the Excursion, Section 79: and why it seldom has or can present itself to Aironauts or *Mountaineers*, at any *considerable* Height above the Region or Level of Clouds, even tho’ Clouds do *not* appear in the Air, either to themselves, or to Spectators *below*.

This Point seems capable of Illustration by Analogy, from the Impossibility of encreasing the *Magnitude*, and at the same Time, *Distinctness* of distant Objects, seen throu’ a *common*

Telescope; on Account of the Quantity of Vapours between them and the Eye Which VAPOURS may be magnified till the Object appears confused and obscure; and even at last become substituted in the Place of the Object, under the Form of Opacity and *Cloudiness*.

219. The *greater* the Height of the Balloon, the more *contracted* was the Circle of Vapour below it; and the more limited the Prospect of the Earth's Surface below the Vapour.

220. It seemed probable that the Sun shone as *bright* on the Countries around the Observer, as on Objects immediately below him: which Objects could not have been illuminated by the Sun's Rays, darting throu' the APPARENT and *contracted* OPENING under him; as the Rays which shone on the Balloon, fell beyond the *Opening*, *obliquely* on Clouds which caught the Shadow of the Balloon.

221. The extreme *Rarity* or *Tenuity* of the Vapours was *evident* from the *progressive* Course of the Balloon, which was *always* in the Center of a *circular* Opening, limiting the lower Prospects; except when the Spectator lost all Sight of the Earth, by dense, watry, intervening Clouds.

This *august* central Situation, ALWAYS CHANGING YET STILL THE SAME, had the most striking Effect on the Senses and Imagination. Yet, however pleasing the Recollection of this GLORIOUS APPEARANCE; however *strongly* impressed, accurately described, or richly painted; it must fall infinitely short of the original SENSATION. Unity and Sameness were there contrasted with *perpetual Variety*: Beauty of Colouring; Minuteness, and consummate Arrangement;—with *Magnificence* and *Splendor*: *actual* Immensity;—with *apparent* Limitation:—all which were *distinctly* conveyed to the Mind, at the *same* Instant, throu' the Intervention of the Organs of Sight: and, to complete the Scene, was added the Charm of NOVELTY.

CHAPTER XXXIX.

CONJECTURES ON THE CAUSES OF THE CIRCULAR TRANSPARENCY TO A CERTAIN DISTANCE BELOW THE BALLOON, AND OF THE RED LIGHT FROM THE SEA AND RIVERS, WHEN SEEN ABOVE THE LEVEL OF THE SUPERIOR CLOUDS.

Section 222. **Q**UERE: As Red is the heaviest and Blue the lightest Colour; and as *red* Rays blended at a certain Angle with *blue* Rays, produce Opacity: further; as RED is the *predominant* Colour reflected from Water, while in the Form of *dense* Cloud, for Instance at the Rising and Setting of the Sun; and BLUE the Colour always reflected from the light Medium of Air or Sky; Does not this Mixture of least and most refrangible Rays, which, when aided with the intermediate primary ones, causes a *Transparency* near and round the Eye of a Spectator placed either on Earth or among the Clouds; produce, at a greater Distance and different Angle, such a Degree of Opacity, as actually to give the Idea of Clouds surrounding him at a Distance?

The latter Part at least is true, that Vapour and Air, which are *naturally* qualified to *transmit* RED and BLUE, rather than any other Light, will, at a certain Angle, when *blended*, produce an OPACITY. (See the Letter sent by NEWTON from Cambridge to Dr. Derham, in order to be presented to the Royal Society,—in “Miscellanea Curiosa, Vol. 1, Page 109.”)

Quere: May not the Rivers below act as a Prism; as Clouds, about Sun-set or Sun-rise, do to a Spectator on Earth, and reflect only the primary Colour RED, the *heaviest* and least refrangible Ray?

It being also considered that Refraction cannot change the primary Colour: nor are Rays, in the Direction from below to the Zenith, refracted; tho’ seen from a rarer into a denser Medium.

Possibly, a Pencil of Rays, in coming up from the River below may be stripped or drained by the double Absorption of the Atmosphere and River, and the Colour RED only, suffered to reach the Eye: “being the last to quit its Basis the Water.” (See Morgan’s Observations on the Light of Bodies, &c. &c. Phil. Trans. for the Year 1785, Part 1, Vol. 75, Chap. 91.)

CHAPTER XXXX.

ON THE EXCESSIVE DIMINUTION OF OBJECTS ON THE SURFACE OF THE EARTH, TO A SPECTATOR SITUATED ABOVE THE REGION OF CLOUD, AT THE BAROMETRIC HEIGHT OF NEAR A MILE AND HALF, PERPENDICULAR.

Section 223. **T**HE Earth's Surface was presented to the Eye throu' a *circular* Opening as already described.

This Opening discovered a *Plain*, smooth and level as a Die: a Sort of *shining* Carpet, enriched with an endless Variety of Figures depicted *without* Shadow, as on a Map: what was really Shadow forming a separate Colour, and not considered at the Time, as *Shadow*. The Objects were distinctly marked, and perfectly known to be Miniatures of the Face of Nature.

All was *Colouring*: no Outline: yet each Appearance curiously defined by a striking Contrast of simple Colours, which served to distinguish the respective Boundaries with most exact Precision, and inconceivable Elegance.

RED Rivers, YELLOW Roads, Enclosures YELLOW and *light* GREEN, Woods and Hedges *dark* GREEN, were the only Objects clearly distinguishable, and their Colouring extremely vivid. The Sun's Rays reflected from the Surface of the Sea, and other Waters, dazzled the Sight.

ALL living Creatures were invisible.

224. The Area of each Inclosure, computed to contain a certain Number of Acres, was seen from above under the Form of a Miniature Picture of a certain Magnitude or visible Extension, perpetually diminishing, as the Eye recedes to a greater Distance.

And the Case is similar, whether the Miniature be seen from *above*, or *along* the Ground.

The Miniature also lessens as the Distance encreases, according to a certain Proportion so exactly,^[49] That,

1. If the *Distance* and *Magnitude* of a tangible Object be known by Mensuration; a Judgment is formed, and Laws laid down, for its corresponding *Miniature* on the Eye.

2. If the *Miniature* be seen, and *Distance* known by Mensuration; the Mind forms a Judgment of its tangible *Magnitude*.

3. And lastly, if the *Miniature* be seen, and *Magnitude* of a tangible Object is known by Mensuration; the Mind makes an Effort, to the Estimation of its Distance from the Eye.

These are some, among many Modes of Comparison, by which the Mind acquires a tolerable Degree of Proficiency, in estimating *Distances* of familiar Objects, *known* from the Appearance of their respective Miniatures on the Fund or Bottom of the Eye.

And so far most Theories agree.

But such *ocular* Test is only true, while the Comparison is made in *nearly* the same Medium.

For an Object, if seen at the same Distance *along* the Ground, will appear less as it rises above it; and least in the Zenith; as the Sun and Moon, at Setting or Rising, appear *large and oval*; but at their greatest Elevation, are *small and round*: because being seen, when passed out of a Medium impregnated with Vapours, which in some Measure intercept the Rays of Light: for the FAINTER^[50] a distant Object appears, the *greater* it is apprehended to be.^[51]

Possibly indeed an Object at the same Distance, if brighter at one Time than another, will *contract the Pupil* in Proportion to its Brightness: which may have the same Effect, as if the Object had made a *smaller* Miniature on the Retina; and will regularly strike the Mind with an Idea of *Magnitude*, *only* equal to its corresponding *Contraction*; i. e. less, when the Object is bright, and greater when faint.

225. If a like Reasoning be applied to the Ascent of Balloons; and it be said that they do not rise so high as is imagined, because their Magnitude is diminished, merely from being elevated into a Portion of the Atmosphere *least* impregnated with Vapours; it will follow, that to a Spectator in the Balloon; known Objects on the Surface of the Earth below,—being seen from a rarer into a denser Medium, also into one which contains a great Quantity of Vapours;—shoud appear *larger*, than when seen along the Ground, at a Distance equal to its Height in the Balloon: all which is contrary to Matter of Fact: particularly if the Barometer gives a proper Estimate of the Height, of which there is little Doubt: a proper Allowance being made, *in certain Cases*, on Account of the Refraction: for, as before mentioned, ([Section 44](#)) Objects seen from the Balloon at a Mile and Half *barometric* Height, continued, with invariable Uniformity, to suggest the Idea of at least seven Miles.

226. By a general Comparison of Enclosures, and of separate Buildings when they could be distinguished from the Balloon above the Region of Cloud, with the most distant Extremities, (on the horizontal Level) of Fields or Houses situated along the Sides of Hills or Mountains, at a known Distance by Miles, making Allowance for their being seen in a straight Line;—the latter seemed at least five Times *larger* than the *former*: supposing them at equal Distances.

To give an Instance. Supposing the most distant Extremities of a known Building or Enclosure, situated on the Side of a Hill or Mountain, presented a Miniature of a *familiar* Magnitude to the Eye of the Spectator on the Ground, at the known Distance of a Mile and Half; the same Object when seen from the Balloon at the same *barometric* Height, appeared full five Times less.

This Comparison was made by Memory, the Morning after the Excursion, tho' suggested while in the Balloon, from the wonderful Minuteness of all Objects then presented to the Eye.

The Author being likewise familiarized to judge of Heights; having been on several of the chief Mountains in Europe: also, of comparative Distances, from his Situation near a large City, in a populous, enclosed Country; on a high Plain, within View of the Sea,

Mountains, Hills, Enclosures, Buildings, and Objects whose Magnitude and Distances were known.

227. The Balloon itself, a Globe twenty-five Feet in Diameter, was seen in the Air on the Day of Ascent, at the Distance of 19 Miles.

228. The Reason already given, for the Solution of the famous Question concerning the apparent Magnitude of the horizontal Moon, seems no less applicable to Objects on the Earth's Surface, when seen from the Balloon: which *Diminution* of Objects *below* confirms the Defect of Dr. Smith's Hypothesis.

For, as they appeared *extremely bright*; being shone on by the Sun, and seen throu' the Air in a perpendicular Line, containing the least possible Quantity of Vapour; the Brightness must have exceeded that of the same Objects, when seen along the Ground: and consequently the Miniatures of the former must have been less than the latter, and also their respective Distances *seem greater*.

CHAPTER XXXXI.

CONJECTURES ON THE CAUSES WHICH INFLUENCE THE DESCENT OF BALLOONS IN THEIR PASSAGE OVER WATER.

1. **C**onjectures concerning the regular Tendency of the Balloon to *descend* on its *Approach* towards WATER.
2. Its *greatest* Descent, when in the Zenith, over the Middle of Rivers.
3. Recovery and *Re-ascent* to the former Level, as it *recedes* from them.

Section 229. Article 1. On the first Ascent in the Castle-Yard, Chester, the Balloon gently moved towards the River Dee, and the Sea.

And woud probably have gone out to Sea, if the ascensive Power had not presently raised it above the Influence of the Water; into an upper Current of Air, which was visible at that Time, and for two Hours before the Ascent, by the Motion of superior Clouds in a safe Direction towards the Land.

229. 2. The Balloon was *affected* in passing across the River Goway, and Trafford Meadows, which are a Mile wide: first moving Westward, and again towards the Sea; making several Curves: then resting and *lingering* between Great and Little Barrow: as the Aironaut was *well* informed by Persons of *Veracity*, who observed it: his Attention being engaged at that Time by other Objects.

229. 3. A proportionable Effect was observed in *crossing* a small Brook near Alvanley.

229. 4. The River Wever and its broad Meadows above Frodsham-Bridge actually stopped the farther Progress of the Balloon: tho' its Course was *merely* ACROSS the River.

The Deviation was gently tho' *invariably* towards the SEA: and, if not *timely* prevented, the Balloon must have fallen in the Middle of the Channel.

229. 5. The same Case woud have happened on the Re-ascent at Bellair; if the *levitating* Force had *not* as at first, overcome the Influence of the WATERS, and lifted the Balloon into the *same* upper Current, which continued to move in its former safe Direction.

229. 6. Different Branches of the Duke of Bridgewater's Canal near Preston-Brook might *possibly* affect it in a small Degree: and, tho' Clouds a little afterwards, secluded the Aironaut from a Sight of the Earth; yet the Balloon was known to hang, for some Time, over the Mersey near Warrington.

229. 7. The Balloon descended and alighted on the Middle of a large Tract of wet Moss Ground.

The Writer saw Sadler's Balloon rise at Manchester, the 11th May, 1785, and descend near Blencow-Bridge, at the Conflux of *two* Rivers.

The above Facts give sufficient Indications of the constant Tendency which Balloons have, to descend on Water.

CHAPTER XXXXII.

Section 230. **T**HREE Causes seem generally to concur in producing the Effect of Descent, over Water.

1. The Water itself.
2. The Air above it.
3. Change of Temperature,

Section 231. Article 1. So long as Gass escapes from the Balloon; it will be instantly and *reciprocally* attracted, throu' the *Crevices*, by the Moisture contained in the *Air*, particularly over *Rivers*: its specific *Gravity* within the Balloon, woud be encreased,^[52] and consequently the Balloon itself rendered less buoyant:

The Gass woud, on the contrary, be repelled by *electric* Air: which woud lessen its Tendency to escape, throu' the Pores of the Silk.

But it is *presumed* that Air-tight Balloons will be little affected by *external* Moisture.

231. 2. Moist Air over Water being generally cooler than over the adjacent Land, will, so long as the Gass continues at its former Temperature, assist and raise the Balloon *thus* moving into a *denser* Stratum: but no sooner is the Balloon contracted by the external Cold, than it descends into a Medium of Air, whose specific Gravity is proportionable to the contracted Bulk of the Balloon, and rests when equal to it.

231. 3. Water is also a Conductor of Electricity, tho' a feeble one: and there is moreover a strong chemical Affinity between WATER, inflammable Air, Gasses, Floguiston, and Electricity.^[53]

231. 4. Water will therefore CONDUCT the Gass to itself: i. e. will draw the Balloon *downwards*, and with accelerating Velocity; as the Attraction is stronger, the nearer the Water.

231. 5. But if the Air over the Water be warmer than that over Land; then the Balloon, moving into a warmer Medium, as over the Sea in frosty Weather, most undoubtedly descends: till the included Gass has received the

additional Encrease of Temperature from that of the Air, at which Time it will have a Tendency to re-ascend, and will rest suspended in Equilibrio, as in the former Case.

The above Causes however may be considered, as *trivial*.

The first may be avoided by making the Balloon *Air-tight*: and the second easily guarded against by throwing out a little *Ballast*.

The *only* formidable one, if any, is

THE DEPRESSION OF THE ATMOSPHERE.

This it will be necessary to consider with some Degree of Attention.

CHAPTER XXXXIII.

Section 232. **W**HOEVER consults Antiquity,^[54] or is acquainted with modern Mèeteorism, will ascent to the Truth of the Facts there recited, viz. That the Storms of DISPERSION called *Prester-John*, and *Ox-Eye* over Table Bay at the Cape of Good-Hope (not to mention those of COLLECTION, as *Whirlwinds*^[55] and *Waterspouts*;) *descend* on Sea and Land from the *middle* Regions of the Air, often *perpendicularly* DOWNWARDS: and then blow violently from a Center, to all Parts of the Compass at once: a necessary Consequence of their beating *forcibly* upon the Land or Water.

The Ancients maintained that the Origin, of Wind was a mere *Depression* and *Percussion* from the Cold of the middle *Region*: and it should be remarked that their Observations were made on the *Continent*, and in *warm* Climates.

Now what is seen to Excess in the *hottest* and *coldest* Climates;^[56] most probably takes Place, in a less Degree, in temperate ones.

Therefore, on a Change of Weather, the upper Atmosphere *descends*: whether its Effects are *Cold*, as in Winter; *Warmth*, as in Spring; *Wind* or *Wet*; at the proper Seasons of the Year.

233. The Balloon, with which Dicker Junior ascended at Bristol, April 19, 1784, on a WINDY Day, proved the Truth of the Conjecture: for tho' the Aironaut threw out most of his Ballast; yet after each Ascent and Recovery, he was repeatedly darted *downwards* EVEN with the Ground.^[57]

234. A similar Event happened to Crosbie, in his Passage over the Sea from Dublin to England; for, tho' he too discharged his Ballast, the Wind kept him *down* and EVEN with the Water.

The Weather at that Time seems to have been an Εκνέφιας, Procella, Percussion, Squall, or Tornado, i. e. a Storm of DEPRESSION, and DISPERSION.

235. The Eknèfiar Winds come from cool Points on each Side the North.

Bacon also observes that all BOISTEROUS Winds, as Procella, Typho, and Turbo, have the evident Direction of a Precipice, or Projection *downwards*, more than other Winds: they seem to rush down like a Torrent or Cascade: and are then reverberated or beat back from the Earth, in all Directions.

Stubble, Corn, or Hay in the Meadows are raised, and spread around in the Form of an EXTENDED CANOPY, (*inverted Cone, elliptic Solid, and hyperbolic Curve.*) See “Bacon’s Historia Ventorum”, Pag. 43, ad Articulum 10.^[58]

236. If then it be allowed to reason from that Analogy which took Place in most of the Cases already mentioned; the *gentler* Depression. of Balloons over Water in *milder* Weather, may be owing to a Cause somewhat similar, tho’ not so evidently an immediate Object of the Senses, viz. *an actual tho’ invisible Descent of Air upon the Water.*

237. Blanchard in his Passage over the Sea from Dover to Bologne in France, when near the Middle of the Channel, suffered an unexpected Depression, and at the same Time was nearly BECALMED.

A CALM also took Place on the Irish Sea: which must have prevented Crosbie from landing,—without *Wings*, or some *propulsive* Machinery, connected with the Balloon.

238. Lunardi rose from Liverpool when the Wind blew *boisterously*: yet was *becalmed* twenty Minutes over the *broad* Turn of the Mersey near Ince, when above the Level of the Wind: and, descending into the same Stream of Wind, was hurried along towards Beeston-Castle in Cheshire.

CHAPTER XXXIV.

Section 239. **T**HE Existence of depressing Columns of Air was well known to a People more ancient than either Romans or Greeks.

240. The sultry Climate of Egypt, whose Situation is that of an extensive Meadow watered by a *broad* River, and enclosed by Mountains to the East and West; consequently not subject to general horizontal Currents of Air, except along the Line of its Meridian,—is *the Country*, wherein Columns of COOL Air descending on the Water, would be soon observed.

And they, in Fact, were almost the only People who applied the Observation to common Life: having, according to Herodotus, as well as later Writers, built lofty Structures OPEN AT THE TOP. By which Means the cool Air RUSHING downwards greatly refreshed the Inhabitants.

The ancient Pantheon, at present called All Saints Church, now standing at Rome; built in the lowest Situation of a Street named the Piazza di Navona is on this Construction: and the Hint probably taken from an Egyptian Model.

241. In all inland Countries, whose Lakes are frequently surrounded by Mountains, as Bala-Pool in North-Wales; those of Westmoreland and Cumberland; the Lake of Geneva in Swisserland;—the Air rushes FORCIBLY on the Surface of the Water in descending Torrents: this the Writer has frequently observed. ^[59]

(In other Languages, the Words applicable to Wind on a Lake, or the Ocean, signify Descent: as, Καταβαίνω, and Επικεῖμαι also the Northerly or *descending* Wind corresponded to the Εκνέφιας while the Southerly or *ascending* Wind answered to the Απογη.)

All this, which may be allowed to take Place in *bad* Weather, may perhaps be excepted to, in *fine*, and still more so, in the *finest* Weather.

As the slightest Change is first observable on the Surface of Water, whether on Lakes or the Ocean, the *Descent of Air* in the finest Weather is familiar to Mariners by the Appellation of LIGHT AIRS, playing in Eddies: and particularly in the *variable* Latitudes; i. e. between 32 and 42: to these the Writer can also witness: as well as on small and large inland Lakes, by partial *Dimplings* and *Rufflings* of the Surface.

OBJECTION TO THE THEORY REMOVED.

242. It may be objected to the above Theory, that the Wind plainly blows in an horizontal Direction, as may be seen from the Motion of Clouds and Trees.

To which it may be answered, that if Clouds are not beside the Question; as it is not asserted that a single Column of Air presses from so great a Height to the Earth; (tho' it be the Case in Squalls;) yet it is extremely difficult to determine whether Clouds move in a

Direction exactly parallel to the Plane of the Horizon: and it is much more probable that they are in a perpetual Change, *encreasing* or *melting*; rising or falling, according to the *Pressure* and specific Gravity of the *Medium* in which they float; its Tendency to Moisture or Driness, Cold or Heat; also the different Combinations and Decompositions, with Respect to which, the Atmosphere is in perpetual Variation.

The Motion of Trees, if carefully attended to, seldom shew Effects of a regular horizontal Current.

And since the more *powerful* the Wind; the more evident and accurate may be the Observation; it will be found, that the *first* general Effect is an oblique Depression, succeeded by a Recovery or instant Exaltation: then a momentary Pause, or actual Retreat of the Wind; and in a few Seconds, a Return of the depressing Torrent.

But the strongest, and, at the same Time, an irrefragable Proof, is by *Appeal* to Men of *Science* in the Navy, or to skilful Pilots, who are conversant with Winds and Waves; who have weathered Storms off Cape Hatteras in Latitude 36; (where probably the Wind is perpetual;) or have made an East-India Voyage:—whether, if a Gale blew in an horizontal Direction ONLY; the Ocean could produce such an Inequality of Surface: or whether when the Sea runs MOUNTAINS *high*; the tremendous Surges must not arise from the *violent* Action of Winds repeated at Intervals, sometimes *descending* perpendicularly; but oftener in forcible elastic Torrents of oblique DEPRESSION, and instant *Resilition*?

CHAPTER XXXV.

Section 243. **I**ntimations of depressing Columns in moderate Weather, are the *sluggish* Clouds, which often make their *first* Appearance, and remain longest, nay almost continually, *over* and *along* great Rivers, and Chains of Mountains, both during a Calm, or from whatever Point the Wind blows.

And hence the greater Quantity, Violence, and Continuance of Wind and Rain, which then *descend*:^[60] also of the *greater* Purity of the Air *during* such Descent.

244. As, therefore, it is plain that atmospheric Air *DESCENDS* *frequently*, both in bad and fine Weather; if a Cause can be assigned so general, as to make it probable, that such *DEPRESSION* does almost continually take Place:—tho' at present the Effect is only evident to the Senses, by actual Experiment in the Passage of Balloons throu' such Columns;—it will be sufficient to put Balloonists on their Guard against the Effects of such *Depression*.

245. In order to investigate the Theory of Depression; it may not be unacceptable, particularly to those who have not had Leisure to peruse the Experiments on Air, by Dr. Priestley, or the Collection on the same Subject by Cavallo;—just to extract a few short Quotations, on the chemical Affinities of Air and Water.

246. Article 1. “Water, as Rain, imbibes only the pure Air of the upper Regions, leaving the lighter and floguisticated Air to ascend.”^[61]

246. 2. Felicè Fontana says, “Common Air receives an Encrease of Bulk and *Elasticity* from being shaken in Water.”^[62]

246. 3. Air absorbs Water, and Water absorbs Air:^[63] and the Absorption of Air by Water is promoted by Agitation: it also absorbs twice as much *defloguisticated* Air, as common Air:^[64] the whole Bulk of the Air absorbed being equal to one-twelfth of the Bulk of the Water: yet the Bulk of the Water seems but *little* encreased: the Air being contained within the Interstices of the Water.

247. The following is a pretty and an easy Experiment, to shew how the *ABSORPTION OF WATER BY AIR* takes Place, under the immediate Inspection of the Observer.

Admitting the Sun's Light into a Room, throu' one Window only; pour a Pint of *boiling* Water into a large Bason: hold the Bason, which will not be half full, next the Light, in such a Manner, that the Sun may shine on the Water and Bason; yet the Eyes be shaded by the Top of the Window Frame.

Incline the Side of the Bason towards the Light, so that the Water may rise even with the Top.

The Eye being placed just above the upper Side of the Bason, farthest from the Light; look on the Water.

You may then observe the Surface of the Water next the Light, refract the Sun's Rays, and produce the primary Colours, particularly the *RED* and *GREEN*: which tho' *transient*,

continue to be *seen* in Succession; as Vapours rise above the Surface of the Water. Their *first* Ascent is plainly discoverable: remaining above its Surface, in the Form of *small Dust*, gently agitated, not *separately* but as a *whole*. Nor do they seem to rise into Steam, till assisted by the Action, and Contact of *dry Air*, which like *dry Spunges*, *licks off* and absorbs the small Dust already accumulated by the Force of the Heat from below, and then becomes visible under the Appearance of Steam, flying off in distinct hollow Vesicles.

The more *still* the Air of the Room, the more slowly will the Spunges of Air come in Contact with the Body of small Dust.—Besides the small Dust already mentioned; the Heat will detach solid Globules of Water; which will remain floating on the Surface of the Body of Water: till the dry Air descends and transports them with it; the Air at the same Instant dissolving the solid Globules into hollow Vesicles.

But the most extraordinary Phenomenon, and which cannot be mistaken, is, that as soon as a Sponge of Air has dipped into the Surface of Water, and received its Lading; the Vesicles continue to accumulate, till another fresh Sponge descends in a similar Form, which may be traced upon the Surface of the Water, and seen in its Shadow, or rather in Beams of Light at the Bottom of the Bason, at the Instant it has flown off with its Burden: for that Part of the Surface of the Water transmits new Rays of Light, on Removal of the Vapour carried away by the Dip and Play of Air.

248. The Removal of the Vapour, likewise exhibits a curious Appearance on the Surface of the Water: which seems as if divided into irregular Parcels detached from each other; like the reticular Daplings visible on the under Side of Clouds elevated to the highest Stratum of the Atmosphere, and there evaporating or dissolving.

249. So powerful is the Attraction between Air and Water; that, while the Steam is rising above and round the Sides of the Bason; *Waves of fresh Air*, by Intervals, press the exterior Parts of the Steam *inwards*, in order to get at the Surface by descending into the Bason.

This Operation is best discovered, when the Bason is held *even*. And the whole Process may be observed more distinctly, if the Bason is raised and fixed on a Frame, near the Height of the Eye of the Observer, standing upright: who will then be able to trace minutely the exact Form of the Steam, and Insinuation of the Waves of Air into the Center of each Curl, or rising Curvature: an Appearance, similar to which, may be seen in *Water* flowing from a small Orifice in a close Vessel; the fresh Air forcibly entering in an opposite Direction; forming a visible Cavity and Curvature in the Center of the Stream. See Halley's Experiments on Evaporation in the open Air, and in a close Room, in Lowthorp's Abridgement of the Phil. Trans. Vol. 2, P. 108.

Having once remarked the foregoing Process at Leisure; the same may be seen over any open Vessel of Water just warm enough to emit visible Steam: but the Air should be as *still* and *calm* as possible: the Steam never rising from all Parts of the Surface at once; but a depressing Sponge of Air always descends to the Surface, the Instant a Lamina of Vapour has been detached.

Such is the regular and invariable Process of Evaporation.

The same Process may be distinctly traced over the Surface of a Piece of Water or River, the Air being perfectly calm, in a gentle Frost, at Sunrise, particularly in Autumn, while the Water retains a Warmth superior to that of the Air.

250. Hence it follows that *as much light*^[65] and *warm* Air as is raised with the Steam by Evaporation from the Surface of any Water; *so much heavy* and *cool* Air is INSTANTANEOUSLY, constantly, and forcibly DEPRESSED upon its Surface, in order to supply the Vacancy, restore the Equilibrium, and continue the Evaporation.^[66]

251. Now, besides the mutual Affinity that Water has to almost all Kinds of Air, and to Floguiston; added to its Power of Absorption; and as the SEA, particularly in Summer, also RIVERS and *damp* MEADOWS are generally *cooler* than the Lands and Countries bordering on them; Currents of *damp cool* Air press forwards to supply the Defect or Vacancy caused by Heat, Rarefaction and Elevation of *dry warm* Air, which is necessarily, and almost constantly rising into the Atmosphere, from heated Lands, Plains, and gentle Eminences *long shone* on by the Sun.

252. Consequently the pure, cool, defloguisticated Atmosphere, is almost continually descending from above; sometimes imperceptibly, often forcibly, on the Surface of the Sea, the Channels of Rivers, Meadows, and all wet Land. Which Depression acts, in Proportion to its Strength, on the Balloon; and always with a sensible Effect: for, being in Equilibrio with the Air at all stationary Heights; the *least* Depression of the Atmosphere makes the Balloon descend, considerably.

253. This Reasoning is, in many Cases, applicable to the Air, and consequently the Weather and Cold of Mountains.

Nor can it otherways be accounted for, why the Snow is perpetual, and the Cold so intense, on Mountains under the Equinoctial, and between the Tropics: but which admits an easy Solution on the above Hypothesis.^[67]

CHAPTER XXXXVI.

Section 254. **T**HE Subject of DEPRESSING TORRENTS requires an accurate Investigation: as it will serve to point out the proper Time of Day or Night, when an Aironaut ought so to calculate his Voyage, as to arrive over the Middle of the Channel, or Arm of the Sea, at some particular Hour: in order to wait for a Sea Breeze which may waft him to the other Side.

A Point not difficult to be ascertained.

Also, this Idea of DEPRESSION, if properly considered and digested; may prove a sufficient Foundation on which to establish a new Theory of the *Weather*, so ill determined at present, from its *aggregate Weight* or *Elasticity* only, as indicated by the Barometer.

255. If a Conjecture may be formed on a Subject, material in itself, yet of which so little is actually known; woud not *the proper Time* of undertaking a Voyage over the Channel be such, that the Aironaut shoud find himself three Parts of the Way across, by NINE o’Clock in the Morning?

256. In *warmer* Climates, where the Seasons are more regular; the *Land-Breeze* blows to Sea from Midnight till X. in the Morning: at which Time, the *Sea-Breeze* blows to Land; continues till V. or VI. in the Evening; and is succeeded by a CALM, which lasts till Midnight.

Whence it follows, that during the Time of the Sea-Breeze, there is a constant Tendency towards a GULPH OF AIR, *along the Middle* of the Channel: the Equilibrium of which is as constantly supplied by a *Depression* of the upper and in general cooler Strata of Air; and therefore a *dangerous* Time for the Passage of Balloons.

On the contrary, during the Night, and till ten in the Morning, there is an *Accumulation* of Air, *along the Middle of the Channel*: which consequently is a proper Time to ensure a *safe* Passage; by the Assistance of WINGS, or some PROPULSIVE Machinery.

257. The Deficiency or Vacuity being supplied from the etherial Regions; it might be taken for granted, that such Ether must be *considerably* lighter than the adjacent common Air on an equal Level, and therefore *proportionably* dangerous for the Passage of Balloons.

But if it be considered that such Air, acting as a WEDGE, or more probably in the Form of an hyperbòlic Solid,^[68] to fill up the Vacuity, descends with Rapidity from a *colder* Atmosphere impregnated with aqueous Vapours *invisible from below*; and that both the Air and Vapour have reciprocal Affinities and Attractions, electric and mechanical, with the Body of Water beneath them; and are often rendered still cooler by its constant Agitation and *Evaporation*; also, that the Supply being immediate and cotemporary, with the DOUBLE TIDE OF AIR flowing from the *middle* over the *opposite* Shores;—there possibly may be little or no Difference between the aggregate or *barometric* Gravity of *such* Columns, and those which are formed by the Sea-Breeze on either Side of them: therefore the Descent of

Balloons is owing, among other Causes, to an almost perpendicular actual Depression of the superincumbent Atmosphere.^[69]

Following up the Idea of a Sea-Breeze, blowing, at a Medium, for 20 Miles over Land; altho' the Stratum of the LOWER CURRENT of Air, or Sea Breeze, may not exceed HALF A MILE IN DEPTH, measuring from the Ground upwards; nearly equal to 26 Inches of the Barometer *above*, the Thermometer also *above* being at 55, i. e. *Temperate*:—yet this Observation may prove of essential Service, while the UPPER CURRENT of Air, i. e. the general Wind blows TOWARDS the Sea, (which will be found to take Place more *frequently* than is, at present, imagined;) or while the Balloon is influenced that Way; as was the Case with Sadler and his Companion when over the Nore: who, on his accidental and sudden Descent, fortunately found Safety in the SEA-BREEZE.

Which Breeze was sought for, and made Use of by the Author, when in the Balloon, near Frodsham, in Cheshire.

For, as the Sea-Breeze is pretty general, Aironauts should not be too apprehensive: as they have it in their Power, by proper Management, to drop into the Breeze—for EITHER SHORE: if they are provided with a Machinery to waft themselves across the intermediate *depressing* or *accumulating* MEDIOCÈANAL COLUMN OF AIR: which Space, between the two Shores, is, as before hinted, frequently BECALMED.

258. Further: as the above Theory of a *mediocèanal* Depression seems to receive additional Confirmation from *each* Balloon Experiment; Lunardi *descending* on the 5th of October last, when near the Middle of the Bay of Edinburgh or Firth of Forth;—it may be found *prudent*, to keep the Balloon continually rising, till the Aironaut is *one-third* of the Passage over.

258. 2. For if the general Wind in the upper Current be not strong; the Aironaut may expect to be *becalmed*, with Respect to the horizontal Direction of the Current, the Instant he finds, by the Rise of the Barometer, that the Balloon *descends*; i. e. when it is acted upon by the depressing Column: in which Case, the *higher* he has soared, the *safer*: as he will have more Room and greater *Latitude* for Exertion by Means of the Machinery: which Machinery will be greatly *aided* by the Force of the descending Column or Gravity; and will act on a similar Principle with the Ferry-Boats over the River Po in Italy; which are a Sort of horizontal Pendulum. For the Aironauts will continue to *descend*, at the same Time that their *Wings* furnish the Means of a progressive Motion.

Therefore, before the Time that the Balloon has reached the Surface of the Water; they will have crossed the depressing Column; and find themselves wafted *gently* by the *new* Sea-Breeze setting in towards the opposite Shore.

259. If the Aironaut *rises up* to Sea with a Wind blowing from the Land on each of the opposite Sides of the Channel, and arrives above the Middle of the Channel, while the same Wind remains; it is probable that the Balloon will continue to rise higher as he proceeds towards the Middle, *where* the MEDIOCÈANAL ACCUMULATION has for some Hours taken Place; and therefore he need not be under any Apprehension of falling: but, as before, it being probable he will also be *becalmed*; the Necessity of propulsive Machinery is equally

urgent, in order to pass the Center of the *Accumulation*: after which, the Balloon will ride Home to the opposite Shore in the new Sea-Breeze, by *that* Time, just beginning to set in.

260. With the Assistance of propulsive Machinery, it is imagined the Aironaut may be enabled in a few Minutes to force throu' the calm mediocèanal Accumulation, or Depression: after which, he will have little Occasion to make Use of it.

261. SUNRISE is, probably, the SAFEST Time of all, to ascend towards the Sea, with an *Air-tight* Balloon: arriving with the Assistance of the Wings, throu' the *calm* mediocèanal Accumulation: and there waiting till the new *Sea-Breeze* sets in to the *opposite* Shore.

CHAPTER XXXXVII.

Section 259. **I**T may be observed here, that the two Difficulties proposed by Sauffure, are, in a great Measure, removed; in admitting the Doctrine of mediocèanal *Depression*, and consequent alternate *Accumulation*,

In a distinct Chapter, treating of the Variation of the Barometer, which he allows has Need of farther Explanation; he asks (Page 308) what Reasons can be assigned, why the *East Winds*, which are *cold and dry*, make the Barometer *descend*, in England and Holland: yet, the *West Winds*, which are *moist and temperate*, make it *rise*?

The East Winds *here* blow chiefly in Spring.

Now it is universally agreed, that the Sea, is sooner heated by the Sun than the Land: and on Account of the marine Acid exhaled,^[70] is also less cold,^[71] during that Season, in the same Latitude.

In Spring, therefore, the great Atlantic or Western Ocean, being *less cold* than England, Holland, and Eastwards; the Air pendent over the most extensive Tract of *dry and cool Land* in the World, rushes Westwards to supply the Equilibrium of *warm light* Air rising upwards, and causing a temporary mediocèanal Accumulation: which (altho' the specific Gravity of the cold Air is greater) must produce an actual Deficiency in the aggregate Weight of the Atmosphere over England and Holland: consequently the Barometer falls.

Again: the West Winds which blow at other Seasons; if, in Winter; are not frequent, except about Noon after frosty Nights which have equalized the Air for the Transmission of vigorous Sunshine: and should be looked upon as (what they are really observed to be) *low partial Sea-Breezes*, or *EDDY Currents*, insinuating themselves near the Surface, and setting Eastwards frequently against the upper and more general Winds; and therefore produce a temporary Accumulation.

If, in Summer; the Supply of cool Air to the heated Land, being made not only from the *Northern Ocean*, and lofty *Mediterranean Mountains*; but also from the *Atlantic Breezes*; the latter, tho' *moist and temperate*, must also tend towards an Accumulation of the Atmosphere over England and Holland: and therefore the Barometer rises.

CHAPTER XXXXVIII.

Section 260. **B**EFORE the Subject of mediocèanal Accumulation and Depression of Air, is wholly quitted; it may be well mention and compare a few Facts and Observations, which will elucidate the Doctrine; and in their Turn, receive Light from it.

261. If, in the Middle of a *hot sunny* Day, Vapours lighter than the Air, were to rise from the Ocean, (which they will continue to do, in hollow Vesicles or Bladders, till the Expansion breaks the Bubble, at which Time the Water woud fall to the Earth, if not drank up by the Attraction of *dry* Spunges of Air;) there woud be a constant Wind blowing from *Land to Sea*, to fill up the Chasm: but at such Time, the Land is more heated than the Sea: therefore hot Air and Vapour arise from both; and the Breeze, on the contrary, blows from *Sea to Land*; consequently if the Vacuities were not *continually* supplied from the ethereal Regions, and from the Ocean, all Animals woud actually die, for Want of Air, as in a *hot close* Room.

Such Supply is therefore constantly made, by Depression of the Atmosphere, and Absorption of the Water.

262. What happens on a great Scale, above the Ocean, as *before* hinted; probably, happens on a smaller, over Channels or Arms of the Sea: and on a still smaller; over and along Rivers, Brooks, wet Meadows, and damp Grounds.

263. In the variable Latitudes on the Atlantic Ocean; *cool fresh* Air is supplied from above, by descending Vortices of Wind and Showers: i. e. *Storms* of COLLECTION.^[22]

264. It may be remarked, in Confirmation of the above Doctrine, that triangular or Latteen Sails are used, and more useful, in a Mediterranean Sea, surrounded by high Lands, from which the Wind suddenly descends in Squalls; than in the open Atlantic, where the Wind is more equal.

264. 2. Perhaps there cannot be a better Account of the depressing Torrent of Air, than that which Bacon has given, in describing the Motion of Wind on the Sails of Ships, in a *Squall*.

“All Wind acting on the Sails of a Vessel, tends to depress or sink it. Wherefore *in strong Gales*, they first haul down the Yards, and take in the Topsails: afterwards all the Sails: cut away the Masts: throw the Lading overboard, the Guns, &c. to lighten the Vessel, and keep her above Water.”^[23]

CHAPTER XXXXIX.

Section 265. **W**ITH Respect to Mountains: on reading what Travellers have written, particularly Ullòà,^[74] they seem to answer the Intention of supplying cool Air to the surrounding Plains, or Continents; by Depression and Condensation: and also, if on Islands; to the Sea itself.

266. Brydone, in his Tour throu' Sicily and Malta, in 1773;^[75] giving an Account of his Ascent to the Top of Etna, says, that at the Foot of the Crater, the Snow was frozen hard and solid:^[76] and that the Crater was so hot; it was impossible to descend into it.

Further: "that the Smoke rolled down from the Sides, like a Torrent: till of equal Gravity with the Air, when it shot off horizontally; forming a long Track, according to the Direction of the *Wind*: which there rose to a VIOLENT Degree: so that it was with Difficulty he could settle the Barometer for an Observation."

He also adds "that *Clouds* began to *gather* round the *Mountain*; but were *dispelled* by the *Wind*."

Now from the foregoing Theory is it not probable to suppose, that a *Torrent of Air rushed* continually down from the etherial Regions, not only to supply the Fire of the Crater; but also the Vacuity caused by the perpetual Elevation of Vapours and heated Air from below: the Torrent likewise *depressing* into the Track with itself, the Volumes of Smoke which were seen to roll directly down the Sides of the Mountain: that this descending Torrent of Air, in its Progress, dispelled the Clouds forming round the Sides of the Mountain, by the Ascent of warm Vapours condensing, as they rose, on their Approach to the cold Mountain: the Smoke shooting *horizontally*, from that Height *only*, at which an *horizontal Current of Air* began to take Place? For it can hardly be imagined that the Air at the Top of Etna, found to be "*electrical*," and which must have been replete with a Mixture of Floguiston, inflammable Air, Gasses, and other aërial Fluids highly rarefied, heated, DRY, (and consequently lighter,) *at the Instant* of rising out of the glowing Cauldron, became so condensed as to fall like Water, without partaking of the Motion of a *violent* Wind, supposed to blow in an horizontal Direction.

267. Glas, in his Account of Teneriffe,^[77] reports, that the Clouds are generally half as high as the Peak, above the Sea,^[78] i. e. according to him, near the Height of a Mile and Half: "*below* which Clouds, the *North Easterly Winds* GENERALLY prevail: and, at the same Time, *above them*, we find a *fresh Westerly GALE*: which I believe to be the Case *in every Part of the World when the TRADE WIND* blows."

In Page 253, he says, that in ascending above the Level of the Clouds, he found the Air sharp, cold and piercing: and the Wind blew strong from South West, and West South West: so that the Wind blew towards the Mountain from three different Points at least, viz. the Trade Wind, from North East below the Clouds; just above them, from South West: and still higher, a fresh Gale, from West.

“The Air on the Top of the Pike was thin, cold, piercing; and of a dry parching Nature, like the South Easterly Winds which I have felt in the great Desert of Africa, or the Levanters in the Mediterranean: or even not unlike those dry easterly Winds which are frequent in the Northern Parts of Europe, in clear Weather, in the Months of March or April,” Page 257.

This dry Wind answers to the Eknèfiar (before mentioned) i. e. *Wind descending* FROM THE CLOUDS.

Glas further observes (Page 250) that the Clouds, in fine Weather, descend gradually towards Evening, and rest on the Woods till Morning: when they re-ascend, and remain suspended above them, till the succeeding Evening.

Here then a nocturnal Depression of the Atmosphere is obvious. But this Appearance will not prove that the Air does not descend below the Level of the Clouds: for, tho’ the Clouds descend with the Air; Vapour-Air, of which they are composed, becomes *transparent* both by Dissolution, in a warmer Stratum, and Proximity to the Earth, as before mentioned.

268. From the Variety of Winds experienced at different Heights, not only on *Teneriffe*, but in different Places; it is plain, that if Balloons can be made durable and Air-tight; they may be wasted between the Tropics by an East or West Current at Pleasure: and also throu’out the Globe; the Occasion being made, in some Respect, subservient to the Time. ^[79]

CHAPTER L.

CORROBORATING PROOFS OF A DEPRESSION.

Sect. 268. Art. 1. **T**HE Author is well informed, that, during an Engagement at Sea;—in *ten* Minutes after the Action has commenced;—tho' it blew a *Gale* before; (that is, tho' it blew *violently*;) the Agitation of the Air, arising from the Explosion of the *great Guns*, and small Arms, woud counteract the Wind, and produce a dead Calm.

268. 2. Quere: does not the *new elastic* Air, produced from the Nitre,^[80] give an instantaneous Compression and Dilatation to the *incumbent* atmospheric Air, round the Place of Action, while the *lighter floguisticated Air* passes throu' it, raising, and affecting to its highest Limit, the *whole* Atmosphere. And does not the Effect of a sudden Calm, suppose the Wind to *descend from above* with a Kind of *saltatory* Motion, instantly counteracted by the *new elastic* Air?—For if the Wind be supposed to blow sideways or horizontally, *to any considerable Height above* the Water, woud not the fresh *lateral* Air glide away, and prevent the Continuance of the Calm?

269. When a Squall happens, or only Rain falls; Air will *rush* from all Sides, and from *above*, to supply the Vacancy of the fallen Cloud and Vapour.

The Air immediately *above* must fall: the lateral Air gravitating towards other Places. Hence *Cold*, and a bright Sky after Rain.

270. The Theory of Accumulation may account for the frequent *warm* Rains in Winter, and during the Night.

For the preceding diurnal Accumulation over the Sea, may *circulate* during the Night, at a great Altitude, to restore the Equilibrium and Loss of *cold* Land Air sent by a low or Ground-Wind to Sea, during the Day-Time: particularly, as the *Accumulation* over the Sea, during Winter, is almost *continual*.

271. The *Wind* would more frequently be perceived to *descend* and *rebound upwards*, (Trials of which might be made by holding an Umbrella, extended at right Angles with its Axis, upright in the Hand;) if the same Opportunity offered, of opposing as great a Surface to it in a perpendicular, as is every Day done, in an horizontal Direction: for in walking, the whole Height of the Body, and half its Surface, is opposed horizontally to the Wind: but the Head only, which is covered, is opposed to the perpendicular Pressure.

272. As every Circumstance in the Order of Nature is so admirably contrived that each apparent Inconvenience rectifies itself; in *heavy* Winds continuing to blow from a *COLD* Point; the Construction of the Atmosphere is such, that the *warm light* Air from the opposite Points will necessarily rise up and flow over the cold Stratum, and by their Tendency to an Equilibrium, will produce an Air *less cold*, before the *same* Wind is exhausted.

273. On the one Hand; it is probable, that, as cold Winds are heavy; the Eknèfiar Winds are covered with frequent Waves of the Apogay, or light warm Air rolling over them, frequently from the opposite Points.

274. On the other Hand, as the *Apogay* Winds are naturally light and warm, it is *improbable* that they should be *frequently* covered with Waves of *cold heavy Air*, rolling over them from Eknèfiar Points.

It may therefore be reasonably concluded, that the Eknèfiar Winds, when approaching or opposed to the Apogay, should be considered as *Ground Winds*, (i. e. Winds blowing next the Surface of the Earth, tho' they be supposed at the same Time to descend) which receive the Apogay above them: and that the Apogay being warm light and *MOIST*, (which last will have the same Effect, as if they were more elastic;)^[81] being also more turbulent, and endued with greater Velocity, press back the Eknèfiar from the Surface of the Earth, and upwards; and at the same Time flow above them.

By which means the Eknèfiar partake of their Qualities;—become less *cold*, less *heavy*, and less *dry*.^[82]

CHAPTER LI.

Section 275. **I**F then this Reasoning be allowed; aërial Travellers will not be subject, when, at a considerable Height, even in Winter, to great Degrees of Cold, supposing that the Air does not actually freeze the Waters below; and the Apogay or Southerly Winds have continued for a few Days.

On the Contrary; Aironauts may expect Cold, encreasing with their Ascent, even in Summer, tho' *warm* below; supposing the Eknèfiar or Northerly Winds to have continued but for a Day before the Ascent: they may possibly, indeed by soaring higher, rise into the regular Stratum of the warm Apogay floating above them.

276. From what has been said, there seems a Degree of Probability, that the Air for a Number of Miles, *above warm cultivated Plains* should differ materially in its Temperature, from Air above Mountains, or *even on a Level* with their Summits.

That the former Air, in moderate Weather, should continue *warm and rarefied*: while the latter is *cool and condensed*.

For the same Reason the Air over the Sea, on the Hours of Accumulation; i. e. during the Night, in Summer, and frequently in Winter, should be found *warm and rarefied*: especially during a Continuance of the Apogay Winds.

277. It is likewise probable that the Atmosphere will be found RESPIRABLE at much greater Heights, than is at present imagined: during the Continuance of the Eknèfiar Winds; and also, on Account of the *defloguisticated* Air,^[83] which is *drier* and *less elastic* in Proportion to its Rarity.^[84]

278. The Height of 10 Miles seems not too great to limit human Respiration, should any Attempt be made, to soar with a Balloon in a mild Atmosphere; and particularly between the Tropics.^[85]

But an Objection would be found in the Size of a Balloon sufficiently capacious to contain nearly 6 Times the Bulk to which the Gass would necessarily expand itself, at the Height of 10 Miles.

279. It seems most likely that the primary Cause that will affect the Ascent of Balloons is the Difficulty of encreasing the Dimension of the Balloon: the Second, is from the excessive Cold; if the Wind blows from any Points of the North.

Supposing the Construction of the Atmosphere to be as represented by different Authors, (which, by the Way, is scarcely credible) ten Miles will perhaps be the utmost attainable Height.

280. There is a Circumstance relative to the Motion of the Air, which has not been sufficiently attended to: and bears some Analogy with that of a *Thorough Air*.

This Circumstance may not improperly be called the *Reception and Dispersion of Air*.

In cold Climates, it is an Object of Dread: in warm ones, a most desirable Piece of Luxury.

A gentle Undulation of the Air is perceived in Peru, and other hot Climates, by Persons sitting in *Arbours* sheltered from the Sun.

The surrounding Air is instantly *contracted* by *Condensation*, during the Absence of the Sun's Rays, and therefore occupies a *less Space*: *fresh Air* is *received*, and as instantly *dispersed* by Expansion towards those Parts, which are the warmest, i. e. where there is least Resistance: so that a gentle Breeze is constantly kept up, *probably* by a Depression from *above*.^[86]

281. Analagous to this, are those Winds which generally *rise early* and die away at *Sunset*: the nocturnal Condensation of the Air being sufficient for the RECEPTION: as Air suffers some Compression without Tumult.

To demonstrate the Changes owing also to remote and invisible Causes least suspected; Boyle somewhere speaks of an Instrument he made, which was so nicely contrived, that he could tell, while sitting in his own Apartment, whenever any detached Cloud passed beneath the Sun's Disk. The Principle on which it acted seems to have been that of a Reception and Dispersion of Air that took Place within *the SHADOW* proceeding from the Cloud.

282. An oblique Argument supporting the Doctrine of Depression, asserted to take Place, in fair Weather, is that *Wind* dries up the Moisture from the Ground more than the *Sun*: and that March which is the *windiest*, is also the *most drying*, tho' *not* the *hottest* Month.

Bacon, in his Enquiry into Motions and Undulations of the Air, uses a Metaphor, which tho' somewhat facetious, is strictly philosophical.^[87] "*For when WINDS lead THE DANCE, it would be agreeable to know the* FIGURE."^[88]

And it is probable, that they really press the Earth with a saltatory progressive undulating Motion, *descending* in elastic Steps of sudden Compression; and *rising* with quick alternate ones, of Dilatation and Expansion.

Dicker's Balloon gave Proof of this.

283. Lastly: the *CHILL* of Air which always takes Place over *WATER*, and *moist* Grounds, even in the FINEST WEATHER, strongly favours the *Reception* and *Dispersion* of it, to the surrounding and more heated Lands: (which can only be supplied, as before mentioned, by Torrents of fresh Air *gradually descending* from the etherial or middle Region of the Atmosphere;) and seems to produce the same Effect, viz. a constant Breeze, with that of the Arbor, Shade, or Shelter from the *Sun*: also with that of the *Shadow* from the Cloud passing under his Disk, which affected a complete Thermometer and Hygrometer.

284. On a Change of Weather from Frost to Thaw, the Colour of the *upper Air* FIRST alters from a *clear and deep*, to a *dull and faint* Blue, or to a muddy Haze, not distinguishable into Clouds, but visible above them; a vivid Brightness still remaining, for many Hours, to about 500 Yards above the Surface of the Earth.

Or, soft warm Showers fall gently, without Wind, or any apparent Change in its Direction.

All which seem to favour the Accumulation and Descent of *warm Air*, by Waves of the Apogay rolling over the Eknèfiai Winds.

CHAPTER LII.

Section 285. **A**S the *safest Hour* of the Day has been already pointed out, for the Ascent of those Aironauts, who propose to cross a Channel, or Arm of the Sea, in a Balloon *Air-tight* or nearly so: it may not be useless to throw out a few Hints on the properest Days in *each Month*, for the Ascent of Balloons.

286. It will perhaps be found true, that the more frequent Winds are generated near the Surface of the Earth: but that *Storms* are generated from above. Cold, Heat, Drought, and Moisture produce the more frequent and diurnal Winds: but the Conjunctions and Operations of the Moon and Planets contribute to the Production of Storms and other Inequalities of the Atmosphere: more especially the *Moon*: at the New and Full. These Attractions first affect the *superior Parts* of the Atmosphere.^[89]

287. “We are sure in the calmest Weather, to have some Breeze at Noon, and at full Tide.” Therefore, both are improper Times for Balloons to be at Sea: the Time of low Water and Midnight would be best in those, if equal in other Respects.

Changes of Weather as to Wind or Calm happen about the New and Full Moon.^[90]

288. Varieties of Tide produced by the united or divided Forces of the Sun and Moon, occasion similar Changes in the Atmosphere nearly at the same Time.

For Instance, at the Time of the New Moon or Conjunction, i. e. when the Earth, Moon, and Sun, are *nearly* in a Line; the Moon being between them: also at the Time of the Full Moon; i. e. when the Moon, Earth, and Sun are *nearly* in a Line; and the Earth between them, which is called the Opposition.^[91]

In the first Case, the Moon and Sun attract the Atmosphere of the Earth conjointly, or with united Force: in the second Case; the Earth being between them, they act in Opposition to each other, still nearly in the same Line.

At these Times, the *SPRING Tides* are at the *highest* i. e. once every Fortnight; and in the two interval Weeks are the *NEAP* or *lowest* Tides: for a like Reason.

Because, in the latter Case, a Line supposed to be drawn from the Moon to the Earth, and another from the Earth to the Sun, woud form nearly a right Angle: or in other Words; because the Moon and Sun woud attract the Earth at right Angles to each other, or in a lateral Direction:—the Moon woud draw one Way and the Sun another:—their Forces woud be divided.

Now it is a Fact, that the Ocean is raised considerably twice every twenty-five Hours, by the Attraction of the Moon, when she comes to the Meridian. So that the Surface of the Sea, instead of putting on the Form of a Sphere, or Globe, will be changed into an *oval* Figure, whose longest Diameter being produced, woud pass throu’ the Moon.

In like Manner a similar Elevation must take Place, as often as the Sun is in the Meridian; either above or below the Horizon.

Moreover, this Elevation is *greatest* on the New and Full Moon, because the Moon and Sun do then conspire in their Attractions: and *least* in the Quarters: as they will then draw different Ways; the *Difference* of their Actions only producing an Effect.

Lastly, the Intumescence will be of a *middle* Degree, at the Times between the Quarters, and New and Full Moon.

289. As in the Ocean, so in the Air above it; a Tide of Air must roll along the Atmosphere, throu' the whole Extent of it; and rise upwards twice in about 24 Hours.

And since the Height of the Atmosphere is computed by Halley at 45 Miles, and the Depth of the Ocean at an Average, but half a Mile; the Air will more easily and quickly obey the Attraction of the Moon and Sun, than the Tide of the Ocean: and, as it revolves in a Sphere which is about 100 Times larger than that of the Ocean, the Agitation and the Velocity of its Tide, will be something greater, in Proportion to its Elasticity, and inferior Density to the Water of the Ocean.^[92]

290. The *Weight* of the Air must now be considered.

The Weight of the Atmosphere in England does not exceed 31½ Inches of Mercury in the Barometer: nor does the least Weight fall short of 28½: the greatest Difference in the Weights may be taken at 2 Inches: dividing 30 (nearly equal to the whole Weight) by 2, the Answer is 15. So that the under Parts of the Atmosphere being pressed upon by about a fifteenth Part less Weight at one Time, than at another; the *specific Gravity* of the Air will sometimes be a fifteenth Part lighter.

But the Height of the Atmosphere being estimated at 45 Miles, which is equipoised by about 30 Inches; when equipoised by a fifteenth Part less Weight; (that is, dividing 45 Miles by 15; which amounts to the same as if a fifteenth Part of the whole Height was taken away; the Answer is 3 Miles;) shews that the Atmosphere is 3 Miles higher at one Time than at another, over certain Places; indicated by the Barometer at those Places.

Such an Accumulation of Air, arising only from Pressure or specific Gravity in one Part of the Atmosphere, and not in another; by its Tendency to an Equilibrium; and when to this Tendency is added its *elastic* Force;—must be productive of WINDS, *descending Torrents*, Inundations of Air, or Storms, near the Surface of the Earth: and nearly such a Difference in the Barometer has been known to happen in a few Hours.

Such Accumulation, however, is not properly *the Tide of Air*.

291. At the New and Full Moon, the united Attractions of the Moon and Sun raise the Spring Tides in the Ocean to the average Height of 10 Feet and a half.^[93]

And in the Moon's Quarters, the Moon drawing one Way, while the Sun draws another, viz. at a right Angle, made by Lines from the Sun and Moon to the Earth's Center; the average Height of the Neap Tides in the Ocean will be 6 Feet 7 Inches.

The same Attraction which raises Water 10 Feet and a half, will raise Air, whose Density is 800 Times less, to almost one third of that to which the whole Pressure of the Atmosphere can raise Fluids:^[94] Now it has been before seen, that the Pressure of the Atmosphere raised the Air 45 Miles: so that the Air is raised by the united Actions of the

Moon and Sun, at the New and Full Moon, to one-third Part of 45; i. e. to 15 Miles. And for the same Reason, the Air is raised at the Moon's Quarters to 10 Miles:^[95] the Difference between which is 5 Miles.

There is consequently a real *Tide of Air* five Miles higher at each New and Full Moon, than at her Quarters: which Tide rolls with incredible Velocity along the Verge or highest Limit of the Atmosphere; and is generally productive of Wind below.

292. The Elasticity of the Air must likewise be brought into the Account, as contributing greatly to its Motion: the Spring of Air always increasing as the Pressure encreases.

Considerable Changes must therefore ensue in the inferior Parts of the Atmosphere.

For as the Effect of the Moon's Attraction is to diminish the Weight of the Atmosphere (tho' its Quantity be increased) by elevating the Column of Air in the Line of her Meridian; the Rarefaction of the Air is therefore encreased, first *at the Top* of the Atmosphere; afterwards it gradually descends to the Bottom, or Surface of the Earth: so that the incumbent Weight being diminished, the Air beneath will be greatly *expanded*.

At whatever Height therefore any *Quantity of Vapour* or superior Cloud *rested*, while the Moon was in her Quarter; it woud *gradually descend* at the Approach of the next New or Full: at which Times it woud remain suspended at a Height, where an Expansion took Place equivalent to the former Expansion, at the Moon's Quarter: and, if the Height during the Moon's Quarter was only equal to that of common Clouds; such Vapour woud, at the New and Full Moon, *descend* in Mist, Rain, Snow, or Wind.

293. Little Reliance is to be placed, in these *Northern* Climates, on the aggregate Weight (or *elastic Power*) of the Air, indicated by the Height of the Barometer, near the Times of the New and Full Moons: tho', in general, it will *descend* about those Times.

These Things being so; it woud be improvident to undertake an aërial Excursion, either three Days before, or three Days after the Day, either of the New, or Full Moon: the Ascent should be forborne every other Week; at least till the Art is a little more advanced.

The two remaining alternate Weeks in each Month, viz. when the Moon is in the Quarters, and the Tide of Air flowing throu' the Atmosphere, is checked, counterbalanced, and equalized, by the lateral Attractions of the Moon and Sun, acting at right Angles, i. e. on different Parts of the Air, pendent on the Earth's Surface;—more settled and regular Weather may be naturally expected; and particularly freer from the Extremes of *Wind* and *Cold*.

Moreover, as the Almanack, and Ephèmeris^[96] may be always consulted; the Day fixed on shoud not be *marked* with Conjunctions of the Planets.^[97] The Inequality of their united Attractions greatly deranges the Equilibrium of the upper Parts of the Atmosphere; producing sudden Squalls and Gusts of Wind: which, tho' of short Continuance, perhaps a few Hours, are inauspicious to the successful Inflation and Ascent of a Balloon, during the Infancy of the Science. (See [Section 211.](#))

CHAPTER LIII.

ON THE MEANS OF SUSTAINING A BALLOON ABOVE THE SURFACE OF THE WATER, BY A TEMPORARY LOSS OF BALLAST: AND OF RECOVERING THE BALLAST.

Sect. 294. Art. 1. **T**HE two Inconveniencies arising from a *Discharge* of Ballast, while the Balloon is under the *Pressure* of a mediocèanal Column of Air, are,

1. First, lest the Balloon should rise too *high*; for by opening the Valve in order to descend; Gass escapes: which is an *actual Loss*: and the Balloon is rendered incapable of supporting its Burden at the same Height, as before.

2. The present Impossibility of resuming the Ballast, in order to *descend*, or *check the Elevation*, on approaching either Shore, or at any other Time.

294. 2. These Inconveniencies are to be remedied by the following Methods.

If *Sand* be the Ballast fixed on; put as much of it into a Bladder by Means of a Tin Funnel, as, when *less* than *half* blown, it will contain, without sinking below the Surface of FRESH *Water*.

Prepare the intended Weight of Ballast, in Bladders, after the same Manner.

Also to EACH Bladder *with Ballast*, tye another Bladder *without Ballast*, half blown.

Tye fast each Set of Bladders, so prepared, with a *leathern* Thong; the Ends of which may be left a few Inches to *spare*.

The Grapple may remain in the Car.

294. 3. When the Balloon *begins* to descend over Water; lower out the Cable, by Degrees.

Tye a Pair of Bladders, one of which contains Ballast, very tight, round the End of the Cable.

Then a second Pair, at such a Distance that the intermediate Part of the Cable, will *float*.

Repeat this Process, till the proper Effect is obtained; or the whole Ballast is discharged.

294. 4. The Car and Balloon may be *hauled* or wound *down* to the Surface of the Water: and the Ballast resumed, as the Balloon approaches the Shore.

294. 5. If it be found necessary, the Ballast may be *discharged* by cutting the THONGS, *gradually*: or the CABLE, *at once*.

294. 6. If the Wind be *contrary*, and the Weather *moderate*; the Tide, or Stream may, by *Calculation* and *Foresight*, be made to serve the Purpose of the Aironaut, in towing the Ballast which floats on its Surface: and thus checking, or gently drawing the Balloon after it.

294. 7. In such Cases, the Aironaut would do well in applying his *propulsive* Machinery.

A GENERAL OBSERVATION.

294. 8. To prevent the car of the *Balloon* from being drawn out of the Perpendicular, a Circumstance not infrequent; it is necessary to have some Contrivance, by which the Cable shall run throu' a moveable Pulley, on a Swivel, in the Center above the Car; and that the Aironaut shall be able *instantly*, by a Screw, or otherways, to fasten the Pulley and Cable so tight, that the Stress shall remain on the Center above the Car, however *forcibly* the Cable may be stretched.

CHAPTER LIII.

ANOTHER METHOD OF SUSTAINING A BALLOON OVER WATER, WITHOUT LOSS OF GASS, OR OF BALLAST.

Section 295. **L**et the Ballast consist of that Kind of Rope (wound on a Reel) that is either by Nature or Art, *specifically* lighter than fresh Water: as a *hollow cylindrical* Rope of Silk, in which Corks are thrust: the Silk to be dipped into elastic Varnish, to prevent the Absorption of Water into the Pores: or a common Rope well varnished; or covered over with a cylindric Case of varnished Silk, might answer the same Intention, if Corks or Bladders were tyed at proper Distances: in which Case, the Rope might, at the first Ascent of the Balloon, hang from the Center above the Car, at its full Extent, suppose a Mile or a Mile and half in Length, without the Encumbrance of a Reel.

If Bladders are used; those that hang near the Car should not be more than *half blown*.

By the above Expedient; as soon as the Balloon began to decline, from Evaporation of Gass, or Depression of the Atmosphere, and the lowest Part of the Rope touched the Water; the Balloon would continue to levitate, in Proportion to the Quantity of Rope sustained on the Surface of the Water.

The Aironaut would move less *swift* indeed, but more conveniently; as he would not be obliged to rise *above* the Wind: but be able to *lower*, and *raise* himself at Pleasure: *first*, by pulling up a Part of the Rope into the Car; and having there *made it fast*;

Secondly, by cutting away, as he saw Occasion, the loose End, and Folds of the Rope so drawn into the Car with him.

CHAPTER LV.

ON THE NECESSITY OF ASCERTAINING THE PROPER MODES OF DIRECTION, BY DIFFERENT AND FREQUENT EXPERIMENTS.

Section 296. **T**HE Necessity of making frequent Experiments, in order to prove how far the Balloon is capable of Direction, by different Combinations of the mechanical Powers, is so apparent; that no Balloon should rise a second Time, without the Application of Machinery to that End.

Each Candidate for Fame, as Proprietor of a Balloon for *public Exhibition*, ought to vie in his Pretensions to a Superiority of Manouvres.

Their respective Performances woud appear in the public Papers; and Decisions be made to the Advantage of the Art.

For it is probable, that by such *Comparison* chiefly;—the COMPARISON of *experimental Blunders* and *Mistakes*, and not by an Union of Theory and Practice, cemented by liberal Patronage, the Balloon can arrive to any Degree of Perfection, in a Country, which is the Scene of *perpetual Contention*: where the Sum of Life seems devoted but to PARTY; and where the *precious* Time of the GREAT is sunk in Luxury, and their *exalted* Talents lost in the *Labyrinth* of Politics.

297. *To strive against the Stream* is proverbially impossible: and it woud be literally so, to attempt by any Kind of Machinery to force the large Surface of a Balloon, with any Degree of Velocity, against a *Stream of AIR*. ([Section 201.](#))

Ships, which have the Aid of an Element 800 Times *denser* than the AIR, are obliged to wait *in Port*, till the Wind is favourable. But neither is this considered as an Argument against *maritime Navigation*: nor does the *Perfection* of the Balloon require its Ascent in a Storm: tho' the Preference due to the Balloon, on such Occasion, woud be decisive in its Favour: as the latter woud presently surmount the Wind, and *lie to*, in the *calm Air* above it.

Sect. 298. Art. 1. By Wings, or some propulsive Machinery, acting forcibly in a Direction required, and with Ease to the *Operator*; TWO *useful Manouvres* may be attempted, and will frequently be *found successful*.

298. Art. 2. First, To RETARD the Course of the Balloon during its Descent; in such a Manner, as to prevent the Wind from *damaging the Machine*, or *snapping the Cable*: and thus to land with Safety, and at the *smallest Distance* BEYOND the Place assigned.

298. 3. A *silken*, or other *light Rope* is to be provided: and to run throu' a *snatch Block* fastened to a RUDDER, or to the CAR, as in Crosbie's Balloon.^[98]

Which Rope *alone* woud lessen immediate and unforeseen Danger, by using the Balloon as a Sail, if it actually alighted on the Water.

298. Art. 4. The same Rope being *a Mile*, or *a Mile and Half* in Length; the *Whole*, or a Part of it, might be suffered to run off the Wheel, and, falling on the Surface *below*, in *misty* Weather, woud serve as a Signal to determine whether the Aironaut was over Land, or Water.

Also by winding up his Wheel, he might, if the Weather was moderate, bring himself *down* to the Grapple, which might be so contrived as to *run down* the Rope, and remain at the Bottom, by Means of a Knot, or other Check.

He might also *loose* his Grapple, and *rise* again: or when down; pull the Valve-Cord, and land.

298. 5. With a SECOND short Cable, snatch Block and Grapple, he woud be able to *moor* the Balloon, from which, he might, by procuring the Country People to load the Car with fresh Ballast equal in Weight to himself;—get out, and even leave the Balloon in their Care.

The Precaution of knowing whether he was over a fresh Water-Lake, (for he might hear the Sea) might be useful in misty and low cloudy Weather by Day, or during the Night; without expending Gass in the *exploratory* Descent.

298. 6. To facilitate the landing, the *Signal-Rope* may be used to the greatest Advantage, particularly in windy Weather; by *lowering out* a Part, or the Whole, whether a Mile, or Mile and half, so that the Grapple may take Effect on the Ground, at the Distance of its Length *by Estimation*, *short* of the Place where the Balloon is intended to land.

As soon as the Grapple *holds*; it is in the Option of the Aironaut, to tye Parcels of his Ballast *loosely* round the Cable, to run downwards along with it.

(For *which Purpose*, Iron-Rings with *Spring-Swivels*, which *open* by *Pressure* of the Fingers, and *shut* of themselves, might answer better than the *leathern Thongs*, as the former might be put, in *an Instant*, round the Cable, and woud run down *quicker*.)

These Parcels of Ballast are to be sent down, in Succession, till the Balloon has acquired such Degrees of FALSE LEVITY, as will be sufficient to counteract that Tendency which the Wind will have to *depress* the Car of the Balloon forcibly on the Surface, so long as it is connected with the Grapple *on the Ground*.

298. 7. When this Point is effected, the Balloon will remain suspended in the Air; and being acted upon by the Wind, will be pressed into a Direction approaching to an horizontal Line, in Proportion to the encreasing Power of the Wind.

And here the Necessity of having the Cable fastened to a Center above the Car, in order to retain its Perpendicularity, is most evident.

The Aironaut, in this Situation, may venture to wind up the Cable *gradually*, and descend, to the Grapple.

298. 8. Secondly: When the different Currents of Air, have been tried by Descent and Ascent of the Pioneer-Balloon,^[99] and found to be *all* unfavourable; the Aironaut is to *rise* still higher, into a Calm, pursue his Course horizontally in the BLUE SERENE, by propulsive Machinery: estimating the Velocity, by the *evident Resistance* of the half Mile white Flag

described in Section 12, 13. and 12, 15. hanging at a proper Distance *below*, and of that which hangs loosely at the Side of the Car, to shew a Change in the Direction of the Wind, (then made by a Resistance of the Air): or he may judge of the Velocity and Direction, by the *Flight* of a *Feather*, repeatedly let loose at certain Intervals of Time.

CHAPTER LVI.

NEW MODE OF ASCENT, TO DETERMINE THE INSTANT THE BALLOON IS ARRIVED AT ANY GIVEN HEIGHT: TO MEASURE THE HEIGHTS: AND TO ESTIMATE THE DENSITIES OF THE AIR AT THE GIVEN HEIGHTS.

ALSO, A METHOD OF ASCENDING TO A FIXED BAROMETRIC HEIGHT: THERE TO REMAIN SUSPENDED IN EQUILIBRIO.

Section 299. **P**REVIOUS to the Ascent, provide a Cord, which shall have sufficient Strength to support twice its own Weight, when so great a Quantity of it is *coiled* together, as, if extended, woud measure half a Mile or a Mile.

Weigh the whole *Coil*, or any Number of Yards, so as to obtain the whole Weight.

Mark the whole Length of the Cord, with different *coloured* Worsted, or otherways, at the Distance of every eight Yards: as a *sounding* Line.

Note the Marks in a Pocket-Book.

These Things being done; give the Balloon, by INFLATION, a Power of Levity *at least* equal to the known Weight of the Cord: which may be easily obtained by throwing into the Car, already *ballasted* and prepared, a Weight equal to the Aironaut, together with that of the *Cord*.

The Cord must also, previous to the Ascent, be rolled upon a Reel, (made fast in the Ground) whose Diameter should be two Feet: each Turn of the Wheel may be called two Yards.

A Barometer with an attached Thermometer fixed in the same Frame, also a second or detached Thermometer placed at the Distance of a Yard from the Frame, shoud remain upon the Ground during the Inflation.

The same Apparatus of Barometer with attached and detached Thermometer, shoud be suspended in the Car.

The Instant the Balloon ascends, an Observer below is to note in a Book the *Point* at which the Quicksilver stands in each of the THREE Tubes of the lower Apparatus, also the Time of Ascent: the Aironaut the same.

The Rope is, previous to the Ascent, to be tyed to a Center above the Car: and as soon as the Balloon has elevated the Car 100 Yards; the Observations, as before, are to be set down below, and by the Aironaut: and repeated at the Height of each 100 Yards: a Drum to beat; during the Time each Observation *below* is *noting* down; and the Balloon not suffered to rise, till the Drum has ceased. By such repeated *Notice*, and *Silence*; the Aironaut will know the *exact Height*, at which the Balloon is checked in its Elevation: and the *exact Time* during which its Elevation is impeded.

This Process is to continue, till the Rope is raised to its full Length.

At which Instant a double-barrel Gun is to be fired: the exact Time noted *below*: and the Time of hearing the Sound noted above.

These Notes are to be compared at the Aironaut's arrival on Earth.

300. For such *nice* Experiments the Aironaut should ascend half an Hour before *SUNRISE*, or *Sunset*: and the Day chosen by the foregoing Rules.

The Air must be *QUITE CALM*: but it is not necessary that it should be free from Clouds or Mist.

When the Rope is at its full Extent, the Operator *below* is to shorten it, by winding down the Balloon, 100 Yards: the Signals *below*, being repeated, till the Balloon is arrived within 100 Yards of the Ground.

301. While one Observer *below* is writing down the Observation to be made the Instant the Balloon has risen exactly 100 Yards; another Operator is to weigh, by Hand, with Spring Steel-Yards, the Force of Levity already acquired, which is to be noted down by a third Bystander.

This Process is to be repeated at every 100 Yards.

The Levity, it is true, will encrease as the Balloon rises, (probably in a geometric Progression;) ^[100] yet the Cord, by rising with the Balloon, will greatly check it: if, however, it prove insufficient for that Purpose, and, lest the Cord should be in Danger of breaking; the Bottom of the Balloon must be opened, or the upper Valve drawn.

If the *Cord*, *Rope*, or *Balancer*, be sufficiently strong; there will be no Necessity for the Aironaut to throw out Ballast occasionally; nor for the Observations in the former Part of this Section: the *Densities* will likewise be more easily determined, by the *Weights*; which shew the *Encrease* of Levity and Expansion of the Balloon, at each of the *given* Heights: Allowance being made for the Weight of the *Balance Rope*, *raised* by the Balloon.

302. The Aironaut, may, at any Height, marked by looking at the Barometer, when at 24 Inches for Example, or as soon as he finds his Balloon sufficiently expanded, pull up the Rope over a Pulley; or, wind it upon a Reel of two Feet Diameter, within the Car; and continue to do so; till he finds that the Barometer begins to *rise*, (which is a Sign that the Balloon *descends*), by the additional *Weight* of the Balancer just brought into the Car: on which, by preconcerted Agreement, he may throw out a *WHITE* Flag, prepared to hang a Yard below the Car.

On Sight of the Flag, the Person at the Reel *below* is to cut the Rope: which Rope, or a Part of it, is to be drawn into the Car.

The Balloon will rise no higher; but remain in *Equilibrio* in the Air, at that Height.

CHAPTER LVII.

ON BALLOONS. THEIR DEFECTS AND FARTHER IMPROVEMENTS.

Section 303. **T**hese Defects are best known from the History: a Detail of which is given to the World in an entertaining, elegant, and scientific Manner, by a celebrated Writer on other Subjects, *Mons. Faujas de Saint Fond*, in two Volumes, 12mo. for the two last Years, illustrated with Engravings by the best Masters.

And he promises a Continuation, or annual Register of Experiments and Improvements.

The Title of the Book is, “Description des Experiences de la Machine aërostatique, &c. &c.”

304. Mr. Cavallo has favoured the British Nation with a cursory tho’ clear Account of the same, in his “History of Airostation:” a Continuation of which it were to be wished he woud likewise publish annually.

305. It might contribute greatly to the Improvement of the Art; if Mr. Faujas woud give Engravings on a large Scale, of the different Machinery, already used or invented to direct the Balloon, with their Proportions: particularly the MOULINET of *Blanchard*: as well as that lately tried by Messrs. Auban and Vallet; whose Machinery is still *more distinguished* and EFFECTUAL.

306. The Titles and Sizes of all useful Books written on the Subject, also the Places where they are to be had, might likewise be inserted, at the End of each *annual* Volume.

307. The principal Defects of the British Balloons are, in

1. The Construction.
2. Production of Gass.
3. Mode of Direction, and
4. Security of landing.

First, Defects of the Construction are both in the Form, and Composition.

The Form ought to be that of a RIGHT^[101] *Cylinder*,^[102] by which the *Capacity* is doubled without encreasing the Resistance: ending above and below, each in a Hemisphere. A cylindrical Trunk, 2 Feet in Diameter, being added to convey the Gass *into* the Balloon; and suffer it to escape, when too much expanded in the etherial Regions.

It shoud also be furnished with a Valve, at the Bottom, of equal Diameter with the Trunk: keeping itself Air-tight; and opening outwards by a *given* Resistance, (as that of ten Pounds Troy,) from the inside Gass.

There must be an upper Valve as usual: occasionally to promote a *swift* Descent.

308. The Form will likewise continue to be defective, till an interior Balloon for common Air is adopted, according to the Plan laid down by the ingenious Mons. Meunier, lately appointed by the French Academy of Sciences at Paris, one of the Commissioners for the Improvement of Airostation.

The Use of which interior Balloon by Compression of the surrounding Gass in the external Balloon, prevents, it is said, the Loss of Ballast and of Gass: two very considerable Advantages.

For the actual Sum total of Gass not being diminished; the Balloon will continue longer in the Air, before an Escape of Gass, throu' the Pores of the Silk, makes it descend.

There will, on the same Account, be less Occasion to take in *meer* Ballast, for the Purpose of throwing it *overboard*, to prevent the Descent.

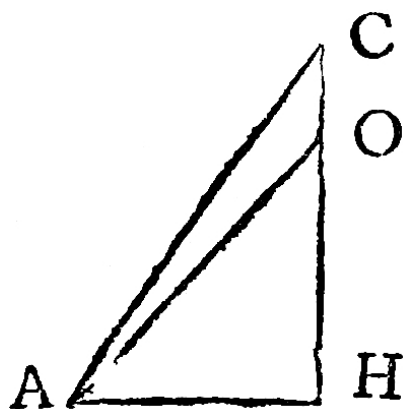
Therefore an equal Weight of Articles necessary to remain in the Car, may be substituted in Place of the Ballast.

309. Art. 1. And, since it is next to *impossible*, the Atmosphere should continue for 24 Hours together, of the *same Density, Weight, and Temperature*; or, in short, without Motion; —the Aironaut will have a Power of seeking, at *different* Heights, for that Current of Air, or *Wind*, which suits him best: or, in a very few Minutes, to rise above all Currents; become stationary, and *lie to* in the *SERENE*, waiting for a *Wind*: which, as before mentioned, he may readily find, by lowering out a Mile of Twine, and his *white* Flag: attending to it, with a small perspective Glass, or Magnifier.

309. 2. Another most *material* Advantage is to be able, in a *high Wind*, to chuse the Spot on which he proposes to alight: or wait for a favourable Opportunity to descend.

310. To compute the Height and Distance of the Balloon, by Means of a *white* Flag, or other *visible* Object, suspended from the Car, at a certain Distance below it.

Let the Observer take the Altitude of the Car with a Quadrant: and also the Altitude of the Object or Flag.



Then by a Case in plain Trigonometry; if the Altitude of the Car be by the Quadrant $59^\circ = HAC$, the Altitude of the Object $55^\circ = HAO$, and the Length of the Line veered out be 200 Yards, or otherwise = CO.

Then the Complement of $HAO = AOH = 35^\circ$; and the Complement of the Angle $HAC = ACH = 31^\circ$; and the Supplement of $OAC + ACO = AOC = 145^\circ$.

Then, $CAO 4^\circ : CO 200 :: AOC 145^\circ : AC$; and Radius : $AC :: CAH 59^\circ : CH 1409$ Yards, the Height of the Balloon taken at the Time.

Next, Radius : $AC :: ACH 31^\circ : AH 846$ Yards, which is the horizontal Distance of the Place on the Earth from the Observer, over which the Balloon was then suspended.

This Method finds the Height truer than the Barometer, and with fewer Circumstances of Confusion.

And if the Balloon Art could be perfected, so as to make them stationary at any Height; this Circumstance would afford excellent Opportunities of proving the Heights by the Barometer: besides which, the Distance also has been obtained: a Point not before attempted.^[103]

CHAPTER LVIII.

OF THE AIR-BOTTLE BALLOON.

Section 311. **T**ILL the Particulars of Meunier's Invention are made public, ^[104] an additional *Air-tight* Balloon, or Air Bottle, at least 15 Feet in Diameter, of a *globular* Form, appended below the Car, and furnished with a *Condenser*, to be worked by *pulling upwards*, or, as the Bellows of an Organ, by the alternate Motion of the Feet of the Aironaut, standing upright in the Car, may be used instead of the interior Balloon; to keep the *great Balloon* at a *given* Height: and consequently prevent the Aironaut *from rising too high*: to atchieve which Purpose, during the *first Ascent*; a Rope or Balancer may be used, a Mile and half long, fastened to the Car, and rising with the Balloon, (to *check* its Power of *Ascent*,) till an Equilibrium is produced: at which Instant, on Sight of the *white* Flag from the Car, the Balance-Rope is to be cut, by the Operator *below*. ([Section 302.](#))

If the Aironaut perceives by the Rise of the *Barometer*, that the Balloon descends; he may throw out a *little* Ballast, (perhaps a Pound or two), and then wind up his Balancer, or suffer it to remain at any Length, at his Option.

312. By keeping the Balloon at a given Height *only*; no Gass is expended in preventing the necessary Tendency of Balloons to a perpetual Elevation: also, during the self Descent of the Balloon; by opening the Air-Bottle, the Aironaut will supersede the Necessity of throwing out Ballast, for a Re-ascent.

313. The Air-Bottle-Balloon shoud be covered by a strong *light* Net, of a Dimension rather less than the Bottle, which will hinder it from bursting: the Resistance of the *condensed* Air within, being then chiefly on the Net, and but little on the Bottle.

The Net may be made of Silk and Cotton Thread; lest the Meshes, by the Pressure of the Knots, shoud eat into the Bottle.

CHAPTER LIX.

SUPERIORITY OF THE AIR-BOTTLE TO AN INTERIOR BALLOON.

Section 314. **T**HE Air-Bottle can be attended with no Sort of Danger. For, if it burst; the only Effect is to raise the Balloon: which is made to descend, at Pleasure, by opening either the *lower* or upper Valve.

Whereas an interior Balloon condensed with common Air, presses against the surrounding exterior Gass: and the Gass, against the INSIDE of the *great Balloon*, when the latter is in an elevated and rarefied Atmosphere; which Atmosphere, in Proportion to its Height, makes *less* Resistance to the *Outside* of the great Balloon: and thereby encreases its Tendency to a Rupture.

By the Application of the Air-Bottle, which will be to a Balloon, what an Air-Bladder, or *Swim* is to a Fish; a concomitant Advantage is derivable.

For the common Balloon and Air-Bottle, which may be called A DOUBLE BALLOON, will, in their *present imperfect* State, be able to remain a Day, or perhaps a Couple of Days in the Air: there being no Loss of Gass: unless by Evaporation, throu' the Pores of the Silk.

And this Advantage of a *double Balloon* may be effected with little EXPENCE (except that of a complete Net) to the different Proprietors, who may make alternate Voyages, with the Balloons *thus* united: one being inflated with Gass; the other occasionally with three or more Atmospheres of common Air *condensed*.

CHAPTER LX.

HINTS FOR THE DIRECTION OF THE BALLOON.

Sect. 315. Art. 1. **I**N the London Chronicle, from the 20th to the 22d of August, 1785, is a Letter from Bury, containing an Account of Mr. Poole's Balloon, with the following Circumstance, viz. "It was found necessary, before the Balloon was liberated, to cut away the Wings, intended to act as Sails, which had been constructed by an ingenious Piedmontese, patronized by LORD ORFORD, and which it was supposed, woud have contributed to *facilitate the Direction of the* Balloon, but were found *greatly to retard the Celerity* of its Motion."

Now if any Credit can be given to Newspaper Accounts, (that of the Beccles Balloon being an entire Fable,) it is to be lamented that the Wings were cut away for the Reason assigned: as it seems the only one that could properly be offered for applying them.

315. 2. Balloons already rise like a Rocket, and press forward almost with the Celerity of the Wind: it is therefore evident, that these Celerities must be *greatly retarded*, in order to *facilitate the Direction*: and consequently that the Wings bid fair to have answered the Intention of their ingenious Projector. And why precipitately cut them away, before the Balloon was left to the Pleasure of the Winds? since no regular or safe Manouvres ought to have been attempted, till that Time.

There appears to have been much the same Reason for rejecting the Piedmontese Wings, that there was for condemning the use of a Parashute, to which a Dog being appended was killed in the Descent: because the Parashute was not let loose at a sufficient Height, nor was it properly distended.

315. 3. It seems, that as the Wings had *greatly* IMPEDED the Balloon; a certain *Addition* to them might have *nearly* STOPPED it in the Air.

For the Balloon having once acquired an uniform Motion, by encreasing the Surface of the resisting Body, or Wings, the Balloon maybe retarded to a certain Point. But the Resistance encreasing woud raise the resisting^[105] Body above its Power of Action, and therefore, in Fact, lessen it; by which Means the Balloon woud continue to be propelled in the Direction of the Wind, with a Force equal to that Diminution.

Suppose, for Instance, that, instead of the half Mile Flag, which evidently checked the progressive Motion of the Balloon ([Section 70](#)) a larger square Surface, of varnished Silk, or a triangular Latteen Sail (like the Αρτεμων of Le Roi^[106]) was substituted, and kept stretched, by a hollow Cane, or Yard.^[107]

315. 4. Also, that by Means of a Fan or small Oar, acting as a Rudder, to be folded and taken back into the Car at Pleasure, the Balloon was compelled to move with a given Side foremost; that the Sail was let down below the Car, by strong silken Cords fastened to each Angle; and lastly, that leaden Weights, (each weighing an Ounce Averdupoise when widely

perforated, and put throu' the Ends of each Cord before it is fastened to the Car), be let down to each Angle; occasionally encreasing the Weights (or Sail) in Proportion to the Wind; which relative Weights (or Sail) will best be determined by repeated Experiments; will not such an Apparatus or Anemometer-Sail, acting as a *Vis Inertiæ* nearly at right Angles against the Force of the Wind, check the Balloon; till the encreasing Resistance raising the Sail upwards towards the Horizon diminishes its Power of Action? With this Sail therefore, which requires little Attention; and with the Assistance of Wings moved by Levers, pressed alternately downwards as the Bellows of an Organ, by the Feet of the Aironaut and mere Weight of his Body, standing upright near the Center of the Car; the Balloon may probably be, in some Respect, subject to Direction, and move obliquely against the Wind, or with Force in a Calm.

The Balloon and Anemometer-Sail, like the Earth and Moon will turn on their common Center of Gravity.

315. 5. It is possible to erect a light hollow Mast throu' the Car, and throu' the Balloon, by Means of a cylindrical Tube of varnished Silk, extending from Top to Bottom, in order to sustain the Balloon in an upright Situation, and make it keep Pace with the Car, when the latter is propelled by the Wings. The Mast should be covered with soft Cotton, to lessen the Roughness of the Friction. It may also contain within it, another slenderer hollow Mast, after the Manner of a Cane Fish-Rod; either to be lowered out, and placed horizontally across or below the Car, to serve as a Guard for the Bottom of the Anemometer-Sail; or to be let down to any Depth occasionally: and other Sails connected, by the usual wooden Rings, and kept tight by Cords running throu' Blocks fastened to any Part of *the equatorial Hoop*, as used at first, by the *gallant Admiral of the Air* BLANCHARD, and afterwards too precipitately rejected; since, in Case of a Rupture of Gass throu' the upper Hemisphere of the Balloon; the equatorial Hoop preserves the Parashute complete: and for Want of which Hoop, young Arnold had certainly lost his Life, if the Water of the Thames had not broke his Fall.

During the Descent of the Balloon, the Sails are to be taken in, and the lower Mast projected into its Socket.

315. 6. Different Trials may be repeatedly made: the Effects of which, whether evidently useful or *apparently otherwise*, being carefully recorded and regularly published *in Detail*, may afford Data for the Prosecution of further Discoveries, and lay the Foundation for a rational Superstructure of *airostatic Navigation*.

Sect. 316. Art. 1. By adding Weights, and encreasing the Surface of Anemometer-Sails; the *Vis Inertiæ* will become so powerful in the Direction of the resisting Medium of the Air; that the Wind in the opposite Direction will force the Balloon out of its Vertical, and incline it to the Horizon. The Car will be a Fulcrum Axis or Center of Motion: on an imaginary Point of which, as on a Pivot, the Balloon and Sails will turn opposite Ways, balancing each other in every Situation.

316. 2. The Balloon must therefore be brought back into the Vertical by a counter Exertion of the Wings: to which the *Vis Inertiæ* must always be made to bear a just Proportion.

The Declination of the Balloon is the only Inconvenience foreseen to result from an Anemometer too large, or too heavily laden: and it is instantly remedied by slacking the Sail.

One Thing still remains to be mentioned.

317. Balloons *durably* Air-tight, and terminating in a *Hemisphere* above, ([Section 307](#)); ought to have their Dimensions such, that there should be no Occasion for more than their upper Hemisphere to be inflated. Under which Form, they may with Ease and Safety be pitched as Tents on the Ground; by Cords fastened at equal Distances to the equatorial Hoop; and on Occasion by the Aironaut himself, while in the Car: who may be provided with Iron Ring Stakes barbed, and fastened or ready to be fastened to each Balloon-Cord: and, as soon as the Balloon is moored by the Anchor, Grapple, and snatch Block, ([Section 298](#), 3) with a light Axe drive down the Stakes round the Car, and regulate them when he alights from it, on the Ground.

CHAPTER LXI.

HINT FOR A VANE-SAIL TO PREVENT THE BALLOON FROM TURNING ROUND, WHILE THE WIND CONTINUES STEADY.

Section 318. **T**O the Block-Pulley in the equatorial Hoop, hoist a Sail, whose Shape is as follows.

From the equatorial Hoop, let fall a Perpendicular: and from the lowest circular Point in the Circumference of the Balloon, draw a Tangent, or horizontal Line, till it meet the former: these Lines, together with that Part of the Circumference intercepted between them, in the Points where they touch the Circle, forms a Space, which is the Shape sought.

The Sail may be kept steady by a hollow Cane or Bowsprit thrust out from the Car, and made fast with the usual Tackling.

319. Hint for an Umbrella-Pendulum or Valve-Swing, to project the Balloon in a Calm in the ethereal Regions, above the Station of Clouds; where the Resistance from the Air is much less than at the Surface of the Earth.

Let the Car of the Balloon be perforated so as to admit a light Gordon Mast, or Pole 18 or 20 Feet long, perpendicularly throu' it. ([315](#), [3](#).)

At the Distance of five Feet from the upper End of the Pole, a light hollow cylindric Tube of Iron, one Foot long, as a Bolt, should be put throu' it, at right Angles: so as to play smoothly in two Iron Bends, fixed in the Car; one Bend so far moveable, as to rise with a Hinge to admit the End of the Bolt; the other Part of the Bend to be perforated: throu' which a hollow Staple is to be fastened, with a spring Cotterel chained: this Apparatus will prevent the Pole from turning round.

Two light Frames of Wood, of a parallelogrammic Form, each twelve Feet by six, and covered with varnished Silk, are to be hooked, one on each of the opposite Sides of the Pole, from its lower End upwards; the Frames to be moveable in such a Manner, that on pressing the Pole one Way on the Axis or Bolt, the Frames shall lie close; but on recovering the Pressure, the Frames shall expand and open, so as to form an obtuse Angle with each other, or to lie almost in the same Plane, when the Recovery is made briskly, and with a Degree of Strength.

A Handle of Wood, the same Size with the Bolt, may be fastened throu' the Substance of the Pole near its upper End.

The Operator is to stand in the Car, and work the Pole backwards and forwards, which will give a progressive Motion to the Balloon in a Calm.

This Method may possibly prove more effectual than the Umbrella-Wheels, on an horizontal Axis, of Mons. Carra;^[108] as the Umbrella-Pendulum is easily unrigged, removed, and brought into the Car, in Case of a Whirlwind; by Means of a *circular* Rope fastened to

the Axis or Bolt, one End being in the Car, and the other put throu' the Aperture at the Bottom, and brought up from the Outside again into the Car.

The Umbrella-Pendulum may be made to turn round horizontally on the Bolt; the Ends of the Bolt being fastened under a circular hinged Socket, or Groove, of Iron.

CHAPTER LXII.

DEFECTS, IN THE COMPOSITION FOR BALLOONS, REMEDIED. ALSO ON THE COCHUC-VARNISH.

Section 320. **B**ALLOONS are defective in the Composition for *the Varnish*; which, till lately, was incapable of rendering the Balloon completely and *durably* Air-tight.

321. It was sometime ago reported at Paris, that Mr. Dutourny de Villiere had undertaken to construct a Balloon so truly *impèrmeable*, that he would warrant the Duration of it, for *several Weeks* in the Air.

And it is *since* known that this *Desideratum* of the Art has been effected, in the Composition for the celebrated Balloon of Messrs. Auban and Vallet, FIRST made subject to Direction.

322. Mr. Berniard, a French Chymist, has made curious tho' unsuccessful Experiments, in order to melt the cochuc or elastic Bottle; as may be seen in the 17th Volume of the "Journal de Physique."

Mr. Faujas and others made similar Trials.

323. The Writer, unacquainted with what had *then* been done in this Matter, could not help remarking the striking Properties of the *Cochuc* in its present Form, to answer every Intention of the best Varnish, if its Price was lower;—viz. *compact, pliant, unadhesive, and unalterable by Weather*;—if it could be dissolved, and afterwards made to recover its present UNADHESIVE Form: an Art in which the East and West-Indians are still *our Masters*.

He has, however, after expensive Trials and Combinations, been able to reduce it into a *limpid Liquor*.

As it may prove a useful Ingredient for *Air-tight* Varnish; the Secret he now discovers to the World: and it is merely this.

324. "Take any Quantity of the Cochuc, as two Ounces Averdupois: cut it into small Bits, with a Pair of Scissars.

Put a strong Iron-Ladle (such as Plumbers or Glaziers melt their *Lead* in) over a common Pit-Coal or other Fire.

The Fire must be gentle, glowing, and *without* Smoke.

When the Ladle is hot, much below a RED *Heat*; put a single Bit into the Ladle.

If *black* Smoke issues, it will presently *flame*, and disappear: or it will evaporate without Flame: the Ladle is *then* too hot.

When the Ladle is less hot, put in a second Bit, which will produce a WHITE *Smoke*.

This WHITE *Smoke* will continue during the Operation, and evaporate the Cochuc: therefore no Time is to be lost: but little Bits are to be put in, a few at a Time, till the whole are melted. It should be continually and gently stirred with an Iron or Brass Spoon.

The Instant the Smoke changes from *white* to BLACK, take off the Ladle; or the whole will break out into a violent Flame, and be spoiled or lost.

(Care must be taken that *no Water* be added: a few Drops only of which, woud—on Account of its superior *specific Gravity*, for the Cochuc swims in Water—make it boil over furiously, with great Noise.)

At this Period of the Process; two Pounds, or one Quart of the BEST DRYING-OIL, (or even of *raw* Linseed-Oil, which, together with a few Drops of Neat's-Foot-Oil, must have stood a Month, or not so long, on a Lump of Quick-Lime, to make it more or less DRYING)—being poured off the Lime-Lees; is to be put into the melted Cochuc, and stirred till hot: and the whole poured into a glazed Vessel, throu' a coarse Gauze, or fine Sieve.

When settled and clear, which will be in a few Minutes; it is fit for Use, either hot or cold.

The Silk should be stretched all Ways horizontally, by Pins or Tenter-Hooks, on Frames; which Frames, the greater they are in Length, the better: and the Varnish poured on COLD, in *hot* Weather; and HOT, in *cold* Weather.

It is *perhaps* best, always to lay it on, when *cold*.

The Art of laying it on properly, consists in making NO INTESTINE Motion in the Varnish, which woud create minute Bubbles. Therefore Brushes of

every Kind are improper.

Each Bubble breaks in drying, and forms a small Hole, throu' which the *Air* will *transpire*.

CHAPTER LXIII.

ON VARNISHES, CONTINUED.

Section 325. **T**O those, who are unacquainted with the Principles of Chemistry, or the Books which teach it; and yet are desirous to make Experiments, which may throw fresh Light on this curious and useful Art, when applied to Varnishes for Umbrellas or Balloons; the following detached Notes are recommended: which were communicated to the Author by *different* Artists; each *eminent* in his Profession.

326. To make copal Varnish.

Procure some bluish Flemish alkaline Ashes, (an Ounce suppose): pound them *very fine*, and lay them before the Fire, till they become *hot* and DRY.

Put them, while hot and dry, into Oil of Turpentine, (a Pint or Pound for Instance): or, into the same Quantity of Spirits of Wine.

For by Means of the Alcaly,^[109] all the Water invisibly contained in the Oil or Spirits will be absorbed, and leave the Oil or Spirits, ALCOHOL, that is, quite pure, and highly rectified: which Process is called *alcalizing* the Turpentine, or Spirits.

Put the Turpentine or Spirits so alcalized, into a Copper Vessel, with half an Ounce of YELLOW COPAL *finely* pounded and sifted.

Stir it, and the Copal will soon melt.

N. B. If you alcalize the Spirit of Turpentine, when the Copal is dissolving, add a little Spirit of Wine: and if you alcalize the Spirit of Wine, when the Copal is dissolving, add a little Spirit of Turpentine.

The SEDIMENT of the Varnish will dry on the Silk, in a few Hours.

The thicker the Varnish, the sooner it dries.

327. Article 1. To make an excellent THIN Varnish.

To one Quart of *cold raw* Linseed-Oil poured off from the Lees made by a Lump of *unslacked* Lime on which the Oil has stood, ten or eight Days, at the least, in order to communicate a drying Quality: (or on *brown Umber* burnt and pounded, which will have the like Effect:)—add half an Ounce of Litharge.

Boil them for half an Hour.

Then add half an Ounce of *the Copal Varnish*.

327. 2. While the Ingredients are on the Fire, in a Copper Vessel; put in one Ounce of Chio Turpentine, or common Resin: and a few Drops of NEAT'S-FOOT-OIL: and stir the whole with a Knife, or any clean Thing.

When *cold*, it is ready for Use.

327. 3. The Neat's-Foot-Oil prevents the Varnish from being sticky, or adhèsive: and may be put into the Linseed-Oil, at the same Time with the Lime, or burnt Umber.

327. 4. To make the above Varnish *transparent*, or *white*; use Mastic and Copal: to make it *brown*, use Seed or Shell-Lac, and *browner still*, use *pounded burnt* Umber.

327. 5. *Rezin*, or *Chio Turpentine* may be added, till the Varnish has obtained the desired *Thickness*.

327. 6. It must likewise be observed, that *Litharge* rots the Silk: therefore Trials must be made without the Use of Litharge.

327. 7. The *longer* the raw Linseed-Oil remains on the unslacked Lime, or Umber, the *sooner* will the Oil dry, after it is used.

If some Months; so much the better. Such Varnish will *set*, i. e. will not run, but keep its Place on the Silk, in four Hours.

The Silk may then be turned, and varnished on the other Side.

328. ON GUM MASTIC, SANDARAC, SEED-LAC, SHELL-LAC, AND COPAL.

328. 1. Gum *Mastic* dissolves, *without pounding*, by adding a few Drops of Oil of Vitriol: so do Gum *Sandarac*, and Gum *Copal*, when finely pounded and sifted.

328. 2. Gum *Sandarac*, and Gum *Mastic* are great Driers of themselves: and may be substituted for Litharge.

328. 3. The Mastic dissolved in the Oil of Vitriol, gives a *sweet* Smell to the Varnish.

328. 4. Sandarac will soon grow *dusk* in the Fire: it melts into a transparent Liquor.

328. 5. Sandarac, Seed-Lac, and Shell-Lac, must be finely pounded and sifted, before they are used.

329. The Author having examined different Kinds of varnished Silks, in different Places, does, from their Excellence, recommend those made by *Fawkner*, Umbrella-Maker, Alport-Street, Manchester; a Person wholly unknown to him, but from the Merit of the Work: which consists not only in the Varnish itself; but in the peculiar Method of *applying* it, which the Author *is not at Liberty to make public*.

Fawkner can warrant his Silk *Air-tight*; *soft* and *unadhesive*; durable, and *unalterable* by that Excess of Heat and Cold, to which the Balloon is, at the same Time, subject; viz. *internally*, to the hot depredating and caustic Fumes, rising with the Gass: and *externally*, to the *Sun*, *Wet*, *Frost*, and *Drought*.

CHAPTER LXIV.

HINTS ON IMPROVEMENT OF THE MACHINERY.

Section 330. **I**N order to make Improvements of the Balloon still more rapid and general; the Society for the Encouragement of Arts, who have given no particular Encouragement, in Imitation of that at Lyons, to the much-wished-for Art of directing the Balloon;—might offer a Premium for different Inventions of a *propulsive Machinery*, the Models of which are to be made at the Expence of the Society, within a certain limited Sum: and, without condemning what cannot be known unless by repeated Trials,—give Encouragement for such Trials: the Models to remain with the Society for public Exhibition.

331. Also, Figures and Explanations of such Machinery as have been tried, viz. the Fly or Moulinet of Blanchard; and of those which have not succeeded for Want of Trial; might be sent by the Inventors, in order to perpetuate the Invention, either to the *Society of Arts*; or to the Editors of creditable Magazines, who would be glad of such ingenious Acquisitions, as it would be a Means of procuring Purchasers, and circulate the Knowledge of this *gigantic* Infant Science.

Improvement woud then go on apace, and in a Chain: each Labourer forging and finishing his respective Link.

Whereas at present every one is obliged to find his own Materials, sink the Foundation, raise and finish the Building. And hence so little Work is done, worthy the Inspection of a skilful Architect.

CHAPTER LXV.

ON THE UTILITY OF BALLOONS: AN INTRODUCTORY CHAPTER.

Sect. 332. Art. 1. **I**T seems a favourite Question, among those who take a Pleasure in objecting to every Thing they neither do nor will understand, to ask, “Of what Use can these Balloons be made?” and without waiting for an Answer, to say—“they pick the Pockets of the Public, risque the Lives of the Incautious, encourage Mobbing and Sharpers, and terrify all the World.” These trite Reasonings are all very true, but little to the Purpose: the Effects above described being merely those arising from Novelty. If, says one in an inferior Station; “they could convert Balloons into common Stage Waggon; Goods might be carried with the greater Expedition:” or, “into Stage Coaches,” says another: or, “into Mail Coaches” says Palmer; “it would be certainly very clever, as I have the Patent:”—“or into comfortable Carriages to step in out of THE WINDOW, at a Moment’s Notice; that would be something,” cries a Nobleman: “it would *save* one a Couple of Sets of Horses, and would eat Nothing: one might ride one’s own Balloon Matches, from one’s Window to Newmarket, and from Newmarket to TOWN; dress for Court as we *do*, and make *Nothing* of it.”

Such are the different Ideas annexed by different Ranks of Men, to the Word UTILITY when applied to Balloons.

332. 2. For once let the feeble Voice of a French Philosopher be heard, the Abbé Bertholon: who may perhaps assert that all this is not impossible.

A Series of Experiments only can determine: and let the following Remarks serve as an Introduction to his Opinions.

332. 3. It is certain that the Progress already made in the Improvement of Balloons, since their Invention only three Years ago, is far superior to the Acquirements in every other Art.

The Antients knew, that excited Amber attracted Straws, and certain other light Substances: but medical Electricity, and a Preservative from Lightening, were notwithstanding reserved for the Moderns.

They likewise attended to some striking Effects of the natural Loadstone: but were totally unacquainted with the artificial Magnet, and the amazing Powers conferrable by it in the Disorders of the Imagination: nor did they know the Polarity of its Needle, or Application of it in the Compass.

They had not combined Nitre and Sulphur with Charcoal: much less had they changed the Mode of War into Science, by establishing Foundries for Cannon, and the Study of Tactics. Yet some Nations with a Knowledge of the Moderns, as the Chinese, have not improved, even in the Construction of their Vessels, according to the European Manner; continuing still in practical Ignorance.

Nor have other Indians improved in Proportion to the Opportunities of Instruction in several Arts.

Those of America, for Example, who continue to hunt, fish, and scalp: neglecting the Plough, and other Arts of Property and Peace.

332. 4. And thus it has been with the British Nation on the Subject of Airostation.

Cavendish, Priestley, and others, had produced inflammable Air, weighed, and found it lighter than common Air: and all that had seen a bright Fire might conclude, if they reasoned at all, that hot Air was lighter than cold.

Yet if Montgolfier had not made, ON A LARGE SCALE the Application of hot Air, in a Bag open at the Bottom, and properly poised; Charles and Roberts woud probably not have thought of applying the Gass of Cavendish: and Mankind woud not yet have soared into the etherial Regions.

332. 5. In this the French are still before the English, and will continue so to be, without a laudable and unlooked-for Emulation in the latter. That the former admire Liberty, Montesquieu's "Spirit of Laws" may determine; but they are not *addicted* to Politics. Their Nobility are endowed with a liberal and enterprizing Spirit. They join and patronize Men of Genius and Talents in the Cultivation of the Arts, and Improvement of every Kind of

experimental Knowledge. Their Pleasure consists in a national Ambition to excel.

They have Leisure, and are sober.

Half that Time which Men of Fortune in France dedicate to Taste, Invention, and Refinement; Britons spend among the Beasts and Birds: the other half, at the Bottle, and in political Cabals.

Present Profit is almost the sole Motive for Excellence in Great-Britain: and Experiments^[110] not made with that View, are seldom repeated; are overlooked and forgotten.

CHAPTER LXVI.

ON THE UTILITY OF BALLOONS.

Section 333. **T**HE Balloon opens a new and unlimited Field for Philosophical Discoveries.

334. The many curious and interesting Conjectures which Mons. de Luc (before the Invention of Balloons) throws out, in the Course of 4 large Volumes, on the Subject and Qualities of the Atmosphere; may now be determined by actual Trial.

335. The Abbée Bertholon wrote in 1784: and has particularly mentioned the following Points, as capable of ample Investigation, and. Discussion.

Sect. 336. Art. 1. *The Temperature of the Air at different Heights.*

Which will determine whether the Atmosphere be *practically Navigable*, at all Times and Places.

336. 2. *The dissolvent Power of the Air by Means of an Atmometer for Evaporation.*

Probably the Height may be determined, to which Clouds commonly ascend in order to find the proper horizontal Level, in which Balloons can move with the greatest Ease, Safety, and Expedition.

336. 3. *Variations of the Barometer.*

This will ascertain the exact Height, without Mensuration.

336. 4. *The DENSITIES at different Heights.*

A principal Object in de Luc's abstruse and scientific Researches: not only useful but necessary to determine the Laws of Refraction; without which, Astronomy, and consequently NAVIGATION, must remain defective.

336. 5. *The different Effects of Tastes and Odors, at different Heights: Experiments on Plants and Animals: also of SOUND.* ^[111]

These may produce new and salutary Effects on the human Body: and determine how far a Change from hot, putrid, and impure, to cool pure Air, impregnated with the invigorating aërial Acid, may contribute, without the Aid of Drugs, to the Recovery of the Sick, and Invalid: or promote Longevity.

336. 6. *The Direction and Velocity of the Wind.*

The different Currents and their different Heights, the Limitation of each Stratum of Wind, together with their different Temperatures at the same Time, will point out the proper Paths for the Balloon to move in, at all Times, and *possibly* without the Necessity of accurate Direction: the Mode of Ascent and Descent being *already* known, and proper Instructions given for a secure Landing.

336. 7. *Electricity of the Air*, METEORS.

This may lead to the Birth Place of Lightening, and Methods how to avoid its Effects in the Air. Tho' it be already known, that little Danger is to be apprehended, on Account of the mutual Repellency between the electric Fluid, inflammable Gass, and oiled Silk.

The Irides, the Coronaes, Haloes, and other Phenomena of Colours: the Generation and Solution of which may be investigated on the Spot.

336. 8. *Geography may become a new Science.*

336. 9. *Use of the Balloon for Signals in the calm Air, above Molestation; above Winds still blowing below: to discover the Positions of an Army, or Navy.* ^[112]

336. 10. *To throw principal Men into a Town: and convey others out of it.*

336. 11. With the Montgolfier Balloon, to try Experiments on Light, and Fire: to transport great Weights: raise them out of the Water: draw up Piles, raise Trees, Vessels, &c.

336. 12. The Parashute to secure a Man from too precipitate a Fall, is to be 5 Yards in Diameter, when extended: the Man,—weighing 140 Pounds, and the Parashute weighing 10 Pounds, with a Surface of 150 square Feet, —woud, in that Case, feel no greater Shock than if he had fallen from the Height of six Feet.

336. 13. *Compass and its Variations: also the different Branches in Astronomy.*

His Hints on the Direction of the Machine are ingenious.

337. 1. Wheels furnished with Wings.

337. 2. Imitations of the Form and Motions of Fish. ^[113]

337. 3. Vessels to condense Air, as the Bladders of Fish.

337. 4. Wind-Guns, Wind-Fountains.

337. 5. Elopile and Vapour Steam.

337. 6. Contrary Currents at different Heights: Proof of.

337. 7. New Hints for Balloons to be raised by Steam.

337. 8. Mons. Gouan's Invention to go THREE HUNDRED MILES A DAY IN A CALM.

338. The general Use to which Balloons seem capable of being applied, with the Assistance of propulsive Machinery, in the Calm which exists above the Level of a CONTRARY Wind; is that of a common Vehicle, not subject to the Inconvenience of Roads and Inns, between distant Places and Countries, for Passengers, properly accommodated in a Boat-shaped covered Car, furnished with Provisions, and occasional Siberian Cloathing: the Car to be surrounded with, and resting on Bladders, one *fourth blown*, and having each a few Drops of Water within, to keep them moist and elastic;—to prevent an *accidental* Shock in alighting on Land; and from sinking, if on Water.

Such a Conveyance (the Balloon being once made *Air-tight*, and furnished with an *Air-Bottle* to ascend and descend *without Loss* of Gass) is ready at all Seasons and Times: both Night and Day: for, as the Aironauts will enjoy continual Sunshine without a Cloud, from his Rising to his Setting: so, during the Night, the Light of the STARS, always intercepted in its Passage to the Earth by Clouds or thick Vapours, will be greatly augmented, when above both: besides the probable Increase of Light *reflected* from the upper Fields of white Clouds shone on continually by the different Planets and Constellations: all which will afford an Illumination

equal, if not greater, than that of a cloudless frosty Night, when the Ground is covered with Snow.

And such Light will be sufficient to read or write by: also to examine the *Barometer*,^[114] in order to know the *Height* and Level of the Balloon above the Surface of the Earth; and the COMPASS for Direction.

If Aironauts propose to ascend by Night, and in the Moon's Quarters; observing likewise the Precautions already given; it may be proper also to consult and take with them the Ephèmeris, in order to know the Time when the Moon rises, and also when she is at the highest, i. e. in the South, or has remained about half her Time above the Horizon.

The plainest Points, on which not only the Success of an Excursion, but the Lives of Aironauts may depend, are too frequently neglected, as unimportant and trivial.

CHAPTER LXVII.

THE PROCESS OF INFLATION.

Sect. 339. Art. 1. **T**HREE cylindric wooden Vessels were sunk more than half their Depth into the Ground: two of them, each, 5 Feet Diameter, and 5 Feet high: the third, 8 Feet in Diameter, and 8 Feet high.

An oblong Hole, 4 Inches by 3, was made in each Vessel: and each Hole was furnished with a solid wooden Plug (made tapering) 6 Inches in Length: throu' these the Vitriol was poured.

Besides which, there was an oblong Opening in each Vessel, large enough to admit a Workman, to distribute the Iron equally over the Bottom, and to pour in Buckets of Water: which Openings were well stopped, as soon as the Iron and Water were poured in.

As the vitriolic Acid is *corrosive*, burning the Skin or Cloaths; the following Precautions were taken.

An occasional moveable Tub was provided, 3 Feet high, and 3 wide: in the Center of whose Bottom was an oblong Aperture, equal to that in each of the Vessels: a corresponding Tin Tube, 6 Inches long, and narrowing to the Bottom, was nailed by its Border on the Inside of the occasional Tub; so as to go easily into any of the oblong Holes.

A Bottle of Vitriol being brought in its Basket by two Men, and made to rest on the Top of one of the fermenting Vessels; a third Assistant held the occasional Tub in his Hands, with the Plug-Staff fastened in the Aperture of the Tin Tube; and the Instant a fourth Person opened the Hole in the fermenting Vessel; the Assistant placed the Tin Tube in the Hole, keeping the Plug tight, to prevent the Escape of Gass.

The Bottle of Vitriol was then immediately poured into the occasional Tub: and the Bottle being removed, the Plug-Staff was taken out, and the Vitriol suffered to run into the fermenting Vessel: the Assistant watching for the Instant when the Vitriol was run out, in order to *force in* the Plug-Staff again, and prevent the Escape of Gass: after which, the Tub was rinsed with a few Quarts of Water, let also into the Vessel.

The same Tub was then removed: the oblong Hole in the fermenting Vessel instantly covered; and, by driving down the solid wooden Plug, continued *Air-tight*; by Means of moist Clay, and a little Water, kept purposely on the Tops of each Vessel, to discover by the Bubbles, whether Gass escaped.

In these Vessels, early on the Morning of the Inflation, were distributed 20 Hundred Weight, at 120lb. Averdupoise to the Hundred, consisting of cast Iron-Filings, and of a Mixture of Cannon-Borings.

The Borings were bright and fresh when thrown into the Water: and any Bits of Wood that swam, were skimmed off.

Rusty Iron emits Gass, that is heavier than common Air, and therefore is improper.

At the same Time, 16 Bottles of concentrated vitriolic Acid, or as it is improperly called Oil of Vitriol, were brought in their Packages near the Place, to be ready for Use: each Bottle at an Average containing 112 Pounds Averdupoise, of Vitriol: each full Bottle and Package together weighing from 136 to 148 Pounds.

339. 2. To the Iron in each Vessel, was then poured a Quantity of Water, which was measured in the Proportion of about 4 to 1: i. e. 4 *Pints* of Water to one *Pound*, of the vitriolic Acid.

The Height of Water and Iron in each Vessel, being then gaged, was about 14 Inches.

In a Line with the two smaller Vessels, and between them, was fixed another wooden Vessel or Cistern, filled with Water.

(N. B. Fresh Water ought to have flowed continually into it, and to have run over the Top of the Cistern: for the same Quantity being once saturated, can no longer absorb the alkaline and fixed Air to be separated from the Gass before the latter enters the Balloon.)

In the Cistern was fixed a Stage, consisting of 4 long Feet, (reaching to the Bottom of the Cistern,) nailed at their upper Ends to the Inside of an inverted Tub or Funnel, so placed over the Center of the Cistern, that 3 Inches of the lower Part of the Rim of the Funnel were under the Surface of the Cistern-Water: the Funnel was *cylindric*, 3 Feet across, and 2 Feet high.

An Open was cut, 1 Foot Diameter, in the Bottom of the inverted Funnel: on the Circumference of which was nailed a Tin-Cylinder or common Conductor, 2 Feet high: and at a *certain* Angle, as most convenient, was soldered a cylindric Arm, of equal Diameter, and 1 Foot long; having a Lip, Ring or Rim, on its outward circular Edge.

Round this Rim was fastened a varnished Linen Tube, of equal Diameter with the Cylinder.

At a small Distance, about a Yard from the Cistern, stood a slender Stillage, 3 Feet high; on which was supported a detached Tin-Cylinder or Connector, 1 Foot long and 1 Foot Diameter, made with a Rim at each End: in the Center of whose lower Side was soldered, at right Angles, another Tin-Cylinder or Evacuatory, 6 Inches long and 6 wide: its Use is to let out any Water, that the Heat of the Mixture might cause to boil and rise up out of the fermenting Vessels: and thus be *evacuated*, without entering the Balloon: or, if condensed in the Balloon, might run out by the same Orifice.

The opposite End of the varnished Linen Tube was fastened round one End of the detached Cylinder on the Stillage: and round the other, was tyed the Neck or Bottom-Opening of the Balloon.

Each of the 2 smaller fermenting Vessels was furnished with a cylindric Tin-Tube; each Tube 4 Inches and a half Diameter, nailed on the Outside of a circular Opening in the Top or Head of each Vessel; communicating by additional rectangular Bends under the Funnel and Water in the Cistern: the great fermenting Vessel had 2 Tubes, each 4 Inches and a half Diameter; communicating with the Funnel.

340. The Process woud have been more complete, if the fermenting Vessels had been sunk till their Tops were even with the Ground: and plaistered round their Outsides with soft moist Clay, six Inches thick, to keep them Air-tight.

Also, if the common Conductor had been only 1 Foot high: its horizontal or rectangular Arm only 6 Inches long: the Linen Trunk but 3 Feet, joining the Connector on the Stillage 1 Foot high, to communicate with the Neck of the Balloon; which Neck shoud be 3 Yards in Length, and its circular Opening 1 Foot, at least in Diameter.

CHAPTER LXVIII.

Section 341. **T**HE Process of inflating the Balloon began about X. in the Morning, by pouring 4 Bottles of Vitriol, immediately one after the other, into the occasional Tub, properly placed over one of the smaller fermenting Vessels: the Tub being instantly rinsed with a few Quarts of Water, which was suffered to fall into the same Vessel.

The oblong Hole was left purposely open for a Minute, till the strong Smell of the Gass was perceived above the Orifice: i. e. till the Gass had pressed out all the common Air that remained floating over the Surface of the Mixture in the fermenting Vessel: which Smell being *plainly perceived*, the *solid* Plug was immediately *driven* down.

And presently the Gass was known to press forward with an elastic Force throu' the Tin Conductor, by the Motion it communicated to the Surface of the Water in the Cistern: thence upwards throu' the common Conductor: at its Departure from both of which throu' the Linen Trunk, and Neck into the Balloon, the Gass makes a guggling obtuse Sound by quick Intervals according to the Quantity of Gass protruded.

And as the Intervals encreased, a Judgment was formed, that the Operation began to be less vigorous: and consequently that it became necessary, either to renew it by an Addition of more Vitriol and Water in the same Vessel, or to set the other small Vessel in Fermentation, the latter of which Mr. Lunardi preferred: this happened about half an Hour after the Vitriol was poured into the first Vessel.

342. After the second half Hour, eight Bottles were poured, by four at a Time, into the great Vessel.

And at one o'Clock, the Balloon, without any farther Trouble was beautifully inflated.

No Iron Rods were used to stir up the Borings or Filings at the Bottom of the Vessels: the Vitriol being found so heavy as to penetrate them as fast as the Iron, contiguous to the Vitriol, had parted with its Gass.

At each of the two former Inflations, a similar Accident happened which may be imputed to the same Cause.

343. During the first Inflation, the solid oblong wooden Plug fell into one of the fermenting Vessels: the hot Vapour, forcibly issuing from the Orifice, was condensed in the Form of a *white* Smoke; which being mistaken by the Company, a Cry was immediately heard of Fire, Fire: on which the Workmen retreated. Mr. Lunardi incautiously thrust his Arm into the Orifice to extract the Plug: at the same Time being much burnt, and failing in the Attempt; the Gass continued to escape, till a new Plug was prepared.

344. During the second Inflation, one of the Plugs being driven too forcibly; it was with Difficulty extricated, by the Strokes of a Hammer against the Sides of it, which tended at the same Time to displace the Boards forming the Top or Head of the Vessel: and, a little

afterwards, occasioned it to burst, unexpectedly INWARDS,^[115] rendering the Vessel useless for the Purpose of Inflation.

Observation. Therefore instead of the solid oblong wooden Plug, a circular Hole, 4 Inches Diameter should be drilled in each Vessel: and a corresponding solid wooden Plug 8 Inches long, 5 Diameter at the upper Part, and tapering to near 3 at the Bottom, should be prepared by the Turner.

In the upper Part of the Solid should be turned an inside Screw, to which an outside Screw of the circular Plug-Staff, made of Oak, Ash, or other heavy Wood, 4 Feet long, and 4 Inches Diameter, should be adapted: the Worm of the Screw to be 5 Inches long.

A wooden Peg of Ash, about a Quarter of an Inch Diameter, may be put throu' a Hole near the Top of the Staff, as a Handle.

A Lever of such a Length and Weight will probably answer every Intention, as no sudden Blows will be required to *fasten or extract it*.

The occasional Tub, Tube, Plug, and Staff, should be fashioned after this Model.

345. *The Price of the Iron and Vitriol for Inflation.*

2000lb. of Iron Filings or Borings ^[116] delivered on the Spot, at 6s. a Hundred,	£. 6 0 0
16 Bottles of Vitriol, at an Average 38s. a Bottle	30 8 0
Concomitant Expences,	3 12 0
£. Total	40 0 0

Observation 1. A great Saving might be made by conducting the Process in a different Manner.

The Author making two Journies to Manchester, purposely to observe the Process by Mr. Sadler; found that his Balloon was inflated in two Hours each Time; by Means only of the two smaller *identical* fermenting Vessels which Mr. Lunardi afterwards purchased; but the Levity procured by the former, tho' he also expended 16 Bottles, was by no Means so great as that gained with the Assistance of the great Vessel.

It has likewise been remarked by the Author, who has made several Experiments to this End, that the Vessels always continued in Fermentation and Ebullition, with a *quick Pulsation*, for at least 24, and commonly during 48 Hours, after the Inflation was completed.

And, that not more than the Depth of *half an Inch* of Filings had been *calcined* during the Operation: the rest being perfectly *bright*, and untouched by the Acid.

Observation. 2. If therefore one Inch in Depth of Filings, be spread over the Bottom of each of the *smaller* Vessels only; the proper Quantity of Water poured in; and *not more* than two Bottles of Acid used at once, in each Vessel; also, as soon as the Fermentation begins to decline; other two Bottles, and a proportionable Supply of Water be added; if suffered to work double, triple, or quadruple the Time;—the Inflation will be as great, if not greater,

for Instance, in six Hours with eight Bottles, and two small *Tubs*, as it woud in three Hours, with 16 Bottles, in the *same Vessels*.

The small conducting Tin Tubes ought instead of four and a half, to be nine Inches Diameter: by which Means there will be no violent Pressure of Gass to endanger the Bursting of the Vessels: particularly if the Gass is not suffered to descend; but, on the contrary, according to Instructions already given, either to rise, or move, in an horizontal Direction, past the Evàcuatory, into the Balloon.

346. The Workmen may begin the Operation at twelve at Night, or at six in the Morning: and the Time previously fixed for the Exhibition, may be eight or ten Hours after the Operation has commenced.

The Necessity of a Current of fresh Water, throu' a Pipe of at least half Inch Bore, the larger the better, to supply the overflowing Cistern, cannot be too much *insisted* on: as the Levity of the Gass almost wholly depends upon so trivial a Circumstance, as that of having a plentiful Supply of *cold fresh* and *soft* Water.

347. *Observation 3.* Supposing the Balloon AIR-TIGHT, near half the Expençe is thus saved in the Inflation.

Besides the greater Probability of *CALM* Weather for the Inflation, if completed before X. in the Morning, more Time is given to remedy Accidents, and rectify Mistakes: the Warmth of the Air likewise encreases.

But above all; if an upper Current carry the Balloon to Sea, the Aironaut may, (as before mentioned) drop into the Sea-Breeze, which will waft him safe back till IV. in the Afternoon, or even later.

CHAPTER LXIX.

MENSURATION OF HEIGHTS.

Section 348. **R**ULES for calculating the Height of Mountains, when applied to those elevated Stations in the Atmosphere *attainable* only by Means of the Balloon, will henceforward become more useful, and be more frequently practised: as the Lives of Aironauts *may* depend on a Knowledge of their *Height* above the Earth; which, not being determinable by *Sight*, in *all Weathers*, or at all Times, must be referred to the *Barometer* and *Thermometers*, they carry up with them.

De Luc, Horseley, Maskelyne, Shuckburgh, and Roy, have each written ABLY on the Subject, in the *Transactions*: tho' few have either Leisure or Inclination to follow them.

Sir George Shuckburgh has made successful Attempts to smooth the Way, by Examples and Tables, yet is still too concise for actual Learners, and the Generality of those who will have Spirit enough to go before the Calculators in exploring the Atmosphere; but cannot dedicate sufficient Leisure to overtake them in their Studies.

Each may therefore assist the other.

349. Whoever is at the Trouble of comparing the Observations made by Shuckburgh, with the Directions here given, will find that the latter contains the *Essentials* of the former, with this material Difference, that the Investigation moves here by Steps, which are all pointed out to the Learner; and not by Strides.

Each Step is self evident: and, by carrying Conviction to the Mind, is just what the Mind itself woud make use of, in the Attainment of any *distant* Truth.

To do every Justice to Sir George, the Merit of whose Performance wants no Eulogium; his three Precepts are copied; tho' rather as a Memorandum for those who understand the Methods; than as plain Directions for such as are yet to learn them.

It will be found likewise, that the first, second, and fourth Tables are greatly enlarged: being calculated for those *extreme* Temperatures, and Heights, which the Balloon *only* can attempt to reach: and the fourth Table, for greater Dispatch in computing the Expansion of the Air.

The Foundation and Construction of each Table, is also methodically traced and elucidated.

CHAPTER LXX.

METHODS TO ASCERTAIN THE TRUE HEIGHT.

Section 350. **M**ETHODS to be pursued on taking and comparing Heights, in order to ascertain the true Height of any Station in the Atmosphere, by the Barometer and Thermometers.

For this Purpose it is necessary, 1st, to provide a Barometer, (whose Bulb or Cistern is *large* enough to contain all the Quicksilver in the Tube;)—into the Frame of which, a Thermometer, on *Fahrenheit's* Scale, is to be fixed or *attached*.

The Use of the *attached Thermometer* is to point out the Temperature of the Barometer.

2d. A second or *detached Thermometer* is also to be provided. ^[117]

This is to be hung in the Shade at the Distance of a Yard (or two) from the other:—to shew the *general* Temperature of the Air at the same Time and Place: and may be called the *Air Thermometer*.

A proper Person, on the Ground, having a good Watch, with Pen Ink and Paper at Hand, is to attend the Instruments *below* every ten Minutes, (or at any other *preconcerted* Intervals of Time,) putting down,

1st. The Time of each Observation.

2d. The Point at which the Quicksilver stands in the Barometer.

3d. The Degree of Temperature of the *attached* Thermometer.

4th, and lastly, the Degree of Temperature of the *detached* or *Air*-Thermometer.

This Employment is to be carefully attended to; during the Time, that *similar* Observations, by *preconcerted* Agreement, are making, with three other *similar* Instruments, on the Top of the Mountain, or any elevated Station in the Atmosphere, by Means of the *Balloon*; and to be written with a *red Lead* Pencil, in a Patent Asses Skin Pocket Book.

Each single Observation, made with one Set of Instruments *below*, is to be compared with each single corresponding Observation, made with the other Set *above*.

And two Observations are said to *correspond*, when both are made *nearly* at the *same* Time, the one *below*, and the other *above*.

351. Take Shuckburgh's first Example, (Ph. Tr. for 1777, 2d Part, Page 577.) viz.

“Let the Point at which the Quicksilver stands in the Barometer, on the Ground, be 29 Inches 4 tenths: the attached Thermometer 50 Degrees of Temperature, and the Air Thermometer, or general Temperature of the Air 45°: at the same Time, that at the Top of the Mountain, or other elevated Station in the Atmosphere, the Barometer stands at 25

Inches 19 Tenths, the attached Thermometer at 46°, and the Air Thermometer at 39° and 1/2: required the upper Height in English Feet.”

352. The Work is divided into three Stages.

The End proposed in this first Stage is to bring the colder Barometer, to the same *Expansion* or *Temperature* with the *other*.

353. 1st. Step. First, write down the Observation made on the Ground, or at the Bottom of the Mountain, thus:

BELOW. Barometer, 29 Inches 4 Tenths. attached Thermometer, 50 Degrees. Air Thermometer, 45°.

354. 2d. Step. Secondly, write down the Observation made at the Top of the Mountain, or upper Station in the Atmosphere, thus:

ABOVE. Barometer, 25 Inches, .19 Tenths. attached Thermometer, 46°. Air Thermometer, 29 1/2.

355. 3d Step. Subtract the *colder attached* Thermometer, from the other attached Thermometer, thus: 46 colder from 50 warmer, and there remains 4° warmer, viz. the Number of Degrees of Temperature to which the *colder* Barometer must be *expanded*, before it becomes equal in Temperature to the *warmer* Barometer: each Barometer being always supposed *equal* in Temperature with its *attached* Thermometer.

356. 4th Step. Give the *colder* Barometer the same *Temperature* with the warmer: or, which amounts to the same, give the *colder* Barometer that *Expansion* which is communicated by the Addition of 4 Degrees of Temperature.

Both Barometers will then have the same *Temperature*, or *Expansion*, viz. an Expansion equal to the warmer Barometer.

This is to be done by referring to the first Table, for the Application of which there are separate Instructions: see the Explanation of the first Table.^[118]

CHAPTER LXXI.

USE AND PRACTICE OF THE FIRST TABLE, IN THE FIRST EXAMPLE.

The USE.

Section 357. **T**O find the Expansion of Quicksilver, and of the barometric Tube in which it is contained: or, in other Words, to find the Point to which the Quicksilver will rise in the Tube, (in Parts of an Inch) with a given additional Temperature, on Farenheit's Scale.

The Question in the first Example is, (Ph. Tr. for 1777, Page 578;)

To find the Expansion that arises, *with* the Addition of 4 Degrees of Heat, *on* the *colder* Barometer resting at Inches 25 .19 Tenths, in order to give it an Expansion equal to that of another Barometer, 4 Degrees warmer than the former: the Temperature of *each* Barometer, being indicated by its respective *attached* Thermometer.

N. B. During the Application of the first Table, the Investigation moves forward two Steps only, viz. the 4th and 5th.

The 4th Step, applied in the first Example.

358. The *Order* to be observed in finding the Expansion of the Quicksilver, with 4 Degrees on Inches 25 .19 Tenths of the Barometer.

1st. Find the Expansion, With 4° on 25 Inches only.

Then in order to obtain with 4° on .19, begin

2d. With 4° on 1 Inch above 25 Inches, i. e. on the 26th Inch.

3d. With 4° on .1, i. e. one Tenth of an Inch above 25 Inches: and lastly,

4th. With 4° on .19, Tenths above 25 Inches.

The PRACTICE.

359. 1st. In the *first* Table, *with* 4 Degrees on the left Hand vertical Column, and with 25 Inches, along the upper Range; at the Point of Meeting, is the Answer .0101^[119] viz. the Expansion, or Rise of the Quicksilver standing at 25 Inches, and receiving an additional Heat of 4°: the Answer .0101 being the Expression for the ten thousand one hundredth Part of an Inch, (viz. in Height, by Expansion.)

360. Add this Number, .0101, Part of an Inch, or Rise by Expansion, to the Barometer resting at Inches 25, .19 Tenths, Units under Units, &c. thus: .0101.

361. 2d. Now, in order to obtain the Expansion *with* 4 Degrees, *on* .19 Tenths i. e. the nine hundred and tenth Part of an Inch of Quicksilver in the Tube (above 25 Inches,) it must be considered, where it ought to be found in the first Table.

Tenths of 1 Inch, above 25 Inches, it must be observed, are at some intermediate Point between 25 and 26 Inches; that is, above 25, yet not so high as 26, or more than 25, yet less than 26.

Therefore, to find the Expansion *with* 4 Degrees, *on* 1 Inch above 25, i. e. on the 26th Inch; look in the Table, first, *with* 4 Degrees on 25 Inches: then *with* 4 Degrees on 26 Inches. The respective Numbers are .0101 and .0105.

And by taking the Expansion *with* 4° on 25 Inches, from the Expansion, *with* 4° on 26 Inches, thus;

$$\text{Expansion} \quad \left\{ \begin{array}{l} .0101 \text{ on } 25 \text{ Inches,} \\ .0105 \text{ on } 26 \text{ Inches,} \end{array} \right.$$

The Remainder .0004 is the Expansion with 4° on 1 Inch, above 25, i. e. on the 26th Inch.

362. 3d. To find the Expansion, with 4° on .1 above 25 Inches; add a Cypher and decimal Point to the former Answer, which then becomes .00004, viz. the Expansion, with 4° on one Tenth, above 25 Inches.

363. 4th. Lastly, to obtain the Expansion *with* 4°, *on* .19, above 25 Inches, say: If one Tenth of an Inch, above 25 Inches, gives this Expansion viz. .00004, what Expansion will nineteen Tenths above 25, give? answer .19 Tenths more; thus:

If .1 : .00004 :: .19?
.19

00036

0004

.00076; then, in order to have

as many decimal Places in the Product as are contained both in the Multiplicand and Multiplier, add a Cypher and Point to the left, and the Product becomes .0000076 which, being divided by .1, gives a Cypher less. viz. the Expansion with 4° on .19 above 25 Inches.

THE FIRST TABLE:
SHEWING THE EXPANSION WITH HEAT
ON INCHES OF THE BAROMETER.

DEGREES OF THE THERMOMETER, FROM 1 TO 40, ON FARENHEIT'S SCALE.								
	9	10	11	12	13	14	15	16
1	.00091	.00102	.00112	.00122	.00132	.00142	.00152	.00162
2	.00182	.00204	.00224	.00244	.00264	.00284	.00304	.00324
3	.00273	.00306	.00336	.00366	.00396	.00426	.00456	.00486
4	.00364	.00408	.00448	.00488	.00528	.00568	.00608	.00648
5	.00455	.00510	.00560	.00610	.00660	.00710	.00760	.00810
6	.00546	.00612	.00672	.00732	.00792	.00852	.00912	.00972
7	.00637	.00714	.00784	.00854	.00924	.00994	.01064	.01134
8	.00728	.00816	.00896	.00976	.01056	.01136	.01216	.01296
9	.00819	.00918	.01008	.01098	.01188	.01278	.01368	.01458
10	.00910	.01020	.01120	.01220	.01320	.01420	.01520	.01620
11	.01001	.01122	.01232	.01342	.01452	.01562	.01672	.01782
12	.01092	.01224	.01344	.01464	.01584	.01704	.01824	.01944
13	.01183	.01326	.01456	.01586	.01716	.01846	.01976	.02106
14	.01274	.01428	.01568	.01708	.01848	.01988	.02128	.02268
15	.01365	.01530	.01680	.01830	.01980	.02130	.02280	.02430
16	.01456	.01632	.01792	.01952	.02112	.02272	.02432	.02592
17	.01547	.01734	.01904	.02074	.02244	.02414	.02584	.02754
18	.01638	.01836	.02016	.02196	.02376	.02556	.02736	.02916
19	.01729	.01938	.02128	.02318	.02508	.02698	.02888	.03078
20	.01820	.02040	.02240	.02440	.02640	.02840	.03040	.03240
21	.01911	.02142	.02352	.02562	.02772	.02982	.03192	.03402
22	.02002	.02244	.02464	.02684	.02904	.03124	.03344	.03564
23	.02093	.02346	.02576	.02806	.03036	.03266	.03496	.03726
24	.02184	.02448	.02688	.02928	.03168	.03408	.03648	.03888
25	.02275	.02550	.02800	.03050	.03300	.03550	.03800	.04050
26	.02366	.02652	.02912	.03172	.03432	.03692	.03952	.04212
27	.02457	.02754	.03024	.03294	.03564	.03834	.04104	.04374
28	.02548	.02856	.03136	.03416	.03696	.03976	.04256	.04536

DEGREES OF THE THERMOMETER, FROM 1 TO 40, ON FARENHEIT'S SCALE.								
	9	10	11	12	13	14	15	16
29	.02639	.02958	.03248	.03538	.03828	.04118	.04408	.04698
30	.02730	.03060	.03360	.03660	.03960	.04260	.04560	.04860
31	.02821	.03162	.03472	.03782	.04092	.04402	.04712	.05022
32	.02912	.03264	.03584	.03904	.04224	.04544	.04864	.05184
33	.03003	.03366	.03696	.04026	.04356	.04686	.05016	.05346
34	.03094	.03468	.03808	.04148	.04488	.04828	.05168	.05508
35	.03185	.03570	.03920	.04270	.04620	.04970	.05320	.05670
36	.03276	.03672	.04032	.04392	.04752	.05112	.05472	.05832
37	.03367	.03774	.04144	.04514	.04884	.05254	.05624	.05994
38	.03458	.03876	.04256	.04636	.05016	.05396	.05776	.06156
39	.03549	.03978	.04368	.04758	.05148	.05538	.05928	.06318
40	.03640	.04080	.04480	.04880	.05280	.05680	.06080	.06480

THE FIRST TABLE CONTINUED:
SHEWING THE EXPANSION WITH HEAT
ON INCHES OF THE BAROMETER.

DEGREES OF THE THERMOMETER, FROM 1 TO 40, ON FARENHEIT'S SCALE.								
	17	18	19	20	21	22	23	24
1	.00172	.00182	.00192	.00203	.00213	.00223	.00233	.00243
2	.00344	.00364	.00384	.00406	.00426	.00446	.00466	.00486
3	.00516	.00546	.00576	.00609	.00639	.00669	.00699	.00729
4	.00688	.00728	.00768	.00812	.00852	.00892	.00932	.00972
5	.00860	.00910	.00960	.01015	.01065	.01115	.01165	.01215
6	.01032	.01092	.01152	.01218	.01278	.01338	.01398	.01458
7	.01204	.01274	.01344	.01421	.01491	.01561	.01631	.01701
8	.01376	.01456	.01536	.01624	.01704	.01784	.01864	.01944
9	.01548	.01638	.01728	.01827	.01917	.02007	.02097	.02187
10	.01720	.01820	.01920	.02030	.02130	.02230	.02330	.02430
11	.01892	.02002	.02112	.02233	.02343	.02453	.02563	.02673
12	.02064	.02184	.02304	.02436	.02556	.02676	.02796	.02916
13	.02236	.02366	.02496	.02639	.02769	.02899	.03029	.03159
14	.02408	.02548	.02688	.02842	.02982	.03122	.03262	.03402
15	.02580	.02730	.02880	.03045	.03195	.03345	.03495	.03645
16	.02752	.02912	.03072	.03248	.03408	.03568	.03728	.03888
17	.02924	.03094	.03264	.03451	.03621	.03791	.03961	.04131
18	.03096	.03276	.03456	.03654	.03834	.04014	.04194	.04374
19	.03268	.03458	.03648	.03857	.04047	.04237	.04427	.04617
20	.03440	.03640	.03840	.04060	.04260	.04460	.04660	.04860
21	.03612	.03822	.04032	.04263	.04473	.04683	.04893	.05103
22	.03784	.04004	.04224	.04466	.04686	.04906	.05126	.05346
23	.03956	.04186	.04416	.04669	.04899	.05129	.05359	.05589
24	.04128	.04368	.04608	.04872	.05112	.05352	.05592	.05832
25	.04300	.04550	.04800	.05075	.05325	.05575	.05825	.06075
26	.04472	.04732	.04992	.05278	.05538	.05798	.06058	.06318
27	.04644	.04914	.05184	.05481	.05751	.06021	.06291	.06561
28	.04816	.05096	.05376	.05684	.05964	.06244	.06524	.06804
29	.04988	.05278	.05568	.05887	.06177	.06467	.06757	.07047
30	.05160	.05460	.05760	.06090	.06390	.06690	.06990	.07290

DEGREES OF THE THERMOMETER, FROM 1 TO 40, ON FARENHEIT'S SCALE.								
	17	18	19	20	21	22	23	24
31	.05332	.05642	.05952	.06293	.06603	.06913	.07223	.07533
32	.05504	.05824	.06144	.06496	.06816	.07139	.07456	.07776
33	.05676	.06006	.06336	.06699	.07029	.07359	.07689	.08019
34	.05848	.06188	.06528	.06902	.07242	.07582	.07922	.08262
35	.06020	.06350	.06720	.07105	.07455	.07805	.08155	.08505
36	.06192	.06534	.06912	.07308	.07668	.08028	.08388	.08748
37	.06364	.06716	.07104	.07511	.07881	.08251	.08621	.08991
38	.06536	.06892	.07296	.07714	.08094	.08474	.08854	.09234
39	.06708	.07078	.07488	.07917	.08307	.08697	.09087	.09477
40	.06880	.07260	.07680	.08120	.08520	.08920	.09320	.09720

THE FIRST TABLE CONCLUDED:
SHEWING THE EXPANSION WITH HEAT
ON INCHES OF THE BAROMETER.

DEGREES OF THE THERMOMETER, FROM 1 TO 40, ON FARENHEIT'S SCALE.								
	25	26	27	28	29	30	31	32
1	.00253	.00263	.00274	.00284	.00294	.00304	.00314	.00324
2	.00506	.00526	.00548	.00568	.00588	.00608	.00628	.00648
3	.00759	.00789	.00822	.00852	.00882	.00912	.00942	.00972
4	.01012	.01052	.01096	.01136	.01176	.01216	.01256	.01296
5	.01265	.01315	.01370	.01420	.01470	.01520	.01570	.01620
6	.01518	.01578	.01644	.01704	.01764	.01824	.01884	.01944
7	.01771	.01841	.01918	.01988	.02058	.02128	.02198	.02268
8	.02024	.02104	.02192	.02272	.02352	.02432	.02512	.0259?
9	.02277	.02367	.02466	.02556	.02646	.02736	.02826	.02916
10	.02530	.02630	.02740	.02840	.02940	.03040	.03140	.03240
11	.02783	.02893	.03014	.03124	.03234	.03344	.03454	.03564
12	.03036	.03156	.03288	.03408	.03528	.03648	.03768	.03888
13	.03289	.03419	.03562	.03692	.03822	.03952	.04082	.04212
14	.03542	.03682	.03836	.03976	.04116	.04256	.04396	.04536
15	.03795	.03945	.04110	.04260	.04410	.04560	.04710	.04860
16	.04048	.04208	.04384	.04544	.04704	.04864	.05024	.05184
17	.04301	.04471	.04658	.04828	.04998	.05168	.05338	.05508
18	.04554	.04734	.04932	.05112	.05292	.05472	.05652	.05832
19	.04807	.04997	.05206	.05396	.05586	.05776	.05966	.06156
20	.05060	.05260	.05480	.05680	.05880	.06080	.06280	.06480
21	.05313	.05523	.05754	.05964	.06174	.06384	.06594	.06804
22	.05566	.05786	.06028	.06248	.06468	.06688	.06908	.07128
23	.05819	.06049	.06302	.06532	.06762	.06992	.07222	.07452
24	.06072	.06312	.06576	.06816	.07056	.07296	.07536	.07776
25	.06325	.06575	.06850	.07100	.07350	.07600	.07850	.08100
26	.06578	.06838	.07124	.07384	.07644	.07904	.08164	.08424
27	.06831	.07101	.07398	.07668	.07938	.08208	.08478	.08748
28	.07084	.07364	.07672	.07952	.08232	.08512	.0879	.09072
29	.07337	.07627	.07946	.08236	.08526	.08816	.09106	.09396
30	.07590	.07890	.08220	.08520	.08820	.09120	.09420	.09720
31	.07843	.08153	.08494	.08804	.09114	.09424	.09734	.10044
32	.08096	.08416	.08768	.09088	.09408	.09728	.10048	.10368

DEGREES OF THE THERMOMETER, FROM 1 TO 40, ON FARENHEIT'S SCALE.								
	25	26	27	28	29	30	31	32
33	.08349	.08679	.09042	.09372	.09702	.10032	.10362	.10692
34	.08602	.08942	.09316	.09656	.09996	.10336	.10676	.11016
35	.08855	.09205	.09590	.09940	.10290	.10640	.10990	.11340
36	.09108	.09468	.09864	.10224	.10584	.10944	.11314	.11664
37	.09361	.09731	.10138	.10508	.10878	.11248	.11618	.11988
38	.09614	.09994	.10412	.10792	.11172	.11552	.11932	.12312
39	.09867	.10257	.10686	.11076	.11466	.11866	.12246	.12636
40	.10120	.10520	.10960	.11360	.11760	.12160	.12560	.12960

The 5th Step, applied in the first Example.

364. Add this, to the former Expansion, thus:

Inches 25.19 Tenths

with 4° on 25 .0101 Expansion

with 4° on .19 .0000076 Expansion

The Answer is 25.2|001076,

viz. the Point at which the Quicksilver woud stand, in the coldest Barometer, when equally *expanded*, i. e. of the same Temperature with the warmer. Reject all but the first Decimal as too minute: this is seen by a Line drawn between the first and second Decimal.

Practice will shew how far to proceed, without computing the decimal Parts of an Inch, to more than 4 Places; but it is always more exact, to follow minutely the above Rules.

CHAPTER LXXII.

Section 365. **H**AVING therefore understood the Foundation, Construction, and Use of the first Table; in the present Case, having also added the decimal Parts of an Inch just found, for the Expansion,—to the Inches and Tenths, expressing the colder Barometer; which will then have the *same Expansion*, or *Temperature* with the warmer, thus;

Inches.

25.19 colder Barometer:

.0101 Expansion *on* the same, in Parts of an Inch with 4° of
Temperature, (rejecting all but the first Decimal as too
minute,)

25.2|001 added; this Sum will express the Point at which the
Quicksilver in the colder Barometer woud stand, when
equally expanded, i. e. in the same Temperature, with the
warmer.

366. 6th Step. Place both Barometers, now of *equal* Temperature with the warmer, together, first, the *upper* Barometer; and under it the *lower*, thus:

Inches 25. 2 Tenths.

29. 4

END OF THE FIRST STAGE.

367. The Ends proposed in the *second Stage* of the Work, (the colder Barometer being *now* brought to the same Expansion or Temperature with the warmer,) are two: First, to find, (by the Application of the second Table) the Heights, in Feet and Tenths, in the Atmosphere, corresponding to the Points at which the Quicksilver stands in both Barometers, which have now the same Temperature, viz. that of the warmer equal to 50°: on a Supposition that they were both exposed to the Temperature of 31°.24, on

Fahrenheit's Scale, which is about the Standard or freezing Point, for which sole Purpose the 2d Table is calculated.

N. B. The *Second Stage* includes two Steps only, viz. the 7th and 8th.

368. 7th Step. The Barometers being placed in one View, as before directed, thus:

Upper Barometer, Inches 25 .2 Tenths.

Lower Barometer, Inches 29 .4; find, with the Temperature of $31^{\circ}.24$, the corresponding Heights in the Atmosphere.

This is to be done by referring to the 2d Table, for the Application of which there are separate Instructions: See the Explanation of the second Table. [\[120\]](#)

CHAPTER LXXIII.

USE AND PRACTICE OF THE SECOND TABLE IN THE FIRST EXAMPLE.

The USE.

Section 369. **T**O find the Heights, in Feet and Tenths, in the Atmosphere, corresponding to the Points at which the Quicksilver stands in both Barometers, which have *now* the *same Temperature*, viz. that of the *warmer* Barometer, on a Supposition that they were both exposed to the Standard-Temperature of $31^{\circ}.24$, on Farenheit's Scale.

The PRACTICE.

The 7th Step applied in the first Example.

370. Look at the first Column, in the *second* Table, for

25.2, and the Answer is 6225.0 in the second Column; and for

29.4, and the Answer is 2208.2. The Answers are the Heights, in Feet and Tenths, in the Atmosphere, at which the Quicksilver stands in both Barometers, with the Temperature of $31^{\circ}.24$: corresponding to their respective Points, for which *sole* Purpose this Table is calculated.

371. 8th Step. Having placed the Barometers and their corresponding Heights in the Atmosphere, shewn by the second Table, at one View: subtract the lesser from the whole Height, and there will remain, *SECONDLY*; (see [Section 367](#)) the greater Height, viz. the Height corresponding to the Barometer in the elevated Station, above the Height corresponding to the Barometer, on the Ground, (both being at the Temperature of $31^{\circ}.24$) thus:

Feet.

Inches 25.2 correspond to 6225.0

Inches 29.4 correspond to 2208.2; subtract:

and the Remainder is 4016.8

viz. a Number in Feet and Tenths corresponding to the Height of the upper above the lower Barometer, both being in the Temperature of 31°.34.

THE SECOND TABLE.

The 1st Column shews the Quicksilver in the barometric Tube standing at each Inch from 1 to 10, and at each Tenth from 10 to 32 Inches.

The 2d Column shews the Height of the barometric Tube, above the *imaginary Level at 32 Inches*,—with the Temperature of 31.24;—in Feet and Tenths, answering to Inches and Tenths of the Barometer in the first Column.

The 3d Column shews the Height in Feet and Tenths, answering to a Tenth of an Inch on the Barometer, being the DIFFERENCE between each two adjoining Heights in the 2d Column.

Inch.	Feet.	Differ- ence.	Inch.	Feet.	Diff.	Inch.	Feet.	Diff.
1	90309.0	18061.8	16.8	16790.4	154.6	24.5	6959.0	106.1
2	72247.2	10565.4	.9	16635.8	153.7	.6	6852.9	105.7
3	61681.8	7496.4	17.0	16482.1	152.9	.7	6747.2	105.3
4	54185.4	5814.6	.1	16329.2	151.9	.8	6641.9	104.9
5	48370.8	4750.9	.2	16177.3	151.1	.9	6537.0	104.4
6	43619.9	4016.8	.3	16026.2	150.2	25.0	6432.6	104.0
7	39603.1	3479.5	.4	15876.0	149.3	.1	6328.6	103.6
8	36123.6	3069.2	.5	15726.7	148.5	.2	6225.0	103.2
9	33054.4	2745.4	.6	15578.2	147.6	.3	6121.8	102.8
10.0	30309.0	259.6	.7	15430.6	146.8	.4	6019.0	102.4
.1	30049.4	256.4	.8	15283.8	146.0	.5	5916.6	102.0
.2	29793.0	254.3	.9	15137.8	145.2	.6	5814.6	101.6
.3	29538.7	251.8	18.0	14992.6	144.3	.7	5713.0	101.2
.4	29286.9	249.3	.1	14848.3	143.6	.8	5611.8	100.8
.5	29037.6	247.0	.2	14704.7	142.8	.9	5511.0	100.6
.6	28790.6	244.7	.3	14561.9	142.0	26.0	5410.4	99.8
.7	28545.9	242.4	.4	14419.9	141.2	.1	5310.6	99.7
.8	28303.5	240.2	.5	14278.7	140.5	.2	5210.9	99.3

Inch.	Feet.	Differ- ence.	Inch.	Feet.	Diff.	Inch.	Feet.	Diff.
.9	28063.3	237.9	.6	14138.2	139.7	.3	5111.6	98.8
11.0	27825.4	235.8	.7	13998.5	139.0	.4	5012.8	98.6
.1	27589.6	233.7	.8	13859.5	138.2	.5	4914.2	98.1
.2	27355.9	231.6	.9	13721.3	137.5	.6	4816.1	97.8
.3	27124.3	229.6	19.0	13583.8	136.8	.7	4718.3	97.4
.4	26894.7	227.6	.1	13447.0	136.1	.8	4620.9	97.0
.5	26667.1	225.6	.2	13310.9	135.3	.9	4523.9	96.7
.6	26441.5	223.7	.3	13175.6	134.5	27.0	4427.2	96.4
.7	26217.8	221.7	.4	13041.1	134.2	.1	4330.8	95.9
.8	25996.1	220.0	.5	12906.9	133.3	.2	4234.9	95.7
.9	25776.1	218.0	.6	12773.6	132.6	.3	4139.2	95.2
12.0	25558.1	216.3	.7	12641.0	131.9	.4	4044.0	95.0
.1	25341.8	214.4	.8	12509.1	131.3	.5	3949.0	94.5
.2	25127.4	212.7	.9	12377.8	130.6	.6	3854.5	94.3
.3	24914.7	211.0	20.0	12247.2	130.0	.7	3760.2	93.9
.4	24703.7	209.3	.1	12117.2	129.3	.8	3666.3	93.6
.5	24494.4	207.7	.2	11987.9	128.7	.9	3572.7	93.2
.6	24286.7	206.0	.3	11859.2	128.0	28.0	3479.5	92.9
.7	24080.7	204.3	.4	11731.2	127.4	.1	3386.6	92.6
.8	23876.4	202.8	.5	11603.8	126.8	.2	3294.0	92.2
.9	23673.6	201.2	.6	11477.0	126.2	.3	3201.8	91.9
13.0	23472.4	199.7	.7	11350.8	125.6	.4	3109.9	91.6
.1	23272.7	198.2	.8	11225.2	125.0	.5	3018.3	91.3
.2	23074.5	196.6	.9	11100.2	124.4	.6	2927.0	90.9
.3	22877.9	195.2	21.0	10975.8	123.7	.7	2836.1	90.7
.4	22682.7	193.7	.1	10852.1	123.3	.8	2745.4	90.3
.5	22489.0	192.4	.2	10728.8	122.6	.9	2655.1	90.0
.6	22296.6	191.0	.3	10606.2	122.0	29.0	2565.1	89.7

Inch.	Feet.	Difference.	Inch.	Feet.	Diff.	Inch.	Feet.	Diff.
.7	22105.6	189.4	.4	10484.2	121.5	.1	2475.4	89.4
.8	21916.2	188.1	.5	10362.7	120.9	.2	2386.0	89.1
.9	21728.1	186.8	.6	10241.8	120.4	.3	2296.9	88.7
14.0	21541.3	185.5	.7	10121.4	119.8	.4	2208.2	88.5
.1	21355.8	184.1	.8	10001.6	119.2	.5	2119.7	88.2
.2	21171.7	182.9	.9	9882.4	118.8	.6	2031.5	87.9
.3	20988.8	181.6	22.0	9763.6	118.1	.7	1943.6	87.6
.4	20807.2	180.3	.1	9645.5	117.7	.8	1856.0	87.3
.5	20626.9	179.0	.2	9527.8	117.1	.9	1768.7	87.0
.6	20447.9	178.0	.3	9410.7	116.6	30.0	1681.7	86.7
.7	20269.9	176.7	.4	9294.1	116.0	.1	1595.0	86.4
.8	20093.2	175.4	.5	9178.1	115.6	.2	1508.6	86.2
.9	19917.8	174.3	.6	9062.5	115.1	.3	1422.4	85.8
15.0	19743.5	173.1	.7	8947.4	114.5	.4	1236.6	85.6
.1	19570.4	172.0	.8	8832.9	114.0	.5	1251.0	85.3
.2	19398.4	170.9	.9	8718.9	113.6	.6	1165.7	85.0
.3	19227.5	169.8	23.0	8605.3	113.0	.7	1080.7	84.7
.4	19057.7	168.6	.1	8492.3	112.6	.8	996.0	84.5
.5	18889.1	167.6	.2	8379.7	112.1	.9	911.5	84.2
.6	18721.5	166.5	.3	8267.6	111.6	31.0	827.3	83.9
.7	18555.0	165.4	.4	8156.0	111.1	.1	743.4	83.7
.8	18389.6	164.1	.5	8044.9	110.6	.2	659.7	83.4
.9	18225.5	163.7	.6	7934.3	110.2	.3	576.3	83.1
16.0	18061.8	162.4	.7	7824.1	109.7	.4	493.2	82.8
.1	17899.4	161.3	.8	7714.4	109.3	31.5	410.4	82.6
.2	17738.1	160.4	.9	7605.1	108.8	.6	327.8	82.4
.3	17577.7	159.3	24.0	7496.3	108.3	.7	245.4	82.0

Inch.	Feet.	Difference.	Inch.	Feet.	Diff.	Inch.	Feet.	Diff.
.4	17418.4	158.4	.1	7388.0	107.9	.8	163.4	81.8
.5	17260.0	157.5	.2	7280.1	107.5	.9	81.6	81.6
.6	17102.5	156.5	.3	7172.6	107.0	32.0	00.0	
.7	16946.0	155.6	.4	7065.6	106.6			

372. Now apply the third Table, or Table for Tenths, *if necessary*; including two more Steps, viz. the 9th and 10th: which, being useless, in the first Example, are, for the present, omitted.

373. An Explanation of the third Table, or Table for Tenths, is, however, for the Sake of *Order*, *here* subjoined. [\[122\]](#)

THE THIRD TABLE, OR TABLE FOR TENTHS:

Serving to compleat the 2d Table, on Expansion of the Barometer, with the Temperature of 31°.24.

1. The upper horizontal Figures shew the Number of Parts into which the Tenth of an Inch has been divided.
2. The Figures in the left vertical Column express the Height in FEET, (above the imaginary Level, at 32 Inches of the Barometer,) or Expansion corresponding to a single Tenth of an Inch of Quicksilver.
3. The FEET in the Place of Meeting are called TENTHS: thus, 90 Feet are 9 Tenths of 100 Feet.

Feet.	Parts into which the Tenth of an Inch is divided.								
	1/10	2/10	3/10	4/10	5/10	6/10	7/10	8/10	9/10
81	8	16	24	32	40	49	57	65	73
82	8	16	25	33	41	49	57	66	74
83	8	17	25	33	41	50	58	66	75
84	8	17	25	34	42	50	59	67	76
85	8	17	25	34	42	51	59	68	76
86	9	17	26	34	43	52	60	69	77
87	9	17	26	35	43	52	61	70	78
88	9	18	26	35	44	53	62	70	79
89	9	18	27	36	44	53	62	71	80
90	9	18	27	36	45	54	63	72	81
91	9	18	27	36	45	55	64	73	82
92	9	18	28	37	46	55	64	74	83
93	9	19	28	37	46	56	65	74	84
94	9	19	28	38	47	56	66	75	85
95	9	19	28	38	47	57	66	76	85
96	10	19	29	38	48	58	67	77	86

Feet.	Parts into which the Tenth of an Inch is divided.								
	1/10	2/10	3/10	4/10	5/10	6/10	7/10	8/10	9/10
97	10	19	29	39	48	58	68	78	87
98	10	20	29	39	49	59	69	78	88
99	10	20	30	40	49	59	69	79	89
100	10	20	30	40	50	60	70	80	90
101	10	20	30	40	50	61	71	81	91
102	10	20	31	41	51	61	71	82	92
103	10	21	31	41	51	62	72	82	93
104	10	21	31	42	52	62	73	83	94
105	10	21	31	42	52	63	73	84	94
106	11	21	32	42	53	64	74	85	95
107	11	21	32	43	53	64	75	86	96
108	11	22	32	43	54	65	76	86	97
109	11	22	33	44	54	65	76	87	98
110	11	22	33	44	55	66	77	88	99
111	11	22	33	44	55	67	78	89	100
112	11	22	34	45	56	67	78	90	101
113	11	23	34	45	56	68	79	90	102
114	11	23	34	46	57	68	80	91	103
115	11	23	34	46	57	69	80	92	103
116	12	23	35	46	58	70	81	93	104
117	12	23	35	47	58	70	82	94	105
118	12	24	35	47	59	71	83	94	106
119	12	24	36	48	59	71	83	95	107
120	12	24	36	48	60	72	84	96	108
121	12	24	36	48	60	73	85	97	109
122	12	24	37	49	61	73	85	98	110
123	12	25	37	49	61	74	86	98	111
124	12	25	37	50	62	74	87	99	112
125	12	25	37	50	62	75	87	100	112
126	13	25	38	50	63	76	88	101	113
127	13	25	38	51	63	76	89	102	114
128	13	26	38	51	64	77	90	102	113
129	13	26	39	52	64	77	90	103	116
130	13	26	39	52	65	78	91	104	117

END OF THE SECOND STAGE.

374. The Ends proposed in the third and last Stage of the Work, are, first, to add the *general* Temperatures of the Air, or detached Air-Thermometers, at each Place of Observation *above* and *below*, into one Sum.

Secondly, to divide that Sum: each Moiety of which is called the *mean Temperature* of the Air.

Thirdly, to apply that Moiety to each Barometer, (both of which have been already brought to the Standard-Temperature of $31^{\circ}. 24;$) in order to prove whether the Moiety (or Quantity of Heat assigned to each Barometer by the *general* Temperature of the Air) *exceeded, fell short of*, or equalled the Standard-Temperature of the Barometers, by the 2d Table.

And fourthly, from the Moiety or mean Temperature of the Air, to find the true Height of the upper Barometer: which Temperature resolves itself into three Cases.

375. 1st. If the Moiety or mean Temperature of the Air is greater than the Standard Temperature, viz. that to which the Barometers are now brought; find the Expansion of Air corresponding to such *Excess* of Temperature by the fourth Table, which Height by Expansion, being added to the Height already found in the 2d Table, shews the true Height, viz. of the upper Barometer.

N. B. The 3d and last Stage includes two Steps only, viz, 11th and 12th.

376. 11th Step. The detached Air-Thermometer	
<i>above</i> was	39 $\frac{1}{2}$ Degrees.
The detached Air-Thermometer <i>below</i> was	45
1st. Add them, for the whole Heat.	2)84 $\frac{1}{2}$
2d. For <i>mean Temperature</i> of the Air-	
Thermometers, or a <i>Moiety</i> of the Heat, divide by 2.	42 $\frac{1}{4}$
3d. Deduct the Standard-Temperature of	31 $\frac{1}{4}$
from either Moiety, and the Remainder	11

is the 11 Degrees of Heat, more than the Standard^[123] for each Barometer.

For $42^{\circ}\frac{1}{4}$, and $42^{\circ}\frac{1}{4}$, equal to $84^{\circ}\frac{1}{2}$, was the whole Height of the Air at both Places of Observation in the upper and lower Stations; of which whole Height the detached or Air-Thermometer *above* received $39^{\circ}\frac{1}{2}$, and the detached or Air-Thermometer *below*, received 45° .

377. 12th Step. Find the Height corresponding to the Expansion of Air, with Excess of Heat or Temperature above the Standard-Temperature of the Barometers: and add it (as in the first Example) to the Height of the upper Barometer, corresponding to the Standard-Temperature already found in the *second* Table, and the Sum is the *true* Height of the upper Barometer.

This is to be done by referring to the 4th Table, shewing Expansion of Air with Heat; for the Application of which there are separate Instructions: see the Explanation of the 4th Table.^[124]

378. The Expansion of Air, in the first Example, is found by the 4th Table to be Feet 107.3 Tenths *higher* than the 4016.8, viz. the Remainder from the 2d Table ([Section 371](#)); which Numbers added give 4124.1 Feet: viz. the true Height of the upper Station required.

CHAPTER LXXIII.

USE AND PRACTICE OF THE FOURTH TABLE, IN THE FIRST EXAMPLE.

The USE.

Section 379. **T**O shew in Feet, and Tenths, what is the Expansion of Air on each thousand Feet, from 1000 to 9000 Feet, *with* each Degree of Temperature from 1 to 100 Degrees, on Farenheit's Scale.

The PRACTICE.

The 12th Step applied in the first Example.

380. For the Expansion of Air with 11 Degrees of Heat on 4016.8 Feet, look in the fourth Table, *with* 11 in the left Hand vertical Column of Temperature, and (first) *on* 4000 Feet, along the upper Line: the Place of Meeting gives the Expansion of the Air, *with* 11 Degrees *on* 4000 Feet: viz. 106.92. [\[126\]](#)

Next; look *with* 11 Degrees, and (as there is a Cypher only in the Place of Hundreds) *on* 10, (viz. of the 16 Feet) call the 10, a 1000; the Place of Meeting, or Answer is 26.73:

Thirdly; *with* 11, *on* 6, (viz. of the 16,) calling it 6000; the Answer is 160.38:

Fourthly; *with* 11, *on* 8, (viz. the .8,) and the Answer is 213.84.

381. Having added the respective Expansions together, thus;

$$\begin{array}{rcll}
 \text{with } 11^\circ, \text{ on } 4016.8 & & \text{Feet. Tenths.} & \\
 \left. \begin{array}{l} \text{with } 11^\circ \\ \text{on} \end{array} \right\} \begin{array}{l} 4000 \\ 10 \\ 6 \\ .8 \end{array} & = & \begin{array}{l} 106.92 \\ 26.73 \\ 160.38 \\ 213.84 \end{array} & \begin{array}{l} 106.92 \\ .2673 \\ .16038 \\ .021384 \end{array} \\
 & & & \hline
 & & \text{Expansion} & 107.369064;
 \end{array}$$

THE FOURTH TABLE,

SHEWING THE EXPANSION WITH **HEAT**, FROM 1 TO 100 DEGREES, ON EACH **THOUSAND FEET** IN THE ATMOSPHERE, FROM 1000 TO 9000 FEET.

DEGREES OF THE THERMOMETER, FROM 1 TO 50, ON FARENHEIT'S SCALE.									
	1000	2000	3000	4000	5000	6000	7000	8000	9000
1	2.43	4.86	7.29	9.72	12.15	14.58	17.01	19.44	21.87
2	4.86	9.72	14.58	19.44	24.30	29.16	34.02	38.88	43.74
3	7.29	14.58	21.87	29.16	36.45	43.74	51.03	58.32	65.61
4	9.72	19.44	29.16	38.88	48.60	58.32	68.04	77.76	87.48
5	12.15	24.30	36.45	48.60	60.75	72.90	85.05	97.20	109.35
6	14.58	29.16	43.74	58.32	72.90	87.49	102.06	116.64	131.22
7	17.01	34.02	51.03	68.04	85.05	102.06	119.07	136.08	153.09

DEGREES OF THE THERMOMETER, FROM 1 TO 50, ON FARENHEIT'S SCALE.									
	1000	2000	3000	4000	5000	6000	7000	8000	9000
8	19.44	38.88	58.32	77.76	97.20	116.64	136.08	155.52	174.96
9	21.87	43.74	65.61	87.48	109.35	131.22	153.09	174.96	196.83
10	24.30	48.60	72.90	97.20	121.50	145.80	170.10	194.40	218.70
11	26.73	53.46	80.19	106.92	133.65	160.38	187.11	213.84	240.57
12	29.16	58.32	87.48	116.64	145.80	174.96	204.12	233.28	262.44
13	31.59	63.18	94.77	126.36	157.95	189.54	221.13	252.72	284.31
14	34.02	68.04	102.06	136.08	170.10	204.12	238.14	272.16	306.18
15	36.45	72.90	109.35	145.80	182.25	218.70	255.15	291.60	328.05
16	38.88	77.76	116.64	155.52	194.40	233.28	272.16	311.04	349.92
17	41.31	82.62	123.93	165.24	206.55	247.86	289.17	330.48	371.79
18	43.74	87.48	131.22	174.96	218.70	262.44	306.18	349.92	393.66
19	46.17	92.34	138.51	184.68	230.85	277.02	323.19	369.36	415.53
20	48.60	97.20	145.80	194.40	243.00	291.60	340.20	388.80	437.40
21	51.03	102.06	153.09	204.12	255.15	306.18	357.21	408.24	459.27
22	53.46	106.92	160.38	213.84	267.30	320.76	374.22	427.68	481.14
23	55.89	111.78	167.67	223.56	279.45	335.34	391.23	447.12	503.01
24	58.32	116.64	174.96	233.28	291.60	349.92	408.24	466.56	524.88
25	60.75	121.50	182.25	243.00	303.75	364.50	425.25	486.00	546.75
26	63.18	126.36	189.54	252.72	315.90	379.08	442.26	505.44	568.62
27	65.61	131.22	196.83	262.44	328.05	393.66	459.27	524.88	590.49
28	68.04	136.08	204.12	272.16	340.20	408.24	476.28	544.32	612.36
29	70.47	140.94	211.41	281.88	352.35	422.82	493.29	563.76	634.23
30	72.90	145.80	218.70	291.60	364.50	437.40	510.30	583.20	656.10
31	75.33	150.66	225.99	301.32	376.65	451.98	527.31	602.64	677.97
32	77.76	155.52	233.28	311.04	388.80	466.56	544.32	622.08	699.84
33	80.19	160.38	240.57	320.76	400.95	481.14	561.33	641.52	721.71
34	82.62	165.24	247.86	330.48	413.10	495.72	578.34	660.96	743.58
35	85.05	170.10	255.15	340.20	425.25	510.30	595.35	680.40	765.45
36	87.48	174.96	262.44	349.92	437.40	524.88	612.36	699.84	787.32
37	89.91	179.82	269.73	359.64	449.55	539.46	629.37	719.28	809.19
38	92.34	184.68	277.02	369.36	461.70	554.04	646.38	738.72	831.06
39	94.77	189.54	284.31	379.08	473.85	568.62	663.39	758.16	852.93
40	97.20	194.40	291.60	388.80	486.00	583.20	680.40	777.60	874.80
41	99.63	199.26	298.89	398.52	498.15	597.78	697.41	797.04	896.67
42	102.06	204.12	306.18	408.24	510.30	612.36	714.42	816.48	918.54
43	104.49	208.98	313.47	417.96	522.45	626.94	731.43	835.92	940.41
44	106.92	213.84	320.76	427.68	534.60	641.52	748.44	855.36	962.28
45	109.35	218.70	328.05	437.40	546.75	656.10	765.45	874.80	984.15
46	111.78	223.56	335.34	447.12	558.90	670.68	782.46	894.24	1006.02
47	114.21	228.42	342.63	456.84	571.05	685.26	799.47	913.68	1027.89
48	116.64	233.28	349.92	466.56	583.20	699.84	816.48	933.12	1049.76
49	119.07	238.14	357.21	476.28	595.35	714.42	833.49	952.56	1071.63
50	121.50	243.00	364.50	486.00	607.50	729.00	850.5	972.00	1093.5
51	123.93	247.86	371.79	495.72	619.65	743.58	867.51	991.44	1115.37
52	126.36	252.72	379.08	505.44	631.80	758.16	884.52	1010.88	1137.24
53	128.79	257.58	386.37	515.16	643.95	772.74	901.53	1030.32	1159.11
54	131.22	262.44	393.66	524.88	656.10	787.32	918.54	1049.76	1180.98
55	133.65	267.30	400.95	534.60	668.25	801.90	935.55	1069.20	1202.85
56	136.08	272.16	408.24	544.32	680.40	816.48	952.56	1088.64	1224.72
57	138.51	277.02	415.53	554.04	692.55	831.06	969.57	1108.08	1246.59
58	140.94	281.88	422.82	563.76	704.70	845.64	986.58	1127.52	1268.46
59	143.37	286.74	430.11	573.48	716.85	860.22	1003.59	1146.96	1290.33
60	145.80	291.60	437.40	583.20	729.00	874.80	1020.60	1166.40	1312.20
61	148.23	296.46	444.69	592.92	741.15	889.38	1037.61	1185.84	1334.07
62	150.66	301.32	451.98	602.64	743.30	903.96	1054.62	1205.28	1355.94
63	153.09	306.18	459.27	612.36	755.45	918.54	1071.63	1224.72	1377.81
64	155.52	311.04	466.56	622.08	767.60	933.12	1088.64	1244.16	1399.68
65	157.95	315.90	473.85	631.80	779.75	947.70	1105.65	1263.60	1421.55

DEGREES OF THE THERMOMETER, FROM 1 TO 50, ON FARENHEIT'S SCALE.									
	1000	2000	3000	4000	5000	6000	7000	8000	9000
66	160.38	320.76	481.14	641.52	791.90	962.28	1122.66	1283.04	1443.42
67	162.81	325.62	488.43	651.24	814.05	976.86	1139.67	1302.48	1465.29
68	165.24	330.48	495.72	660.96	826.20	991.44	1156.68	1321.92	1487.16
69	167.67	335.34	503.01	670.68	838.35	1006.02	1173.69	1341.36	1509.03
70	170.10	340.20	510.30	680.40	850.50	1020.60	1190.70	1360.80	1530.90
71	172.53	345.06	517.59	690.12	862.65	1035.18	1207.71	1380.24	1552.77
72	174.96	349.92	524.88	699.84	874.80	1049.76	1224.72	1399.68	1574.64
73	177.39	354.78	532.17	709.56	886.95	1064.34	1241.73	1419.12	1596.51
74	179.82	359.64	539.46	719.28	899.10	1078.92	1258.74	1438.56	1618.38
75	182.25	364.50	546.75	729.00	911.25	1093.50	1275.75	1458.00	1640.25
76	184.68	369.36	554.04	738.72	923.40	1108.08	1292.76	1477.44	1662.12
77	187.11	374.22	561.33	748.44	935.55	1122.66	1309.77	1496.88	1683.99
78	189.54	379.08	568.62	758.16	947.70	1137.24	1326.78	1516.32	1705.86
79	191.97	383.94	575.91	767.88	959.85	1151.82	1343.79	1535.76	1727.73
80	194.40	388.80	583.20	777.60	972.00	1166.40	1360.80	1555.20	1749.60
81	196.83	393.66	590.49	787.32	984.15	1180.98	1377.81	1574.64	1771.47
82	199.26	398.52	597.78	797.04	996.30	1195.56	1394.82	1594.08	1793.34
83	201.69	403.38	605.07	806.76	1008.45	1210.14	1411.83	1613.52	1815.21
84	204.12	408.24	612.36	816.48	1020.60	1224.72	1428.84	1632.96	1837.08
85	206.55	413.10	619.65	826.20	1032.75	1239.30	1445.85	1652.40	1858.95
86	208.98	417.96	626.94	835.92	1044.90	1253.88	1462.86	1671.84	1880.82
87	211.41	422.82	634.23	845.64	1057.05	1268.46	1479.87	1691.28	1902.69
88	213.84	427.68	641.52	855.36	1069.20	1283.04	1496.88	1710.72	1924.56
89	216.27	432.54	648.81	865.08	1081.35	1297.62	1513.89	1730.16	1946.43
90	218.70	437.40	656.10	874.80	1093.50	1312.20	1530.90	1749.60	1968.30
91	221.13	442.26	663.39	884.52	1105.65	1326.78	1547.91	1769.04	1990.17
92	223.56	447.12	670.68	894.24	1117.80	1341.36	1564.92	1788.48	2012.04
93	225.99	451.98	677.97	903.96	1129.95	1355.94	1581.93	1807.92	2033.91
94	228.42	456.84	685.26	913.68	1142.10	1370.52	1598.94	1827.36	2055.78
95	230.85	461.70	692.55	923.40	1154.25	1385.10	1615.95	1846.80	2077.65
96	233.28	466.56	699.84	933.12	1166.40	1399.68	1632.96	1866.24	2099.52
97	235.71	471.42	707.13	942.84	1178.55	1414.26	1649.97	1885.68	2121.39
98	238.14	476.28	714.42	952.56	1190.70	1428.84	1666.98	1905.12	2143.26
99	240.57	481.14	721.71	962.28	1212.85	1443.42	1683.99	1924.56	2165.13
100	243.00	486.00	729.00	972.00	1215.00	1458.00	1701.00	1944.00	2187.00

382. The decimal Points in the Answer must be changed, thus:

1. For the Place of *Thousands* in the Question, (viz. 4000,) the Answer must remain, viz. 106.92, as in the Table, which is calculated for the Place of *Thousands*.

2. For the Place of *Hundreds*, in the Question, (viz. which in the present Case was a Cypher;) if there had been a Figure or Figures in the Place of hundreds; then the decimal Point in the Answer must have been removed over *one* Figure or Place to the left.

3. For the Place of *Tens*, in the Question, (viz. 10 Feet,) the decimal Point in the Answer, must be removed over *two* Figures, or Places, to the left.

4. For the Place of *Units*, in the Question, (viz. 6) the decimal Point in the Answer, must be removed over *three* Figures, or Places, to the left.

5. For the Place of a *Decimal*, in the Question, (viz. .8) the decimal Point, in the Answer, must be removed over *four* Figures, or Places to the left, by adding a Cypher: and for the Place of each further Decimal in the Question;—*one* Place more in the Answer, by the further occasional Addition of a Cypher, thus: *on*

Feet 4000, the Ans. 106.92 is still 106.92		
10	26.73	becomes .2673
6	160.38	.16038
.8	213.84	.021384
		107.369064

383. Which Sum, by rejecting all but the first Decimal, in the Answer, is Feet 107.3 Tenths equal to the Expansion of Air, *with* 11° of Heat, *on* 4016.8 Feet, the Height of the upper Barometer, with the Temperature of 31°.24, according to the 2d Table.

END OF THE LAST STAGE.

384. The RULE underneath, consisting of 3 Precepts *only*, is laid down by Sir George Shuckburgh, in the Transactions for 1777, Page 574, in order to ascertain the Height of Mountains, &c. (See [Section 349](#)).^[127]

385. Recapitulation for each Step of the Work, in the first Example; referring to the Sections.

Below. Barometer, Inches 29, .4 Tenths.

Attached Thermometer, 50 Degrees, Air-Thermometer 45°.

Above. Barometer, Inches 25, .19 Tenths.

Attached Thermometer 46°, Air Thermometer, 29°1/2.

From 50° subtract
46

and there remains 4 Degrees of Temperature to be added to the colder Barometer.

By Means of the first Table, find the Expansion of the *colder* Barometer, with Degrees of Heat, viz. 4° on Inches 25, .19, *gradually*, thus:

with 4° on 25.	= .0101
with 4° on .19	= .0000076

25.2|

Upper Barometer, Inches 25, .2 Tenths.

Lower Barometer, 29, .4

End of the first Stage.

By Means of the 2d Table, find the corresponding Heights in the Air, at 31°. 24.

25, .2	Answer	6225.0
29, .4		2208.0

The Remainder is 4016.8 Height in Feet, &c.

The 3d Table, or Table for *Heights* in the Atmosphere corresponding to the *Tenth* of an Inch *on* the Barometer, including the 9th and 10th Steps, is useless in this first Example.

End of the Second Stage.

Detached Air-Thermometer, <i>above</i> ,	29 1/2
Ditto <i>below</i> ,	45°

Whole Heat	2)841 1/2
Half Heat or mean Temperature	431 1/4
Deduct Standard	311 1/4

Moiety above Standard	11°
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By Means of the 4th Table, find the
Expansion of Air, with 11° on

	4106.8 Feet
viz.	107.3

which added to the same Height gives 4124.1 for the
true Height, in English Feet, of the *Mountain*, or *upper Station*, sought.

End of the last Stage.

CHAPTER LXXV.

PRACTICE OF THE SECOND EXAMPLE:

With a distinct View of the Work. (Ph. Tr. for 1777, Page 579.)

Section 386. **T**HE Point at which the Quicksilver stood in the Tube of the Barometer on the Mountain, or in the Car of the Balloon, being Inches 24.178 Tenths; its *attached* Thermometer, Degrees 57.2 Tenths, and its Air-Thermometer 56°; while the Barometer on the Ground stood at Inches 28, .1318 Tenths; its *attached* Thermometer, Degrees 61, .8 Tenths, and its Air-Thermometer 63°, .9; what is the Height of the *upper* Station?

387. 1st. Step. Set down the Observation on the Ground, thus:

BELOW, Barometer, Inches 28, .1318 Tenths,

Attached Thermometer, Degrees 61, .8 Tenths.

Air-Thermometer, 63°, .9.

388. 2d. Step. Set down the Observation, on the Mountain, or *in the Car*, thus:

ABOVE, Barometer, Inches 24, .178 Tenths.

Attached Thermometer, Degrees 57, .2 Tenths.

Air-Thermom. 56°.

389. 3d. Step. From the *warmer attached* Thermometer, subtract the colder, thus:

$$\begin{array}{r} 61^{\circ}, .8 \\ 57, .2 \\ \hline 4, .6 \end{array}$$

390. 4th. Step. Give the *colder* Barometer the same Expansion, viz. 4°, .6 with the warmer, by the *first* Table.

CHAPTER LXXVI.

PRACTICE OF THE FIRST TABLE IN THE SECOND EXAMPLE.

4th Step applied in the 2d Example.

Section 391. **T**HE *Order* to be observed in finding the Expansion *with* $4^{\circ}.6$, i. e. with 4 Degrees, .6 Tenths of Heat, on 24.178, i. e. 24 Inches, .178 Tenths of the coldest Barometer.

Find the Expansion required, thus:

Case the 1st.

1st. Part. *With* 4° on 24 Inches.

2d. Part. *With* 4° on .178 Tenths of an Inch above 24 Inches.

Case the 2d.

1st. Part. *With* .6 Tenths of a Degree, on 24 Inches.

2d. Part. *With* .6 Tenths of a Degree, on .178 Tenths above 24 Inches.

SPECIFICALLY, *thus*:

1st. Part of *Case the 1st.* To find the Expansion,

With 4° on 24 Inches.

2d. Part of *Case the 1st.*

With 4° , on .178 Tenths of an Inch above 24 Inches; begin thus:

With 4° , on 24 Inches: then,

With 4° , on 25: then,

With 4° , on 1 Inch above 24, i. e. on the 25th Inch: then,

With 4° , on .1 Tenth above 24: then,

With 4° , on .178 Tenths above 24.

1st Part of *Case the 2d.* To find the Expansion,

With .6 above 4° on 24; begin thus:

With 4° on 24 Inches: then,

With 5° on 24: then,

With 1° above 4° , on 24, i. e. the 5th $^{\circ}$: then,

With .1 Tenth above 4° , on 24: then

With .6 Tenths above 4°, on 24.

2d Part of *Case the 2d*. To find the Expansion,

With .6 Tenths above 4° of Heat on .178 Tenths above 24 Inches: to be done thus:

The EXPANSION with 4°, on .178 Tenths above 24 Inches, being once found; divide IT by 4: and the Quotient is the Expansion with 1° above 4°, on .178 Tenths of an Inch above 24 Inches.

Then for the Expansion with .1 Tenth above 4°, on .178 Tenths above 24 Inches; add a Cypher and decimal Point to the left of the same Quotient.

Then for the Expansion with .6; multiply that Sum into .6, and add a Cypher and decimal Point.

The Answer is the PART of an Inch, to which .6 Tenths of a Degree above 4° of Heat, on .178 Tenths of an Inch above 24 Inches, raises the Barometer.

It is true, the PART is so minute as to be rejected: yet the Mode of Proceeding, in order to investigate the Expansion with Precision, is proper to be retained.

392. PRACTICE of the first Part of *Case the 1st*.

For the Expansion with 4°, on 24 Inches; look, in the first Table, ([Sect. 363](#)) and in the left vertical Column, with 4 Degrees of the Thermometer; and along the upper horizontal Line, on 24 Inches of Quicksilver in the Tube of the Barometer: the Point of Meeting gives the Expansion .0097;^[128] which, preparatory to Addition, is to be placed under the 24, .178 thus,

.0097

PRACTICE of the 2d Part of *Case the first*.

393. In order to obtain the Expansion, with 4°, of Heat on .178 Tenths of an Inch above 24 Inches of the Barometer; let it be considered where it ought to be found in the Table: for, Tenths of 1 Inch above 24 Inches, are at some intermediate Point between 24 and 25; that is, above 24, yet not so high as 25: or more than 24, yet less than 25.

Look therefore in the Table, with 4 Degrees of Heat, on 24 Inches; then with 4° on 25 Inches: and the respective Numbers are .0097 and .0101.

And by taking the Expansion with 4° on 24 Inches, from 4° on 25; the Remainder will be the Expansion with 4° on 1 Inch above 24 Inches, viz. on the 25th Inch, thus:

With 4° on } 25 = .0101 from;
 } 24 = .0097 subtract:

—————
.0004 :

This therefore is the Expansion with 4°, on 1 Inch above 24 Inches.

Then with 4°, on .1 Tenth of an Inch above 24 Inches.

The Answer is the same as the former, viz. .0004, with the Addition of a Cypher and decimal Point to the left, thus; .0004 becomes .00004, viz. the Expansion with 4°, on .1

Tenth of an Inch above 24 Inches.

Then for the Expansion *with* 4°, *on* .178 Tenths, say,

If the Expansion *with* 4°, *on* .1 Tenth above 24 Inches gives .00004 Part of an Inch, what will the Expansion *with* 4°, *on* .178 give?

Thus; .1 : .00004 :: .178?

Multiply the two last Terms, thus:

$$\begin{array}{r} .00004 \\ .178 \\ \hline 00032 \\ 00028 \\ 00004 \\ \hline \end{array}$$

0000712:

and, as in Multiplication of Decimals, the Product must have as many decimal Places, as are in the Factors; a Cypher must be added to the left Hand, thus: .00000712: but having divided that Product by the first Term .1, viz. a Decimal, the Answer is a Cypher less; viz. .0000712.

This Answer is the Expansion *with* 4°, *on* .178 Tenths of an Inch above 24 Inches: prepare it for *Addition*, as the former,

$$\begin{array}{r} 24.178 \\ .0097 \\ .0000712 \end{array}$$

PRACTICE of the first Part of *Case the 2d*.

394. For the Expansion of .6 Tenths of a Degree of Heat, (more than the 4 Degrees) on 24 Inches of the *coldest* Barometer; it should be considered where such Tenths can lie in the Table.

Now .6 Tenths of 1 Degree, (more than the 4°) are at some intermediate Point of the Thermometer between 1 and 2 Degrees: above 1; yet not so high as 2: or more than 1; yet less than 2.

Therefore .6 Tenths of 1 Degree above 4 Degrees, are somewhere between the 4th and 5th Degree: above 4; yet not so high as 5: or more than 4; yet less than 5.

Look in the Table ([Section 363](#)); first *with* 4 Degrees of Heat, *on* 24 Inches, and then *with* 5 Degrees of Heat *on* 24 Inches; and the respective Numbers are .0097 and .0121: and by taking the Expansion *with* 4 Degrees *on* 24 Inches, from the Expansion *with* 5 Degrees *on* the same 24 Inches; the Remainder will be the Expansion *with* 1 Degree above 4° *on* 24 Inches: viz.

$$\text{with } \left\{ \begin{array}{l} 5^\circ = .0121 \\ 4^\circ = .0097 \end{array} \right\} \text{ on 24 Inches, as in whole Numbers.}$$

Remainder, .0024

This therefore is the Expansion *with* 1 Degree of Heat, above 4, viz. *with* the 5th Degree, on 24 Inches of the Barometer.

Then say, if 1 Degree of the Thermometer (above 4, viz. the 5th Degree) gives by Expansion, a certain additional Height, or Part of an Inch, viz. .0024, on 24 Inches of the Barometer; what Height will 6 Degrees give? Answer 6 Times *more*.

Multiply the 2d and 3d Terms, and divide by the first, thus;

$$\begin{array}{r} 1 : .0024 :: 6? \\ 6 \\ \hline .0144 \end{array}$$

is the Expansion, or Height, in Parts of an Inch, for 6 Degrees.

And farther, to proportion for the Decimal; say as .1 Tenth of a Degree gives a certain Tenth of the former .0024, in additional Height, viz. .00024; what Height will .6 Tenths give? Answer, .00144.

Prepare this *Height* for Addition to the Numbers already found.

PRACTICE of the 2d Part of *Case the 2d*.

395. To find the Expansion of .6 above 4° on .178 above 24 Inches.

The Expansion *with* 4° on .178 is already found to be .0000712: divide it by 4, and the Answer is .0000178, viz. the Expansion *with* 1° on .178 above 24 Inches:

And, for the Expansion with .1 Tenth; the Answer, with the Addition of a Cypher and decimal Point to the left, becomes .00000178.

Lastly, for the Expansion with .6, say,

If .1 : .00000178 :: .6?

Multiply the 2d and 3d Terms, and divide by first:

$$\begin{array}{r} .00000178 \\ .6 \\ \hline .000001068. \end{array}$$

The Answer is a Decimal less, viz. .00001068; i. e. the Decimal of an Inch, to which .6 Tenths of a Degree above 4 Degrees of Heat, on .178 Tenths of an Inch above 24 Inches, raises the Barometer: which, after all, is so inconsiderable, that it may be fairly rejected.

Yet the Rules by which these Deductions are made, may be useful in other Cases.

Prepare for Addition, as before.

The Decimals, in the Answers, may be omitted, when they exceed four Places.

396. 5th Step. To proceed with the second Example.

Place the different Expansions now found, above each other, Units, Tens, &c. under Units, Tens, &c. preparatory to Addition, thus;

For the Expansion *with* 4°, .6 *on* 24, .178:

1st. <i>with</i>	4°,	<i>on</i> 24,	.0097
2d. <i>with</i>	.6	<i>on</i> 24,	.00144
3d. <i>with</i>	4°,	<i>on</i> .178	.0000712
4th. <i>with</i>	.6	<i>on</i> .178	.00001068

The Expansions with 4°, .6 added = .01122188

To the Sum add the Height of the *colder* Barometer
24.178

24.1892|

The Answer is Height of the *colder* Barometer, now equal in Temperature to the *warmer*: (rejecting all but the four first Decimals.)

397. 6th Step. Place the Barometers *now* of the same Temperature, i. e. *equal* to the warmer, in one View, thus:

1st. the *upper* Barometer, 24.1892

2d. the *lower* Barometer, 28.1328

The 7th Step applied in the second Example.

398. Find the Height, in Feet, in the 2d Column of the 2d Table, corresponding to Inches and Tenths of the *upper* barometric Tube, in the 1st. Column of the same Table, thus: ([Sect. 371.](#))

The Barometer standing at 24.1892; it must be considered where, in the 2d Column of the 2d Table, a Height corresponding to *such* Inches and Tenths can lie: and the Answer is, somewhere *above* 24 Inches .1 Tenth, but not so high as 24 Inches .2 Tenths: 24 Inches .1892 Tenths, being *more* than 24 Inches .1 Tenth, but *less* than 24 Inches .2 Tenths.

First then, look in the 1st Column for Inches 24, .1 Tenth; and the corresponding Height in Feet is 7388.0: but the Height for 24, .2, in the 2d Column, beneath the former Number, is *only* 7280.1.

399. 8th Step. Subtract the latter from the former and the Remainder is 107.9, the same as in the 3d Column: viz. the Height, in Feet and Tenths, corresponding to one Tenth only, namely, the 1st Tenth above Inches 24, .1 Tenth: with the Temperature of 31.24 of Farenheit, for which sole Purpose the 2d Table is calculated.

A new Question *then* arises, viz. what are the Heights in Feet and Tenths, corresponding to the remaining Tenths or Decimals of an Inch above

Inches 24, .1 Tenth,
 viz. .08
 .009
 .0002?

which is to be resolved, by Application of the 3d Table, or Table for *Tenths*, which see, ([Section 373.](#))

400. *9th Step applied in the 2d. Example.*

First for the *upper* Barometer.

Look in the Table for Tenths, in the left vertical Column with 107, (rejecting the .9, as too minute;) and along the horizontal Line at the top, with 8: and find the Answer *gradually*, thus:

1st. With 107, and 8, (as a whole Number,) answering to .08: which, in the Place of Meeting, gives 86 Feet.

2d. With 107, and 9, (as a whole Number,) answering to .009: which, in the Place of Meeting, gives 97.

3d. With 107, and 2, (as a whole Number,) answering to 0002: which, in the Place of Meeting, gives 21.

Place them in View, and add, and bring them back again into Decimals, thus:

With 107 and 8, answering to .08	giving 86. Feet
and 9,	to .009 9.7
and 2,	to .0002 .21
	95.9 1

(Next: with the 9, *if required*; which was before rejected;) but there being no .9 Tenths in the left Vertical, call it 90, and allow for it in each Answer by moving the decimal Point two Places to the left, thus:

with 90, and 8, answering to .08	giving 72 = .72
and 9,	to .009 81 = .081
and 2,	to .0002 18 = .0018

	To .8 00 28
Add the former Sum 95.9	
	Total = 96.7)

Which 95.9 is the *Height* in Feet and Tenths corresponding to .0892 Decimals of an Inch above Inches 24 .1 Tenth: and 24 .1 gave Feet 7388.0 in *Height*; therefore an additional *Height*, of so many Tenths of an Inch of Quicksilver in the Tube of the Barometer, must

give in Feet, a *less* Height of the Barometer elevated above the *imaginary* Level indicated at 32 Inches.

401. 10th. Step. Subtract the *Height* in Feet, corresponding to the *Expansion* on .0892 Tenths of an Inch, (*less* than Inches 24.2 Tenths, of the *upper* barometric Tube,) from the *Height*, in Feet, corresponding to *the Expansion* on Inches 24.1 Tenth of the same barometric Tube, continuing at the Standard Heat, ^[129]

viz. 7388.0

95.9

The Remainder 7292.1

gives the real, viz. the *less* Height of the *upper* Barometer, at 24.1892 with the Standard Temperature.

Repeat the same Process, viz. the 9th. and 10th. Steps, for the *lower* Barometer, thus:

For the lower Barometer in the 2d. Example.

First, Find the Height, in *Feet*, of the lower Barometer, standing at Inches 28.1318 Tenths, in the 2d. Column of the 2d. Table, corresponding to Inches and Tenths of the Quicksilver in the barometric Tube, in the first Column of the same Table, thus:

The lower Barometer standing at 28.1318; it must be considered, where in the 2d. Column of the 2d. Table, a Height corresponding to such Inches and Tenths can lye: and the Answer is, somewhere above 28 Inches, .1 Tenth, but not so high as 28 Inches .2 Tenths: 28.1318 Tenths being more than 28 Inches .1 Tenth, yet less than 28 Inches .2 Tenths.

First, then, look, in the first Column for	28.1,
and the corresponding Height, in Feet, is	3386.6:
but the Height for 28.2, is only	3294.0:

subtracting the less from the greater; the Remainder is 92.6,
the same as in the 3d. Column, viz. the Height, in Feet and Tenths, corresponding to *one Tenth only* above 28.1.

Having therefore found that Feet 92.6 Tenths, are the Height, corresponding to one Tenth only above Inches 28.1 Tenth, of the lower Barometer, with the Temperature of freezing; for which *sole* Purpose, the 2d Table is calculated;—a new Question arises, viz. what are the Heights, in Feet and Tenths, corresponding to the remaining Decimals above 28.1, viz.

.03

.001

.0008; to be resolved by Application of the third Table, or Table for Tenths, which see, (in Section 373.)

Look in the 3d. Table, with 92, (omitting the .6 as too minute) and with

3 answering to .03, which gives 28 = Feet 28.

1	to .001,	9 =	.9
8 to	.0008,	74 =	.74
			29.6 4

Which 29.6 is the *Height* in Feet and Tenths corresponding to .0318 Tenths above Inches 28.1 Tenth: and Inches 28.1 Tenth gave Feet 3386.6 Tenths in Height: therefore an additional Height of so many Tenths or Decimals of an Inch of Quicksilver in the Tube of the Barometer, must give in Feet, a *less* Height of the *lower* Barometer, elevated above the *imaginary* Level indicated by the Quicksilver resting in the Tube at 32 Inches. [\[130\]](#)

402. Therefore subtract the *Height*, in Feet, corresponding to the *Expansion on* .0318 Tenths of an Inch (*less* than Inches 28.2 Tenths of the *lower* barometric Tube,) from the Height, in Feet, corresponding to the *Expansion on* 28.1 Tenth of the same Barometer, viz.

3386.6
29.6

and the Remainder - 3357.0,
gives the *real* Height in Feet of the lower Barometer, at 28.1318 when above the *imaginary* Level, and with the Temperature of *freezing* by the second Table.

403. Then, by taking the Number of Feet and Tenths *above* the imaginary Level, (indicated by the Quicksilver, in both Tubes, resting at 32 Inches) answering to the *Expansion on* Inches and Tenths of the *lower* Tube, from the Number of Feet, &c. by the former Process, answering to that of the *upper* Tube; viz.

<i>upper</i>	7292.1
<i>lower</i>	3357.0

and the remaining Feet 3935.1
Tenth is the *Height*, by which the *Station* of the *upper* Barometer exceeds the *Station* of the *lower*; both being at the Temperature of 31°.24 on Farenheit's Scale. See [Section 371](#).

END OF THE SECOND STAGE

Section 404. 11th Step.
(See the Practice in the 1st Example, Sect. 376.)

<i>Air</i> -Thermom. ABOVE was	56°.
<i>Air</i> -Thermom. BELOW was	63.9

Whole Heat	119.9 (0 adding a Cypher)
------------	---------------------------

Half Heat	59.95
Standard-Heat	31.24
which deduct; and there	_____
remains each Moiety,	28.71
above the Standard-Heat.	

405. 12th Step. (See the Practice in the first Example, Section 377.)

By the fourth Table, find the Expansion of Air, *with* 28.71, (more than the Standard-Temperature) *on* Feet 3935, .1 Tenth, gradually, thus:

406. *First with* 28° *on* Feet 3000 = 204.1^[131]
 900 as 9000 = 612.3
 30 3000 = 204.1
 5 5000 = 340.1
 .1 1000 = 68.0

Note: 1st. The decimal Point in the Answer corresponding to the Place of *Thousands*, in the Question, is to remain, as taken from the Table calculated for thousand Feet, thus: 204.1.

2d. For *Hundreds* in the Question, remove the decimal Point *one Place* in the Answer, thus: 612.3 becomes 61.23:

3d. For *Tens*, *two* Places, thus: 204.1 becomes 2.041:

4th. For *Units*, *three* Places, thus: 340.1 becomes .3401:

5th. And for each *Decimal*, a Place more, by adding Cyphers to the left, if wanted, thus: 68.0 becomes .00680.

407. Place the plain and decimated Answers, in one View, and add the latter together, thus:

204.1 = the same	204.1
612.3 = becomes	61.23
204.1 =	2.041
340.1 =	.3401
68.0 =	.00680

viz. Expansion of Air *with* 267.7|179
 28° *on* 3935.1

408. *Second, with* .71° *on* Feet 3000 = 517.5
 900 as 9000 = 1552.7
 30 3000 = 517.5
 5 5000 = 862.6
 .1 1000 = 172.5

In order to decimate these Answers, it must be observed that the Expansion was not *with* 71 Degrees, but with .71 *Tenths* of a Degree of Heat; therefore the decimal Point corresponding to 3000 Feet in the Question, must in the Answer be removed *two* Places to the left, thus: 517.5

becomes 5.175: for the 100, three Places: for the 10, *four* Places: and so on.

$$\begin{array}{r} 1.5527 \\ .05175 \\ .008626 \\ .0001725 \\ \hline 6.7|882485 \end{array}$$

The Expansion with .71 being found, viz. Feet 6.7 Tenths; add it to the Expansion on 28 Feet already found, viz.

$$\begin{array}{r} 267.7 \\ \hline 274.4 \text{ Answer.} \end{array}$$

Which *Height* in Feet and Tenths, corresponding to the *Expansion* of Air with 28°.71 Tenths of a Degree of Heat more than the Standard 31°.24, being added to the *Height* in Feet and Tenths, corresponding to the *Expansion on Inches* of the Quicksilver in the *upper* Barometer, with the Standard-Heat, already found, viz.

	3935.1
gives the <i>real Height</i> of the <i>Mountain</i> , or <i>upper Station</i> , sought.	274.4
	4209.5

END OF THE THIRD STAGE.

The second Example briefly stated: referring to the Sections.

409. Below: Barometer 28.1318.

Attached Thermometer 61°.8; Air ditto 63.9.

Above: Barom. 24.178.

Attached Thermometer 57°.2; Air ditto 56°. Degrees of Heat, viz. 4°.6 to be added to the <i>colder</i> Barometer at Inches	24.178 Tenths,
by the first Table, viz.	.0112

Parts of an Inch of the Quicksilver in the Barometer, raised by 4°.6 of Heat.

The Sum

24.1892

is the POINT, in Inches and Tenths of an Inch, at which the upper Barometer *now* rests, being of *equal* Heat with the lower.

End of the first Stage.

By the 2d. Table, find the *Height*, in Feet and Tenths, corresponding to the *said* POINT when at the Standard-Heat; gradually, thus: the *Height* corresponding to Feet 24.1 is 7388.0: then with the Difference 107.9, (rejecting the .9).

Find the Height by the 3d. Table corresponding to

$$\left. \begin{array}{r} .08 \quad 86.0 \\ .009 \quad 9.7 \\ .0002 \quad .2 \end{array} \right\} = \text{Feet } 95.9 \text{ Tenths.}$$

Which Height subtract from 7388.0

$$\begin{array}{r} 7388.0 \\ 95.9 \\ \hline \end{array}$$

And there remains, in Feet, 7292.1

The Height corresponding to Inches 24.1892 Tenths of the *upper* Barometer, with the Standard Temperature of 31.24; for which sole Purpose the 2d. Table is calculated.

Repeat the last Process with the *lower* Barometer, resting at 28.1318, gradually, thus:

By the 2d. Table, find the *Height* corresponding to 28.1, which is 3386.61; then with the Difference 92.6 (rejecting the .6) find the corresponding *Height*, by the 3d. Table for the remaining Tenths or Decimals of an Inch, above 28.1, viz.

$$\left. \begin{array}{r} .03 \quad 28.0 \\ .001 \quad .9 \\ .0008 \quad .7 \end{array} \right\} = \text{Feet } 29.6 \text{ Tenths.}$$

Which Height subtract from 3386.6

$$\begin{array}{r} 3386.6 \\ 29.6 \\ \hline \end{array}$$

And there remains, 3357.0

viz. the *Height* in Feet corresponding to Inches 28.1318 Tenths of the lower Barometer, with the Standard Temperature of 31.24, for which sole Purpose the 2d. Table is calculated.

Subtract the *Height* in Feet, corresponding to Inches of Quicksilver in the upper Barometer,

viz. 7292.1 from ditto in *lower* Barometer,

viz. 3357.0 and there remains the *Height* in Feet

———— of the upper Barometer at the Standard-Temperature

viz. 3935.1 of 31.24.

End of the second Stage.

On which Number of Feet, viz. 3935.1, by the
4th Table, find the *Height*, with 28°.71 of Heat:
With 28°. on Feet 3935.1 = 267.7 and
With .71 on the same = 6.7

Sum 274.4 : which
Height, more than the Standard-Heat,
being *added* to 3935.1
the Height, with the Standard, _____
gives the true Height, viz. 4209.5.

End of the third Stage.

CHAPTER LXXVII.

PRACTICE OF THE THIRD EXAMPLE,

REFERRING TO THE SECTIONS. [\[132\]](#)

Section 410. *B*ELOW: Barom. Inches 30, .0168:

Attached Therm. 60°.6; Air-ditto, 60°.2:

Above: Barom. - - Inches 29, .5218:

Attached Therm. 56°.6; Air-ditto, 57°.

Subtract the *colder* ——— from the *warmer*,

and there remains 4° of Heat to be added

to the *colder* Barometer; to give it an *equal* Temperature: which is to be done by *the 1st Table*, thus:

To find the Expansion *with* 4° of Heat, *on* the *colder* Barometer; (which, as before, is the *upper* Barometer) standing at Inches 29, .5218 Tenths.

First, with 4° *on* 29 Inches = .0117:

2d, with 4° *on* .5218 Tenths *above* 29 Inches:

In order to obtain which, begin

with 4° *on* 29 = .0117

then *with* 4° *on* 30 = .0121

Subtract for the Expansion *with* ———

4° *on* 1 Inch *above* 29, and there remains .0004.

Then for the Expansion *with* 4° *on* .1 Tenth of an Inch *above* 29 Inches; add a Cypher and decimal Point,

viz. .00004 :

Then for the Expansion *on* .5128, multiply

the two last Terms, and divide ———

the Product by the first Term

.1: the Answer is .0002 |0872

Add the Expansion *with* 4° *on* .0117

29 Inches, just found,

to the Inches of the *colder* 29.5218

Barometer, viz.

Answer; Inches 29.5337 Tenths of

the *colder* Barometer, are *now* expanded equally with the *warmer*: (rejecting the Decimals as in Section 395.)

Place the Barometers, thus:

Upper Barometer, 29.5337
Lower Barometer, 30.0168

End of the first Stage.

411. By the 2d Table, and in the 2d Column, find the *Height* of each Barometer, with the *Standard-Heat*, in Feet and Tenths, corresponding to the Inches and NEAREST Tenth *above* and *below* the Point required: and

First of the *upper*, at 29.5337:

The Inches and *nearest Tenth* is above Feet.

29.5, corresp. to	2119.7	} Difference between .5 and .6 above 29 Inches.
and below 29.6, cor. to	2031.5	
	88 .2	

412. By the 3d Table, with the *Difference* 88 Feet, find the *Expansion* on the remaining Decimals, above 29.5, viz. on .0337, thus:

on 03 = 26 decimated 26.

003 = 26	2.6
0007 = 62	.62

Feet 29.22

From the *Height* corresponding to 29.5
viz. Feet 2119.7 Tenths,
subtract the 29.22, i. e. Height cor. to .0338
and there _____
remains 2090.4|8, the *Height* cor. to 29.5338
with Expansion of the Standard-Heat.

413. Repeat the 4 last Steps for the *lower* Barometer, at 30.0168.

1st. The Inches and *nearest Tenth* is *above*

30. corresp. to Feet	1681.7	} Difference of .1 above 30 Inches.
and below 30.1 cor.	1595.0	
	86 .7	

2d. Then with 86 Feet, find the *Expansion* on the remaining Decimals, above 30,

viz. .0168, thus: on 01 = 9 9.
 006 = 52 5.2
 0008 = 69 .69

Feet 14.89

414. (3d.) From the *Height* corresponding to
 30 Inches, viz. Feet 1681.7 Tenths,
 subtract the Height 14.89 corresp. to .0168,

and there remains 1666.81, the Height corresp.
 to 30.0168, with *Expansion* of the Standard-Heat.

4th. From the *upper* Height, at 2090.48
 Subtract the *lower* Height, at 1666.81

And there remains the Height 423.67 in Feet
 and Tenths of the upper Barometer, with the
 Standard Temperature.

End of the second Stage.

415. Detached Therm. *above* 57°
 Detached ditto, *below* 60.2

Whole Heat 117.2
 Half Heat 58.6 (0 adding a
 Standard Heat 31.24 Cypher)

which being deducted, leaves 27°.36, viz. Degrees
 of Heat more than the *Standard*, for each Barometer.

416. By the 4th Table, find the Expansion of Air, with 27°.36, on Feet 423.67 Tenths.

First, with 27°, on 423.67, thus:

viz. on 400 as 4000 = 262.4 decimated 26.24

20 as 2000 = 131.2	1.312
3 as 3000 = 196.8	.1968
.6 as 6000 = 393.6	.03936
.07 as 7000 = 459.2	.004592

Expansion = 27.692752

Second, with .36 on the same, thus:

on 400 as 4000 = 349.9 decimated	.3499
20 as 2000 = 174.9	.01749
3 as 3000 = 262.4	.002624
.6 as 6000 = 524.8	.0005248
.07 as 7000 = 612.3	.00006123

Expansion =	.37050003
Add the former	27.692752

Height in Feet 28.06325203

417. Which Height for Expansion of Air, *with more than* the Standard Heat, being ADDED^[133] to the Height, for Expansion of the Barometer, *with* the Standard-Heat, gives the true Height of the upper Barometer, at the given Heat.

For *Expansion* of Air above Standard Heat,

Height in Feet	28.0
----------------	------

For *Expansion* of Barometer,

with Standard: Height in Feet	423.6
-------------------------------	-------

418. True Height of the <i>upper</i> Barometer	451.6
Lower Barometer 1 Foot above the Water	1.0
Height of the Top of the Cross above the Gallery	50.0

Height of the Top of the Cross above the Tyber	502.6
Height of the same, measured the same Day geometrically, was	Feet 502.9

End of the last Stage.

CHAPTER LXXVIII.

PRACTICE OF THE FOURTH EXAMPLE,^[134] FOR MEASURING SMALL HEIGHTS.

Section 419. Attached Therm. *below*, 71°.
Attached Therm. *above*, 70 .5

Subtract, and there remains .5

Tenths of a Degree of Heat to be added to the *colder* Barometer (which in the present Case is the *upper*, but might possibly have been otherwise) by the 1st Table.

First, with 0°.5 on 29 Inches. To obtain which, begin

with 1° on 29 Inches = .002:

with $0^\circ.1$ above 1° , on $29 = .0002$: then

with $0^\circ.5$ above 1° , on 29 = .001.

Prepare it for Addition to the *colder* Barometer.

colder Barometer 29.985

Expansion *with* .5 above 1°, on 29 .001

29.986

Secondly, with .5 Tenths above 1°, on .985 Tenths above 29 Inches. To obtain which, (having already found the Height from Expansion *with* .5 above 1°, on 29 Inches, to be .001;) since the Expansion on .985 Tenths above 29 Inches, is somewhere above 29, yet below 30 Inches; find the Expansion *with* .5 above 1°, on 30 Inches, thus:

first, with 1° , on 30 = .003

2d. *with* $0^\circ.1$ above 1° , on 30 = .0003

3d. *with* $0^{\circ}.5$ above 1° , *on* 30 = .0015

Subtract the Expansion *with .5 Tenths above 1°, on 29 Inches*, from the Expansion *with .5 Tenths above 1°, on 30 Inches*:

viz. $on\ 30 = .0015$

$on\ 29 = .001$

The Answer is .0005,

the Height from Expansion, *with* .5 Tenths above 1°, *on* 1 Inch above 29, i. e. on the 30th Inch: Then, if 1 Inch above 29 gives .0005;

.1 gives	.00005:
and	985
multiplied	00025
as whole	00040
Numbers,	00045
give	.0004 925
add the former Number 29.986	
and, for the three remaining Decimals,	
may be substituted 1 Decimal	
in the fourth Place	1
colder Barometer of equal Heat	29.9865
with the warmer	

420. When the Quicksilver in each Barometer indicates the same Number of Inches, differing but one or two Tenths at the most; (which will frequently be the Case, in levelling flat Countries, or measuring small Heights;—instead of the usual Method, (to find the Height of each Barometer separately, with the Standard-Heat, by the 2d Column of the 2d Table, as in Section 411;)—it will be more convenient,

1st. To subtract the lower Barometer from the upper. Then,

2dly. By the 3d Column of the same Table, find the DIFFERENCE, (viz. of one or two Tenths at the most) below the Inches and nearest Tenth of the lower Barometer.

And lastly, with that DIFFERENCE, find by the 3d Table, the Height at the Standard-Heat, corresponding to the remaining Decimals above the upper Barometer.

421. (1st.) From the lower Barom. viz.	30.082
Subtract the upper	29.9865

Remaining Decimals above the upper	.0955
------------------------------------	-------

2d. Find, by the 2d Table, the Height corresponding to the Inches, and nearest Tenth above and below the Point at which the Quicksilver rests in the lower Barometer.

The Inches and nearest Tenth is	
above 30 Inches, correspond. to Feet	1681.7
and below 30.1, corresponding to	1595.0
	86.7

which is the DIFFERENCE of .1 below 30.1.

Lastly. Find, by the 3d Table, *with* the DIFFERENCE, viz. 86 Feet, *on* the remaining Decimals, for the Height, in Feet, corresponding to the Standard-Heat.

.09	77	=	77. Feet.
.005	43	=	4.3
viz. .0005	43	=	.43

Answer, Height in Feet 81.73
corresponding to .0955 above Inches 29.9865 Tenths of an Inch, of Quicksilver in the upper Barometer thus brought to the Standard-Heat.

422. Prepare for Expansion of Air from Excess above Standard-Heat, on the same Number of Feet:

Detached Thermom. <i>above</i>	76°.
Detached Thermom. <i>below</i>	68.0

Whole Heat	144.0
Half Heat	72.0 (0 adding a Cypher)
Standard-Heat	31.24

which deduct, and there remains	40.76:
---------------------------------	--------

with which, by the 4th Table, find the Expansion of Air on Feet 81.73:

First, with 40°, on 81.73, thus:

on 80.	as 8000	777.6 = 7.776
1.	as 1000	97.2 = .0972
.7	as 7000	680.4 = .06804
.03	as 3000	291.6 = .002916

7.944156

Second, with .76 on 81.73, thus:

on 80.	as 8000	1477.4 = .14774
1.	as 1000	184.6 = .001846
.7	as 7000	1292.7 = .0012927
.03	as 3000	554.0 = .0000554

Expansion .1509341

add the former Expansion 7.944156

Sum of the Expansions, viz. 8.0950901
Height in Feet

from Excess of Heat above Standard, <i>with</i> 40°.76 <i>on</i> 81.73,	
ADDED to the Height at the	81.73
Standard-Heat, in Feet	

gives, in Feet and Tenths, the true 89.8|2.
Height of the Tarpeian Rock

CHAPTER LXXIX.

A CALCULATION TO ASCERTAIN THE HEIGHT OF THE BALLOON ON THE DAY OF ASCENT: ONE BAROMETER AND ONE THERMOMETER ONLY, BEING TAKEN UP INTO THE CAR.

Section 423. **T**HE Question is stated from Section 36: and the Mode of Operation taken from the *Recapitulation* of the second Example, Section 409.

Observation before the Ascent:

Below: Barometer 29.8; attached Thermometer 0; detached Thermometer 65°.

Above: Barometer $23\frac{1}{4} = 23.25$ ^[135] attached Thermom. 0; detached Thermom. 65°.

There being no attached Thermometers; the *first* Table is useless: the Barometer below is therefore supposed to be of the same Temperature as when above; the detached Thermometer remaining at the same Degree, viz. 65°.

State the Barometer, thus: when *below*, at 29.8
when *above*, at 23.25.

End of the first Stage.

424. Find the Height (at the Standard-Heat) corresponding to the Inches and *nearest Tenth* above and below 23.25: i. e. above 23.2, and below 23.3: by the 2d Table.

Now 23.2 corresponds to 8379.7: and the Difference of .1 above, i. e. to 23.3, is in Feet = 112|.1: by the 3d Column of the same Table.

With this Difference, consult the 3d Table: i. e. with 112, (omitting the .1 as too minute) on the remaining Decimals above 23.2, viz. on 05, as on 5, or 5/10; and the Answer is 56 Feet: which Number being subtracted from

8379.7, the Remainder 8323.7, is the Height in Feet of the Barometer in the Car, at the Standard-Heat.

Repeat the last Process for the Barometer on the Ground.

Now 29.8, by the 2d Table, corresponds to 1856.0; and there being no Parts or Decimals more minute than a Tenth, viz. .8, there is no Occasion for the 3d Table.

Subtract the Barometer in the Car, from the same when on the Ground; and, by the 2d Table,

upper Barom. 23.25, corresp. to 8323.7, and the
 lower Barom. 29.8, to 1856.0: the
 Remainder is the Height in Feet ——— of the
 Barometer in the Car viz. 6467.7, with the Standard-Heat.

End of the second Stage.

425. Detached Therm. above, at 65°

Detached Therm. when below, at 65

Whole Heat	130
Half Heat	65. (00 adding Cyphers)
Standard-Heat	31.24

which deduct, and there remains 33.76 Degrees
 more than the Standard-Heat, for each Barometer.

Then for the Expansion of Air, with such Heat more than the Standard, consult the 4th Table: viz. *with 33°.76 on Inches 6467.7*, the Height of the Barometer in the Car with the Standard-Heat, thus:

426. *First, with 33°, on 6467.7*

on 6000 as 6000 = 481.1,	481.1
400 as 4000 = 320.7	32.07
60 as 6000 = 481.1 decimated	4.811
7 as 7000 = 561.3	.5613
.07 as 7000 = 561.3	.05613

$$\text{Expansion} = 518.59843$$

427. *Second, with .76 on 6467.7:*

on, as before,	6000 = 1108. decim.	11.08
	4000 = 738.7	.7387
	6000 = 1108.	.1108
	7000 = 1292.7	.012927
	7000 = 1292.7	.0012927

$$\begin{aligned} \text{Expansion} &= 11.9437197 \\ \text{Add the former } 518.59843 \end{aligned}$$

$$\text{Total Expansion} = 530.5|542197$$

viz. Height *by Expansion* in Feet,
with more than the Standard-Heat,
add to Height in Feet at
the Standard-Heat

$$6467.7$$

428. The true Height, in Feet and
Tenths, of the Barometer in the
Car

$$6998.2$$

Feet in a Yard 3)

$$\text{Yards in a Mile } 1760) \quad 2332.2 \text{ Feet.}$$

$$1760 \quad (1 \text{ Mile.}$$

$$\begin{aligned} \text{Yards in a Quarter of a Mile } 440) \quad & 572 \text{ (1 Qr.} \\ & 440 \end{aligned}$$

$$32 \text{ Yards.}$$

The Height of the Balloon 1 Mile, 1 Quarter, 32 Yards, and 2 Feet.

*End of the last Stage,
and of the Mensuration of Heights.*

N. B. A *thermometric* sliding Rule, for the Expansion of Quicksilver, and of Air, may possibly, from the foregoing Tables, be so contrived and adapted to the Barometer, as to tell the Height by Inspection, while in the Car of the Balloon.

CHAPTER LXXX.

HINTS, ON THE CHEAPEST METHOD OF INFLATING BALLOONS, WITH DESCRIPTIONS OF DIFFERENT MODELS FOR A GASS-STEAM-ENGINE.

Section 429. **T**HE *Expen*ce attending the Inflation of Balloons is a solid Objection to their frequent Use.

A Check is thereby given to every Improvement that might otherwise be expected from a Repetition of Experiments.

It is, in short, the chief Difficulty under which the AIRONAUTIC ART at present labours.

This Difficulty, however, if once overcome, (and of which there is little Doubt) will probably bring those extraordinary Machines, into general Estimation.

What *now* costs fifty Pounds, may *then* be done for five: abating the Expence of the preparatory Engine.

Mons. Lavoisier, by the Application of Steam to Iron Filings enclosed in a Copper Retort, has generated inflammable Air, or light Gass:^[136] and Dr. Priestley, by converting a Gun-Barrel into a Steam-Engine, has produced a Gass 13 Times lighter than common Air;^[137] whereas by the present expensive Method, with Metal and Acid, the Gass for Inflation is seldom more than six Times lighter.

What has hitherto been atchieved on a small Scale, is here meant to be extended.

As no Particulars are made public, or at least, have yet come to the Author's Knowledge, relative to the Construction of such a Gass-Steam-Engine, as may, with Safety and Effect, be applied to the Inflation of Balloons; the following Descriptions of different Models may deserve some Notice:—may possibly excite the Attention of the ingenious; and put them on contriving *easier* Means to obtain the *same* End.

I.

430. Let there be an Iron *Hot-hearth*, one Yard square, and two Inches thick. Let it be *set* on a Common Brick Stove, built as near the Ground as possible, (or even below it) in the open Air. Its Chimney to consist of malleable Iron, flat at the Top, and strong enough to support a Tea-Kettle or Boiler to produce Steam: and extending at least one Yard from the End of the Hearth horizontally, before it turns up. It may rise three or four Yards high, slanting farther from the Hearth: the Form a hollow Cylinder: with a Turn-Cap at the Top, two Feet long, set on at right Angles; for the Management of the Smoke.

Supposing then the Fire-Place to face the West; the Chimney may project Eastward. The North Side is to be appropriated to the Iron-Borings or Turnings; and on the South Side is to be deposited the Dross or Calx.

A Muffle or Mould of malleable Iron is to be screwed and luted over the hot Hearth. The four Sides of the Muffle next the Hearth are to have horizontal Lips or Rims projecting half an Inch: and Screws are to be driven, throu' Holes drilled at proper Distances, into the Hearth. The Sides are to rise upright a Couple of Inches: closing, as they rise, in the Form of a hollow Cylinder, one Foot in Diameter, and perhaps a Yard above the Hearth: which is now converted into a *Gass-Steam-Engine*.

It is proposed to strew over the *Hot-hearth* a thin Layer of Borings, one Tenth of an Inch thick; to which Layer when *red* hot, the boiling Steam is to be applied. The extricated Gass is to be conveyed from the Top of the Cylinder, by Means of an extended Trunk of Tin, and varnished Linen, into a Tub of cold Water kept *continually* flowing over, into which a few Lumps of quick Lime are thrown: and from thence the Gass is to rise into the Balloon.

431. The Iron, whether Filings or Turnings, proper for Inflation, must be *bright*; wholly free from Chips, Bits of Wood, and all heterogeneous Particles: but particularly RUST, and GREASE: *less* than a cubic Inch of the latter, woud spoil a Ton of the brightest, and otherwise the best prepared Materials. ([Section 339.](#))

A Day or two *only*, before a Balloon is inflated; the proper Quantity of bright Iron shoud be heated RED HOT in *Charcoal*, and suffered to go cold.

For Want of this simple Preparation of the Iron, the Gass has proved defective in Point of LEVITY: altho' the Balloon appeared fully inflated.

This Misfortune happened at Birmingham, and other Places.

432. The *Desideratum* is, *quickly* to apply, and *remove* the Borings, keeping the Machine *nearly* Air-tight. For, it is now well known, that the Gass will *explode*, if one-third Part of common Air be introduced: or, if less; it may *unite* with the Gass, and detract from *its* Levity.

433. The following *Particulars* may likewise be considered as an Improvement.

II.

1. To lay a Plate of Iron, Brass, or Copper, over the Hearth; which, if made of *cast* Iron, will be apt to crack, in Contact with the Steam; and will also unite with and concrete the Iron Turnings or Gun-Borings into a solid Mass, that woud be separated with Difficulty.

2. To make the Dross-Pit in the Form of a hollow Wedge, narrow at the Top: screwing and luting it to the South Side of the Hearth. It shoud hold the Dross arising from a Ton of Borings; which will be sufficient for the Inflation of a Balloon, to carry one Person.

3. On the North Side is to be erected a Platform of Brick, a Yard square, *floored* with a Plate of Iron: the inside Surface to be even with the Bottom of the Hearth.

4. The Ton of Borings is to be placed on the *Floor*, and covered with another Muffle, secured and luted to the Side of the Hearth: having a Communication of two Inches high, and one Yard wide, with the Bottom of the Hearth: as the Dross-Pit has.

5. A Brass or Copper Rake is to remain within the two Muffles: to press forward the Borings, spread them over the Hearth; stir them frequently;—by turning the Instrument, scrape them into the Dross-Pit; and apply fresh from the *Deposit*.

6. To perform these manual Operations within the Machine kept Air-tight; it will be necessary, at the exterior End of the Muffle, to fasten a

strong leathern Case, made very wide and pliant, and two Yards long: into which the End of the Rake-Handle is to be inserted.

III.

434. *The Mode of Operation.*

The Borings being spread on the Hearth, and *red* hot; the Steam Pipe is to be opened, and *instantly* shut. The Gass being *suddenly* extricated; the Pipe is to be opened, and shut again as before: the Borings pushed into the Dross-Pit, and a fresh Supply spread. This Process to be renewed, till the Inflation is completed.

If it be thought necessary to prevent the Steam from communicating with the whole Depôt of Borings, and so evolve too much Gass; a little Brass Door with Hinges of the same, might be made to hang from the Top of the Communication between the two Muffles: which Door opening inwards, and hanging vertically, woud by the Pressure of the Gass, stop up the Open: and yet, if made strong, *not* prevent the Operations of the Rake, at proper Times.

III.

435. The Machine woud be less complex, with one large Muffle, somewhat longer North and South than the Hearth; furnished with leathern Case and Rake. Put in the Borings at one End: keep the Steam-Pipe always open; with a *Hand* at the Rake; pushing away the Dross, and pressing forwards fresh Borings.

V.

436. Further: it has since occurred, that a Machine in the Form of a GUN-BARREL, *extended in all its Dimensions*, will probably answer *every* Intention.

And of this Kind are the hollow cylindrical Tubes, of *different* Lengths, and about a Foot in Diameter,^[138] which are *cast*, for the Conveyance of Steam, from the Boiler of a Steam-Engine.

Such a one, (previously lined with a Cylinder of Copper, or malleable Iron, to prevent the Adhesion of the Borings, when reduced to a Calx by the Admission of Steam;) might be placed horizontally over a Stove, (with or without a Chimney) and surrounded with *red* hot Coals.

The Ton of Borings might be deposited at one End of the Tube; and, by Means of the Air-tight flexible leathern Case, be pressed with a Rake, *gradually* into the Fire, and *beyond* it when calcined.

Care must be taken to make the *Apparatus* nearly Air-tight.

The Steam should pass into the Tube, from *below*: and the Gass be conducted towards the Balloon throu' another Iron Cylinder, nearly equal in Diameter and at right Angles with the first; lying also in an horizontal Direction; along the Ground.

The Tubes might be *forged* or *cast*, so as to form but one rectangular Piece.

The further End of the second Tube should communicate with a *third*, made of Tin, and bent downwards about a Foot; thence at right Angles, for six Inches: then to rise up, also at right Angles, the Length of six Inches more.

The Tin Tube is to descend into a Cistern of cold Water, made to flow over continually, by a fresh Supply; and into which, a few Lumps of Quicklime should be thrown.

The Gass, which will press upwards throu' the Water, is to be received into an inverted Funnel, and thence (as in Section 339, Art. 2.) conveyed to the Balloon.

VI.

437. The following Alterations would supersede the Use of the Rake, and *leathern* Cases: the latter of which, by any accidental Crack or Flaw in the Leather, might admit a sufficient Quantity of common Air to produce an Explosion.

The cylindric Form of the Copper, or malleable Iron (to be used as a Lining for the Tube) is to be changed, into that of a half Cylinder, or inverted Muffle: and to be perforated with small Holes.

This Muffle is to be *nearly* filled with a Ton of Iron Borings: (the Ends to be made up, to prevent the Borings from falling out into the Tube;) the Muffle itself is to be supported by a Cradle^[139] of the same Form, made of STRONG Copper Wire,^[140] like the *open Iron-Wire-Fenders*: and the whole is to be thrust into the Tube.

The Length of the Muffle depends on the Quantity of Borings that are intended to be used.

The Ends of the Tube should not be made so strong as the Tube itself: that, if an Explosion happens, they *may* give way first, and prevent a Rupture of the Tube: not that any Danger is to be apprehended, that such an Event will take Place, so long as the Steam-Pipe is attended to, by a proper Person: the above Caution being only given, to prevent a Possibility of Rupture.

Each End should be cast, or forged with a hollow Handle; and should screw into the Tube.

The Length of the Tube should be such, that the Person who attends the Steam-Pipe, should feel no Inconvenience from the Heat of the Fire.

Nine Feet would therefore be a proper Length: the conducting Tube the same.

Within six Inches from each End of the Tube which holds the Borings, a Hole, half an Inch in Diameter should be drilled across the Middle of the Tube, in an horizontal Direction.

Into these, an Iron Axis is to be fitted, (so as to take out *occasionally*) and pass throu' the Tube: each End of the Axis is to project outwards a Couple of Inches, and to be made *square*, for the Socket of a strong Iron Winch or Handle.

Each Axis to be furnished with a strong Chain, of equal Length with the Tube; one End of which Chain is to be riveted, or otherwise fixed, to the Middle of the Axis; and the other, to be fastened *occasionally* to one Extremity of the Cradle and Muffle: the second Axis and Chain in like Manner, to the other Extremity.

The Muffle is to be placed in the Cradle: both are then to be thrust into the Tube, and fastened to the Chain at the farther Axis: in which Position

the Muffle may be filled with Borings, and gradually drawn into the Tube; till the same End has reached the Center of the Fire. The nearer End is then to be hooked by the nearer Chain, already wrapped round the nearer Axis: and the light Iron Caps to be screwed on each End of the Tube.

438. The Boiler for Steam may be fixed on any Part of the Tube near the Fire, and near the opposite Axis; so that one Person may attend both the Steam-Pipe, and Axis. The Steam to be conveyed throu' a small Orifice made in the Bottom of the Tube, between the same Axis and the Fire.

439. As soon as the Materials, above the Center of the Fire, are supposed to be *red* hot, the Steam-Pipe is to be opened for a Moment and SHUT AGAIN. The extricated Gass will be instantly HEARD, rushing throu' the Vessel of *cold* Water; and as instantly SEEN to swell the varnished Linen-Trunk as it passes into the Balloon.

The Steam-Pipe is to be regulated by these infallible Signals: and the Process continued, till that Quantity of Borings, that was in the Center of the Fire, and consequently *red* hot, is supposed to be calcined.

At which Time, the Handles are to be applied to the Axis, and the Cradle and Muffle drawn 5 or 6 Inches forward into the Fire.

When drawn too far; Recourse must be had to the second Axis.

440. If great Expedition is required, two or three Conductors from the same Tube may be used: and, at the Distance of six or seven Feet from the Fire, *Tin-Conductors* may be added; taking Care that they are *made, applied, and continued Air-tight.*

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FOOTNOTES:

[1] Ποιησον δ' Αιθρην, δος δ' Οφθαλμοῖσιν ιδεσθαι·
 Ἐν δε Φαει και ολεσσον, επει νυ τοι ευαδεν ούτως.
 Homer's Iliad, Book 17, Line 646.

[2] Phil. Trans. Vol. LXVII, for 1777, Part II, Page 513, containing Sir G. Shuckburgh's Rules for the Mensuration of Heights with the Barometer. Also Vol. LXVIII, for 1778, Part II, Page 681:

[3] And Page 688.

[4] It were to be wished that the Divisions of the Thermometer by Farenheit were become general throughout Europe, in preference to those by Reaumur yet retained *abroad*; which Divisions of Reaumur are not sufficiently minute to mark the least sensible Change in the Temperature, are subject to frequent Mistakes, and the Inconvenience of adding in the Notation, the Words *above* or *below* the Cypher, zero, or Point of Congelation: besides their being in Conversation not easily compared with those of Farenheit; each Degree of the latter having to that of the former nearly the

Proportion of 18 to 11: since Farenheit from the freezing Point upwards to boiling Water has $212 - 32 = 180^\circ$, and Reaumur to the same Height, 110° Divisions: Mr. Saussure says as 4 to 9; in which there is an evident Oversight: see his curious and philosophic Investigation of the Atmosphere in “Essais sur l’Hygrometrie.” 4to. A Neuchatel, 1783.

Frequent Mention being made of the Thermometer graduated according to Farenheit’s Scale, in different Parts of the following Account; it may not be amiss to shew the corresponding Points according to Reaumur, taken from “Thermometre universel de Comparaison, extrait du Journal de Physique de M. L’Abbé Rozier.”

Farenheit. Reaumur.

54	13 & 4-9ths above the Cypher.
55	14 ditto, nearly.
57	15 2-9ths ditto, nearly.
59	16 4-9ths ditto, nearly.
60	17 1-9th ditto.
65	20 1-9th ditto, nearly.

[5] The Strength of the Rope, or Cable, if its Length does not exceed 10 or 12 Yards, ought to be such as to support a weight, greater than the Weight of the Balloon and it’s Appendages, for the Resistance made by the Grapple against the Balloon acted on by the Wind is immediate: The Rope ought therefore to be made of Indian-Gut, as most elastic, or Silk, as lightest. But if the Rope be half a Mile, or a Mile long; the Resistance is gradual: the Balloon descending for some Minutes; and having an open Space to move in through the Air: the Rope or Cable acting as a Radius, and the Levity of the Balloon and Opposition of the circumambient Air preventing it from falling with any Violence.

The shorter Cable may be used at the Height of 10 Yards; in aid of the longer, to prevent it from rising; or to moor it, by winding the Reel, and hauling down the Balloon close to the Ground.

[6] The Resistance being as the Square of the Velocity; therefore if the Velocity be increased 3 Times, the Resistance will be as $3 \times 3 = 9$, i. e. will be increased 9 Times.

[7]

	Pounds Averdupois.
Weight of the Aironaut	160
Provisions and Articles calculated at	20
Sand-Ballast prepared in Bags	44
Levity for Ascent	10
<i>Sum total,</i>	234

[8] Ancient Warriors among the Arabs, Spaniards, Romans, Gauls, and Germans, being frequently obliged to pass deep Rivers, never undertook a Campaign without them. For the above Anecdote, and many curious Experiments on Air, see Sam. Reyheri, *Dissertatio de Aëre*, tertium edita. Kiliae. 1673.

[9] Equal Time with a Regulator corrected by an Observation.

[10] Being a Dial-Compass, the Dipping of the Needle was frequently checked by the Glass at the Top. A Mariner’s Compass is the best.

[11] The Loop should have been furnished with a SWIVEL: or the Lath or Reel should have been a Kind of Pulley, a Foot in Diameter, and two Inches wide. The Hook of which having also a Swivel might have been held in the Hand: and thus the Twine would have run off in a short Time with the greatest Readiness; the Swivel conforming to the circular Motion of the Balloon. The D
Reel r

[12] SLATE (according to Cronstedt) is the WHETSTONE of fine Particles, composed of Glimmer, Quartz; and, in some Species, of a martial argillaceous Earth, See “Essay on Mineralogy” by

Mendes Da Costa, Sect. 264.

[13]

To know whether the Air is hazy, tho' the Sun continues shining.

Metho
discov
round
bright

The Method taken for that Purpose was by placing the Hand so as to cover his Disk or Body, and then observe the Glory blazing round him; which may, in general, be seen to issue in great Abundance, in Rays of a *golden Colour*: occasioned by a Haziness or Vapour which pervades the *lower* Regions of the Air, most frequently in the hottest and calmest Weather, and in the hottest Climates. The Accumulation of these Vapours, before they are formed into Clouds, are often so great as to intercept the Sun's Rays, or dye them the Colour of Blood: an Appearance frequent in Virginia, and also throughout the torrid Zone.

In the *Campania* of Rome, for Instance, the Italians have a peculiar Name for such Kind of Weather, when the Sun is neither *visible nor invisible*: Il Sole si vede, e' non si vede.

By Degrees the Hand is to be removed so as just to have a Glance of the Sun's Limb. And it frequently happens that the Air is exceedingly hazy; tho' not a Cloud appears above the Horizon.

[14] Esse in IMAGINIBUS quâpropter *Causa* videtur *Cernendi*, neque posse SINE HIS Res ulla videri.

Lucretius de Rerum Natura. L. 4. V. 238.

[15] Notwithstanding what has been said; THIS, to the great and to the sordid Vulgar, woud still appear a solitary, helpless, and deplorable Situation. But such are not captivated with the golden Lines of EPICETUS, (Chap. 13. Line 3. see Mrs. Carter's Translation.)

“ΠΑΝΤΑ ΘΕΩΝ μεστα και ΔΑΙΜΟΝΩΝ.—Βλεπων τον ΗΛΙΟΝ και Σεληνην, και Άστρα, και ΓΗΣ απολαυων και ΘΑΛΑΣΣΗΣ, έρημος εστιν ου μαλλον ή και άβοηθητος.” Nor are they PRACTICALLY influenced by the better Words of a much finer Writer: “The Earth is full,” &c. &c. And “If I take the Wings of the Morning,” &c. &c.

[16] There being, at first, no Clouds, as usual, to occupy the Place of the lowest Stratum.

[17] It has been said that the *apparent* Height from the Balloon to the Ground was 7 Miles, viz. 4 to the Summit of the Clouds, and 3 below: and the *barometric* Height was about a Mile and half, viz. 2332 Yards, *a Calculation of which will be given.*

If then we divide that Height or Distance into 2 such Parts, that the greater shall be to the less as 4 to 3; we obtain the Length of each Part; i. e. the barometric Height from the Balloon to the Summit of the Clouds, and thence to the Earth; which is done thus:

Suppose the whole Distance to be any Line, as A. B. to be divided in C. Then, as 7 is the whole Line, and 4 the greater Part; say, as the whole 7 is to the greater Part 4, so is the whole Distance to a fourth Term proportional, which will be equal to the greater Distance sought:

Whole Distance in Yards.	Greater Distance in Yards.
-----------------------------	----------------------------

Thus 7, : 4 ::	2332 : 133247 Ans.
----------------	--------------------

	4
--	---

	7)9328

2332 the whole. 133247

133247 being the greater Distance found; take the greater from the whole, and then will remain the lesser Distance wanted, viz. 99937: the 133247 = the greater Distance, and 99937 = the lesser Distance: and adding the Fractions $\frac{4}{7} \frac{3}{7} = 1$ to the 999; we have 1332 Yards for the greater Distance, or Height of the Balloon above the Summit of the superior Clouds: and 1000 Yards for the less Distance, or Height from the Earth to the Summit of the superior Clouds.

Note. *The Line A. B. here selected is the famous Measure of (half) a MATHEMATICAL Rhinland and Roman FOOT, according to Snellius. (See Geographia Generalis of Varenius, published by Newton. Lib. 1. Cap. 2. De variis Mensuris.)*

[18]

P R O B L E M.

To find the circular Boundary of the *celestial* Prospect over the Tops of the superior Clouds, from the Balloon at the Height of near a Mile and half above the Surface of the Earth, viz. 2332 Yards. The Height from the Earth to the upper Surface or Floor of Clouds being 1000 Yards; and the Height above the Floor to the Balloon being 1332 Yards.

On the Curvature of the Earth and Clouds, and Elevation of the Eye above their circular Horizon.

Rule. To the Earth's Diameter, equal to 7940 geographical Miles, add *the Height* of the Eye above its Surface: multiply the Sum by that Height: then the square Root of the Product gives the Distance at which an Object on the Surface of the Earth can be seen by an Eye so elevated. Note the Diameter of the Earth, in Feet, is 41798117, according to Newton. (See Practical Navigator, by J. Moore, 7th Ed. Page 251.)

F I R S T.

Double 1000 Yards, the Height from the Earth to the Clouds for an Addition to the Diameter of the Earth, whose Surface is now considered, as extended to the concentric Floor of Cloud.

$$\begin{array}{r} 1000 \\ 1000 \\ \hline 2000 \end{array}$$

S E C O N D.

13932702(1/3) Diameter of the Earth in Yards.
2000 Addition to the Diameter.

13934702	Sum, to which add
1332	the Height of the Eye or of the
13936034	Balloon above the Floor of Cloud.
1332	Sum, which multiply into
27872068	the Height of the Eye above the
41808102	Floor.
41808102	
13936034	

Extract the Square Root	 18562797288 1 — 23) 85 69 — 266) 1662 1596 — 2722) 6679 5444 — 27244) 123572 108976 —	1760) Yards in a Mile. (136245 (77 Miles. 12320 — 13045 12320 — Yards 440) 725 (1 Quarter of a Mile. 440 — 285 Yards. Ans. 77 Miles, 1 Qu. 285 Yards.
------------------------------------	--	---

$$\begin{array}{r}
 272485) 1459688 \\
 \underline{1362425} \\
 97263
 \end{array}$$

P R O B L E M.

Circuli
of the
Prospect
Balloon
Day.

To find the circular Boundary of the *terrestrial* Prospect, on a clear Day, from the Balloon at the Height of near a Mile and half, viz. 2332 Yards: the Earth's Diameter being

equal to	$ \begin{array}{r} 1393270523 \\ \underline{\text{add } 2332} \\ 13935037 \\ \underline{2332} \\ 27870074 \\ 41805111 \\ 41805111 \\ \underline{27870074} \\ \hline . \quad . \quad . \quad . \quad . \\ 32496506284 \\ 1 \end{array} $	Yards, the Height of the Eye or Balloon. the Sum, multiply into the Height of the Eye, &c. 1760) Yards in a Mile. (180267(102, 1760 say 102 1/2 Miles, Ans.
Extract the square Root	$ \begin{array}{r} 28)224 \\ \underline{224} \\ 3602)9650 \\ \underline{7204} \\ 36046)244662 \\ \underline{216276} \\ 360527)2838684 \\ \underline{2523689} \\ \hline 314995 \end{array} $	4267 3520 747 Yards, Remainder. Remainder.

[19] See his "Minute Philosopher."

[20] Ullòà in his voyage to South-America relates, that in passing over the DESERTS, Írides are frequently seen by Travellers round *their own Heads* as the Center of the *Iris*; and visible only to themselves. But what Analogy the *Balloon Iris* bears to them, Time and future Experiments may discover. See his "Voyage to South America, Vol. 1. Pa. 442."

[21]

As Sound travels	1142	Feet in a
Second, it must have moved in	30	Seconds
	$ \begin{array}{r} 3)34260 \\ \hline 1760)11420 \\ \hline 10560 \end{array} $	= Feet (6 Miles
Yards in a Quarter of a Mile	440)860	(1 Quarter
	440	
Answer 6 Miles, 1 Quarter, and	420	Yards.

[22] Equal to 2085 Yards; or 1 Mile, 325 Yards.

[23] Long's Astronomy. Pages 227, 229.

[24] Also called the *Horsham Stone*, from a Place so named, in Surrey, where great Quantities are found.

[25]

P R O B L E M.

To find the *Length* of the *Shadow* from a Person of *middle* Stature, (five Feet and a half High) viz. at XII o'Clock, on the 8th Day of September, 1785, at Chester, whose North Latitude is $53^{\circ} 12'$; (and $3^{\circ} 11'$ West Longitude from London.)

F I R S T,

To find the Sun's Altitude at XII.

From $90^{\circ}.00''$ Subtract

The Latitude 53.12

The Remain. 36.48 is the Complement of Latitude,
to which add (from the Tables)

Sun's N. Decl. 5.29

The Remain. 42.17 is the Sun's Altitude (viz. at XII.)

S E C O N D,

For the Shadow say,

As the Sine of the Sun's Altitude $42^{\circ} 17'$

To the Person's Height, viz. 66 Inches,

So is the Co-Sine of the Sun's Altitude,

To the Length of the Shadow.

For the Sine of the Sun's Altitude $42^{\circ} 17'$ in the Table of artificial Sines, is the Logarithm 9.82788, which, subtracted from the arithmetic Complement, viz. 9.99999 (supposing the last Figure a 10) becomes,

.17212

Then for the Person's Height, viz. 66 Inches: in the Table of Logarithms is the corresponding Number,

1.81254

And for the Co-Sine (had by subtracting the Altitude 42.17 from 90.00) viz. 47.43 : among the artificial Sines is the Logarithm,

9.86913

The above Sums added, are

11.86079

which logarithmic Number (deducting the *Initial* 1 as useless) viz. 1.86079, in the Table of Logarithms, corresponds to 72.57, equal to 72 Inches, for the Length of the Shadow at XII.

Reducing then the Numbers 66 and 72, to the lowest Denomination, thus $6)66/72 = 11/12$ the Proportion which the *Length* of the *Shadow* bears to the *Height* of the *Object* is thereby obtained: that is

[26] If the *Length* of the Shadow be divided into 12 Parts, the Height of the Object would be 11 of those Parts.
See Moore's Practical Navigator.

P R O B L E M.

An EASY Way to find the Proportion which the *Length* of the Shadow bears to the *Height* of an Object is, AT ANY TIME WHEN THE SUN SHINES, to fix a Plummet Line and FRAME *upright* in the Ground; measure the *Length* of its *Shadow*, and compare it with the *Height* of the FRAME.

[27] Equal to 3 Quarters of a Mile and 121 Yards.

[28] i. e. When the Barometer *below* is at 30 Inches, and Thermometer *below* at 60° viz. about 1000 Yards high in *fine* Weather, and 500 in *changeable*.

[29] Being 1083 Yards, i. e. half a Mile, and 203 Yards.

[30] It was High Water at Chester and Frodsham-Bridge, at 38 Minutes past I.

[31] Articles parted with, to check the *first* Descent at Bellair, near Frodsham: and to ascend the *second* Time.

To check the <i>first</i> Descent.	Pounds. Ounces.	
Ballast, at twice:	24	0
To clear Trees and Hedges, and <i>re-ascend</i> :		
Barometer and Frame,	0	121/2
Basket with Tunning Dish and Bottles (except the Flask with Brandy and Water)	4	10
Half Mile of Twine on the Reel	1	0
Speaking Trumpet	0	81/2
Woollen Gloves	0	1
	31	0
	24	0
Remains for Re-ascent	7	0

[32] The Sun's Azimuth from the North Point *Westward*, being 118.26': its Supplement to 180° is 61°.34' South westerly: i. e. South West by West, half West *nearly*.

[33] The *Length* of the Shadows being more than *double* the Height of the OBJECTS: see [34].

[34] To find the Length of the Shadow at half past III.
(See Section 84, [Note \[25\]](#).)

Given $\left\{ \begin{array}{l} \text{Lat. of Chester, } 53^{\circ} 12' \\ \text{Sun's Dec. } 5 \ 29 \\ \text{Hour III, 30M. } 52 \ 30 \end{array} \right\}$ To find Sun's Alt.

This is the Case of an oblique spheric Triangle, wherein are two Sides and one Angle between them given, to find the Sun's Azimuth, and the Sun's Co-Alt.

Side 84. 31	}	Sum of Sides 121. 19
Side 36. 48		Diff. of Sides 47. 43
(31/2 Hour) Angle contained 52. 30		
Half ditto 26. 15	}	63. 45
Half Sum of Sides 60. 39		Co. 29. 22
Half Difference ditto 23. 51		66. 9

THE FIRST PREPARATIVE PROPORTION.

As Sine of 1/2 Sum of Sides	60. 39	0.05966 Co-Ar.
To Sine of 1/2 Difference of Sides	23. 51	9.60675
So Co-Tangent 1/2 contained Angle	63. 45	10.30703

To T. of 1/2 Diff. of the other two Angles 43. 15 9.97344

SECOND PREPARATIVE PROPORTION.

As Co-Sine 1/2 Sum of Sides	29. 21	0.30968 Co-Ar.
To Co-Sine 1/2 Diff.	66. 9	9.96123

So Co-Tangent 1/2 contained Angle 63. 45 10.30703

To T. 1/2 Sum of other Angles	75. 11	10.57794
Half Diff. before found	43. 15	

Sum, is greater Angle	118. 26 = Sun's Azim.
Diff. is lesser Angle	31. 56 = S's right Asc.

Then by first Axiom in Trigonometry, to know the Sun's Altitude say,

As Sine Sun's right Asc.	31. 56	0.27659
To Sine Co-Lat.	36. 48	9.77744
So Sine of the contained Angle	52. 30	9.89947

To Co-Sine of the Sun's Alt.	63. 57	9.95350
from 90.		

Sun's Alt.	26. 3
------------	-------

Having Sun's Alt. to find the Shadow,

As Sine Sun's Alt.	26. 3	0.35738 Co-Ar.
To Person's Height,	66 Inches,	1.81954
So Co-Sine of the Sun's Alt.	63. 57	9.95350

To Length of Shadow,	135 Inches,	2.13042
----------------------	-------------	---------

Then $6(66/135 = 11/22 - \frac{1}{2} - \frac{3}{6}$ or $\frac{1}{2}$, i. e. as 22 to 45: supposing the Length of the Shadow divided into 45 Parts; the *Height* of the Object would be 22 of those Parts; or not quite *half* the *Length* of the Shadow, at half past III.

[35] See "Priestley on Electricity."

[36] Εὔροια.

[37] The Breath is said to become *visible* at Sea or Land at any Temperature of the Thermometer not *exceeding* 60°: tho', in Latitude 41°, and Westward of the Azores Islands, being in Sight of the Peak of ST. GEORGE, (which probably equals, if not exceeds, the Height of Teneriffe) the Observer has seen his *own Breath*, and *that* of the Sailors on Deck, when the Thermometer in the *Shade* was at 61: the Air (in January) being *then remarkably* damp.

An Ac
Breath
visible
when
Therm
was at

[38] This Assertion may seem to contradict what was said in Section 44: When—"every Thing, that could be seen at all, was seen DISTINCT:" but it only proves that the Balloon had attained a greater Altitude during the Re-ascent, and that the SHADOWS were *much lengthened*, as the Evening advanced.

[39] Angelica Kauffman.

[40] It consists of a Frame, made by placing two strong Posts, moveable at Pleasure, each nine or ten Feet high, upright in the Ground, at the Distance of two Yards: the Posts being well secured by broad Pedestals, to keep them firm: a strong horizontal Iron Axis goes throu' the Top of the Posts; and throu' the Centers of four Arms or Levers at their Junction.

Between the four corresponding Ends of each two Arms, (which Arms are also strengthened by Beams from one to the other), are fixed four Seats or Boxes, well secured, each holding three or four Persons, and moving on Iron Pivots, near the Top of the Boxes, so as always to preserve the *vertical* Equilibrium.

[41] Why not recommend the Use of that Machine to Invalids? who would find Refreshment in the OPEN Air: as its Rotation communicates a gentle Motion to the System, [42] without the least Fatigue; rather encreasing the *Animal* Spirits. Recor
Invalid

[42] Particularly the Stomach and Diaphragm. See “Berdoe’s Enquiry.”

[43] *Talis Aër qualis Spiritus*. See “*Health’s Improvement*,” by Dr. Moffet, Chapter 3, Of Air, Page 79.

[44] Or Solid of *least* RESISTANCE, see Chambers’s Dictionary, with the Supplement.

[45] It will be found, that, on comparing the two Calculations in Section 52, Note [Note \[18\]](#), corrected; the circular *Distance* from the Eye, above the Clouds, was 102 Miles, 1 Quarter, 320 Yards: while *that* above the Earth, seen from the same elevated Situation, (supposing the Day to have been CLEAR for such a View,) was 102 Miles, 1 Quarter, 307 Yards: whose Difference is only 13 Yards: that is, the *Distance* above the Clouds to the *nebulous* Horizon, was *rather* more extensive, than *that* above the Earth to the *terrestrial* Horizon.

It may not, to some Readers, be deemed either unentertaining, or foreign to the Subject; if the Distance of the *Prospect* from the Balloon at its greatest *barometric* Altitude, viz. 2332 Yards, or a Mile and Half within 33 Yards, be compared with the Distance which may be seen from the *Summit* of the principal Mountains in different Parts of the Globe.

1. Cotopàzy, a Mountain in the Province of Quito, in America, and under the equinoctial Line, is *said* by Ullòa (Vol. 1. Page 422) to be 3126 Toizes or Fathom, i. e. 6252 Yards, or 3 Miles and a Half and 92 Yards in Height.

2. White Mountain, called by the French Mount Blanc, near Geneva, is considered by Sir G. Shuckburgh (Phil. Trans. Vol. 67, Part 2d, Page 598, for the Year 1777) as the highest Land in Europe, Asia, or Africa (known to Europeans) and calculated by him at 5220 Yards, or 3 Miles within 60 Yards above the Level of the Mediterranean Sea.

Mons. Bourit just returned from his last Tour, see his “Description de Glacieres” in 1773, makes the White Mountain but 5102 Yards in Height, (which is 30 Yards lower than Teneriffe) including the 410 Yards for the Level of the Lake of Geneva above the Mediterranean.

3. The Peak of Teneriffe in the Canary Islands, which, *in approaching towards it*, Authors agree, may be seen at the Distance of 120 Miles at Sea, if the Weather is clear; (Modern History, Vol. 14th, Page 451;) and, *in returning from it*, is discoverable at the Distance of 150 Miles, according to Glas’s History of the Canaries (Page 234);—has been estimated by Dr. Heberden in Madeira (Guide to the Lakes, Page 187) at 5132 Yards, or 3 Miles within 148 Yards.

Glas remarks farther, that in sailing from Teneriffe, the Peak, at the Distance of 150 Miles is very little darker than the *azure* Sky, on Account of the great Quantity of Vapour intercepted between the Eye and the Mountain: and *not* because it ceased to be an Object too small for the Sight; or was in Fact, below the Horizon, and only raised by Refraction of the Vapour.

With Respect to the Peak of St. George, situated in the Island called *Pico*, one of the Azòres; the Writer of this Account asserts, from the Mouth of an able and experienced Officer in his em some Weeks off those Islands; that the latter has frequently observed the Peak, at the Distance of 120 Miles, and could then distinguish a *third* Part of its Height *down* the Mountain. Section 126, [Note \[37\]](#)), see also [\[46\]](#) below.

4. Etna is 3877 Yards above the Mediterranean: (according to Brydone’s Tour throu’ Sicily and Malta, Vol. 1. Page 211) or 2 Miles and 357 Yards.

5. Blue Ridge, the highest Mountain in the Island of Jamaica, is, according to Dr. Clark, who measured it in November last, 3080 Yards, or 1 Mile and three Quarters, above the Level of the Ocean.

The DISTANCE to be SEEN is considered as terminating the Radius of a Circle, whose Center is the EYE of the Observer, on *each* Mountain.

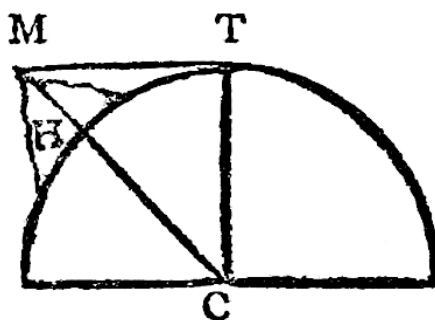
Height of the Mountains.

DISTANCE *to be SEEN from them* *in Miles.*

Cotopàzy 3 Miles and a Half and 92 Yards, (for the Process, see Section 52, Note [18])	167 1/2 and 405 Yards.
White Mountain 3 Miles within 60 Yards.	153 1/4 and 13 Yards.
Peak of Teneriffe 3 Miles within 148 Yards.	152 within 72 Yards.
Mount Etna 2 Miles and 357 Yards.	132 and 127 Yards.
Blue Ridge 1 Mile and 3 Quarters.	117 3/4 and 30 Yards.
Balloon 1 Mile and half within 33 Yards.	102 1/4 and 307

As it is well known that Objects of the *greatest* Magnitude appear but as BLUE AIR at even a *less* Distance than 100 Miles; to which add the Difficulty of Journies, and Ascent to the Summit of these astonishing Mounds of Earth; and all this for the Sake, not of a complete DOWN PROSPECT, subject to a *perpetual Variety*, but merely an *imperfect Side-View*: the PLEASURE and EASE of attaining still *more* stupendous Heights at *any* Place and Time, by Means of the BALLOON, are strikingly in Favor of that Invention. And, notwithstanding the confessed Merit of Dr. Black's Project with the *Farciminàlis* of a Calf, and Mr. Cavallo's Soap Bubbles with inflammable Air; (see his History of Aerostation, Page 34;) if the Emperor had been alive who offered a Reward for the Invention of a *NEW PLEASURE*; the *first* Prize had been due to the Brothers Montgolfier, and a *second* to the Brothers Roberts.

[\[46\]](#) As therefore it may be supposed that the Peak of St. George, in *receding* from it, woud *vanish* at the Distance of 150 Miles; its Height may *easily* be ascertained geometrically thus:



See the Figure annexed.

Let M be the Summit of the Mountain: and let the Line M T drawn to the Circumference of the Circle at T, be the *evanescent* Distance of the Mountain in the Horizon, viz. 150 Miles.

Join T C, viz. a Line drawn from the *Tangent* to the Center of the Circle, which Line will therefore represent the Semidiameter of the *Earth*, viz. 3958 Miles, according to Newton.

Draw a Line from C to M, which will pass throu' some Point of the Circumference as H, the Base of the

Mountain.

Then, in the Triangle M T C, as the Angle at T is a right Angle (Euclid's Elements, Book 3, Proposition 18;) and the Sides M T, and T C, containing the right Angle, are *known*; the *third* Side C M is readily found: (being a Corollary to the 47th Prop. 1st Book Euclid:) viz. having the two Sides of a right Angle Triangle given to find the *third*. Therefore

R U L E.

Multiply the Sides containing the right Angle, each into itself: viz. 150 and 3958: add the Products into one Sum: from which extract the *square Root*; *equal* to the Length in Miles, of the *third* Side required.

From the *third* Side, subtract that Part, viz. C H, which is equal to the Semidiameter T C already found: and the Remainder H M is the *Height of the Mountain*.

Thus:	150 Miles.	3958 Miles in the Semidiameter of
	150	3958 the Earth
	7500	31664
	15	19790

22500	35622	
	11874	
Square of the greatest visible Distance.	15665764	Square of the Semidiameter add 22500 of the Earth.
Extract the sq. Root,	15688264	(3960.84 Square Root.
	9	3958 subtract.
	69) 668	
	621	
	786) 478.2	Rem. 2.84 Answer in Miles.
	471 6	
	79208) 6664.00	continued to 2 Decimals.
	6336 64	
	792164) 32736.00	ditto.
	31686 56	
	104944	

To find the .84 Part of a Mile; multiply

Decimal Parts of a Mile to be reduced	1760 Yards in a Mile, .84 into Yards.
	7040
	14080
	1760) 1478.40 (0
Subtract	1478
	282

Answer: the Height of the Mountain is 2 Miles 282 Yards.

[47] Rays flowing from the Sun seem to be RED ORANGE or YELLOW, according to the Quantity of Vapours floating in the Atmosphere, which absorbs the most refrangible ones: and the fewer the Vapours the more does the Sun's Light approach to a perfect and intense WHITE, according to the Doctrine of Newton: which seems to receive Confirmation from the Purity of the Solar Light, when seen *above Clouds and Vapours*, in the Balloon: where the Sun shines not so much with a *golden* as with a *sparkling* SILVER Light.

[48] SOUNDS IMMEDIATELY UNDER the Balloon, seemed, as if originated *near* the Ear, and *louder* than they would have been heard, at the Distance of some Yards *only*, when on a Level with themselves: augmenting rather than decreasing, during the *Ascent* of the Balloon, till it arrived to a Height indicated by the Barometer at 27 Inches. Presently afterwards, the Balloon still rising; the Sounds *died away*: much sooner indeed than was expected.

The like was observed in *descending* from a State of perfect Tranquillity and Silence: *Sounds* from *below*, when about the same Height, *suddenly rushing* on the Ear.

It must be considered that by *this* Time, the SHADOWS were much encreased; tho' at half past II, they were *more* than double in Length to the Height of each Object.

The Trees woud therefore spread a SHADE *across* the Road.

The TOPS of the *Houses* likewise, being Part of them in the Shade; and either *thatched* with Straw, or covered with Slates of a *dusky* Hue; woud prevent their *throwing off* any *striking* Colour.

Possibly the *Encrease* of Shade *alone*, might give the Face of the Country *below*, a *dark-green* Cast.

It is certain that the Height of the Balloon must have been very great, to prevent the Sight of public and *Turnpike-Roads*, *above* which it *frequently* passed, and which had been PLAINLY seen *before* the *Re-ascent*.

For suppose the Road but 5 Yards wide, which is less than the Truth; if it be allowed that an Object may be distinguished by a *sharp-sighted* Person, when its *Distance* from the Eye does not *exceed* 5156 Times the Diameter of the Object; i. e. when the Object does not subtend a *less* Angle at the Eye than 30 *Seconds* of a Circle, (Smith's Optics, Article 97) which is the *smallest* visible Point, and equal to the 8000th Part of an Inch on the *Retina*;—by multiplying 5 Yards, viz. the Diameter of the public Road, into 5156 (or, in round Numbers, into 5000) Times its Distance from the Eye in the Balloon; the Product is 25000 Yards: which Product being divided by 1760, the Number of Yards in a Mile, amounts to 14 Miles, and 360 Yards.

Supposing farther, that a *common* Eye can *only* see an Object at *half* that Distance; the Height would *then* be 7 Miles.

The *Improbability*, therefore, (on Account of the *Warmth* of the Air at that Height, viz. 60°;) of having *soared* to so great an Altitude, seems to point out, that the SHADOWS must have contributed a *principal* Share, in preventing a Sight of the public and *Turnpike* Roads.

[49] The MAGNITUDE of an Object *decreases*, as the SQUARES of its Distance from the Eye *increase*.

At whatever Distance, for Example, the Eye can see any Object clearly; as at the Distance of a Foot, or a Yard, if the Object be removed to *twice* that Distance; it will appear 4 Times smaller than it did before: 2 multiplied into 2, equals 4, which is the Square of 2: in the same Manner, if the Object be removed to thrice the Distance from the Eye, it will appear 9 Times as small, as at the first Distance: for 3 into 3 gives 9, the Square of 3: and so of any farther Distance.

[50] See “Berkeley’s New Theory of Vision, Section 67.”

[51] Dr. Smith having Recourse to *intervening* Objects; the Writer cannot assent to the Validity of his Argument, illustrated by a well-known Figure, to solve the Appearance of the *horizontal Moon*. See “Priestley’s History of Light and Colours, Page 712.”

[52] Phil. Trans. for 1785, Part 1, Page 287.

[53] Cavallo’s Treatise on Air, Page 576. Vitriolic Acid Air, Alkaline Air, and other elastic Fluids, are instantly ABSORBED by *Water*; (Page 673.) Inflammable Air, and fixed Air, are likewise ABSORBED by WATER. (Page 434).

[54] Nam fit, ut interdum tanquam demissâ Columnâ In Mare de Cœlo descendat.—Lucr. L. 6. V. 425.

Una Eurus Notusque ruunt, creberque Procellis Africus. Also
Omnia Ventorum concurrere Prælia vidi. VIRGIL.

[55] Franklin’s Account of Whirlwinds and Waterspouts, in his Miscellaneous Tracts. Lowthorp’s Abridgement of Phil. Trans. Vol. 2. Page 103. Varenus Geogr. Gen. C. 21, Pag. 265. A clear Account of the Effects of a DEPRESSION is to be met with in “the History of Jamaica, in 3 vols. vol. 3. Page 800, on *Trade and Land Winds*.”

[56] Mons. Maupertius has found, that the extreme Cold at Tornea, in the northern Regions beyond the Artic Circle, came directly from *above*: see “La Figure de la Terre,” Page 59. Il semble que le vent souffle—de tous Côtés à la Foix; et il lance la *Neige* avec une telle Impetuosité, qu’en un Moment tous les Chemins font perdus. “It seems that the Wind blows from all Points of the Compass at once,” &c.

[57] The Doctrine of smokey Chimnies distinctly treated of under the Article SMOKE, in the Encyclopædia Britannica, may receive some Improvement, from Circumstances which ascertain the sudden Descent, Elevation, and quick Depression of *Columns* or rather *Torrents* of Air, viz. by widening the Tubes, and covering their Tops.

[58] It is thought more *candid*, and will to *many* be more *satisfactory*; to make occasional References to different Authors who have treated distinctly on a Subject, and leave the Reader to draw his own Conclusions by applying to their *express* Words;—than, either to insert abundant Quotations; or *weave* their Thoughts into the *Texture* of the Work: which must encrease its Bulk, without producing any Thing either new or instructive.

[59] Once, particularly, in the Month of January, at Lausanne: Farenheit's Thermometer at 7 only: the Country covered with Snow; and a North Wind beating *VIOLENTLY* on the Lake, which continued liquid without Ice: owing, perhaps, in *Part*, to subterranean Heat, and Exhalations.

[60] The Depression and Reverberation of the Wind near Rivers, and its Descent from Mountains, *a Point to be discussed*, may furnish a Hint and Reason, why Rain falls more in one Place, than in another not far distant: and why in the same Place it falls in different Quantities, at different Heights, irregularly.

[61] Cavallo's Treatise on Air, Page 446.—

[62] 442.—

[63] 441.—

[64] 442.

[65] It is *light* in Consequence of its *Warmth*, when compared with the *cooler condensed Air* above it.

[66] In the same Manner that Curls and Streams of Air *descended* into the Bason over the rising *Steam*, and interrupted the Regularity of its Elevation; in the *larger* Towns, during Winter (*the Weather being moderate*) the Pressure of Air on all Sides, from without, produces a constant Breeze towards the Center of the Town: as may be discovered, not only by the Smoke in its Deviation from the Perpendicular, as it issues from the Chimneys; but by all who are inclined to make the Trial; for, on leaving the Town, they will *meet* the Breeze.

In calm Weather, during Summer, the contrary Event happens: but more particularly in *hot* Climates. For the Country being hotter than the Town; a *Depression* of the Atmosphere takes Place, and scatters the Smoke on all Sides round the Town.

The Cities in Italy, and other hot Climates, on Account of the Buildings, and *desirable* Narrowness of the Streets, form *one* contiguous *Shelter*, *Arbor*, or grand *Parasol*: For which Reason, the Nobility leave the Country, and reside in the Towns during Summer: there finding a Coolness and Refreshment unknown on the *scorching* Plains.

A Reception and Dispersion of Air takes Place; as will presently be mentioned.

The same ocular Proof and Process in the Evaporation of Steam, accounts at once, for a curious Phenomenon constantly observable on all Waters; viz. *a narrow SMOOTH irregular Surface of considerable Length, nearly in the Direction of the Wind, yet unaffected by it*: all which is probably nothing more than *rising* Volumes of *elastic invisible* Steam; *resisting* the *two* nearest *descending* Waves of AIR; and preventing them from approaching the *Surface* of Water, over which the Steam is compressed; and *there* producing a *temporary* CALM.

[67] Phil. Trans. for 1777, Page 470. Thibet in Lat. 31, *cold* with Snow and Frost.

See Ullòà's Voyage to South-America, Book 6, Chapter 7; where he describes the snowy Mountains, under the Equator.

As the Weather, near the Equinoctial, is more regular, its Changes closely following those of the Moon; and also the Winds and Hurricanes more violent; the Truth of the foregoing Theory will receive the strongest Confirmation by tracing the Effects of *DEPRESSING TORRENTS OF AIR*, in the Island of Jamaica, extracted from the Author already mentioned.

"The cool Vapour *rushes* from the Mountains towards the hot dry Air, which hovers over the Savannas or Vallies.

The Rain falls heaviest in the Mountains. Vol. 3, Page 600.

The *Land-Wind* after Rain, proceeds from that Quarter whence the Rain has fallen *heaviest*; and seems to *rush* from above.

In Spain and North-America, the Wind *rushes down*. Page 601.

When the *Land* is *most* heated, the Sea-Breeze blows almost *all* Night. Page 602.

The Barometer subsides from 1 Inch to 1 1/2 *at* the full Moon, or just *after* it.

Wind blows from the Mountains all round the Island: and still a Sea-Breeze over the Mountains: to the Low-Lands, none, 604.

(In Jamaica likewise the Wind blows off the Island *every way* at once, so that no Ship can any where come in by Night, or go out but early in the Morning, before the Sea-Breeze sets in. See Abr. Phil. Tr. Vol. 3, P. 548.)

Mountain Air rushes down in a continual Current to every Part of the Coast, the Stream descending incessantly throu' the Night: while heavy cold Air descends to the Mountain Tops, 604.

With a West Wind below there is an *East Scud* above, 605.

Mountains CLOUDY, *low Lands* SUNNY. 606.

In ALL the River-Courses of Jamaica, there is a sensible Current of Air. Rain never comes without some Wind: and the Showers almost invariably follow the very Meanders of the larger Rivers, 608.

Rain always cools: the Thermometer falling, after a Shower, from 6 to 8 Degrees, 610.

(And Iron rusts least in rainy Weather: [the Air being then DRIEST,] descending from the *upper* Regions. Abr. Ph. Tr. V. 3, P. 546.)"

It is said also that "in Jamaica the Clouds gather, and *shape* according to the Mountains: so that *old Seamen* will tell you each *Island* towards Evening, by the *Shape* of the Cloud over it."

The Sea-Breeze, being counterpoised by *Descent* of the *etherial Air*, produces a CALM.

The same Author likewise says, that "the Clouds begin to gather about 2 or 3 o'Clock in the Afternoon *at the Mountains*, and do not *embody* first in the Air, and after settle there, but *settle* first and *embody* there: the rest of the Sky being clear till *Sun-set*. So that they do not pass *near* the Earth in a *Body*, and only *stop* where they meet with Parts of the Earth elevated *above the rest*; but PRECIPITATE *from a very great Height*, and in Particles of an *exceeding rarified Nature*; so as not to *obscure the Air or Sky at all*: that great Variety of beautiful Colours in the Canopy of Heaven being raised to a much greater Distance [he means Height] in Jamaica than it is here." Abr. Ph. Tr. V. 3, P. 557.

(Prognostics of Weather, at certain Periods of the Moon, are mentioned by Captain Langford. Lowthorp's Abr. Phil. Trans. Vol. 2, Page 105.)

[68]The Depression of a *Torrent* of Air in the Form of an hyperbolic Solid, *contracting* as it *descends* to the Earth, in Proportion as its *Density* encreases; may furnish a Hint towards the Solution of a Difficulty how to account for the Augmentation of vesiculous Vapours into large solid Drops, frequent during *Summer-Showers*.

[69]Mons. Saussure's valuable "Essais sur L'Hygrometrie," throw new Light on the Doctrine of Rarefaction and Condensation not unfavourable to the Hypothesis here advanced. Page 260.

[70]Ice, when exposed to marine acid Air, is dissolved by it, as fast as if it touched a red hot Iron. See Cavallo's Treatise on Air, Page 727. Also Priestley's Experiments and Observations, Vol. 1, Page 148.

[71]"The WATER remains TRANSPARENT or colourless, tho' saturated with marine acid Air, and by a very gentle Degree of Heat, the Gass may be again expelled from it, as it is expelled from Spirit of Salt."

This Observation is applicable to the Transparency of Vapours, in the Air, tho' mixed with the marine Acid exhaled from the Sea: for when the acid or Sea Air is mixed with Alkaline or Land Air,

they instantly *combine*; lose their Elasticity, and form a *white* visible Substance or *Cloud*. Cavallo, Page 728. Priestley's Exp. and Obs. Vol. 2, Page 293.

[72]On the Descent of Air in *Thunder-Gusts*, see "Chalmer's Account of the Weather in South-Carolina, Vol. 1, Page 1, to 39."

[73]"Historia Ventorum, Pag. 54, Art. 34."

[74]Book V. Chapter 2d.

[75]Vol. 1. Page 184.

[76]Page 195.

[77]History of the Canary Isles, Page 252.

[78]As the superior Clouds, during the Balloon Excursion, did not much exceed the Height of 1000 Yards; supposing then the Clouds at an equal Height above the Sea, near Teneriffe; one ought to conclude, either, that the Peak was not so high as Glas represents it; or, that the Level of the Clouds was less than half the Height of the Mountain.

[79]See "Royal Astronomer, by R. Heath, Page 321, on *Trade Winds and Monsoons*."

[80]One Pound of Nitre only, producing by mere Heat, 6 cubic Feet of Air. "Cavallo, Page 332, and 811, Experiments on Gun-Powder."

[81] "See Recherches sur les Modifications de l'Atmosphere. No. 715." Ph. Trans. Part 2, for 1777. Col. Roy's Experiments, Sect. 2d, Page 689, 744, 753, 764.

[82]The different Phenomena of the *Aurora Borealis* may be owing to the Ascent and Motion of the Apogay, in the *middle* Region, over the Stratum of Eknèfiar or *Ground-Winds*.

The Effects of *Tides in the Air yet to be* mentioned, must not, however, be wholly excluded.

The Aurora Borealis is seen in *Spring, Autumn, and Winter*: sometimes *culminating*, sometimes moving in *Streams and Waves* in the *superior* Regions of the Atmosphere: when *culminating*; as if rising out of Clouds in the North.

This Appearance may be owing to warm moist Air perpetually generating between the Tropics, and rolling over the cold *dry* Stratum of Eknèfiar Winds, which cut off its Communication with the Earth: till accumulating over the Poles, it enlightens the Atmosphere, converting a six *Month's Night* into Day; and returns to the Surface silently: or in Lightning, whenever it is communicated to the Earth, throu' *Vapour descending* by its own specific Gravity; or along with *depressing* Torrents of Air, known to be accompanied by frequent FLASHES.

When the Vapour is *condensed* in its Descent, by passing throu' a Stratum of the Eknèfiar Winds; it becomes *overcharged* with the electric Matter, *surrounding* and *adhering* to it; and deposits the Overplus in Lightning, on its Approach to *other* Clouds, or to the *Earth*.

It is visible in the Form of a Vapour, when the Vapour to which it adheres, becomes overcharged with electric Matter, by Descent into a *cool* Eknèfiar Stratum below: there forming a luminous and transparent Atmosphere: the Particles of Light and Vapour being repelled to great Distances from each other at so *rare* a Height.

It culminates above the Vapour, because less heavy than the circumambient Air: and may be subject to the Attraction of other Planets.

The Aurora Borealis is also seen to issue in Streams and Waves of Light, with inexpressible Velocity, on its Return to the South, in a lower Stratum, as it *passes throu'* Interstices, between the Vesicles of warm Vapour, raised and dispersed by the turbulent Apogay Winds, in the middle Region.

During Summer, the middle Region becomes blended with the lower, throu' Defect of Cold: and the electric Matter is supposed to be communicated to the Earth, silently, and continually; but by Lightning, when a lower and colder Atmosphere condenses and overcharges the Vapour, and cuts off the Communication.

It cannot be seen but in escaping from Vesicle to Vesicle: nor, during Summer, after Sunset, on Account of the Twilight.

[83] Air is not unfit for Respiration, by having lost its *vital* Principle, but because it has imbibed *Floguiston*, which cannot *easily* be separated from it, but by Agitation in Water. Cavallo, on Air, Pages 479, 670.

[84] For if Moisture be one Cause, which keeps the Particles of Air at greater Distances from each other; this Cause decreases at *great* Altitudes.

If also the *Elasticity* decreases in Proportion, not only to the Height, but the Driness; its Particles must, on both Accounts, approach each other, at great Altitudes: tho', from the Altitude only; they woud separate according to the Rule, viz. that the Rarity of the Air is proportionable to the Relaxation of the Force compressing it.

So that at the Height of 8 or 10 Miles, a Quantity of Air taken from the Surface of the Earth, woud occupy 6 Times its former Space: supposing the Air both below and above to be of the *same Kind*, as well as of the *same mean* Temperature of 55, on the Thermometer. See "Martin's Philosophical Grammar, Page 178."

[85] Chalmer describing a Whirlwind, which is a *Storm* of COLLECTION and *Ascent* of HOT Air, &c. by Rarefaction, says, "as the Wind ceased, presently after the Whirlwind passed, the BRANCHES and Leaves of various Sorts of Trees, which had been carried into the Air, continued to FALL for HALF AN HOUR; and, in their Descent, appeared like Flocks of Birds of different Sizes."

This Circumstance proves that Columns of HOT Air must have been raised in a Body, in Succession, to so considerable a Height, that *Branches* of Trees carried up by them, took *half an Hour* in falling.

[86] It may be from this Principle, that in the East, Liquids are kept *cool* by being hung in the Shade, in the *open* Air, suspended in *wet Cloths*: there being a continual Breeze and Succession of COOL DRY *Spunges* (as it were) of Air, in Contact with the *wetted* Cloths, whose Moisture will thus be more quickly evaporated.

[87] *Historia Ventorum*, Pag. 48, Art. 33.

[88] "Cum enim (Venti) Choreas ducant, Ordinem Saltationis nosse jucundum fuerit. Art. 18."

[89] On the Action of the Sun and Moon over Animal Bodies, by Dr. Mead, *Miscell. Cur.* Vol. 1. P. 372, 373.

[90] For these Observations see Gassendus's Natural Philosophy. De Chales's Navigator. And Astro-Meteoro-Logica, per J. Goad.

[91] See Maclaurin's Newton, Page 376.

[92] Air at a Medium is 800 Times *rarer* than Water: so that if 800 Times the Quantity of Air *naturally* contained in a Vessel whose Dimensions are those of a cubic Foot, were pressed into it by a Syringe or *Condenser*, the Air woud differ nothing from Water in Density.

[93] See Wilson on Climate, Chap. 15. Pages 46, 54.

[94] 55.

[95] By reducing 10 Feet 6 Inches, and 6 Feet 7 Inches, into Inches, and dividing by common Divisors, as 3 and 2; it is found that 10 Feet 6 Inches, will be to 6 Feet 7 Inches, as 3 to 2 nearly: that is, as 15 Miles to 10 Miles.

[96] White's Ephèmeris, Page 38, for the Speculum Phenomenorum, or Mirror of the Heavens.

[97] See the Book which gives an Account of Walker's Eidouranion.

The *intelligent* Reader will easily distinguish the Effects, attributed to the Planets, viz. their mutual Attractions, owing to natural Causes only;—from the futile Ravings of judicial Astrology.

[98] See London Chronicle, 26th July, 1785.

[99]To find the Direction of an upper Current, without the Inconvenience of rising above the Level which the Aironaut has fixed on.

This the Abbé Bertholon has hinted at, by Means of a smaller Balloon.

The Dimensions of which, must however be so large; that, allowing for the Evaporation of Gass, it will *just* rise with the Weight of a Quantity of Cord, a Mile and half, for Instance, in Length: and have sufficient Room left within, to admit of the Expansion of Gass without Rupture.

The Pioneer-Balloon may be taken up, *empty*, and filled with Gass necessarily escaping from THE MOUTH of the *great Balloon*, when stationary: and may be sent up with a Cord, fastened to the Center above the Car of the *great Balloon*, to reconnoitre the *superior* Currents: or it may be only filled *in Part*; and made to *descend*, and *discover* the *lower* Currents.

See “Des Avantages de Ballons, &c. Page 72.”

[100] As the *Heights* of the Atmosphere encrease in an *arithmetical* Progression; the Densities are said to encrease in a *geometrical* Progression: which is a mathematical and pedantic Mode of Expression.

For *arithmetical* Progression *here* means no more than the Height of 1, 2, 3, 4, 5, 6, &c. &c. Yards, Fathoms, Roods, or any other equal Interval.

If then at the Height of one Yard, the Balloon has acquired (suppose) the Levity of 1 Pound; then, if this Levity encreases in geometrical Progression; (as twice 1 is 2,) it will, at the Height of 2 Yards, have encreased to 2 Pounds: and, as twice 2 is 4;) it will, at the Height of 3 Yards, have encreased to 4 Pounds: and, as (as twice 4 is 8;) it will, at the Height of 4 Yards, have encreased to 8 Pounds: and, (as twice 8 is 16;) it will, at the Height of 5 Yards, have encreased to 16: and, (as twice 16 is 32;) the Levity will, at the Height of 6 Yards, have encreased to 32 Pounds; and so on, *doubling* the preceding Number; at the Height of each Yard, Fathom, Rood, Mile, &c. &c.

[101]Whiston’s Tacquet’s Euclid, Book XI. Definition of a *right* Cylinder, Art. 3, Page 166.

[102]Archimedes’s Theorems. Proposition 33, 34; at the End of Whiston’s Euclid, Page 42.

[103]Inferred in the Chester Chronicle, Sept. 30, 1785.

[104]The Writer not having yet been able to procure it from the London Booksellers.

[105]See Chambers’s Dictionary under the Article RESISTENCE.

[106]See his “Navires des Anciens.”

[107]See “Gordon’s Principles of Naval Architecture.”

Also the Balzaes and Guaraes, in Ullòà’s Voyage to America, Book 4, Chapter 9, Vol. 1, Page 183.

[108]Mons. Carra proposed to ascend with two Balloons. One, a seventh Part less than the other, is to be connected by a Rope, throu’ a Pulley fixed in the equatorial Hoop of the great Balloon, to a Reel in the Center of the Car: in descending, the Reel is to be unwound: the great Balloon and Car will therefore descend, while the small Balloon remains in the Air. The Scheme is certainly practicable. See the Cut in the London Magazine for June, 1784.

[109]See “Lewis’s Commerce of the Arts.”

[110]See Priestley’s numerous Experiments: and that Library of *curious Investigation*, the Philosophical Transactions.

[111]And *Magnitude* of distant Objects.

Bacon says that Objects are more *visible* in an East Wind, and Sounds more *audible* in a West Wind; being heard at a *greater* Distance. “Historia Ventorum, P. 37, Art. 31.”

[112]See Le Roi’s Uses of the airostatic Globe *at Sea*, in his “Navires des Anciens, Page 225.”

[113]The *natural Figure* of the *Diodon-Globe-Fish*, a coloured Print of which is given in “Martyn’s new and elegant Dictionary of natural History:” where it is described as follows: “The Form of the

Body is usually oblong; but when the Creature is alarmed, it possesses the Power of *inflating* its Belly to a globular Shape of great Size;”—seems to furnish a Hint for the proper Figure of a Balloon, when the Art is more improved.

The Balloon, as far as it is meant to resemble the upper Part of the Fish, is to be made stiff, with Pasteboard or *Papier-mâché* varnished; for, being strong, and in a permanent Form, it is more capable of continuing Air-tight: the lower Parts being *flaccid*, will be inflated, as the Balloon rises, and deflated during the Descent.

Rowers, and propulsive Machinery, are to be fixed within the Fish, in Place of the Fins: and Goods of GREATER Weight placed in a covered Car below: the Air-Bottle-Balloon being fixed between both.

[114] And by *Kunckel's* or *Canton's* Phosphorus, See “*Priestley's History of LIGHT*. Pages 585, 370.”

[115] This was owing to the cool Air rushing in to supply the Tendency to a Vacuum by the Expansion of hot Steam, with the extricated Gass.

The Accident proves that no Danger is to be dreaded from EXPANSION of the Gass.

[116] From *Bersham-Forge* near *Wrexham*, where there is always a sufficient Quantity.

[117] The *detached* Thermometer might be protected from the *Sun*, by being swung a few Inches *below* the Car of the Balloon by means of an *Opening* made purposely throu' the Center of the Car.

[118]

Foundation of the first Table.

(Ph. Tr. for 1777, Part 2d, Page 567.)—It was found by

Experiment that the Decimal	.000262
was the Expansion on 30 Inches of Quicksilver, with each Degree of Temperature from freezing to boiling Water: also, the Decimal	.000042
was the Expansion on 30 Inches of the Glass Tube (containing the Quicksilver), with each Degree of	_____
Temperature: therefore by Addition,	.000304
or by taking only 4 Decimals,	.0003

is the Expansion on 30 Inches of Quicksilver, and the Glass Tube containing it, with each Degree of Temperature.

Construction of the first Table.

Thus any vertical Number, shewing the Expansion, may be readily *formed*, by *doubling*, *first*, the Number immediately under each Inch for the Expansion below it: and *afterwards*, by adding the Number immediately under each Inch, to the Expansion last found.

Note: The vertical Columns, below each Inch of Quicksilver shew the Expansion on that Inch, with corresponding Degrees of Temperature indicated by the Thermometer in the Column to the left Hand. Example: to find the Expansion on 30 Inches of Quicksilver with 1 Degree of Temperature: the Answer in the Table is .003: i. e. such Expansion raises the Quicksilver the 3000th Part of an Inch.

[119] There is seldom Occasion to take more than the four first Decimals out of the Table, the Remainder being of *little value*.

[120]

The Foundation of the second Table.

This Table is calculated from Briggs's Logarithms: each Number, in the second Column, being nothing more than the Logarithm—corresponding to the Point, (in the *first* Column,) at which the Quicksilver stands in the barometric Tube,—subtracted from the Logarithm of 32 Inches multiplied by 6.

Construction of the second Table.

This Table consists of three *vertical* Columns only: tho' *here tripled*, for the greater Convenience of Inspection.

The first or left Hand Column shews, in Inches and Tenths (from ten Inches) the Gradations of the Quicksilver in the barometric Tube, beginning as low as one Inch above the Surface in the Cistern, and proceeding throu' all the intermediate Points, to the unusual Extent of 32 Inches:^[121] supposing likewise that the Tube is elevated in the Atmosphere, so that the contained Quicksilver, when exposed to the Temperature of 31°.24 of Farenheit, rests at each Point in the Table.

The second vertical Column gives the different Heights in Feet and Tenths, to which the barometric Tube must be raised above its Level at 32 Inches, in order that the contained Quicksilver, if exposed to the Temperature of 31°.24 of Farenheit, may stand at each Point indicated in the first Column.

The third vertical Column, gives, likewise in Feet and Tenths, the DIFFERENCE between each two adjoining Heights in the second Column, corresponding to a single Tenth (of Quicksilver): which single Tenth is the Difference between each two adjoining Tenths of an Inch in the first Column.

For Example: Suppose the Quicksilver in the barometric Tube, in the first Column, stands at

Inches	16.1	answering to	19570.4	} Height in Feet in the Atmosphere.
And again at	16.2	answering to	19398.4	

Difference of .1 in Feet: remaining = 172.0

which sixteen Inches two Tenths, is a single Tenth more than sixteen Inches one Tenth, and will therefore answer to a *less* Height in the Atmosphere by that single Tenth; considering that the lower the Quicksilver falls in the Tube, the higher must the Barometer itself be raised in the Atmosphere, in order that the Quicksilver may rest at the lower Points of the Tube. If therefore a *less* Height in the Atmosphere be required which shall answer to one Tenth more than 16 Inches two Tenths; subtract the Height answering to 16.2 from the Height answering to 16.1, i. e. subtract the *less* Height from the *greater*, and the Remainder gives that *less* Height in the third Column, answering to the Height of one Tenth more than 16 Inches 2 Tenths, of the Barometer.

^[121] The Barometer, (to which the Scale of Heights is applied, in the 2d Column of the 2d Table) is supposed to be sunk within the Surface of the Earth, till the Quicksilver rests at 32 Inches, as appears from the last Article in the table, viz. 32 Inches, 0.00 Feet. 32 Inches is therefore the Foundation of the Table, and corresponds, according to Shuckburgh, to 1647 Feet, under the Surface of the Sea, at low Water.

This Depth then being the imaginary Level pointed out by the Quicksilver, at the unusual Extent of 32 Inches; each interior Inch and Tenth of Quicksilver will correspond to a superior Elevation of the Instrument, in Feet and Tenths above that Level, and will include the Mensuration of the deepest Mines.

For the mean Pressure of the Barometer, at low Water, from 132 Observations in Italy and England, is 30.04 Inches: the Temperature of the Barometer being at 55°, i. e. Temperate, and that of the Air at 62°.

^[122]

Foundation of the Table for Tenths.

The Height, in Feet, corresponding to the Expansion on the Tenth of an inch of Quicksilver with the Temperature of 31°.24 (as in the 3d Column of the 2d Table) are reduced by this Table into a ten Times less Number of Feet; and the Tenth of an Inch (of Quicksilver) is also again divided into *ten* more Parts: in order to shew, in a ten Times less Number of *such* Feet, the Expansion corresponding to any of those Parts into which the *Tenth* of an Inch (of Quicksilver) has been divided.

Construction and Use of the Table for Tenths.

1. The Figures in the left vertical Column shew the Height in Feet, (from 81 to 130) corresponding to a single Tenth of an Inch of Quicksilver, viz. to the higher of two adjoining Tenths, as in the 3d Column of the 2d Table.

2. The Figures, along the upper horizontal Line, shew the Number of Parts into which the Tenth of an Inch has been divided.

3. The Figures, at the Point of Meeting, express, in a ten Times less Number, of *the Feet* in the left vertical Column, the Expansion corresponding to any of those Parts, into which the Tenth of an Inch (of Quicksilver) has been divided.

Thus: 90 is a *Number of Feet* called 9 Tenths of 100: but the *Tenths* are *Feet*, and not Tenths of a Foot.

[123] The Standard Temperature was $31^{\circ}.24$, which not being exactly 1 Quarter, another Decimal is added, (for Ease in Computation,) by which 31.24 becomes 31.25 , i. e. by dividing one Degree of Heat into 100 Parts, and taking 25 of those Parts, or dividing the 100 by 25, the Answer is 4, i. e. $\frac{1}{4}$ of the whole 100: or $(31)\frac{1}{4}$.

[124]

The Foundation of the fourth Table.

(Ph. Tr. for 1777, Part 2d, Pages 564, and 566,)—From the *Mean* of a Series of Experiments with a Manòmeter, or Instrument to measure the *Rarity* and Density of the Atmosphere, depending on the Action of *Heat* and Cold, it was found, that when the *Portion of a Tube* containing Air (at the Temperature of freezing by Farenheit, and Pressure of $30\frac{1}{2}$ Inches^[125] by a common Barometer) was divided into 1000 Parts; the Volume of Air within it, encreased *nearly* in a certain Proportion, as each Degree of Temperature encreased; viz. at a Mean, 2.43, or simply (by rejecting the 2d Decimal as too minute) 2.4: that is, a 1000 Parts of Air became by Expansion with one Degree of the Thermometer, equal to 1002.43: i. e. the Portion of Air occupying 1000 Parts, did, with the Addition of one Degree of Heat, occupy 1002.43 Parts: that is (by rejecting the 2d Decimal 3 as too minute) occupied two Parts and 4 Tenths more than the thousand.

Construction of the fourth Table.

Supposing therefore that the Portion of the Tube containing Air, was one Foot in Length of Height, divided also into a thousand Parts; one Degree of Heat would encrease or expand it two Parts and four Tenths more than the thousand Parts into which the Foot was divided.

CAUTION.

The fourth Table properly consists of only nine horizontal Columns of thousands, in Breadth; which Columns are extended in Length to one hundred Lines, corresponding to 100 Degrees of Heat.

The Table is here divided, in order that it may conform to the Size of the Pages: by which Means the Formation of each vertical Number by the following Rule, (which renders the Table self-evident) might without this Caution, have been attended with some Difficulty.

The vertical Columns *below* the Figures expressing each thousand, shew the Expansion of Air *on* each respective thousand, *with* the corresponding Degrees of Temperature indicated by the Thermometer in the vertical Column to the left Hand.

Example the first: to find the Expansion of Air *on* one thousand Feet, *with* one Degree of Temperature; the Answer in the Table is 2.4, or 2.43: i. e. 2 Feet and 4 Tenths of a Foot, rejecting the 2d Decimal as too minute.

Example the second: to find the Expansion *on* 8 thousand Feet, *with* 99 Degrees of Heat: the Answer is 1924.56: and so of the Rest.

Thus *any* of the *vertical Numbers* shewing the Expansion, may be readily *formed*, by *doubling*, *first*, the Number immediately under each thousand in the horizontal Line, for the nine first thousands, (of which the Breadth of the Table properly consists, exclusive of the thermometric Column) for the Expansion below it: and, *afterwards*, for each Expansion immediately below the former, by adding, to the Expansion *last* found, the Number immediately under its respective thousand.

First Example: to find the vertical Number for the Expansion under the first thousand, viz. 1000, with 2 Degrees of Heat: the Number under 1000 is 2.43: double this: and the Answer is 4.86.

Second Example: suppose the Expansion *last* found be that on one thousand Feet with 24 Degrees of Heat; viz. 58.32: and the Expansion on the same thousand, with one Degree of Heat more, viz. on 25 Degrees, be required; add the Expansion

on one thousand Feet, with 24 Degrees, viz.	58.32
to the Expansion on the same 1000, with 1 Degree, viz.	2.43

and the Answer is, by Addition,	60.75
---------------------------------	-------

Third Example: supposing the Expansion *last* found to be the Expansion on 9000 Feet with 99 Degrees of Heat, which in the Table is 2165.1.

It is required to find the Expansion on the same 9000 Feet, with 100 Degrees of Heat; add to the Expansion last found,

viz. 2165.13, the Expansion on the same 9000 Feet,
viz. 21.87 with one Degree of Heat, and

2187.00 is the Answer by Addition.

Any vertical Number shewing the Expansion may likewise be found, first, by multiplying the first Figure, or Number, of the given thousand Feet (in the horizontal Line,) into the Answer or Expansion on the first thousand Feet, with one Degree of Heat: for Example;

To find the Expansion on 9000 Feet with one Degree of Heat.

The Expansion on 1000 Feet, with 1 Degree of Heat (from whence, all the other Expansions are derived) being 2.43; multiply that Number by 9, the first Figure of the given thousand Feet, and the Answer or Expansion with 1 Degree of Heat, is 21.87: hence all the Answers or Expansions, immediately under the horizontal Line of thousands, are formed.

Then 2dly, any other vertical Number or Expansion may be formed by multiplying the Expansion immediately under the given thousand Feet in the horizontal Line, into the given Number of Degrees: for Example;

To find the Expansion on 9000 Feet, with 50 Degrees.

The Expansion with one Degree on 9000, is 21.87: therefore the Expansion with 50°, is 50 Times more, viz. 1093.50, and so of the Rest.

These different Methods serve to prove the Answers, and to elucidate the Table.

[125] These Experiments were made with the Manòmeter when the Atmosphere was half an Inch heavier than in the Experiments to prove the Expansion of Quicksilver, the Barometer then standing at 30 Inches only.

[126] There is seldom Occasion to take more than the first Decimal out of the Table.

[127]

“R U L E.

“Precept the 1st. With the Difference of the two Thermometers that give the Heat of the Barometer (and which for Distinction sake, are called the attached Thermometers) enter Table I, with the Degrees of Heat in the Column on the left Hand, and with the Height of the Barometer in Inches, in the horizontal Line at the Top; in the common Point of Meeting of the two Lines will be found the Correction for the Expansion of the Quicksilver by Heat, expressed in decimal Parts of an English Inch; which added to the coldest Barometer, or subtracted from the hottest, will give the Height of the two Barometers, such as would have obtained, had both Instruments been exposed to the same Temperature.

“Precept the 2d. With these corrected Heights of the Barometers enter Table II, and take out respectively the Numbers corresponding to the nearest Tenth of an Inch; and if the Barometers,

corrected as in the first Precept, are found to stand at an even Tenth, without any further Fraction, the Difference of these two tabular Numbers (found by subtracting the less from the greater) will give the approximate Height in English Feet. But if, as will commonly happen, the correct Height of the Barometers should not be at an even Tenth, write out the Difference for one entire Tenth, found in the Column adjoining, intitled Differences; and with this Number enter Table III, of proportional Parts in the first vertical Column to the left Hand, or in the 11th Column; and, with the next Decimal, following the Tenths of an Inch in the Height of the Barometer (viz. the hundredths) enter the horizontal Line at the Top, the Point of meeting will give a certain Number of Feet, which write down by itself; do the same by the next decimal Figure in the Height of the Barometer (viz. the thousandths of an Inch,) with this Difference, striking off the last Cypher to the right Hand for a Fraction; add together the two Numbers thus found in the Table of proportional Parts, and their Sum subduct from the tabular Numbers, just found in Table II; the Differences of the tabular Numbers, so diminished, will give the approximate Height in English Feet.

“Precept the 3d. Add together the Degrees of the two detached or Air Thermometers, and divide their Sum by 2, the Quotient will be an intermediate Heat, and must be taken for the mean Temperature of the vertical Column of Air intercepted between the two Places of Observation: if this Temperature should be $31^{\circ}1/4$ on the Thermometer, then will the approximate Height before found be the true Height; but if not, take its Difference from $31^{\circ}1/4$, and with this Difference seek the Correction in Table IV, for the Expansion of Air, with the Number of Degrees in the vertical Column on the left Hand, and the approximate Height to the nearest thousand Feet in the horizontal Line at the Top; for the hundred Feet strike off one Cypher to the right Hand; for the Tens strike off two; for the Units three: the Sum of these several Numbers added to the approximate Height, if the Temperature be greater than $31^{\circ}1/4$, subtracted if less, will give the correct Height in English Feet. An Example or two will make this quite plain.”

[128] There is no Occasion to take more than four Decimals out of the Table.

[129] See Section 368, [Note \[120\]](#).

[130] Section 368, [Note \[121\]](#) on Note [120].

[131] Taking one Decimal only out of the Table.

[132] THE QUESTION: In the upper Gallery of the Dome of St. Peter’s Church at Rome, and 50 Feet below the Top of the Cross, the Barometer, from a Mean of several Observations, stood at Inches 29.5218 Tenths: the attached Thermometer being at Degrees 56.6 Tenths; and the Air-Thermometer at 57 Degrees: at the same Time that another, placed on the Banks of the River Tyber, one Foot above the Surface of the Water, stood at 30.0168, the attached Thermometer at $60^{\circ}.6$, and the Air-Thermometer at $60^{\circ}.2$: what, was the Height of the Building above the Level of the River?

[133] See [Section 375](#). 2dly. If the Moiety, *Half-Heat*, or mean Temperature of the Air, is equal to the Standard-Temperature, to which the two Barometers are brought, by the 2d Table; the fourth Table, for *Expansion of Air*, is needless: the Height already found, in the 2d Table, being the true Height of the upper Station.

3dly. If the Moiety, *Half-Heat*, or mean Temperature of the Air, is less than the Standard-Temperature of $31^{\circ}.24$; subtract the mean Temperature from 31.24 ; and with the Remainder find the Expansion, as usual, by the 4th Table: subtract the Sum, (which is a corresponding Height in Feet and Tenths) from the Height in Feet and Tenths of the upper Barometer, at the Standard-Temperature, in the 2d Table: and the Remainder will be the true Height of the Mountain or upper Station. Section 384, Note a.

[134] THE QUESTION: Near the Convent of St. Clare, in a Street called *La Strada dei Specchi*, at Rome, the lower Barometer stood at 30.082, its attached Thermometer 71 Degrees, and detached ditto at 68 Degrees: on the Tarpeian Rock, or West-End of the famous Hill called The Capitol, the upper Barometer was at 29.985, its attached Thermometer $70^{\circ}.5$, and detached ditto 76° : what was the Height of the Eminence?

[135] Sadler’s *Practical Arithmetic*, Page 293.

[136] The Writer has not hitherto been so fortunate as to meet with the original Memoir, containing the Particulars of this curious Experiment by Mons. Lavoisier.

[137] Dr. Priestley's Experiments and Observations relating to Air and Water. Ph. Tr. for 1785, Vol. 75, Part 1, Page 279.

[138] The Diameter may be enlarged.

[139] By Means of the Cradle, *both* are more easily moved: the Muffle is prevented from adhering to the Tube; and Steam is admitted to the Borings.

[140] Copper sustaining a *red* Heat, better than Iron; the latter of which, *calcines* with Steam, or, in cooling.

Transcriber's Notes:

- New original cover art included with this eBook is granted to the public domain.
- Table of contents added.
- Obvious typographical errors have been silently corrected.
- Archaic language and spelling is left as-is, except "AERIAL" was printed with dots above the 'A' and 'E', this was assumed to be a typesetter's limitation and replaced with "AËRIAL", to match the lower case usage.
- Errata have been applied, as much as I understood them.
- Numbers for sections 259–261 are repeated.

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